K-UNSTABLE VARIETIES WITH LARGE ALPHA INVARIANTS

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Abstract. We answer a question of Liu and Zhuang on the sharpness of Tian's criterion for K-stability: in any dimension at least two, there exists a K-unstable $\mathbb{Q}$-Fano variety whose alpha invariant is larger than the threshold of the criterion in one dimension lower.

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1. Introduction

We work over the field of complex numbers $\mathbb{C}$.

Tian introduced the alpha invariant of a Fano manifold and proved that a Fano manifold X of dimension d with $\alpha(X) > \frac{d}{d+1}$ admits a Kähler–Einstein metric [Tia87]. Demailly and Kollár extended Tian's criterion to Fano orbifolds [DK01]. In algebraic terms, the alpha invariant of a $\mathbb{Q}$-Fano variety X is

$$\alpha(X) = \inf\{\mathrm{lct}(X; D) \mid D \geq 0 \text{ is a } \mathbb{Q}\text{-divisor and } D \sim_{\mathbb{Q}} -K_X\}$$

(see Definition 2.10 for $\mathbb{Q}$-Fano varieties and alpha invariants). Let X be a $\mathbb{Q}$-Fano variety of dimension d. Odaka and Sano proved that $\alpha(X) > \frac{d}{d+1}$ implies that X is K-stable, and $\alpha(X) \geq \frac{d}{d+1}$ implies that X is K-semistable [OS12, Theorem 1.4]. K-stability was introduced by Tian [Tia97] and Donaldson [Don02], and we refer the reader to [Fuj19a, Section 2.1] for the definitions of K-stability and K-semistability. Fujita proved that a Fano manifold X of dimension $d \geq 2$ with $\alpha(X) = \frac{d}{d+1}$ is K-stable [Fuj19b, Theorem 1.2]. See also [LZ22, Theorem 1.2]. Conversely, Fujita and Odaka proved that $\alpha(X) \geq \frac{1}{d+1}$ if X is K-semistable [FO18, Theorem 3.5].

K-semistability can be tested on prime divisors over X. By the valuative criterion of Fujita and Li [Fuj19a, Corollary 1.5(2)], [Li17, Theorem 3.7] (see Theorem 2.12), X is K-semistable if and only if $\beta_X(E) \geq 0$ for any prime divisor E over X, where $\beta_X(E)$ is Fujita's beta invariant (see Definition 2.11). Blum and Jonsson described the alpha invariant in terms of prime divisors over X: $\alpha(X)$ is the infimum, over all prime divisors E over X, of the ratio of the log discrepancy of E with respect to X to the pseudo-effective threshold of $-K_X$ along E [BJ20, Theorem C].

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We say that a $\mathbb{Q}$-Fano variety is *K-unstable* if it is not K-semistable. Liu and Zhuang asked the following question.

Question 1.1 ([LZ22, Question 1.6])**.** Let $d \geq 2$ be an integer.

(1) Does there exist a K-semistable $\mathbb{Q}$-Fano variety X of dimension d such that
$$\frac{1}{d+1} < \alpha(X) < \frac{1}{d}?$$
(2) Does there exist a K-unstable $\mathbb{Q}$-Fano variety X of dimension d such that
$$\frac{d-1}{d} < \alpha(X) < \frac{d}{d+1}?$$

See also [LZ22, Question 5.7].

Jiang conjectured that a K-semistable Fano manifold X of dimension d satisfies $\alpha(X) < \frac{1}{d}$ if and only if $X \cong \mathbb{P}^d$ [Jia17, Conjecture 1.6]. Liu and Zhuang asked Question 1.1 in connection with Tian's criterion and Jiang's conjecture.

We now explain the examples of Liu and Zhuang, which show how far the bound $\frac{d}{d+1}$ in the criterion of Odaka and Sano is from being optimal [LZ22, Theorem 1.3].

(1) For $d = 4$ or $d \geq 7$, [LZ22, Proposition 5.2] constructs a hypersurface X of degree $d+1$ in $\mathbb{P}^{d+1}$ with one singular point such that $\alpha(X) = \frac{d}{d+1}$ and X is K-semistable but not K-stable. In particular, the smoothness assumption in [Fuj19b, Theorem 1.2] cannot be removed. See [Fuj17, Section 9] for an example with a boundary.
(2) For integers $e \geq 2$ and $d \geq 10 + e^2$, [LZ22, Proposition 5.5] constructs a complete intersection X of degrees e and $d+2-e$ in $\mathbb{P}^{d+2}$ such that X is not K-semistable and
$$\alpha(X) = 1 - \frac{1}{d+2-e}.$$
For $e = 2$, the assumption $d \geq 10 + e^2$ means $d \geq 14$, and then $\alpha(X) = \frac{d-1}{d}$ [LZ22, Theorem 1.3(3)].

Neither family enters the open interval $(\frac{d-1}{d}, \frac{d}{d+1})$ of Question 1.1(2). Indeed, the hypersurfaces of [LZ22, Proposition 5.2] satisfy $\alpha(X) = \frac{d}{d+1}$, the upper bound of the interval. For the complete intersections of [LZ22, Proposition 5.5], we have
$$\frac{d-1}{d} < 1 - \frac{1}{d+2-e} < \frac{d}{d+1} \quad \text{if and only if} \quad 1 < e < 2,$$
which is impossible for an integer e.

Question 1.1(1) has been answered affirmatively for any $d \geq 2$. Liu and Zhu constructed, for any $d \geq 2$, a K-semistable toric $\mathbb{Q}$-Fano variety X of dimension d with $\alpha(X) = \frac{2}{2d+1}$ [LZ26, Theorem 1.4]. Wu proved that any rational number in $[\frac{1}{d+1}, \frac{1}{2}]$ is the alpha invariant of a K-semistable toric $\mathbb{Q}$-Fano variety of dimension d [Wu26, Theorem 1.2]. Toric varieties cannot answer Question 1.1(2) [Wu26, Remark 2.4]. Indeed, by [Wu26, Theorem 2.3], any toric $\mathbb{Q}$-Fano variety X satisfies $\alpha(X) \leq \frac{1}{2}$, while $\frac{d-1}{d} \geq \frac{1}{2}$ for $d \geq 2$. More generally, a $\mathbb{Q}$-Fano variety X with $\alpha(X) > \frac{1}{2}$ has finite automorphism group $\mathrm{Aut}(X)$ [LZ22, Corollary 3.6]. Thus any $\mathbb{Q}$-Fano variety X of dimension $d \geq 2$ with $\alpha(X) > \frac{d-1}{d}$ has finite automorphism group.

We recall what is known about Question 1.1(2) for several well-studied classes of $\mathbb{Q}$-Fano varieties. The alpha invariants of smooth del Pezzo surfaces take only the values 1, $\frac{5}{6}$, $\frac{3}{4}$, $\frac{2}{3}$, $\frac{1}{2}$, and $\frac{1}{3}$ [Che08, Theorem 1.7]. No alpha invariant of a smooth del Pezzo surface belongs to $(\frac{1}{2}, \frac{2}{3})$, so for $d = 2$ an example for Question 1.1(2) must be singular. For del Pezzo surfaces with Du Val singularities, the alpha invariants were computed in [PW10, CK14]. In dimension three, quasismooth Fano threefold hypersurfaces of index one are K-stable [CO24, Theorem 1.5], so none of them is K-unstable. The threefold X of Theorem 6.4 is not covered by [CO24, Theorem 1.5]:

$\omega_X \cong \mathcal{O}_X(-2)$ by Proposition 6.5, and X is not quasismooth at the point $[0:0:0:0:1]$, where all partial derivatives of the defining equation of X vanish.

We answer Question 1.1(2) affirmatively for any $d \geq 2$.

Theorem 1.2. *Let $d \geq 2$ be an integer. Then there exists a $\mathbb{Q}$-Fano variety X of dimension d such that*

$$\frac{d-1}{d} < \alpha(X) < \frac{d}{d+1}$$

and $\beta_X(E) < 0$ for some prime divisor E over X. In particular, X is not K-semistable.

For any $d \geq 2$, Table 1 lists the values of $\alpha(X)$ and $\beta_X(E)/\operatorname{vol}(-K_X)$, where X is our example of dimension d and E is the prime divisor over X with $\beta_X(E) < 0$ given by the result in the last column. For $d = 2$, the example is a complete intersection of two hypersurfaces in the weighted projective space $\mathbb{P}(3,4,5,6,7)$ (see Definition 2.15). For $d = 3$, the example is a hypersurface in $\mathbb{P}(3,4,4,3,3)$. For $d \geq 4$, the examples are hypersurfaces in weighted projective spaces whose equations are linear in one of the coordinates (see Theorem 4.3).

TABLE 1. Examples

d	$\alpha(X)$	$\beta_X(E)/\operatorname{vol}(-K_X)$	Reference
2	$\frac{7}{12}$	$-\frac{1}{14}$	Theorem 7.3
3	$\frac{7}{10}$	$-\frac{1}{6}$	Theorem 6.4
4	$\frac{15}{19}$	$-\frac{1}{20}$	Proposition 5.1
5	$\frac{1734}{2165}$	$-\frac{421}{2586}$	Proposition 5.2
6	$\frac{875}{1028}$	$-\frac{43}{1071}$	Proposition 5.2
7	$\frac{9240}{10609}$	$-\frac{343}{10952}$	Proposition 5.3
8	$\frac{2310}{2633}$	$-\frac{274}{2907}$	Proposition 5.6
9	$\frac{252}{281}$	$-\frac{9}{290}$	Proposition 5.6
10	$\frac{360}{397}$	$-\frac{10}{407}$	Proposition 5.6
$d \geq 11$	$\frac{d(d-1)(d-2)}{(d+1)(d-1)(d-2)+2}$	$-\frac{2d}{(d+1)(d^2-3d+4)}$	Proposition 5.6

Sketch of the proof. By the valuative criterion (Theorem 2.12), Theorem 1.2 follows from Theorems 7.3 and 6.4 for $d = 2, 3$ and from Propositions 5.1, 5.2, 5.3, and 5.6 for $d \geq 4$, which check the conditions of Theorem 4.3 directly or through Corollaries 4.19 and 4.20.

In Theorem 4.3, $X = (zh + y^e + G = 0) \subset \mathbb{P}(a_1, \ldots, a_d, 1, c)$ is the hypersurface of Set-up 4.1, where h and G are weighted homogeneous polynomials in $x_1, \ldots, x_d$ of degrees $k = \sum_i a_i$ and $e = k + c$ respectively, $S = (h = 0)$ is a quasismooth hypersurface in $W = \mathbb{P}(a_1, \ldots, a_d)$, and $p = [0 : \cdots : 0 : 1] \in X$. Theorem 4.3 follows from three steps, where E is the exceptional divisor of the plt blow-up at p (Construction-Proposition 4.9) and $D \geq 0$ is any $\mathbb{Q}$-divisor with $D \sim_{\mathbb{Q}} -K_X$.

(1) $(X, \frac{A_X(E)}{T_X(E)}D)$ is lc near p (Proposition 4.13), by Proposition 4.12 and Lemmas 3.1 and 2.13.
(2) (X, D) is lc away from p, by Lemmas 4.17, 4.15, and 4.16.
(3) A monomial basis adapted to E (Definition-Lemma 4.10) computes $\operatorname{vol}(-K_X - tE)$, which implies $\beta_X(E) < 0$ (Proposition 4.11).

Theorems 6.4 and 7.3 are proved by analogous steps. In each example, the prime divisor E which computes $\alpha(X) = \frac{A_X(E)}{T_X(E)}$ satisfies $\beta_X(E) < 0$. Step (2) is the main new step: the projection from p is crepant for a pair (Lemma 4.17), and on the cone over S, the image of D is a convex combination of a $\mathbb{Q}$-divisor controlled by the condition on S in Theorem 4.3 and a $\mathbb{Q}$-divisor controlled by the multiplicity bound of Theorem 2.28. Steps (1) and (3) reduce to Theorem 2.28 and to counting monomials.

For $d \geq 3$,

$$\mathrm{vol}(-K_X - tE) = \mathrm{vol}(-K_X)\Big(1 - \Big(\frac{t}{T_X(E)}\Big)^d\Big) \quad \text{for } 0 \leq t \leq T_X(E),$$

and $T_X(E) = A_X(E) + 1$ (Propositions 4.11 and 6.7, Remark 7.4). Thus

$$\alpha(X) = \frac{A_X(E)}{A_X(E)+1} \quad \text{and} \quad \frac{\beta_X(E)}{\mathrm{vol}(-K_X)} = \frac{A_X(E) - d}{d+1},$$

so the conclusion of Theorem 1.2 for X and E is equivalent to $d-1 < A_X(E) < d$. For $d = 2$, the example of Theorem 7.3 is a member X_λ of the family of Set-up 7.1, and the prime divisor F over X_λ of Construction-Proposition 7.6, extracted by a plt blow-up at the point $p = [1:0:0:0:0]$ of Set-up 7.1, plays the role of E. The hypothesis of Lemma 3.1 at p and the estimates away from p hold with room to spare, and the difficulty is the volume of $-K_{X_\lambda} - tF$: a curve of negative self-intersection splits it into two quadratic pieces (Lemma 7.7), and $\beta_{X_\lambda}(F) < 0$ reduces to the inequality $\frac{7}{12} < \frac{5}{8}$ (Proposition 7.8).

(a) Weighted hypersurfaces. The examples of [LZ22, Propositions 5.2 and 5.5] miss the interval of Question 1.1(2) for an arithmetic reason. In [LZ22, proofs of Propositions 5.2 and 5.5], the exceptional divisor of a weighted blow-up of the singular point (in the case of Proposition 5.2, of the point $[1:0:\cdots:0]$ of the special degeneration X_0 of [LZ22, Lemma 5.1]) has an integer log discrepancy A and pseudo-effective threshold $A+1$, and $\alpha = \frac{A}{A+1}$. We make A fractional with weights.

The hypersurfaces X of Set-up 4.1 are weighted analogues of the hypersurfaces of [LZ22, Proposition 5.2]. Near p, X is the quotient by μ_c of $(h + y^e + G = 0) \subset \mathbb{A}^{d+1}$, and a weighted blow-up of $(h+y^e+G=0)$ has an exceptional divisor with log discrepancy $\frac{k}{\ell}$ and ramification index $\frac{c}{\ell}$ under the quotient, where $\ell = \gcd(k, c)$. Thus $A_X(E) = \frac{k}{c}$ (Construction-Proposition 4.9(1)), and the condition $\frac{k}{d} < c < \frac{k}{d-1}$ of Set-up 4.1(4) is $d-1 < A_X(E) < d$. If all a_i are 1, then $k = d$, the value $c = 1$ corresponds to the hypersurfaces of [LZ22, Proposition 5.2], with $A = d$, and no integer c satisfies the condition. The polynomial G cannot be omitted: for $G = 0$, the $\mathbb{Q}$-divisor $\frac{1}{c}(z = 0)$ shows $\alpha(X) < \frac{1}{d}$ (Remark 4.7). The blow-up at the singular point $[0:\cdots:0:1]$ of a weighted projective space $\mathbb{P}(1,\ldots,1,a)$ was used in [CFST26, Corollaries 7.12 and 7.13] to construct K-unstable quasismooth weighted Fano hypersurfaces [CFST26, Example 7.14].

(b) The threshold at p. Lemma 3.1, a variant of [LZ22, Lemma 5.3], needs $\alpha(E, \Delta_E) \geq 1$, a condition on all $\mathbb{Q}$-divisors $B \geq 0$ on E with $B \sim_{\mathbb{Q}} -(K_E + \Delta_E)$. In [LZ22, proof of Lemma 5.4], the analogous bound follows from properties of general complete intersections of large dimension. In our case, $(E, \Delta_E) \cong (W, (1 - \frac{1}{e})S)$ (Construction-Proposition 4.9(2)). We use the bound of Johnson and Kollár on the multiplicity of any subvariety of a weighted projective space, in the form given by Totaro (Theorem 2.28, [JK01, Proposition 11], [Tot22, Theorem 4.3]). Since $-(K_W + (1 - \frac{1}{e})S) \sim_{\mathbb{Q}} \frac{k}{e}\mathcal{O}_W(1)$ and $\frac{k}{e} < 1$, it suffices to bound the multiplicities of $\mathbb{Q}$-divisors of class at most $\mathcal{O}_W(1)$. By $a_{(1)}a_{(2)} \geq k$ (Set-up 4.1(3)), where $a_{(1)} \leq a_{(2)}$ are the smallest and the second smallest numbers among $a_1, \ldots, a_d$, counted with multiplicity, any $\mathbb{Q}$-divisor $C \geq 0$ on S of class at most $\mathcal{O}_S(1)$ has multiplicity at most 1, and inversion of adjunction along S (Theorem 2.6) implies $\alpha(W, (1 - \frac{1}{e})S) \geq 1$ (Proposition 4.12).

(c) Away from p. Away from p, X is not quasismooth over the singular points of $(G = 0)|_S$ (Subsection 4.4). The projection π from p to $\mathbb{P}' = \mathbb{P}(a_1, \ldots, a_d, 1)$ satisfies $K_{X\setminus\{p\}} = \pi^*(K_{\mathbb{P}'} + C_S)$, where $C_S \subset \mathbb{P}'$ is the cone over S, and π sends the points where X is not quasismooth to

quasismooth points of C_S (Lemma 4.17). Eliminating z by $zh = -g$, where $g = y^e + G$, shows that $D \le \pi^* D'$ on $X \setminus \{p\}$ for a $\mathbb{Q}$-divisor $D' \ge 0$ on $\mathbb{P}'$ not containing C_S, of class at most $\frac{e}{c}\mathcal{O}_{\mathbb{P}'}(1)$. Thus it suffices to show that $(\mathbb{P}', C_S + D')$ is lc away from the vertex of C_S (**Steps 2 and 3** of the proof of Theorem 4.3 in Subsection 4.5). Outside C_S, this follows from Theorem 2.28, since y has weight 1 and $a_{(1)} \ge \frac{e}{c}$ (Lemma 4.15(1)).

On C_S, $D'|_{C_S}$ is a convex combination of $\frac{1}{c}(g = 0)|_{C_S}$, which corresponds to $\frac{1}{c}(z = 0)$ on X and is lc by the condition on S in Theorem 4.3 (Lemma 4.16), and a $\mathbb{Q}$-divisor of class $\mathcal{O}_{C_S}(1)$, which is lc by Theorem 2.28 and $a_{(1)}a_{(2)} \ge k$ (Lemma 4.15(3)). Inversion of adjunction along C_S concludes step (2). Bounding the multiplicity of $D'|_{C_S}$ by Theorem 2.28 alone would need the condition $a_{(1)}a_{(2)} \ge \frac{ke}{c}$ of Corollary 4.19, which fails for Propositions 5.1 and 5.3, for example $a_{(1)}a_{(2)} = 30 < \frac{285}{2} = \frac{ke}{c}$ for $d = 4$.

(d) Dimension three. Set-up 4.1 requires $d \ge 3$, and its other conditions cannot hold for $d = 2$ or $d = 3$: well-formedness forces $k = 2$ if $d = 2$, quasismoothness and well-formedness force $k \in \{3, 4, 6\}$ if $d = 3$, and in both cases no integer c satisfies $\frac{k}{d} < c < \frac{k}{d-1}$ (see the beginnings of Sections 6 and 7). The equation of the threefold X of Set-up 6.2 is linear in z with coefficient $h = x^4 + u^3 + v^3$, but x, u, and v have the weights 3, 4, and 4, so that $\mathbb{P}(3, 4, 4)$ is not well formed, and y has weight 3. At p, X behaves like the hypersurfaces of Set-up 4.1, but on $E \cong \mathbb{P}(1, 1, 3)$, Δ_E has multiplicity $\frac{3}{4} + \frac{4}{5} = \frac{31}{20} > 1$ at three points, and Lemma 6.8 passes to a triple cover by $\mathbb{P}^2$.

Away from p, Theorem 2.28 only bounds the multiplicity of $\frac{7}{10}D$ by $\frac{7}{3}$ (Remark 6.11), and X has quotient singularities along two curves Γ and Λ. We restrict to surfaces containing Γ or Λ, where the differents $\frac{2}{3}\Gamma$ and $\frac{3}{4}\Lambda$ leave only $\frac{1}{3}$ and $\frac{1}{4}$ for the restriction of $\frac{7}{10}D$, and use the inequality of Han, Jiang, and Luo (Theorem 2.9, [HJL22, Theorem 1.11]), with no room to spare (Remark 6.18). By Lemma 6.19 and Propositions 6.14 and 6.17, the points near which $(X, \frac{7}{10}D)$ is not lc are finitely many smooth points of X, and the Kawamata–Viehweg vanishing theorem [KM98, Theorem 2.64], applied on a log resolution to $H - (K_X + c_0 D) \sim_{\mathbb{Q}} (3 - 2c_0)H$, where $\mathcal{O}_X(H) \cong \mathcal{O}_X(1)$ and $c_0 = \operatorname{lct}(X; D)$, which is less than $\frac{7}{10}$ if such points exist, excludes them, since $H^0(X, \mathcal{O}_X(1)) = 0$ (cf. [Zhu20, Theorem 1.6]).

Structure of the paper. In Section 2 we recall preliminary material and fix notation. In particular, we introduce plt blow-ups, the valuative criterion of Fujita and Li (Theorem 2.12), orbifold charts of weighted projective spaces, and a multiplicity bound of Totaro (Theorem 2.28). In Section 3 we prove a criterion for a pair to be lc near a point in terms of a plt blow-up at the point, describe weighted blow-ups of germs of hypersurfaces, and prove a lemma on monomial bases of graded hypersurface rings. In Section 4 we study hypersurfaces in weighted projective spaces whose equations are linear in one of the coordinates, and prove Theorem 4.3. In Section 5 we construct the examples with $d \ge 4$. In Sections 6 and 7 we treat the cases $d = 3$ and $d = 2$, respectively. In Section 8 we prove Theorem 1.2.

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2. Preliminaries

We adopt the standard notation and definitions for the MMP from [Sho92, KM98, BCHM10] and use them freely.

2.1. Pairs and singularities.

Definition 2.1. A *pair* (X, B) consists of a normal quasi-projective variety X and an $\mathbb{R}$-divisor $B \ge 0$ on X such that $K_X + B$ is $\mathbb{R}$-Cartier. If we do not require that $B \ge 0$, then we say that

(X, B) is a *sub-pair*. Let $h : X' \to X$ be any log resolution of (X, B) and E a prime divisor on X', and write $K_{X'} + B' := h^*(K_X + B)$. Then we define

$$a(E, X, B) := 1 - \operatorname{mult}_E B'$$

as the *log discrepancy* of E with respect to (X, B). Such h exists for any prime divisor E over X (cf. [KM98, Theorem 0.2 and Lemma 2.45]), and $a(E, X, B)$ does not depend on h (cf. [KM98, Remark 2.23 and Definition 2.25]).

For any (sub-)pair (X, B), we say that (X, B) is *(sub-)lc* if $a(E, X, B) \geq 0$ for any prime divisor E over X, and *(sub-)klt* if $a(E, X, B) > 0$ for any prime divisor E over X. An *lc place* of (X, B) is a prime divisor E over X such that $a(E, X, B) = 0$. An *lc center* of (X, B) is the center of an lc place of (X, B) on X.

Definition 2.2. Let X be a normal variety, E a prime divisor over X, and D an $\mathbb{R}$-Cartier $\mathbb{R}$-divisor on X. We define $\operatorname{mult}_E D := \operatorname{mult}_E h^*D$, where $h : X' \to X$ is a projective birational morphism from a normal variety X' such that E is a prime divisor on X' (this does not depend on h).

Definition 2.3. Let (X, B) be a (sub-)pair and $Z \subset X$ a closed subset. We say that (X, B) is *(sub-)lc near Z* if there exists an open neighborhood U of Z such that $(U, B|_U)$ is (sub-)lc, and similarly for (sub-)klt and for plt. For a pair (X, B) and an $\mathbb{R}$-Cartier $\mathbb{R}$-divisor $D \geq 0$ on X, we define the *log canonical threshold* of D with respect to (X, B) as

$$\operatorname{lct}(X, B; D) := \sup\{t \geq 0 \mid (X, B + tD) \text{ is lc}\},$$

and we write $\operatorname{lct}(X; D) := \operatorname{lct}(X, 0; D)$.

Lemma 2.4. *Let (X, B_1) and (X, B_2) be pairs and $Z \subset X$ a closed subset such that (X, B_1) and (X, B_2) are lc near Z. Then $(X, tB_1 + (1-t)B_2)$ is lc near Z for any real number $t \in [0, 1]$.*

Proof. Let U_1 and U_2 be open neighborhoods of Z such that $(U_1, B_1|_{U_1})$ and $(U_2, B_2|_{U_2})$ are lc, and let $U := U_1 \cap U_2$ and $B_t := tB_1 + (1-t)B_2$. Then $B_t \geq 0$, and $K_X + B_t = t(K_X + B_1) + (1-t)(K_X + B_2)$ is $\mathbb{R}$-Cartier. Let E be a prime divisor over U and $h : U' \to U$ a log resolution of $(U, B_1|_U + B_2|_U)$ such that E is a prime divisor on U'. Since h^* is linear, the log discrepancy is affine in the boundary:

$$a(E, U, B_t|_U) = ta(E, U, B_1|_U) + (1-t)a(E, U, B_2|_U) \geq 0.$$

Thus $(U, B_t|_U)$ is lc. □

Definition-Proposition 2.5 ([Kol13, Definition 4.2, Proposition 4.5(4), Lemma 4.8, and Theorem 4.9])**.** Let X be a normal quasi-projective variety, S a reduced divisor on X with normalization S^ν, and $B \geq 0$ a $\mathbb{Q}$-divisor on X such that S and B have no common component and $K_X + S + B$ is $\mathbb{Q}$-Cartier. We denote by $\operatorname{Diff}_{S^\nu}(B)$ the *different* of B on S^ν, so that

$$K_{S^\nu} + \operatorname{Diff}_{S^\nu}(B) \sim_{\mathbb{Q}} (K_X + S + B)|_{S^\nu},$$

and we write $\operatorname{Diff}_S(B) := \operatorname{Diff}_{S^\nu}(B)$ when S is normal. Then the following hold.

(1) If X and S are smooth near a closed point $s \in S$, then $\operatorname{Diff}_{S^\nu}(B) = B|_S$ near s.
(2) $(X, S + B)$ is plt near S if and only if S is normal and $(S, \operatorname{Diff}_S(B))$ is klt.

Proof. The different and the $\mathbb{Q}$-linear equivalence are [Kol13, Definition 4.2, (4.2.9)]. Near s, S is a smooth divisor on the smooth variety X, so S is Cartier and $S^\nu = S$ near s. Thus (1) follows from [Kol13, Proposition 4.5(4)] applied to a neighborhood of s. If $(X, S + B)$ is plt near S, then S is normal and $(S, \operatorname{Diff}_S(B))$ is klt by [Kol13, Lemma 4.8] and the remark after it. Conversely, if S is normal and $(S, \operatorname{Diff}_S(B))$ is klt, then $(X, S + B)$ is plt near S by [Kol13, Theorem 4.9(1)]. □

Theorem 2.6 ([Kaw07, Theorem]). *Let X be a normal quasi-projective variety, S a reduced divisor on X with normalization S^ν, and $B \geq 0$ a $\mathbb{Q}$-divisor on X such that S and B have no common component and $K_X + S + B$ is $\mathbb{Q}$-Cartier. Then $(X, S + B)$ is lc near S if and only if $(S^\nu, \mathrm{Diff}_{S^\nu}(B))$ is lc.*

Lemma 2.7 ([Kol13, Claim 2.10.4]). *Let X be a smooth quasi-projective variety and $B \geq 0$ a $\mathbb{Q}$-divisor on X such that $\mathrm{mult}_x B \leq 1$ for any closed point $x \in X$. Then (X, B) is lc.*

Proof. By [Kol13, Claim 2.10.4] (by upper semicontinuity of multiplicity, $\mathrm{mult}_p B \leq 1$ also holds at the non-closed points p of X), (X, B) is canonical, that is, $a(E, X, B) \geq 1$ for any exceptional prime divisor E over X. For any prime divisor P on X and any closed point $x \in P$, we have $\mathrm{mult}_P B \leq \mathrm{mult}_x B \leq 1$, so $a(P, X, B) = 1 - \mathrm{mult}_P B \geq 0$. Thus (X, B) is lc. □

Lemma 2.8 ([KM98, Proposition 5.20]). *Let $g : X' \to X$ be a finite surjective morphism between normal quasi-projective varieties and B a $\mathbb{Q}$-divisor on X such that $K_X + B$ is $\mathbb{Q}$-Cartier. Let B' be the $\mathbb{Q}$-divisor on X' such that $K_{X'} + B' = g^*(K_X + B)$. Then the following hold.*

(1) For any prime divisor E' over X', there exist a prime divisor E over X and a positive integer r such that

$$a(E', X', B') = r a(E, X, B) \quad \text{and} \quad \mathrm{mult}_{E'} g^* D = r \,\mathrm{mult}_E D$$

for any $\mathbb{Q}$-Cartier $\mathbb{Q}$-divisor D on X.

(2) If P' is a prime divisor on X' and r is the ramification index of g along P', then

$$\mathrm{mult}_{P'} B' = 1 - r\big(1 - \mathrm{mult}_{g(P')} B\big).$$

(3) For any closed point $x \in X$, (X, B) is sub-lc (resp. sub-klt) near x if and only if (X', B') is sub-lc (resp. sub-klt) near $g^{-1}(x)$.

Proof. We first prove (2). Let $P := g(P')$. Since X and X' are normal, they are smooth near the generic points of P and P' respectively. By the Hurwitz formula [Kol13, 2.41], since the ramification of g along P' is tame, $K_{X'} = g^* K_X + (r - 1)P'$ near the generic point of P', and $\mathrm{mult}_{P'} g^* B = r \,\mathrm{mult}_P B$. Taking the coefficients of P' in $K_{X'} + B' = g^*(K_X + B)$, we obtain

$$\mathrm{mult}_{P'} B' = r \,\mathrm{mult}_P B - (r - 1) = 1 - r(1 - \mathrm{mult}_P B).$$

Let $f : Y \to X$ be a projective birational morphism from a normal variety Y, let Y' be the normalization of the irreducible component of $Y \times_X X'$ dominating X', and let $h : Y' \to Y$ and $f' : Y' \to X'$ be the induced morphisms. Then h is finite and surjective, f' is projective and birational, and $g \circ f' = f \circ h$ (h and f' are the composites of the normalization, a closed immersion, and the base changes $Y \times_X X' \to Y$ and $Y \times_X X' \to X'$ of the finite morphism g and the projective morphism f respectively, normalization is finite, $\dim Y' = \dim Y$, and f is an isomorphism over a dense open subset of X). Let B_Y and $B_{Y'}$ be the $\mathbb{Q}$-divisors such that $K_Y + B_Y = f^*(K_X + B)$ and $K_{Y'} + B_{Y'} = f'^*(K_{X'} + B')$. Then

$$K_{Y'} + B_{Y'} = f'^* g^*(K_X + B) = h^*(K_Y + B_Y).$$

Let E' be a prime divisor on Y', $E := h(E')$, and r the ramification index of h along E'. By (2) applied to h, $\mathrm{mult}_{E'} B_{Y'} = 1 - r(1 - \mathrm{mult}_E B_Y)$. Moreover,

$$\mathrm{mult}_{E'} f'^* g^* D = \mathrm{mult}_{E'} h^* f^* D = r \,\mathrm{mult}_E f^* D$$

for any $\mathbb{Q}$-Cartier $\mathbb{Q}$-divisor D on X. Thus

$$a(E', X', B') = r a(E, X, B) \quad \text{and} \quad \mathrm{mult}_{E'} g^* D = r \,\mathrm{mult}_E D. \tag{2.1}$$

We prove (1). Let E' be a prime divisor over X' and $f'_0 : Y'_0 \to X'$ a projective birational morphism from a normal variety Y'_0 such that E' is a prime divisor on Y'_0. The generic point of E' is a codimension one regular point of Y'_0 which does not dominate X. By [Kol13, Lemma 2.22] applied to $g \circ f'_0 : Y'_0 \to X$, there exists a sequence of blow-ups $X_n \to X$ such that the induced rational map $Y'_0 \dashrightarrow X_n$ is defined at the generic point of E' and maps it to a codimension one

regular point of X_n. Let Y be the normalization of X_n, $f : Y \to X$ the induced projective birational morphism, and E the closure of the image of the generic point of E' in Y. Then E is a prime divisor on Y, since $Y \to X_n$ is an isomorphism near a regular point. Let Y', h, and f' be constructed from f as in the second paragraph of the proof. Since $f \circ h = g \circ f'$, h maps the center of E' on Y' onto E. Since h is finite, the center of E' on Y' is a prime divisor on Y', and so E' is a prime divisor on Y' (the local ring of Y' at the generic point of the center of E' is a discrete valuation ring dominated by the valuation ring of E', hence equal to it). Thus (1) follows from (2.1).

We prove (3). Suppose that (X, B) is sub-lc (resp. sub-klt) on an open neighborhood U of x, and let $U' := g^{-1}(U)$. Then U' is an open neighborhood of $g^{-1}(x)$, and (1) applied to the finite surjective morphism $U' \to U$ induced by g implies that $(U', B'|_{U'})$ is sub-lc (resp. sub-klt). Conversely, suppose that (X', B') is sub-lc (resp. sub-klt) on an open neighborhood V of $g^{-1}(x)$. Since g is a closed map, $U := X \backslash g(X' \backslash V)$ is an open neighborhood of x, and $U' := g^{-1}(U) \subset V$. Let E be a prime divisor over U and $f : Y \to U$ a projective birational morphism from a normal variety Y such that E is a prime divisor on Y. Let Y', h, and f' be constructed from f and the morphism $U' \to U$ induced by g as in the second paragraph of the proof. Since h is finite and surjective, there exists a prime divisor E' on Y' such that $h(E') = E$. By (2.1), there exists a positive integer r such that

$$a(E, U, B|_U) = \frac{1}{r} a(E', U', B'|_{U'}) \geq 0 \quad (\text{resp.} > 0).$$

Thus $(U, B|_U)$ is sub-lc (resp. sub-klt). □

In Section 6, we need the log canonicity of a pair $(X, B + tC)$ at a point of a smooth curve C on a smooth surface X, where t is close to 1 and the multiplicity of B at the point is bounded only by a constant smaller than 1. At such a point x, the multiplicity $\operatorname{mult}_x B + t$ of $B + tC$ at x can exceed 1, so Lemma 2.7 does not apply to $B + tC$. The following bound uses both the multiplicity of B and the local intersection number of B and C.

Theorem 2.9 ([HJL22, Theorem 1.11])**.** *Let X be a smooth quasi-projective surface, $x \in X$ a closed point, $B \geq 0$ a $\mathbb{Q}$-divisor on X, and C a curve on X such that C is smooth at x and C is not a component of B. Let $m := \operatorname{mult}_x B$ and $I := (B \cdot C)_x$, and assume that $m \leq 1$ and $I > 0$. Then $(X, B + tC)$ is lc near x for any real number $0 \leq t \leq \min\{1, 1 + \frac{m}{I} - m\}$.*

In [HJL22, Theorem 1.11], the conclusion is a lower bound for $\operatorname{lct}(X \ni x, B; C)$. The threshold is attained by [HJL22, Lemma 2.8 and Remark 2.11].

2.2. **Alpha invariants and K-semistability.**

Definition 2.10. We say that a normal projective variety X is a $\mathbb{Q}$-*Fano variety* if X is klt and $-K_X$ is an ample $\mathbb{Q}$-Cartier divisor. Let (X, B) be a klt pair such that X is projective and $-(K_X + B)$ is an ample $\mathbb{Q}$-Cartier $\mathbb{Q}$-divisor. We define the *alpha invariant* of (X, B) as

$$\alpha(X, B) := \inf\{\operatorname{lct}(X, B; D) \mid D \geq 0 \text{ is a } \mathbb{Q}\text{-divisor and } D \sim_{\mathbb{Q}} -(K_X + B)\},$$

and for a $\mathbb{Q}$-Fano variety X we write $\alpha(X) := \alpha(X, 0)$ (cf. [LZ22, Definition 1.1]).

Definition 2.11. Let X be a $\mathbb{Q}$-Fano variety of dimension d, E a prime divisor over X, and $f : Y \to X$ a projective birational morphism from a normal variety Y such that E is a prime divisor on Y. We set $A_X(E) := a(E, X, 0)$. Following [Fuj19a, Definitions 1.1 and 1.3], for any real number $t \geq 0$ we define

$$\operatorname{vol}(-K_X - tE) := \limsup_{m \to +\infty} \frac{\dim H^0\big(Y, \mathcal{O}_Y(\lfloor m(f^*(-K_X) - tE) \rfloor)\big)}{m^d/d!},$$

where m runs over the positive integers such that $-mK_X$ is Cartier. We define

$$T_X(E) := \sup\{t \geq 0 \mid \operatorname{vol}(-K_X - tE) > 0\}, \quad S_X(E) := \frac{1}{\operatorname{vol}(-K_X)} \int_0^\infty \operatorname{vol}(-K_X - tE) dt,$$

and

$$\beta_X(E) := A_X(E)\operatorname{vol}(-K_X) - \int_0^\infty \operatorname{vol}(-K_X - tE)dt = \operatorname{vol}(-K_X)\big(A_X(E) - S_X(E)\big).$$

We call $T_X(E)$ the *pseudo-effective threshold* of $-K_X$ along E.

The invariants in Definition 2.11 depend only on E and not on the choice of f [Fuj19a, Remark 1.2]. We refer to [Fuj19a, Section 2.1] for the definition of K-semistable $\mathbb{Q}$-Fano varieties.

Theorem 2.12 ([Fuj19a, Corollary 1.5(2)], [Li17, Theorem 3.7])**.** *A $\mathbb{Q}$-Fano variety X is K-semistable if and only if $\beta_X(E) \geq 0$ for every prime divisor E over X.*

Lemma 2.13 turns the global invariant $T_X(E)$ into a local bound on the multiplicity along E of any $\mathbb{Q}$-divisor $\mathbb{Q}$-linearly equivalent to $-K_X$. Together with the local criterion of Lemma 3.1, it bounds from below the log canonical thresholds of these $\mathbb{Q}$-divisors at the point blown up in Sections 4, 6, and 7.

Lemma 2.13. *Let X be a $\mathbb{Q}$-Fano variety, E a prime divisor over X, and $D \geq 0$ a $\mathbb{Q}$-divisor on X such that $D \sim_{\mathbb{Q}} -K_X$. Then $\operatorname{mult}_E D \leq T_X(E)$.*

Proof. Let $d := \dim X$ and $c := \operatorname{mult}_E D$, and let $f : Y \to X$ be a projective birational morphism from a normal variety Y such that E is a prime divisor on Y. Since $f_*\mathcal{O}_Y = \mathcal{O}_X$ (as $f_*\mathcal{O}_Y$ is a coherent $\mathcal{O}_X$-algebra contained in the function field of X, and X is normal) and $-K_X$ is ample, the asymptotic Riemann–Roch theorem [Laz04, Corollary 1.4.41] implies $\operatorname{vol}(-K_X) = (-K_X)^d > 0$, so $T_X(E) \geq 0$. Thus we may assume that $c > 0$. Let t be a rational number with $0 \leq t < c$, let $s := t/c$, and let m be a positive integer such that $-mK_X$, $-msK_X$, and msD are Cartier and $msD \sim -msK_X$. Then $mt = msc$ is an integer, $f^*(msD) - mtE \geq 0$, and

$$m\big(f^*(-K_X) - tE\big) \sim f^*\big(-m(1-s)K_X\big) + \big(f^*(msD) - mtE\big).$$

Hence, for any positive integer k,

$$\dim H^0\big(Y, \mathcal{O}_Y(km(f^*(-K_X) - tE))\big) \geq \dim H^0\big(X, \mathcal{O}_X(-km(1-s)K_X)\big).$$

Dividing by $(km)^d/d!$ and letting $k \to +\infty$, we obtain $\operatorname{vol}(-K_X - tE) \geq (1-s)^d \operatorname{vol}(-K_X) > 0$. Thus $T_X(E) \geq t$ for any rational number $0 \leq t < c$, and so $T_X(E) \geq c$. □

2.3. **Plt blow-ups.**

Definition-Proposition 2.14 (cf. [Pro00, Definition 3] and [Kud01, Definition 1.1])**.** Let X be a normal quasi-projective variety and $x \in X$ a closed point such that X is klt near x. We say that a projective birational morphism $f : Y \to X$ is a *plt blow-up* of X at x if Y is normal, the exceptional locus of f is a prime divisor E with $f(E) = \{x\}$, $-E$ is a $\mathbb{Q}$-Cartier f-ample divisor, and (Y, E) is plt near E. We call E the *exceptional divisor* of f. Let $f : Y \to X$ be a plt blow-up of X at x with exceptional divisor E, and let $\Delta_E := \operatorname{Diff}_{E^\nu}(0)$, where E^ν is the normalization of E. Then E is normal, (E, Δ_E) is klt, and $-(K_E + \Delta_E) \sim_{\mathbb{Q}} -a(E, X, 0)E|_E$ is ample.

Proof. Since $f(E) = \{x\}$, the preimage under f of any open neighborhood of x is an open neighborhood of E. Possibly replacing X with an open neighborhood of x, we may assume that X is klt and $lK_X \sim 0$ for some positive integer l. Let $U \subset Y$ be an open neighborhood of E such that (U, E) is plt. Then (U, E) is dlt, and so E is normal [KM98, Proposition 5.51]. Thus $E^\nu = E$, $\Delta_E = \operatorname{Diff}_E(0)$, and (E, Δ_E) is klt [Kol13, Lemma 4.8]. Since E is the only f-exceptional prime divisor,

$$K_Y + E = f^*K_X + a(E, X, 0)E.$$

Since $lK_X \sim 0$, $(f^*K_X)|_E \sim_{\mathbb{Q}} 0$, and so

$$K_E + \Delta_E \sim_{\mathbb{Q}} (K_Y + E)|_E \sim_{\mathbb{Q}} a(E, X, 0)E|_E.$$

Since X is klt, $a(E, X, 0) > 0$. Since $-E$ is f-ample and $f(E) = \{x\}$, $-E|_E$ is ample. Thus $-(K_E + \Delta_E) \sim_{\mathbb{Q}} -a(E, X, 0)E|_E$ is ample, so $\alpha(E, \Delta_E)$ is defined (Definition 2.10). □

2.4. **Weighted projective spaces.**

Definition 2.15. Let $a_0, \dots, a_N$ be positive integers, and let $\mathbb{P} := \mathbb{P}(a_0, \dots, a_N)$ be the weighted projective space $\operatorname{Proj} \mathbb{C}[x_0, \dots, x_N]$, where $\deg x_i = a_i$. We say that $\mathbb{P}$ is *well formed* if $\gcd(a_0, \dots, a_{i-1}, a_{i+1}, \dots, a_N) = 1$ for any i [IF00, Definition 5.11]. We call μ_r the *stabilizer* of a point $[x_0 : \cdots : x_N] \in \mathbb{P}$, where $r := \gcd\{a_i \mid x_i \neq 0\}$. For a closed subvariety $V \subset \mathbb{P}$ with homogeneous ideal $I_V \subset \mathbb{C}[x_0, \dots, x_N]$, let $R := \mathbb{C}[x_0, \dots, x_N]/I_V$, so that $V = \operatorname{Proj} R$, and for an integer m we denote by $\mathcal{O}_V(m)$ the sheaf associated to the graded R-module $R(m)$. For any $0 \leq j \leq N$, the group μ_{a_j} acts on $U_j := (x_j = 1) \subset \mathbb{A}^{N+1}$ by $\zeta \cdot x_i = \zeta^{a_i} x_i$, and we call the quotient morphism $\rho_j : U_j \to (x_j \neq 0) \subset \mathbb{P}$ by μ_{a_j} the j-th *orbifold chart* of $\mathbb{P}$.

Definition-Lemma 2.16. Let $\mathbb{P} = \mathbb{P}(a_0, \dots, a_N)$, let $V \subset \mathbb{P}$ be a closed subvariety with homogeneous ideal I_V, and let $R := \mathbb{C}[x_0, \dots, x_N]/I_V$. Assume that R is normal and that the points of V with non-trivial stabilizer form a closed subset of codimension at least 2 in V. Then there exists a Weil divisor H_V on V, unique up to linear equivalence, such that $\mathcal{O}_V(m) \cong \mathcal{O}_V(mH_V)$ for any integer m, and we also denote by $\mathcal{O}_V(m)$ the linear equivalence class of mH_V.

Proof. Let $C_V := \operatorname{Spec} R \subset \mathbb{A}^{N+1}$ be the affine cone over V and $C_V^* := C_V \backslash \{0\}$. The group $\mathbb{G}_m$ acts on C_V^* by $\lambda \cdot x_i = \lambda^{a_i} x_i$, and V is the quotient of C_V^* by $\mathbb{G}_m$. For an open subset $W \subset V$ with preimage $\tilde{W} \subset C_V^*$, $H^0(W, \mathcal{O}_V(m))$ is the space of regular functions f on $\tilde{W}$ such that $f(\lambda \cdot y) = \lambda^m f(y)$ for any $\lambda \in \mathbb{G}_m$ and $y \in \tilde{W}$: for $W = D_+(g)$ with $g \in R$ homogeneous of positive degree, both spaces are equal to $(R_g)_m$, and the general case follows by gluing. Since R is normal, C_V satisfies Serre's condition (S_2) [Mat86, Theorem 23.8], so a regular function on $\tilde{W} \backslash \tilde{Z}$ extends to $\tilde{W}$ for any closed subset $\tilde{Z} \subset \tilde{W}$ of codimension at least 2. The fibers of $C_V^* \to V$ are curves, so the preimage of a closed subset of codimension at least 2 in W has codimension at least 2 in $\tilde{W}$. Thus the restriction map

$$H^0(W, \mathcal{O}_V(m)) \to H^0(W \backslash Z, \mathcal{O}_V(m))$$

is bijective for any integer m, any open subset $W \subset V$, and any closed subset $Z \subset W$ of codimension at least 2 in W.

Let $V^\circ \subset V$ be the open subset of points with trivial stabilizer. For a point $v \in V^\circ$, let $I := \{i \mid x_i(v) \neq 0\}$ and $g := \prod_{i \in I} x_i$. Since $\gcd\{a_i \mid i \in I\} = 1$, there exist integers n_i with $\sum_{i \in I} n_i a_i = 1$, and $u := \prod_{i \in I} x_i^{n_i}$ is a unit of degree 1 in R_g. Thus $(R_g)_m = u^m (R_g)_0$ for any integer m. Hence $\mathcal{O}_V(1)$ is invertible on $D_+(g) \ni v$, and the multiplication map $\mathcal{O}_V(1)^{\otimes m} \to \mathcal{O}_V(m)$ is an isomorphism on $D_+(g)$. In particular, $\mathcal{O}_V(1)|_{V^\circ}$ is invertible and

$$\mathcal{O}_V(m)|_{V^\circ} \cong \mathcal{O}_V(1)^{\otimes m}|_{V^\circ}$$

for any integer m.

The variety V is normal, because $(R_{x_i})_0$ is the ring of $\mathbb{G}_m$-invariants of the normal domain R_{x_i} for any i. Let H° be a Cartier divisor on V° with

$$\mathcal{O}_{V^\circ}(H^\circ) \cong \mathcal{O}_V(1)|_{V^\circ},$$

and let H_V be the closure of H° in V. Then

$$\mathcal{O}_V(mH_V)|_{V^\circ} \cong \mathcal{O}_V(m)|_{V^\circ}$$

for any integer m. Since V is normal and $V \backslash V^\circ$ has codimension at least 2 in V, the sections of $\mathcal{O}_V(mH_V)$ on an open subset $W \subset V$ are the sections on $W \cap V^\circ$, and the sections of $\mathcal{O}_V(m)$ on W are the sections on $W \cap V^\circ$ by the bijectivity of the restriction maps above. Therefore the isomorphism on V° extends to an isomorphism $\mathcal{O}_V(m) \cong \mathcal{O}_V(mH_V)$. If H' is a Weil divisor on V with $\mathcal{O}_V(H') \cong \mathcal{O}_V(1)$, then $\mathcal{O}_V(H') \cong \mathcal{O}_V(H_V)$, so $H' \sim H_V$ because V is normal. □

Remark 2.17. The assumption on the stabilizers excludes the case that a prime divisor on V consists of points with non-trivial stabilizer, and in that case $\mathcal{O}_V(m)$ need not be isomorphic to $\mathcal{O}_V(1)^{\otimes m}$. For example, for $V = \mathbb{P}(1, 2, 2)$, whose points with $x_0 = 0$ have stabilizer μ_2,

$(R_{x_i})_1 = x_0(R_{x_i})_0$ for any i, so $\mathcal{O}_V(1) \cong \mathcal{O}_V$. On the other hand, $\mathcal{O}_V(2)$ corresponds to $\mathcal{O}_{\mathbb{P}^2}(1)$ under the isomorphism $\mathbb{P}(1,2,2) \to \mathbb{P}^2$, $[x_0 : x_1 : x_2] \to [x_0^2 : x_1 : x_2]$, so no Weil divisor H_V on V satisfies $\mathcal{O}_V(m) \cong \mathcal{O}_V(mH_V)$ for $m = 1, 2$.

Definition 2.18. Let $r \geq 1$, a, and b be integers. We say that the germ $(V \ni x)$ of a normal surface V at a closed point x is a *cyclic quotient singularity of type* $\frac{1}{r}(a,b)$ if it is analytically isomorphic to the germ at the image of the origin of the quotient of $\mathbb{A}^2$ by the action of μ_r given by $\zeta \cdot (u,v) = (\zeta^a u, \zeta^b v)$. In this case we also say that x is a cyclic quotient singularity of type $\frac{1}{r}(a,b)$.

Definition 2.19. Let $\mathbb{P} = \mathbb{P}(a_0, \dots, a_N)$. We say that a weighted homogeneous polynomial $F \in \mathbb{C}[x_0, \dots, x_N]$ is *quasismooth* if the affine hypersurface $(F = 0) \subset \mathbb{A}^{N+1}$ is smooth outside the origin. We say that a locally closed subvariety $V \subset \mathbb{P}$ is *quasismooth* if the preimage of V in $\mathbb{A}^{N+1} \backslash \{0\}$ is smooth.

Definition-Lemma 2.20. Let $\mathbb{P} = \mathbb{P}(a_0, \dots, a_N)$ with orbifold charts $\rho_0, \dots, \rho_N$, and let $\mathbb{G}_m$ act on $\mathbb{A}^{N+1}$ by $\lambda \cdot x_i = \lambda^{a_i} x_i$. Then the following hold for any $0 \leq j \leq N$.

(1) The morphism
$$\mathbb{G}_m \times U_j \to (x_j \neq 0) \subset \mathbb{A}^{N+1}, \quad (\lambda, u) \to \lambda \cdot u,$$
is the quotient by a free action of μ_{a_j}. In particular, it is étale.

(2) Let $V \subset \mathbb{P}$ be a locally closed subvariety, and let $\widehat{V} \subset \mathbb{A}^{N+1} \backslash \{0\}$ be the preimage of V. Then the morphism in (1) restricts to an étale surjective morphism
$$\mathbb{G}_m \times \rho_j^{-1}(V) \to \widehat{V} \cap (x_j \neq 0).$$
In particular, $\rho_j^{-1}(V)$ is smooth if V is quasismooth.

(3) Let $V \subset \mathbb{P}$ be a quasismooth locally closed subvariety and s a weighted homogeneous polynomial of degree m that does not vanish identically on any irreducible component of V. Then there exists a unique $\mathbb{Q}$-divisor on V, denoted by $(s = 0)|_V$, whose restriction to $V \cap (x_i \neq 0)$ is $\frac{1}{a_i} \operatorname{div}(s^{a_i}/x_i^m)$ for any $0 \leq i \leq N$. Moreover, $\rho_j^*\big((s = 0)|_V\big) = \operatorname{div}(s_j)$ on $\rho_j^{-1}(V)$, where s_j is the restriction of s to U_j.

Proof. (1) Let $\zeta \in \mu_{a_j}$ act on $\mathbb{G}_m \times U_j$ by $\zeta \cdot (\lambda, u) = (\lambda\zeta^{-1}, \zeta \cdot u)$, where $\zeta \cdot u$ is the action of Definition 2.15. The action of μ_{a_j} on $\mathbb{G}_m \times U_j$ is free, because λ is multiplied by ζ^{-1}, and the morphism $(\lambda, u) \to \lambda \cdot u$ is invariant under it. The preimage of a point $x \in (x_j \neq 0)$ is
$$\{(\lambda, \lambda^{-1} \cdot x) \mid \lambda^{a_j} = x_j\},$$
which is one μ_{a_j}-orbit. Thus the morphism induces a finite bijective morphism $(\mathbb{G}_m \times U_j)/\mu_{a_j} \to (x_j \neq 0)$, and it is an isomorphism, since $\mathbb{C}[\lambda^{\pm 1}, u_i]^{\mu_{a_j}}$ is generated by $\lambda^{\pm a_j} = x_j^{\pm 1}$ and $\lambda^{a_i} u_i = x_i$, where u_i with $i \neq j$ are the coordinates on U_j. The quotient morphism by a free action of a finite group is étale.

(2) The subset $\widehat{V}$ is $\mathbb{G}_m$-invariant, so by (1) the preimage of $\widehat{V} \cap (x_j \neq 0)$ in $\mathbb{G}_m \times U_j$ is $\mathbb{G}_m \times \rho_j^{-1}(V)$. If V is quasismooth, then $\widehat{V}$ is smooth, so $\mathbb{G}_m \times \rho_j^{-1}(V)$ is smooth, being étale over $\widehat{V} \cap (x_j \neq 0)$. Hence $\rho_j^{-1}(V)$ is smooth.

(3) The variety $V \cap (x_i \neq 0) = \rho_i^{-1}(V)/\mu_{a_i}$ is the quotient of a smooth variety by a finite group, so it is normal, and s^{a_i}/x_i^m is a regular function of degree 0 on it. On $V \cap (x_i x_k \neq 0)$,
$$\frac{1}{a_i} \operatorname{div}(s^{a_i}/x_i^m) - \frac{1}{a_k} \operatorname{div}(s^{a_k}/x_k^m) = \frac{m}{a_i a_k} \operatorname{div}(x_k^{a_i}/x_i^{a_k}) = 0,$$
because $x_k^{a_i}/x_i^{a_k}$ is invertible there. Thus these $\mathbb{Q}$-divisors glue to a unique $\mathbb{Q}$-divisor $(s = 0)|_V$ on V. Since $x_j = 1$ on U_j, $\rho_j^*(s^{a_j}/x_j^m) = s_j^{a_j}$. The pullback of a principal divisor is the divisor of

the pulled back function, so

$$\rho_j^*\big((s=0)|_V\big) = \frac{1}{a_j}\operatorname{div}(s_j^{a_j}) = \operatorname{div}(s_j). \quad \square$$

Lemma 2.21. *Let $m \geq 3$ and let $F \in \mathbb{C}[x_1, \dots, x_m]$ be a quasismooth weighted homogeneous polynomial of positive degree with respect to positive integer weights. Then F is irreducible.*

Proof. A factor of a weighted homogeneous polynomial is weighted homogeneous, as one sees by comparing the components of lowest and highest degree. Suppose that $F = F_1F_2$ with F_1 and F_2 non-constant. Then F_1 and F_2 are weighted homogeneous of positive degree, so the origin is contained in $(F_1 = 0) \cap (F_2 = 0) \subset \mathbb{A}^m$. Since $\dim Y \geq m - 2 \geq 1$ for any irreducible component Y of $(F_1 = 0) \cap (F_2 = 0)$, there exists a point $y \neq 0$ with $F_1(y) = F_2(y) = 0$. Then $F(y) = 0$ and

$$\frac{\partial F}{\partial x_i}(y) = F_1(y)\frac{\partial F_2}{\partial x_i}(y) + F_2(y)\frac{\partial F_1}{\partial x_i}(y) = 0$$

for any i, so $(F = 0)$ is singular at $y \neq 0$, a contradiction. Here F_1 may be a repeated factor of F. $\square$

Lemma 2.22. *Let $\mathbb{P} = \mathbb{P}(a_0, \dots, a_N)$ be well formed, let $F_1, \dots, F_r \in \mathbb{C}[x_0, \dots, x_N]$ be weighted homogeneous polynomials of degrees $e_1, \dots, e_r$ forming a regular sequence, and let $R := \mathbb{C}[x_0, \dots, x_N]/(F_1, \dots, F_r)$ and $X := \operatorname{Proj} R \subset \mathbb{P}$. Assume that R is a normal domain, that $\dim X = N - r \geq 2$, and that the locus of points of $\mathbb{P}$ with non-trivial stabilizer intersects X in a closed subset of codimension at least 2 in X. Then $\omega_X \cong \mathcal{O}_X(\sum_i e_i - \sum_j a_j)$, $\mathcal{O}_X(1)^{N-r} = \frac{e_1 \cdots e_r}{a_0 \cdots a_N}$, and $H^0(X, \mathcal{O}_X(m)) = R_m$ for any integer m.*

Proof. Let $n := N - r$. The variety X is normal, because $(R_{x_i})_0$ is the ring of $\mathbb{G}_m$-invariants of the normal domain R_{x_i} for any i. By Definition-Lemma 2.16, there exists a Weil divisor H_X on X such that $\mathcal{O}_X(m) \cong \mathcal{O}_X(mH_X)$ for any integer m.

We prove the formula for ω_X. Let $U \subset \mathbb{P}$ be the open subset of points with trivial stabilizer. For a point $v \in U$, let $I := \{i \mid x_i(v) \neq 0\}$ and $g := \prod_{i \in I} x_i$. Since $\gcd\{a_i \mid i \in I\} = 1$, there exists a Laurent monomial u in the x_i with $i \in I$ of degree 1, and u is a unit of $\mathbb{C}[x_0, \dots, x_N]_g$. Thus

$$\mathbb{C}[x_0, \dots, x_N]_g = (\mathbb{C}[x_0, \dots, x_N]_g)_0[u, u^{-1}],$$

so $(g \neq 0) \subset \mathbb{A}^{N+1}$ is isomorphic to $D_+(g) \times \mathbb{G}_m$, and $D_+(g)$ is smooth. Moreover, $\mathcal{O}_{\mathbb{P}}(m)$ is freely generated by u^m on $D_+(g)$, and the ideal of $X \cap D_+(g)$ is generated by $F_1u^{-e_1}, \dots, F_ru^{-e_r}$. Hence U is smooth, $\mathcal{O}_U(m) := \mathcal{O}_{\mathbb{P}}(m)|_U$ is invertible, $\mathcal{O}_X(m)|_{X\cap U} = \mathcal{O}_U(m)|_{X\cap U}$, and $X \cap U$ is the zero locus in U of the sections $F_1, \dots, F_r$ of $\mathcal{O}_U(e_1), \dots, \mathcal{O}_U(e_r)$. Since $\dim X = N - r$, $X \cap U$ is a local complete intersection of codimension r in U, and by the adjunction formula [Har77, Proposition II.8.20] on the smooth locus of $X \cap U$, whose complement has codimension at least 2 in X,

$$\omega_{X\cap U} \cong \omega_U \otimes \mathcal{O}_{X\cap U}\Big(\sum_i e_i\Big).$$

Since $\mathbb{P}$ is well formed, and quasismooth because the preimage $\mathbb{A}^{N+1}\backslash\{0\}$ of $\mathbb{P}$ is smooth, $\omega_{\mathbb{P}} \cong \mathcal{O}_{\mathbb{P}}(-\sum_j a_j)$ by [IF00, 6.14] applied with no equations, so

$$\omega_{X\cap U} \cong \mathcal{O}_{X\cap U}\Big(\sum_i e_i - \sum_j a_j\Big).$$

By assumption, $X\backslash U$ has codimension at least 2 in X. Since X is normal and both ω_X and

$$\mathcal{O}_X\Big(\sum_i e_i - \sum_j a_j\Big) \cong \mathcal{O}_X\Big(\Big(\sum_i e_i - \sum_j a_j\Big)H_X\Big)$$

are sheaves associated to Weil divisors, the isomorphism on $X \cap U$ extends to an isomorphism

$$\omega_X \cong \mathcal{O}_X\Big(\sum_i e_i - \sum_j a_j\Big).$$

We prove that $H^0(X, \mathcal{O}_X(m)) = R_m$. The Čech complex of the open cover $\{D_+(x_i)\}_{0\le i\le N}$ of X implies that $H^0(X, \mathcal{O}_X(m))$ is the kernel of

$$\prod_i (R_{x_i})_m \to \prod_{i<j} (R_{x_i x_j})_m.$$

Thus $\bigoplus_{m\in\mathbb{Z}} H^0(X, \mathcal{O}_X(m))$ is the ring of regular functions on $\operatorname{Spec} R\backslash\{0\}$, where 0 is the closed point defined by $(x_0, \dots, x_N)$. Since R is normal and $\dim R = n+1 \ge 3$, R satisfies (S_2) [Mat86, Theorem 23.8], so any regular function on $\operatorname{Spec} R\backslash\{0\}$ extends to $\operatorname{Spec} R$. Hence $H^0(X, \mathcal{O}_X(m)) = R_m$ for any integer m.

We prove that $\mathcal{O}_X(1)^n = \frac{e_1\cdots e_r}{a_0\cdots a_N}$, that is, $(H_X^n) = \frac{e_1\cdots e_r}{a_0\cdots a_N}$ (Definition-Lemma 2.16). Let $L := \operatorname{lcm}(a_0, \dots, a_N)$. Then $\mathcal{O}_{\mathbb{P}}(L)$ is invertible, generated by x_j^{L/a_j} on $(x_j \ne 0)$, and

$$\mathcal{O}_{\mathbb{P}}(mL) = \mathcal{O}_{\mathbb{P}}(L)^{\otimes m}$$

for any integer m. The morphism $\phi : \mathbb{P} \to \mathbb{P}^N$ defined by $x_0^{L/a_0}, \dots, x_N^{L/a_N}$ is finite, because $\mathbb{C}[x_0, \dots, x_N]$ is a finite module over $\mathbb{C}[x_0^{L/a_0}, \dots, x_N^{L/a_N}]$, and

$$\phi^*\mathcal{O}_{\mathbb{P}^N}(1) \cong \mathcal{O}_{\mathbb{P}}(L).$$

Thus $\mathcal{O}_X(L) = \mathcal{O}_{\mathbb{P}}(L)|_X$ is an ample line bundle [Har77, Exercise III.5.7(a)(d)]. Since $\mathcal{O}_X(L) \cong \mathcal{O}_X(LH_X)$, LH_X is an ample Cartier divisor, and H_X is $\mathbb{Q}$-Cartier. By Serre vanishing for the ample line bundle $\mathcal{O}_X(L)$ [Har77, Proposition III.5.3] and the Riemann–Roch theorem [KM98, Theorem 1.36],

$$\dim R_{Lm} = h^0(X, \mathcal{O}_X(mLH_X)) = \frac{(H_X^n)}{n!}(Lm)^n + O(m^{n-1}).$$

A general point of X has trivial stabilizer, so the weights a_i of the coordinates x_i with $x_i|_X \not\equiv 0$ have greatest common divisor 1. Hence there exists an integer s_0 such that any integer $k \ge s_0$ is a non-negative integral combination of the weights a_i with $x_i|_X \not\equiv 0$, and $R_k \ne 0$ for $k \ge s_0$, since R is a domain. Fix an integer w with $Lw \ge s_0 + L$. Let $k = Lm + q$ with $0 \le q < L$. Multiplication by non-zero elements of R_{Lw+q} and of R_{Lw-q} is injective, so

$$\dim R_{L(m-w)} \le \dim R_{Lm+q} \le \dim R_{L(m+w)}.$$

Thus $\dim R_k = \frac{(H_X^n)}{n!}k^n + O(k^{n-1})$. Since $\sum_{k\ge 0}\binom{k+n}{n}t^k = (1-t)^{-n-1}$, we get

$$(H_X^n) = \lim_{t\to 1}(1-t)^{n+1}\sum_{k\ge 0}\dim R_k t^k.$$

Since $F_1, \dots, F_r$ is a regular sequence, the Hilbert series of R is

$$\sum_{k\ge 0}\dim R_k t^k = \frac{\prod_{i=1}^r(1-t^{e_i})}{\prod_{j=0}^N(1-t^{a_j})},$$

and the limit above is $\frac{e_1\cdots e_r}{a_0\cdots a_N}$. □

Definition 2.23. Let $\mathbb{P} = \mathbb{P}(a_0, \dots, a_N)$ with orbifold charts $\rho_0, \dots, \rho_N$. Let $V \subset \mathbb{P}$ be a quasismooth locally closed subvariety and $B \ge 0$ a $\mathbb{Q}$-divisor on V. We say that (V, B) is *lc on orbifold charts* if $\big(\rho_j^{-1}(V), \rho_j^*B\big)$ is lc for any j. For an irreducible closed subvariety $M \subset \mathbb{P}$ with homogeneous ideal I_M and a closed point $v \in (x_j \ne 0)$, we define the *multiplicity* of M at v as the multiplicity at a point of $\rho_j^{-1}(v)$ of the closed subscheme $(I_M = 0) \cap U_j$ of U_j, which does not depend on the choice of the point of $\rho_j^{-1}(v)$ or of j [Tot22, Section 4]. We extend the multiplicity linearly to cycles with rational coefficients.

Here $(I_M = 0) \cap U_j$ is reduced, because the affine cone over $M \cap (x_j \neq 0)$ is the quotient of $\big((I_M = 0) \cap U_j\big) \times \mathbb{G}_m$ by a free action of μ_{a_j}. Thus the multiplicity of $(I_M = 0) \cap U_j$ at a point is the multiplicity of the corresponding cycle [Ful98, Example 4.3.4]. It does not depend on the choice of the point of $\rho_j^{-1}(v)$, because $\rho_j^{-1}(v)$ is a μ_{a_j}-orbit and $(I_M = 0) \cap U_j$ is μ_{a_j}-invariant.

A point of U_j is fixed by a non-trivial element of μ_{a_j} if and only if the image of the point in $\mathbb{P}$ has non-trivial stabilizer. For an open subset of a well formed $\mathbb{P}$, Lemma 2.24 below compares log canonicity with log canonicity on orbifold charts. We work on orbifold charts because the multiplicity of Definition 2.23, Theorem 2.28, and Lemma 2.7 are all used on the smooth varieties $\rho_j^{-1}(V)$.

Lemma 2.24. *Let $\mathbb{P} = \mathbb{P}(a_0, \dots, a_N)$ be well formed with orbifold charts $\rho_0, \dots, \rho_N$, let $V \subset \mathbb{P}$ be an open subset, and let $B \geq 0$ be a $\mathbb{Q}$-divisor on V. Then for any $0 \leq j \leq N$ and any closed point $v \in V \cap (x_j \neq 0)$, (V, B) is lc (resp. klt) near v if and only if $\big(\rho_j^{-1}(V), \rho_j^* B\big)$ is lc (resp. klt) near $\rho_j^{-1}(v)$. In particular, (V, B) is lc (resp. klt) if and only if $\big(\rho_j^{-1}(V), \rho_j^* B\big)$ is lc (resp. klt) for any j.*

Proof. Fix j and let $\zeta \in \mu_{a_j}$ with $\zeta \neq 1$. The fixed locus of ζ on U_j is the linear subspace defined by $x_i = 0$ for all $i \neq j$ with $\zeta^{a_i} \neq 1$. If the fixed locus of ζ contains a hyperplane $(x_i = 0)$, then $\zeta^{a_k} = 1$ for any $k \neq i$, where $\zeta^{a_j} = 1$ holds since $\zeta \in \mu_{a_j}$. Since $\gcd(a_0, \dots, a_{i-1}, a_{i+1}, \dots, a_N) = 1$, this implies $\zeta = 1$, a contradiction. Thus μ_{a_j} acts freely on U_j outside a closed subset of codimension at least 2, so ρ_j is étale in codimension 1 and the ramification index of ρ_j along any prime divisor on U_j is 1. Since $\mathbb{P}$ has only quotient singularities, $K_V + B$ and B are $\mathbb{Q}$-Cartier [KM98, Proposition 5.15]. By Lemma 2.8(2) applied to $\rho_j : \rho_j^{-1}(V) \to V \cap (x_j \neq 0)$,

$$K_{\rho_j^{-1}(V)} + \rho_j^* B = \rho_j^*(K_V + B).$$

By Lemma 2.8(3), for any closed point $v \in V \cap (x_j \neq 0)$, (V, B) is lc (resp. klt) near v if and only if $\big(\rho_j^{-1}(V), \rho_j^* B\big)$ is lc (resp. klt) near $\rho_j^{-1}(v)$. Thus the first equivalence holds. Since the open subsets $(x_j \neq 0)$ cover $\mathbb{P}$, the second equivalence follows. □

Definition 2.25. Let $\mathbb{P} = \mathbb{P}(a_0, \dots, a_N)$. For an irreducible closed subvariety $M \subset \mathbb{P}$ of dimension r with homogeneous ideal $I_M \subset \mathbb{C}[x_0, \dots, x_N]$, we define the *degree* of M as

$$\deg M := \lim_{t \to 1} (1 - t)^{r+1} \sum_{m \geq 0} \dim_{\mathbb{C}} \big(\mathbb{C}[x_0, \dots, x_N]/I_M\big)_m t^m,$$

and we extend the degree linearly to cycles with rational coefficients.

Lemma 2.26. *Let $M \subset \mathbb{P}(a_0, \dots, a_N)$ be an irreducible closed subvariety of dimension r. Let*

$$\mathcal{P}(a_0, \dots, a_N) := [(\mathbb{A}^{N+1} \backslash \{0\})/\mathbb{G}_m]$$

be the weighted projective stack, where $\mathbb{G}_m$ acts by $\lambda \cdot x_i = \lambda^{a_i} x_i$, with its line bundle $\mathcal{O}(1)$ [Tot22, Section 1], and let $\mathcal{M} \subset \mathcal{P}(a_0, \dots, a_N)$ be the closed substack defined by the homogeneous ideal of M. Then

$$\deg M = \int_{\mathcal{M}} c_1(\mathcal{O}(1))^r,$$

that is, $\deg M$ is the degree of $\mathcal{M}$ in the sense of [Tot22, Section 1].

Proof. Let $R := \mathbb{C}[x_0, \dots, x_N]/I_M$, let s be the greatest common divisor of the weights a_i of the coordinates $x_i \notin I_M$, and let $L := \operatorname{lcm}(a_0, \dots, a_N)$. A general point of M has stabilizer μ_s, so μ_s is the stabilizer of a general point of $\mathcal{M}$. Since R is generated by the images of the $x_i \notin I_M$, $R_k = 0$ unless $s \mid k$. Let $\mathcal{O}_M(L) := \mathcal{O}_{\mathbb{P}}(L)|_M$, which is an ample line bundle on M, since $\mathcal{O}_{\mathbb{P}}(L)$ is the pullback of $\mathcal{O}_{\mathbb{P}^N}(1)$ by the finite morphism $\phi : \mathbb{P} \to \mathbb{P}^N$ defined by $x_0^{L/a_0}, \dots, x_N^{L/a_N}$ [Har77, Exercise III.5.7(a)(d)].

We compute the right-hand side. Let $\pi : \mathcal{M} \to M$ be the morphism from $\mathcal{M}$ to its coarse moduli space, which is M. The line bundle $\mathcal{O}(L) = \mathcal{O}(1)^{\otimes L}$ on $\mathcal{M}$ is $\pi^* \mathcal{O}_M(L)$. Since $\mathcal{M}$ is

integral with generic stabilizer μ_s, $\pi_*[\mathcal{M}] = \frac{1}{s}[M]$, and by the projection formula [Vis89, Section 1 and Proposition 6.1],

$$\int_{\mathcal{M}} c_1(\mathcal{O}(1))^r = \frac{1}{L^r}\int_{\mathcal{M}} \pi^* c_1(\mathcal{O}_M(L))^r = \frac{(\mathcal{O}_M(L)^r)}{sL^r}.$$

We compute the left-hand side. Let $T := \mathbb{C}[y_0, \dots, y_N]$ with $\deg y_j = 1$, and let $R^{(L)} := \bigoplus_{m\geq 0} R_{Lm}$, graded with R_{Lm} in degree m, which is a graded T-module via $y_j \to x_j^{L/a_j}$. Via $y_j \to x_j^{L/a_j}$, R is a finite T-module and $R^{(L)}$ is a direct summand of R as a T-module, so $R^{(L)}$ is a finitely generated T-module, and the coherent sheaf on $\mathbb{P}^N = \operatorname{Proj} T$ associated to $R^{(L)}$ is $\phi_*\mathcal{O}_M$. Thus, for $m \gg 0$, $\dim R_{Lm} = h^0(M, \mathcal{O}_M(L)^{\otimes m})$ [Har77, Exercise II.5.9(b)], and by Serre vanishing for the ample line bundle $\mathcal{O}_M(L)$ [Har77, Proposition III.5.3] and the Riemann–Roch theorem [KM98, Theorem 1.36],

$$\dim R_{Lm} = \frac{(\mathcal{O}_M(L)^r)}{r!}m^r + O(m^{r-1}).$$

The weights a_i/s of the coordinates $x_i \notin I_M$ have greatest common divisor 1, so there exists an integer k_0 such that $R_{sk} \neq 0$ for any $k \geq k_0$, since R is a domain. Fix an integer w with $Lw \geq sk_0 + L$. Since $s \mid L$, for any integer $n \geq 0$ we may write $sn = Lm + q$ with $0 \leq q < L$ and $s \mid q$. Multiplication by non-zero elements of R_{Lw+q} and of R_{Lw-q} is injective, so

$$\dim R_{L(m-w)} \leq \dim R_{sn} \leq \dim R_{L(m+w)}.$$

Thus $\dim R_{sn} = \frac{\kappa}{r!}n^r + O(n^{r-1})$, where $\kappa := \frac{s^r}{L^r}(\mathcal{O}_M(L)^r)$. Since $\sum_{n\geq 0}\binom{n+r}{r}\tau^n = (1-\tau)^{-r-1}$, we get

$$\lim_{\tau\to 1}(1-\tau)^{r+1}\sum_{n\geq 0}\dim R_{sn}\tau^n = \kappa.$$

Let $\tau := t^s$. Then $(1-t)^{r+1} = (1-\tau)^{r+1}\left(\frac{1-t}{1-t^s}\right)^{r+1}$ and $\frac{1-t}{1-t^s} \to \frac{1}{s}$ as $t \to 1$, so

$$\deg M = \lim_{t\to 1}(1-t)^{r+1}\sum_{n\geq 0}\dim R_{sn}t^{sn} = \frac{\kappa}{s^{r+1}} = \frac{(\mathcal{O}_M(L)^r)}{sL^r}.$$

This is the right-hand side computed above.

For example, the j-th coordinate point has $R = \mathbb{C}[x_j]$ and degree $\lim_{t\to 1}\frac{1-t}{1-t^{a_j}} = \frac{1}{a_j}$. An irreducible hypersurface $(F = 0)$ with $\deg F = e$ has $R = \mathbb{C}[x_0, \dots, x_N]/(F)$ and degree

$$\lim_{t\to 1}(1-t)^N\frac{1-t^e}{\prod_{j=0}^N(1-t^{a_j})} = \frac{e}{a_0\cdots a_N}. \quad \square$$

Lemma 2.27. *Let $F_1, \dots, F_r \in \mathbb{C}[x_0, \dots, x_N]$ be weighted homogeneous polynomials of degrees $e_1, \dots, e_r$ forming a regular sequence, where $\deg x_i = a_i$ and $0 \leq r \leq N$. Let $R := \mathbb{C}[x_0, \dots, x_N]/(F_1, \dots, F_r)$, and let*

$$Z := \sum_{\mathfrak{p}} \ell(R_{\mathfrak{p}})V(\mathfrak{p}),$$

where $\mathfrak{p}$ runs over the minimal prime ideals of R, $\ell(R_{\mathfrak{p}})$ is the length of the Artinian local ring $R_{\mathfrak{p}}$, and $V(\mathfrak{p}) \subset \mathbb{P}(a_0, \dots, a_N)$ is the closed subvariety defined by $\mathfrak{p}$. Then

$$\deg Z = \frac{e_1\cdots e_r}{a_0\cdots a_N}.$$

Proof. Since $F_1, \dots, F_r$ is a regular sequence, the Hilbert series of R is

$$\sum_{k\geq 0}\dim R_k t^k = \frac{\prod_{i=1}^r(1-t^{e_i})}{\prod_{j=0}^N(1-t^{a_j})},$$

so

$$\lim_{t\to 1}(1-t)^{N+1-r}\sum_{k\geq 0}\dim R_k t^k = \frac{e_1\cdots e_r}{a_0\cdots a_N}.$$

The ring R is Cohen–Macaulay [BH98, Theorem 2.1.3(a)], so (localize at the homogeneous maximal ideal and use [BH98, Theorem 2.1.2(a) and Lemma 1.5.6(b)(ii)]) the minimal prime ideals $\mathfrak{p}$ of R are homogeneous and satisfy $\dim R/\mathfrak{p} = \dim R = N+1-r \geq 1$. In particular, $V(\mathfrak{p})$ is a closed subvariety of dimension $N-r$ whose homogeneous ideal is the preimage of $\mathfrak{p}$ in $\mathbb{C}[x_0,\dots,x_N]$.

Since the associated prime ideals of a finitely generated graded R-module are homogeneous and are annihilators of homogeneous elements [BH98, Lemma 1.5.6(b)(ii)], there exists a filtration $0 = Q_0 \subset Q_1 \subset \cdots \subset Q_\ell = R$ by graded R-submodules such that $Q_k/Q_{k-1} \cong (R/\mathfrak{q}_k)(-n_k)$ for homogeneous prime ideals $\mathfrak{q}_k$ and integers n_k. Localizing at a minimal prime ideal $\mathfrak{p}$, the number of k with $\mathfrak{q}_k = \mathfrak{p}$ is $\ell(R_\mathfrak{p})$. Any other $\mathfrak{q}_k$ strictly contains a minimal prime ideal, so $d_k := \dim R/\mathfrak{q}_k < N+1-r$. If $d_k = 0$, then $\dim_\mathbb{C} R/\mathfrak{q}_k < +\infty$. If $d_k \geq 1$, then $R/\mathfrak{q}_k$ is a finite module over a polynomial ring in d_k homogeneous elements of positive degree [BH98, Theorem 1.5.17], so $\dim(R/\mathfrak{q}_k)_m = O(m^{d_k-1})$ and $\sum_m \dim(R/\mathfrak{q}_k)_m t^m = O\big((1-t)^{-d_k}\big)$ as $t\to 1$. In both cases,

$$\lim_{t\to 1}(1-t)^{N+1-r} t^{n_k}\sum_{m\geq 0}\dim(R/\mathfrak{q}_k)_m t^m = 0.$$

Since the Hilbert series is additive in the filtration and $t^{n_k}\to 1$, Definition 2.25 implies

$$\lim_{t\to 1}(1-t)^{N+1-r}\sum_{k\geq 0}\dim R_k t^k = \sum_{\mathfrak{p}}\ell(R_\mathfrak{p})\deg V(\mathfrak{p}) = \deg Z$$

(cf. [BH98, Corollary 4.7.8]). Comparing with the first limit, we obtain $\deg Z = \frac{e_1\cdots e_r}{a_0\cdots a_N}$. □

By Lemma 2.26, the degree in Definition 2.25 is the degree of the corresponding substack. In Theorem 2.28 and Corollary 2.29, the multiplicity is that of Definition 2.23 and the degree is that of Definition 2.25. Theorem 2.28 is the only tool in this paper that bounds the multiplicity of an arbitrary cycle of a weighted projective space, not only of a coordinate hyperplane.

Theorem 2.28 ([Tot22, Theorem 4.3])**.** *Let $a_0 \geq \cdots \geq a_N$ be positive integers and $M \subset \mathbb{P}(a_0,\dots,a_N)$ an irreducible closed subvariety of dimension r. Then the multiplicity of M at any closed point of $\mathbb{P}(a_0,\dots,a_N)$ is at most $a_0\cdots a_r \deg M$. If moreover $M \not\subset (x_N = 0)$, then the multiplicity of M at any closed point of $\mathbb{P}(a_0,\dots,a_N)$ is at most $a_0\cdots a_{r-1}a_N \deg M$.*

The first bound goes back to [JK01, Proposition 11].

Corollary 2.29. *Let $a_0 \geq \cdots \geq a_N$ be positive integers, and let $M = \sum_i c_i M_i$ be a cycle on $\mathbb{P}(a_0,\dots,a_N)$ with rational coefficients $c_i \geq 0$ whose components M_i have dimension r. Then the multiplicity of M at any closed point of $\mathbb{P}(a_0,\dots,a_N)$ is at most $a_0\cdots a_r \deg M$. If moreover no M_i is contained in $(x_N = 0)$, then the multiplicity of M at any closed point of $\mathbb{P}(a_0,\dots,a_N)$ is at most $a_0\cdots a_{r-1}a_N \deg M$.*

Proof. The multiplicity at a point and the degree are linear in the cycle, and $c_i \geq 0$ for any i. Thus both bounds follow by Theorem 2.28 for each M_i. □

3. A local criterion, weighted blow-ups, and monomial bases

In this section, we prove two lemmas on singularities of pairs and a lemma on graded hypersurface rings, which we use in Sections 4, 6, and 7.

To check that (X, B) is lc near a closed point x for all divisors B in a family, one needs in general to control all prime divisors over x. The following lemma shows that if E is the exceptional divisor of a plt blow-up at x and the pair (E, Δ_E) has alpha invariant at least one, then one inequality along E suffices.

Lemma 3.1. *Let X be a normal quasi-projective variety and $x \in X$ a closed point such that X is klt near x. Let $f : Y \to X$ be a plt blow-up of X at x with exceptional divisor E, and assume that $\alpha(E, \Delta_E) \geq 1$. Then for any $\mathbb{Q}$-divisor $B \geq 0$ on X such that B is $\mathbb{Q}$-Cartier near x and $\operatorname{mult}_E f^*B \leq a(E, X, 0)$, the pair (X, B) is lc near x.*

Proof. After shrinking X to an open neighborhood of x, we may assume that X is klt and B is $\mathbb{Q}$-Cartier. Let $a := a(E, X, 0)$ and $b := \operatorname{mult}_E f^*B$. Then $a > 0$ and $0 \leq b \leq a$. Let B_Y be the strict transform of B on Y. Since the exceptional locus of f is E, we have

$$f^*B = B_Y + bE \quad \text{and} \quad K_Y + (1-a)E = f^*K_X.$$

Thus

$$K_Y + B_Y + qE = f^*(K_X + B), \quad \text{where } q := 1 - a + b \leq 1.$$

In particular, $\operatorname{Diff}_E(B_Y) = \Delta_E + B_Y|_E$ [Kol13, (4.2.10)], since $B_Y = f^*B - bE$ and $K_Y + E = f^*K_X + aE$ are $\mathbb{Q}$-Cartier.

By Definition-Proposition 2.14, E is normal, (E, Δ_E) is klt, and $-(K_E + \Delta_E) \sim_{\mathbb{Q}} -aE|_E$ is ample. Since f is projective and $f(E) = \{x\}$, E is projective and $(f^*B)|_E \sim_{\mathbb{Q}} 0$. Thus

$$B_Y|_E \sim_{\mathbb{Q}} -bE|_E \sim_{\mathbb{Q}} t\big(-(K_E + \Delta_E)\big), \quad \text{where } t := \frac{b}{a} \in [0, 1].$$

We show that $(E, \Delta_E + B_Y|_E)$ is lc. If $t = 0$, then $B_Y|_E \geq 0$ is a $\mathbb{Q}$-divisor on the projective variety E such that $B_Y|_E \sim_{\mathbb{Q}} 0$, so $B_Y|_E = 0$, and (E, Δ_E) is klt. If $t > 0$, then $D := t^{-1}B_Y|_E \geq 0$ and $D \sim_{\mathbb{Q}} -(K_E + \Delta_E)$. Since $\alpha(E, \Delta_E) \geq 1$, Definitions 2.3 and 2.10 imply that $(E, \Delta_E + sD)$ is lc for any $0 \leq s < 1$. Since the log discrepancies $a(F, E, \Delta_E + sD)$ are affine functions of s, $(E, \Delta_E + sD)$ is lc for any $0 \leq s \leq 1$, and in particular for $s = t$. Thus $(E, \operatorname{Diff}_E(B_Y))$ is lc.

Since E and B_Y have no common component and $K_Y + E + B_Y$ is $\mathbb{Q}$-Cartier, Theorem 2.6 implies that $(Y, E + B_Y)$ is lc near E. Since $q \leq 1$ and E is $\mathbb{Q}$-Cartier, $(Y, B_Y + qE)$ is sub-lc near E. Since f is proper, after shrinking X to an open neighborhood of x, $(Y, B_Y + qE)$ is sub-lc. Since $f_*(B_Y + qE) = B$, [KM98, Lemma 2.30] implies that

$$a(F, X, B) = a(F, Y, B_Y + qE) \geq 0$$

for any prime divisor F over X. Thus (X, B) is lc near x. □

Lemma 3.1 is a variant of [LZ22, Lemma 5.3], where the singularity at x is assumed to be weakly exceptional in place of the bound on the alpha invariant of (E, Δ_E). We assume the bound on the alpha invariant since it is the condition that we check in our examples. By [LZ22, Theorem 2.7], which goes back to [Pro00] and [Kud01, Theorem 2.1], the bound on the alpha invariant implies that the singularity at x is weakly exceptional, but the proof above does not use [LZ22, Theorem 2.7].

For positive integers $w_1, \dots, w_m$, we use the weighted blow-up of $\mathbb{A}^m$ with weights $w_1, \dots, w_m$ as defined in [Pro01, Section 3.2]. The next lemma shows that when the lowest weight part of the equation of a hypersurface singularity is quasismooth, the weighted blow-up is a plt blow-up whose exceptional divisor and log discrepancy can be read off from the weights.

Lemma 3.2. *Let $m \geq 3$, let $w_1, \dots, w_m$ be positive integers with $\gcd(w_1, \dots, w_m) = 1$, and let $F \in \mathbb{C}[x_1, \dots, x_m]$ be a polynomial with $F(0) = 0$. For a non-zero polynomial $\varphi \in \mathbb{C}[x_1, \dots, x_m]$, let $w(\varphi)$ be the minimum of the weights $\sum_i \gamma_i w_i$ of the monomials $x_1^{\gamma_1} \cdots x_m^{\gamma_m}$ of φ, and let φ_w be the sum of the terms of φ of weight $w(\varphi)$. Let $\delta := w(F)$, and assume that F_w is quasismooth, that F_w is not of the form cx_i with $c \in \mathbb{C}$ and $1 \leq i \leq m$, and that $w_1 + \cdots + w_m > \delta$. Let $X := (F = 0) \subset \mathbb{A}^m$, and let $f : Y \to X$ be the morphism from the strict transform Y of X in the weighted blow-up of $\mathbb{A}^m$ with weights $w_1, \dots, w_m$. Then, after shrinking X to an open neighborhood of 0, the following hold.*

(1) X is normal and klt, and f is a plt blow-up of X at 0.
(2) The exceptional divisor E of f is isomorphic to the hypersurface $(F_w = 0)$ in $\mathbb{P}(w_1, \dots, w_m)$.

(3) $a(E, X, 0) = w_1 + \cdots + w_m - \delta$.
(4) *For any polynomial* $\varphi \in \mathbb{C}[x_1, \ldots, x_m]$ *such that* φ_w *is not divisible by* F_w, *we have* $\varphi|_X \neq 0$ *and* $\mathrm{mult}_E(\varphi|_X = 0) = w(\varphi)$.

Proof. Let $g : W \to \mathbb{A}^m$ be the weighted blow-up with weights $w_1, \ldots, w_m$, and let E_W be the exceptional divisor of g. Then g is a projective morphism which is an isomorphism over $\mathbb{A}^m\backslash\{0\}$, Y is the closure of $g^{-1}(X\backslash\{0\})$ in W, and $f = g|_Y$. We use the following description of W [Pro01, Section 3.2]. For $1 \leq i \leq m$, let $U_i := \mathbb{A}^m$ with coordinates $y_1, \ldots, y_m$, on which $\zeta \in \mu_{w_i}$ acts by $\zeta \cdot y_i = \zeta^{-1} y_i$ and $\zeta \cdot y_j = \zeta^{w_j} y_j$ for $j \neq i$. Then W is covered by open subsets $W_1, \ldots, W_m$ such that W_i is the quotient of U_i by μ_{w_i} with quotient morphism $q_i : U_i \to W_i$, and $h_i := g \circ q_i : U_i \to \mathbb{A}^m$ is given by

$$x_i = y_i^{w_i} \quad \text{and} \quad x_j = y_j y_i^{w_j} \text{ for } j \neq i.$$

Moreover, $E_W \cong \mathbb{P}(w_1, \ldots, w_m)$, $E_W = g^{-1}(0)$ as sets, $q_i^{-1}(E_W) = (y_i = 0)$, and the morphism $(y_i = 0) \to E_W \cap W_i$ induced by q_i is the i-th orbifold chart ρ_i of $\mathbb{P}(w_1, \ldots, w_m)$ after identifying $(y_i = 0)$ with $(x_i = 1) \subset \mathbb{A}^m$ by $x_j = y_j$ for $j \neq i$: for $y \in U_i$ with $y_i \neq 0$, the point $h_i(y) = (y_1 y_i^{w_1}, \ldots, y_i^{w_i}, \ldots, y_m y_i^{w_m})$ is on the $\mathbb{C}^*$-orbit of $(y_1, \ldots, 1, \ldots, y_m)$ for the action with weights $w_1, \ldots, w_m$, and as $y_i \to 0$, the point $q_i(y) \in W$ tends to the point

$$[y_1 : \cdots : 1 : \cdots : y_m] \in \mathbb{P}(w_1, \ldots, w_m).$$

For any positive integer b divisible by $\mathrm{lcm}(w_1, \ldots, w_m)$, bE_W is Cartier and

$$\mathcal{O}_{E_W}(-bE_W) \cong \mathcal{O}_{\mathbb{P}(w_1,\ldots,w_m)}(b)$$

is ample. Since E_W is the only positive-dimensional fiber of g, $-E_W$ is a $\mathbb{Q}$-Cartier g-ample divisor. An element $\zeta \in \mu_{w_i}$ fixes the generic point of $(y_i = 0)$ if and only if $\zeta^{w_j} = 1$ for any $1 \leq j \leq m$, that is, if and only if $\zeta = 1$, as $\gcd(w_1, \ldots, w_m) = 1$. Thus $q_i^* E_W = (y_i = 0)$.

Step 1. In this step, we describe Y on the charts U_i.

For a monomial $x^\gamma = x_1^{\gamma_1} \cdots x_m^{\gamma_m}$, we have

$$x^\gamma \circ h_i = y_i^{\sum_j \gamma_j w_j} \prod_{j \neq i} y_j^{\gamma_j}.$$

Thus for any non-zero polynomial φ,

$$\varphi \circ h_i = y_i^{w(\varphi)} \big(\varphi_w(y_1, \ldots, 1, \ldots, y_m) + y_i \Phi_i\big) \tag{3.1}$$

for some polynomial Φ_i, where 1 is in the i-th place. Two distinct monomials of weight $w(\varphi)$ remain distinct after substituting $x_i = 1$, so $\varphi_w(y_1, \ldots, 1, \ldots, y_m) \neq 0$. Applying (3.1) to $\varphi = F$, we write $F \circ h_i = y_i^\delta G_i$, where G_i is a polynomial such that

$$G_i|_{(y_i=0)} = F_w(y_1, \ldots, 1, \ldots, y_m) \neq 0.$$

Thus the hypersurface $(G_i = 0) \subset U_i$ has no component contained in $(y_i = 0)$. Since $h_i^{-1}(0) = (y_i = 0)$, we have $h_i^{-1}(X\backslash\{0\}) = (G_i = 0)\backslash(y_i = 0)$. Since q_i is open and surjective, $q_i^{-1}(Y \cap W_i)$ is the closure of $(G_i = 0)\backslash(y_i = 0)$ in U_i, hence $q_i^{-1}(Y \cap W_i) = (G_i = 0)$ as sets. Let $Y_i' := (G_i = 0) \subset U_i$ and

$$E_i' := Y_i' \cap (y_i = 0) = \big(y_i = F_w(y_1, \ldots, 1, \ldots, y_m) = 0\big).$$

We show that Y_i' is smooth at any point $P' \in E_i'$. Let $Q \in \mathbb{A}^m$ be the point with $x_i = 1$ and $x_j = y_j(P')$ for $j \neq i$. Then $Q \neq 0$ and $F_w(Q) = 0$. Since F_w is quasismooth, $\frac{\partial F_w}{\partial x_l}(Q) \neq 0$ for some l. By the Euler formula $\sum_l w_l x_l \frac{\partial F_w}{\partial x_l} = \delta F_w$, we have

$$w_i \frac{\partial F_w}{\partial x_i}(Q) + \sum_{j \neq i} w_j y_j(P') \frac{\partial F_w}{\partial x_j}(Q) = 0,$$

so $\frac{\partial F_w}{\partial x_j}(Q) \neq 0$ for some $j \neq i$. Since

$$\frac{\partial G_i}{\partial y_j}(P') = \frac{\partial F_w}{\partial x_j}(Q) \quad \text{for } j \neq i,$$

the differentials of y_i and G_i at P' are linearly independent. Thus Y_i' is smooth near E_i', E_i' is a smooth divisor on Y_i', and $y_i|_{Y_i'}$ vanishes to order one along any component of E_i'.

Step 2. In this step, we prove (2) and show that $q_i|_{Y_i'}$ is unramified along any component of E_i'.

By **Step 1**, $q_i^{-1}(Y \cap E_W \cap W_i) = E_i'$. Since ρ_i maps the hypersurface $(F_w(y_1, \dots, 1, \dots, y_m) = 0)$ in $(y_i = 0)$ onto

$$(F_w = 0) \cap (x_i \neq 0) \subset \mathbb{P}(w_1, \dots, w_m),$$

we have $Y \cap E_W = (F_w = 0)$ as subsets of $E_W \cong \mathbb{P}(w_1, \dots, w_m)$. By Lemma 2.21, F_w is irreducible, so $(F_w = 0) \subset \mathbb{P}(w_1, \dots, w_m)$ is a variety and

$$\dim(F_w = 0) = m - 2 = \dim Y - 1.$$

Since g is an isomorphism over $\mathbb{A}^m \backslash \{0\}$, the exceptional locus of f is the prime divisor $E := Y \cap E_W$ with the reduced structure, $f(E) = \{0\}$, and (2) holds.

If $(F_w = 0) \subset (x_l = 0)$ for some $1 \leq l \leq m$, then x_l vanishes on the affine hypersurface $(F_w = 0) \subset \mathbb{A}^m$, so $x_l \in (F_w)$ as F_w is prime, and $F_w = cx_l$ for some $c \in \mathbb{C}$, which is excluded. Thus $(F_w = 0) \not\subset (x_l = 0)$ for any $1 \leq l \leq m$. In particular, $E_i' \neq \emptyset$ for any i.

Let ℓ be a prime number. By Definition 2.15, the points of $\mathbb{P}(w_1, \dots, w_m)$ whose stabilizer contains μ_ℓ form the subset

$$\Lambda_\ell := \bigcap_{\ell \nmid w_l} (x_l = 0),$$

and $\ell \nmid w_l$ for at least one l since $\gcd(w_1, \dots, w_m) = 1$. If $\ell \nmid w_l$ for at least two indices l, then $\operatorname{codim} \Lambda_\ell \geq 2$, so $E \not\subset \Lambda_\ell$. If $\ell \nmid w_l$ for exactly one index l, then $\Lambda_\ell = (x_l = 0)$ and $E \not\subset \Lambda_\ell$ by the previous paragraph. Thus the generic point of E has trivial stabilizer. For a point of $(y_i = 0)$, the stabilizer in μ_{w_i} is the stabilizer of the image under ρ_i in the sense of Definition 2.15. Thus the generic point of any component of E_i' has trivial stabilizer in μ_{w_i}, and $q_i|_{Y_i'}$ is unramified along any component of E_i'. Moreover, since $E_i'/\mu_{w_i} \cong E \cap W_i$ is irreducible, μ_{w_i} permutes the components of E_i' transitively.

Step 3. In this step, we shrink X and show that Y is normal and $-E$ is f-ample.

Since the Jacobian determinant of h_i is $w_i y_i^{\sum_j w_j - 1}$, h_i is étale over $(y_i \neq 0)$. Since $F \circ h_i = y_i^\delta G_i$, a point $P' \in Y_i' \backslash E_i'$ is a smooth point of Y_i' if and only if $h_i(P')$ is a smooth point of X. Let Σ be the closure in X of the singular locus of $X \backslash \{0\}$, and suppose that $0 \in \Sigma$. Since g is proper, the closure of $g^{-1}(\Sigma \backslash \{0\})$ in W contains a point of $E_W \cap W_i$ for some i. Since q_i is open and surjective, the closure of $h_i^{-1}(\Sigma \backslash \{0\})$ in U_i contains a point $P' \in (y_i = 0)$. Since $h_i^{-1}(\Sigma \backslash \{0\})$ is contained in the singular locus of Y_i', P' is a singular point of Y_i' and $P' \in E_i'$, which contradicts **Step 1**. Thus $0 \notin \Sigma$. Possibly replacing X with $X \backslash \Sigma$, we may assume that $X \backslash \{0\}$ is smooth, so Y_i' is smooth for any i.

Since X is a hypersurface, X is Cohen–Macaulay. Since $X \backslash \{0\}$ is smooth and $\dim X = m - 1 \geq 2$, X is regular in codimension one. By Serre's criterion, X is normal. Possibly shrinking X further, we may assume that X is irreducible. Since $Y \cap W_i = Y_i'/\mu_{w_i}$ and Y_i' is smooth, Y is normal and has only quotient singularities, so Y is $\mathbb{Q}$-factorial [KM98, Proposition 5.15]. By **Step 2**, $q_i^* E = E_i'$. By **Step 1**, $q_i^*(E_W|_Y) = (y_i = 0)|_{Y_i'} = E_i'$. Thus $E_W|_Y = E$, and $-E$ is f-ample.

Step 4. In this step, we prove (3).

Let ω be the residue on X of $dx_1 \wedge \cdots \wedge dx_m / F$. By the adjunction formula [Har77, Proposition II.8.20], ω generates ω_X on the smooth locus $X \backslash \{0\}$. Since X is normal and

$\dim X \geq 2$, ω generates ω_X, so $K_X = 0$. Let $a := a(E, X, 0)$. Then $K_Y + (1-a)E = f^*K_X = 0$. Since h_i is étale over $(y_i \neq 0)$, we have

$$h_i^*\omega = w_i y_i^{\sum_j w_j - 1 - \delta}\omega_i \quad \text{on } Y_i' \backslash E_i',$$

where ω_i is the residue on Y_i' of $dy_1 \wedge \cdots \wedge dy_m/G_i$. By the adjunction formula [Har77, Proposition II.8.20], ω_i generates $\omega_{Y_i'}$. Since $y_i|_{Y_i'}$ vanishes to order one along any component of E_i', the divisor of $h_i^*\omega$ is

$$K_{Y_i'} = \Big(\sum_j w_j - \delta - 1\Big)E_i'.$$

Let $B' := -K_{Y_i'}$. Then $K_{Y_i'} + B' = q_i^*(K_Y + (1-a)E) = 0$, where $q_i|_{Y_i'} : Y_i' \to Y \cap W_i$ is finite and surjective. Let P' be a component of E_i'. Then $\operatorname{mult}_{P'} B' = 1 + \delta - \sum_j w_j$. By **Step 2**, the ramification index of $q_i|_{Y_i'}$ along P' is 1, so $\operatorname{mult}_{P'} B' = 1 - a$ by Lemma 2.8(2) (applied to each connected component of Y_i', each of which maps finitely onto $Y \cap W_i$). Thus $a = w_1 + \cdots + w_m - \delta$, which is (3).

Step 5. In this step, we prove (1).

By **Step 3**, $K_Y + E$ is $\mathbb{Q}$-Cartier. Let B_i' be the $\mathbb{Q}$-divisor on Y_i' such that $K_{Y_i'} + B_i' = q_i^*(K_Y + E)$. By Lemma 2.8(2) (applied to each connected component of Y_i', each of which maps finitely onto $Y \cap W_i$), $\operatorname{mult}_{P'} B_i' = 1$ for any component P' of E_i', and $\operatorname{mult}_D B_i' = 1 - r_D \leq 0$ for any other prime divisor D on Y_i', where r_D is the ramification index of $q_i|_{Y_i'}$ along D. Thus $B_i' \leq E_i'$. Since Y_i' is smooth and E_i' is a smooth divisor on Y_i', $a(F', Y_i', E_i') \geq 1$ for any exceptional prime divisor F' over Y_i' [KM98, Corollary 2.31(3)], so $a(F', Y_i', B_i') \geq 1$. By [KM98, Proposition 5.20(2)] (applied, as Lemma 2.8(2) above, to each connected component of Y_i'),

$$(\deg q_i|_{Y_i'}) \cdot \inf_F a(F, Y \cap W_i, E) \geq \inf_{F'} a(F', Y_i', B_i') \geq 1,$$

where F (resp. F') runs over the exceptional prime divisors over $Y \cap W_i$ (resp. Y_i'). Thus $(Y \cap W_i, E)$ is plt for any i [KM98, Definition 2.34], so (Y, E) is plt.

By (3) and our assumption, $a = a(E, X, 0) > 0$. For any exceptional prime divisor F over Y, $a(F, Y, (1-a)E) = a(F, Y, E) + a \operatorname{mult}_F E > 0$. Moreover, $a(E, Y, (1-a)E) = a > 0$, and any other prime divisor on Y has log discrepancy 1 with respect to $(Y, (1-a)E)$. Since $K_Y + (1-a)E = f^*K_X$, [KM98, Lemma 2.30] implies that $a(F, X, 0) = a(F, Y, (1-a)E) > 0$ for any prime divisor F over X, so X is klt. By **Steps 2 and 3**, f is a projective birational morphism from the normal variety Y whose exceptional locus is the prime divisor E, $f(E) = \{0\}$, and $-E$ is a $\mathbb{Q}$-Cartier f-ample divisor. Thus f is a plt blow-up of X at 0, which is (1).

Step 6. In this step, we prove (4).

Let φ be a polynomial such that φ_w is not divisible by F_w, and let P' be a component of E_i'. We show that $\psi := \varphi_w(y_1, \ldots, 1, \ldots, y_m)$ does not vanish identically on P'. For $\zeta \in \mu_{w_i}$, we have $\zeta^{w_i} = 1$, so

$$\varphi_w(\zeta^{w_1} y_1, \ldots, 1, \ldots, \zeta^{w_m} y_m) = \zeta^{w(\varphi)}\psi,$$

and the zero set of ψ in $(y_i = 0)$ is μ_{w_i}-invariant. Suppose that ψ vanishes on P'. By **Step 2**, μ_{w_i} permutes the components of E_i' transitively, so ψ vanishes on E_i'. Thus φ_w vanishes on $(F_w = 0) \cap (x_i = 1) \subset \mathbb{A}^m$. Since φ_w and F_w are weighted homogeneous, φ_w vanishes on $(F_w = 0) \cap (x_i \neq 0)$. By **Step 2**, $(F_w = 0) \cap (x_i \neq 0)$ is dense in the irreducible hypersurface $(F_w = 0) \subset \mathbb{A}^m$, so $\varphi_w \in (F_w)$ as F_w is prime, a contradiction.

By (3.1) and **Step 1**, the order of $(\varphi \circ h_i)|_{Y_i'}$ along P' is $w(\varphi)$. In particular, $(\varphi \circ h_i)|_{Y_i'} \neq 0$, so $\varphi|_X \neq 0$. Since the ramification index of $q_i|_{Y_i'}$ along P' is 1 (**Step 2**) and $q_i^* f^*(\varphi|_X = 0)$ is the divisor of $(\varphi \circ h_i)|_{Y_i'}$, we have $\operatorname{mult}_E(\varphi|_X = 0) = w(\varphi)$, which is (4). □

In the next lemma, we study the maximal lowest weight of the polynomials representing a homogeneous element of a graded hypersurface ring: (1) shows that the maximal lowest weight of

a product is the sum of the maximal lowest weights of the factors, and (2) provides a monomial basis from which it can be read off term by term.

Lemma 3.3. *Let $P := \mathbb{C}[x_1, \dots, x_N]$ be graded by positive integer degrees, and let $w_1, \dots, w_N$ be non-negative integers. For a non-zero polynomial $s \in P$, let $w(s)$ be the minimum of the weights $\sum_i \gamma_i w_i$ of the monomials $x_1^{\gamma_1} \cdots x_N^{\gamma_N}$ of s, and let s_w be the sum of the terms of s of weight $w(s)$. Let $F \in P$ be a homogeneous polynomial such that F_w is prime, let $R := P/(F)$, and for a non-zero homogeneous element $s \in R$ let $b(s)$ be the supremum of $w(\tilde{s})$ over all homogeneous polynomials $\tilde{s} \in P$ representing s. Then the following hold.*

(1) For any non-zero homogeneous $s \in R$, the supremum $b(s)$ is attained. For all non-zero homogeneous $s, t \in R$, we have $st \neq 0$ and $b(st) = b(s) + b(t)$.

(2) Assume that F and F_w are both monic of degree e as polynomials in x_N. Then the monomials $x^\gamma = x_1^{\gamma_1} \cdots x_N^{\gamma_N}$ with $\gamma_N < e$ form a basis of R, and $b(s) = \min\{\sum_i \gamma_i w_i \mid c_\gamma \neq 0\}$ for any non-zero homogeneous $s = \sum_{\gamma_N < e} c_\gamma x^\gamma \in R$.

Proof. For an integer k, let P_k be the space of homogeneous polynomials of degree k. Let $s \in R$ be a non-zero homogeneous element of degree k. Then any homogeneous representative of s is a non-zero element of P_k. Since $\deg x_i > 0$ for any $1 \leq i \leq N$, $\dim P_k < +\infty$, so the weights of the monomials in P_k form a finite set, and $b(s)$ is attained.

We show that for any homogeneous representative $\tilde{s}$ of s,

$$w(\tilde{s}) = b(s) \quad \text{if and only if} \quad \tilde{s}_w \notin (F_w). \tag{3.2}$$

Suppose that $\tilde{s}_w = gF_w$ for some $g \in P$. Since $\tilde{s}_w$ and F_w are homogeneous for both the degree and the weight, possibly replacing g with the component of g of degree $k - \deg F$ and weight $w(\tilde{s}) - w(F)$, we may assume that g is homogeneous of degree $k - \deg F$ and of weight $w(\tilde{s}) - w(F)$. Then $\tilde{s} - gF = (\tilde{s} - \tilde{s}_w) - g(F - F_w)$ is a non-zero homogeneous representative of s, and any monomial of $\tilde{s} - gF$ has weight larger than $w(\tilde{s})$. Thus $b(s) > w(\tilde{s})$. Conversely, suppose that $w(\tilde{t}) > w(\tilde{s})$ for some homogeneous representative $\tilde{t}$ of s. Then $\tilde{t} - \tilde{s} = gF$ for some non-zero $g \in P$, $w(gF) = w(\tilde{s})$, and $(gF)_w = -\tilde{s}_w$. Since P is a domain, $(gF)_w = g_w F_w$, so $\tilde{s}_w \in (F_w)$. This proves (3.2).

(1) Let $s, t \in R$ be non-zero homogeneous elements, and let $\tilde{s}, \tilde{t}$ be homogeneous representatives of s, t such that $w(\tilde{s}) = b(s)$ and $w(\tilde{t}) = b(t)$. By (3.2), $\tilde{s}_w, \tilde{t}_w \notin (F_w)$. Since F_w is prime, $\tilde{s}_w \tilde{t}_w \notin (F_w)$. Since P is a domain, $(\tilde{s}\tilde{t})_w = \tilde{s}_w \tilde{t}_w$. If $st = 0$, then $\tilde{s}\tilde{t} = gF$ for some $g \in P$, and $(\tilde{s}\tilde{t})_w = g_w F_w \in (F_w)$, a contradiction. Thus $st \neq 0$. By (3.2),

$$b(st) = w(\tilde{s}\tilde{t}) = w(\tilde{s}) + w(\tilde{t}) = b(s) + b(t).$$

(2) Since F is monic of degree e in x_N, for any non-zero $g \in P$ the degree of gF in x_N is at least e. By division by F as a polynomial in x_N, the monomials x^γ with $\gamma_N < e$ form a basis of R. Since F is homogeneous, any non-zero homogeneous $s \in R$ of degree k has a unique representative $\tilde{s} = \sum_{\gamma_N < e} c_\gamma x^\gamma$, and $\tilde{s}$ is homogeneous of degree k. Since F_w is monic of degree e in x_N, for any non-zero $g \in P$ the degree of gF_w in x_N is at least e, so the monomials x^γ with $\gamma_N < e$ are linearly independent modulo (F_w). Since $\tilde{s}_w$ is a non-zero linear combination of such monomials, $\tilde{s}_w \notin (F_w)$. By (3.2) and the definition of $w(\tilde{s})$, we have

$$b(s) = w(\tilde{s}) = \min\Big\{ \sum_i \gamma_i w_i \;\Big|\; c_\gamma \neq 0 \Big\},$$

which completes the proof of (2). □

4. Weighted hypersurfaces

In this section, we study the hypersurfaces in Set-up 4.1 below and prove Theorem 4.3. Set-up 4.1 allows any $d \geq 3$, but it has no member with $d = 3$ (Section 6). Section 5 constructs members with $d \geq 4$ and checks the hypotheses of Theorem 4.3.

Set-up 4.1. Let d, $a_1, \dots, a_d$, W, k, $a_{(1)}$, $a_{(2)}$, c, e, h, G, g, $\mathbb{P}$, R, X, p, H, and S be as follows.

(1) $d \geq 3$ is an integer, $a_1, \dots, a_d$ are positive integers, and $W := \mathbb{P}(a_1, \dots, a_d)$ is well formed.
(2) $k := a_1 + \cdots + a_d$, and $a_{(1)} \leq a_{(2)}$ are the smallest and the second smallest numbers among $a_1, \dots, a_d$, counted with multiplicity.
(3) $a_{(1)}a_{(2)} \geq k$.
(4) c is a positive integer such that $\frac{k}{d} < c < \frac{k}{d-1}$, and $e := k + c$.
(5) $h, G \in \mathbb{C}[x_1, \dots, x_d]$ are weighted homogeneous polynomials of degrees k and e respectively, where $\deg x_i = a_i$, such that h is quasismooth and G is not divisible by h, and $g := y^e + G$.
(6) $\mathbb{P} := \mathbb{P}(a_1, \dots, a_d, 1, c)$ with coordinates $x_1, \dots, x_d, y, z$ of degrees $a_1, \dots, a_d, 1, c$, $R := \mathbb{C}[x_1, \dots, x_d, y, z]/(zh + y^e + G)$, and $X := \operatorname{Proj} R \subset \mathbb{P}$.
(7) $p := [0 : \cdots : 0 : 1] \in X$, $H := \mathcal{O}_X(1)$, and $S := (h = 0) \subset W$.

Since $\deg h = \sum_i a_i$, we have $\omega_X \cong \mathcal{O}_X(-1)$ and $-K_X \sim H$ (Proposition 4.8). Condition (3) is used in Proposition 4.12 and Lemma 4.15: by Theorem 2.28, it implies that any $\mathbb{Q}$-divisor $D \geq 0$ on S of class $t\mathcal{O}_S(1)$ with $0 \leq t \leq 1$ has multiplicity at most 1. Since h is irreducible (Lemma 2.21), the condition $h \nmid G$ in (5) means that G does not vanish identically on S, so $(G = 0)|_S$ is a $\mathbb{Q}$-divisor on S, and Remark 4.7 below shows that G cannot be omitted.

Remark 4.2. Condition (4) of Set-up 4.1 is equivalent to $d - 1 < \frac{k}{c} < d$, and also to

$$\frac{d-1}{d} < \frac{k}{e} < \frac{d}{d+1}.$$

Indeed, since $e = k + c$, the inequality $\frac{d-1}{d} < \frac{k}{e}$ is equivalent to $(d-1)c < k$, and $\frac{k}{e} < \frac{d}{d+1}$ is equivalent to $k < dc$.

Theorem 4.3. *Notation and conditions as in Set-up 4.1. Assume moreover that $a_{(1)} \geq e/c$ and that the pair $\left(S, \frac{1}{c}(G = 0)|_S\right)$ is lc on orbifold charts. Then X is a $\mathbb{Q}$-Fano variety of dimension d such that*

$$\alpha(X) = \frac{k}{e} \quad \text{and} \quad \frac{d-1}{d} < \alpha(X) < \frac{d}{d+1},$$

and there exists a prime divisor E over X such that $\beta_X(E) < 0$. In particular, X is not K-semistable.

We prove Theorem 4.3 in Subsection 4.5. The hypothesis $a_{(1)} \geq e/c$ is used only in Lemma 4.15, and the lc condition on S only in Lemma 4.16, both of which are estimates away from p. Corollary 4.19 states a simpler condition on the weights under which both hypotheses hold, and Corollary 4.20 states conditions on h and G under which the lc condition on S holds.

4.1. **Normality and klt singularities.** In this subsection, we show that X is a $\mathbb{Q}$-Fano variety with section ring R. The singularities of X over the singular points of $(G = 0)|_S$ are not explicit, so we do not check the klt property point by point. Instead, the family of Construction 4.4 degenerates X to $X_0 = (zh + y^e = 0)$, whose singularities are explicit (Lemma 4.6), and the klt property passes from X_0 to X (Proposition 4.8).

Construction 4.4. Notation and conditions as in Set-up 4.1. For any $\tau \in \mathbb{C}$, let

$$X_\tau := (zh + y^e + \tau^{ce}G = 0) \subset \mathbb{P},$$

so that $X_1 = X$.

Lemma 4.5. *Notation and conditions as in Set-up 4.1. Then for any $\tau \in \mathbb{C}$, the locus of points of $\mathbb{P}$ with non-trivial stabilizer intersects X_τ in a closed subset of codimension at least 2 in X_τ.*

Proof. Let $v \in \mathbb{P}$ be a point with non-trivial stabilizer μ_m, where m is the greatest common divisor of the degrees of the non-zero coordinates of v, and let r be a prime factor of m. Since $\deg y = 1$, we have $y(v) = 0$, and $x_i(v) = 0$ for any i with $r \nmid a_i$. If r divided all but at most one of $a_1, \dots, a_d$, say all a_i with $i \neq i_0$, then r would divide $\gcd(a_i \mid i \neq i_0) = 1$, because W is well formed. Thus there exist $i_1 \neq i_2$ with $r \nmid a_{i_1}$ and $r \nmid a_{i_2}$, and $v \in (x_{i_1} = x_{i_2} = y = 0)$. Hence

the locus of points of $\mathbb{P}$ with non-trivial stabilizer is contained in the finite union of the closed subsets $(x_{i_1} = x_{i_2} = y = 0) \subset \mathbb{P}$ with $i_1 \neq i_2$, and

$$\dim(x_{i_1} = x_{i_2} = y = 0) = d - 2.$$

Since X_τ is a hypersurface in $\mathbb{P}$, any irreducible component of X_τ is of dimension d, so the intersection of X_τ with the locus of points with non-trivial stabilizer has codimension at least 2 in X_τ. □

Lemma 4.6. *Notation and conditions as in Set-up 4.1, and let* $R_0 := \mathbb{C}[x_1, \dots, x_d, y, z]/(zh+y^e)$. *Then* R_0 *is a normal domain, and* X_0 *is normal and klt.*

Proof. By Lemma 2.21, h is irreducible. Since h has positive degree and does not involve y, h does not divide y^e. Thus $zh + y^e$ is a primitive polynomial of degree one in z over the factorial ring $\mathbb{C}[x_1, \dots, x_d, y]$, hence irreducible by the Gauss lemma, and R_0 is a domain.

The ring R_0 is a hypersurface ring, hence Cohen–Macaulay [BH98, Theorem 2.1.3(a)], and in particular R_0 satisfies S_2. By the Jacobian criterion, the singular locus of $\operatorname{Spec} R_0$ is contained in

$$(h = y = 0) \cap \Big(z\frac{\partial h}{\partial x_1} = \cdots = z\frac{\partial h}{\partial x_d} = 0\Big).$$

Since h is quasismooth, the partial derivatives of h have no common zero on $(h = 0)$ outside $x_1 = \cdots = x_d = 0$. Thus the singular locus is contained in

$$(h = y = z = 0) \cup (x_1 = \cdots = x_d = y = 0),$$

whose irreducible components are of dimension $d - 1$ and 1. Since $\dim \operatorname{Spec} R_0 = d + 1$ and $1 \leq d - 1$, R_0 satisfies R_1. Hence R_0 is normal [Mat86, Theorem 23.8].

For a coordinate $\xi \in \{x_1, \dots, x_d, y, z\}$ of $\mathbb{P}$, we write $\rho_\xi : U_\xi \to (\xi \neq 0)$ for the corresponding orbifold chart of $\mathbb{P}$ (Definition 2.15). Let $V_{0,\xi} := \rho_\xi^{-1}(X_0) \subset U_\xi$ and let $\psi_\xi : V_{0,\xi} \to X_0 \cap (\xi \neq 0)$ be the restriction of ρ_ξ. By Definition-Lemma 2.20(1)(2), the morphism

$$\mathbb{G}_m \times V_{0,\xi} \to \operatorname{Spec} R_0 \cap (\xi \neq 0)$$

is the quotient by a free action of $\mu_{\deg \xi}$, hence étale and surjective. Since R_0 is normal, $V_{0,\xi}$ is normal. Since $V_{0,\xi}$ is a hypersurface in $U_\xi \cong \mathbb{A}^{d+1}$, $V_{0,\xi}$ is Gorenstein. The stabilizer in $\mu_{\deg \xi}$ of a point of U_ξ is the stabilizer of the image point in $\mathbb{P}$, so by Lemma 4.5 with $\tau = 0$, ψ_ξ is étale outside a closed subset of codimension at least 2. Thus $K_{V_{0,\xi}} = \psi_\xi^* K_{X_0}$, and K_{X_0} is $\mathbb{Q}$-Cartier [KM98, Proposition 5.20(1)]. Moreover, $X_0 \cap (\xi \neq 0) = V_{0,\xi}/\mu_{\deg \xi}$ is normal. By Lemma 2.8(2)(3) applied to ψ_ξ with $B = 0$, it suffices to show that $V_{0,\xi}$ is klt for any ξ.

If $\xi = y$, then $V_{0,y} = (zh + 1 = 0)$ is smooth, because $h \neq 0$ on $V_{0,y}$.

Assume that $\xi = x_i$. Then $V_{0,x_i} = (zh_i + y^e = 0) \subset U_{x_i}$, where $h_i := h|_{x_i=1}$. Let $v \in V_{0,x_i}$. If $h_i(v) \neq 0$, then V_{0,x_i} is smooth at v. Assume that $h_i(v) = 0$. If $\frac{\partial h}{\partial x_l}$ vanished at the point $x(v)$ for any $l \neq i$, then the Euler formula

$$\sum_{l=1}^{d} a_l x_l \frac{\partial h}{\partial x_l} = kh$$

and $x_i(v) = 1$ would imply $\frac{\partial h}{\partial x_i}(x(v)) = 0$, which contradicts the quasismoothness of h. Thus $dh_i(v) \neq 0$, and $u := h_i$ is part of a system of étale coordinates of U_{x_i} at v together with y, z, and $d - 2$ of the coordinates x_l. Hence, étale locally at v, V_{0,x_i} is isomorphic to $(zu + y^e = 0) \times \mathbb{A}^{d-2}$. The surface $(zu + y^e = 0) \subset \mathbb{A}^3$ is the quotient of $\mathbb{A}^2$ by μ_e acting by $(s_1, s_2) \to (\epsilon s_1, \epsilon^{-1} s_2)$, via $z = s_1^e$, $u = -s_2^e$, and $y = s_1 s_2$. Since μ_e acts freely outside the origin, $(zu + y^e = 0)$ is klt by Lemma 2.8(3), and so is V_{0,x_i} near v.

Assume that $\xi = z$. Then $V_{0,z} = (h + y^e = 0) \subset \mathbb{A}^{d+1}$. Since $\frac{\partial}{\partial y}(h + y^e) = ey^{e-1}$ and h is quasismooth, $V_{0,z}$ is smooth outside the origin, that is, $h + y^e$ is quasismooth with respect to the

weights below. Let $\ell := \gcd(k, c)$ and

$$w = (w_1, \dots, w_{d+1}) := \Big(\frac{ea_1}{\ell}, \dots, \frac{ea_d}{\ell}, \frac{k}{\ell}\Big),$$

the weights of $x_1, \dots, x_d, y$, so that $w_{d+1} = k/\ell$ is the weight of y. Since $\gcd(e, k) = \gcd(c, k) = \ell$, we have $\gcd(e/\ell, k/\ell) = 1$. Since W is well formed, $\gcd(a_1, \dots, a_d) = 1$. Thus gcd of the entries of w is 1. The polynomial $h + y^e$ is weighted homogeneous, irreducible by Lemma 2.21, and not of the form $c'x_i$ or $c'y$ with $c' \in \mathbb{C}$, since $e \geq 2$. Moreover, $w(h + y^e) = ek/\ell$, and the sum of the weights minus $w(h + y^e)$ is

$$\sum_{i=1}^{d+1} w_i - w(h + y^e) = \frac{e\sum_i a_i + k}{\ell} - \frac{ek}{\ell} = \frac{k}{\ell} > 0.$$

Thus the assumptions of Lemma 3.2 hold, and $V_{0,z}$ is klt near the origin by Lemma 3.2(1). □

Remark 4.7. The polynomial G in Set-up 4.1 cannot be omitted. Indeed, consider the hypersurface $X_0 = (zh + y^e = 0)$ of Construction 4.4. Since $\mathbb{P}$ is well formed ($\deg y = 1$ and $\gcd(a_1, \dots, a_d) = 1$), Lemmas 4.5, 4.6, and 2.22 imply that X_0 is a $\mathbb{Q}$-Fano variety ($-K_{X_0}$ is ample: for any positive integer m divisible by $a_1, \dots, a_d$, and c, the sheaf $\mathcal{O}_{X_0}(m)$ is invertible, and X_0 is covered by the affine open subsets $X_0 \cap (\xi \neq 0)$ with $\xi \in \{x_1, \dots, x_d, y, z\}$, which are the non-vanishing loci of the sections $\xi^{m/\deg \xi}$ of $\mathcal{O}_{X_0}(m)$ [Gro61, Théorème (4.5.2)(a') and Définition (4.5.3)]) and that $-K_{X_0} \sim \mathcal{O}_{X_0}(1)$. Let $P_0 := (y = z = 0) \subset X_0$, and let $(z = 0)|_{X_0}$ be the divisor of zeros of the section $z \in (R_0)_c$ of $\mathcal{O}_{X_0}(c)$. Near the generic point of P_0, we have $h \neq 0$ and $z = -y^e/h$ on orbifold charts, so

$$(z = 0)|_{X_0} = eP_0.$$

Since $\frac{1}{c}(z = 0)|_{X_0} \sim_{\mathbb{Q}} \mathcal{O}_{X_0}(1)$, we have

$$\alpha(X_0) \leq \operatorname{lct}\Big(X_0; \frac{1}{c}(z = 0)|_{X_0}\Big) \leq \frac{c}{e} < \frac{1}{d},$$

where the last inequality is equivalent to $c < \frac{k}{d-1}$.

Proposition 4.8. *Notation and conditions as in Set-up 4.1. Then R is a normal domain, X is a $\mathbb{Q}$-Fano variety of dimension d, $\omega_X \cong \mathcal{O}_X(-1)$, $-K_X \sim H$, and $H^0(X, \mathcal{O}_X(m)) = R_m$ for any integer m.*

Proof. The proof is long, so we divide it into several steps.

Step 1. In this step, we introduce the total space of the family X_τ and show that it is normal near X_0.

Let τ be the coordinate of $\mathbb{A}^1$ and

$$\mathcal{X} := (zh + y^e + \tau^{ce}G = 0) \subset \mathbb{P} \times \mathbb{A}^1,$$

so that the fiber of $\mathcal{X}$ over τ is X_τ. For a coordinate $\xi \in \{x_1, \dots, x_d, y, z\}$, let

$$\mathcal{V}_\xi := (\rho_\xi \times \mathrm{id}_{\mathbb{A}^1})^{-1}(\mathcal{X}) \subset U_\xi \times \mathbb{A}^1$$

and $V_{0,\xi} := \mathcal{V}_\xi \cap (\tau = 0) = \rho_\xi^{-1}(X_0)$. Since $\mathcal{V}_\xi$ is a hypersurface in $U_\xi \times \mathbb{A}^1 \cong \mathbb{A}^{d+2}$, $\mathcal{V}_\xi$ is Cohen–Macaulay [BH98, Theorem 2.1.3(a)], hence satisfies S_2, and any irreducible component of $\mathcal{V}_\xi$ is of dimension $d+1$. Since R_0 is normal (Lemma 4.6), $V_{0,\xi}$ is normal by Definition-Lemma 2.20(1)(2).

Let $v \in V_{0,\xi}$, let $A := \mathcal{O}_{\mathcal{V}_\xi, v}$, and let $\mathfrak{p} \subset A$ be a prime ideal of height at most one. Suppose that $A_\mathfrak{p}$ is not regular. Since $V_{0,\xi} = (\tau = 0)$ is a Cartier divisor on $\mathcal{V}_\xi$, $\mathcal{V}_\xi$ is regular at any point where $V_{0,\xi}$ is regular. Thus $A/\tau A$ is not regular at any prime ideal containing $\mathfrak{p} + \tau A$, that is, $V(\mathfrak{p} + \tau A) \subset \operatorname{Sing} V_{0,\xi}$ near v. By the Krull principal ideal theorem,

$$\dim A/(\mathfrak{p} + \tau A) \geq \dim A/\mathfrak{p} - 1 \geq \dim A - 2.$$

On the other hand, $V_{0,\xi}$ is normal and $\dim \mathcal{O}_{V_{0,\xi},v} = \dim A - 1$, so $\operatorname{Sing} V_{0,\xi}$ is of dimension at most $\dim A - 3$ near v. This is a contradiction. Thus A satisfies R_1 and S_2, and A is normal [Mat86, Theorem 23.8]. Hence $\mathcal{V}_\xi$ is normal in a neighborhood of $V_{0,\xi}$, and $\mathcal{X}$, which is the quotient of $\mathcal{V}_\xi$ by $\mu_{\deg \xi}$ over $(\xi \neq 0)$, is normal in a neighborhood of X_0.

Step 2. In this step, we show that X and R are normal and that R is a domain.

For $\tau \neq 0$, the graded automorphism $x_i \to \tau^{ea_i} x_i$, $y \to \tau^k y$, $z \to z$ of $\mathbb{C}[x_1, \dots, x_d, y, z]$ sends $zh + y^e + G$ to

$$\tau^{ek}\big(zh + y^e + \tau^{ce} G\big),$$

because $\deg h = k$, $\deg G = e$, and $e^2 - ek = ce$. Thus the automorphisms for $\tau \in \mathbb{G}_m$ define an isomorphism

$$\Phi : \mathcal{X}|_{\mathbb{G}_m} \to X \times \mathbb{G}_m$$

over $\mathbb{G}_m$.

Let $N \subset \mathcal{X}$ be the non-normal locus, which is closed. By **Step 1**, $N \cap X_0 = \emptyset$. Since $\mathcal{X} \to \mathbb{A}^1$ is proper, the image of N in $\mathbb{A}^1$ is a closed subset not containing 0. Via Φ, $N \cap \mathcal{X}|_{\mathbb{G}_m}$ corresponds to $N_X \times \mathbb{G}_m$, where N_X is the non-normal locus of X. If $N_X \neq \emptyset$, then the image of N would be $\mathbb{G}_m$, which is not closed in $\mathbb{A}^1$. Thus $N_X = \emptyset$, and X is normal. Moreover, $\mathcal{X}$ is normal.

The polynomial $g = y^e + G$ is monic in y, so $h \nmid g$. Since h is irreducible (Lemma 2.21), $zh + g$ is a primitive polynomial of degree one in z over $\mathbb{C}[x_1, \dots, x_d, y]$, hence irreducible, and R is a domain.

For a coordinate ξ, let $V_\xi := \rho_\xi^{-1}(X) \subset U_\xi$ and let $\psi_\xi : V_\xi \to X \cap (\xi \neq 0)$ be the restriction of ρ_ξ. By Definition-Lemma 2.20(1)(2), $\operatorname{Spec} R \cap (\xi \neq 0)$ is the quotient of $\mathbb{G}_m \times V_\xi$ by a free action of $\mu_{\deg \xi}$, so $\operatorname{Spec} R \cap (\xi \neq 0)$ is normal if and only if V_ξ is normal. Since V_ξ is a hypersurface in U_ξ, V_ξ satisfies S_2. By Lemma 4.5 with $\tau = 1$, ψ_ξ is étale outside a closed subset of codimension at least 2 in V_ξ. Since X is normal, V_ξ is regular at the generic point of any prime divisor not contained in that subset, so V_ξ satisfies R_1. Thus V_ξ is normal [Mat86, Theorem 23.8], and $\operatorname{Spec} R$ is normal outside the vertex. Since the vertex is of codimension $d + 1 \geq 2$ in $\operatorname{Spec} R$, R satisfies R_1. As a hypersurface ring, R satisfies S_2. Hence R is normal [Mat86, Theorem 23.8].

Step 3. In this step, we compute ω_X and the sections of $\mathcal{O}_X(m)$.

The weighted projective space $\mathbb{P}$ is well formed, since $\gcd(a_1, \dots, a_d) = 1$ and $\deg y = 1$. By **Step 2**, Lemma 4.5 with $\tau = 1$, and Lemma 2.22, we have

$$\omega_X \cong \mathcal{O}_X\big(e - (k + 1 + c)\big) = \mathcal{O}_X(-1) \quad \text{and} \quad H^0(X, \mathcal{O}_X(m)) = R_m$$

for any integer m. By Definition-Lemma 2.16, $-K_X \sim H$. For any positive integer m divisible by $a_1, \dots, a_d$, and c, the sheaf $\mathcal{O}_X(m)$ is invertible, because $\mathcal{O}_X(m)$ is generated by $\xi^{m/\deg \xi}$ on $X \cap (\xi \neq 0)$. For any coordinate ξ, $X \cap (\xi \neq 0)$ is the non-vanishing locus of the section $\xi^{m/\deg \xi} \in R_m$ of $\mathcal{O}_X(m)$, and it is affine. The open subsets $X \cap (\xi \neq 0)$ cover X, so $\mathcal{O}_X(m)$ is ample [Gro61, Théorème (4.5.2)(a') and Définition (4.5.3)]. Thus K_X is $\mathbb{Q}$-Cartier and $-K_X$ is ample.

Step 4. In this step, we show that X is klt, which finishes the proof.

The locus of points of $\mathbb{P} \times \mathbb{A}^1$ with non-trivial stabilizer intersects any fiber X_τ of $\mathcal{X}$ in a closed subset of dimension at most $d - 2$ (Lemma 4.5), so the locus intersects $\mathcal{X}$ in a closed subset of codimension at least 2 in $\mathcal{X}$. Thus the restriction $\mathcal{V}_\xi \to \mathcal{X} \cap (\xi \neq 0)$ of $\rho_\xi \times \mathrm{id}_{\mathbb{A}^1}$ is étale outside a closed subset of codimension at least 2. Since $\mathcal{V}_\xi$ is a hypersurface in $U_\xi \times \mathbb{A}^1$, it is Gorenstein, and it is normal near $V_{0,\xi}$ (**Step 1**), $K_{\mathcal{X}}$ is $\mathbb{Q}$-Cartier near X_0 [KM98, Proposition 5.20(1)]. By **Step 3** and the isomorphism Φ, $K_{\mathcal{X}}$ is $\mathbb{Q}$-Cartier over $\mathbb{G}_m$. Thus $K_{\mathcal{X}}$ is $\mathbb{Q}$-Cartier.

The fiber $X_0 = (\tau = 0)$ is a Cartier divisor on $\mathcal{X}$, so $K_{\mathcal{X}} + X_0$ is $\mathbb{Q}$-Cartier. By Lemma 4.6, X_0 is normal and klt. Thus $(\mathcal{X}, X_0)$ is plt near X_0 [KM98, Theorem 5.50(1)]. Since $a(F, \mathcal{X}, 0) \geq a(F, \mathcal{X}, X_0)$ for any prime divisor F over $\mathcal{X}$, $\mathcal{X}$ is klt near X_0.

Let $Z \subset \mathcal{X}$ be the non-klt locus of $\mathcal{X}$. The set Z is closed, since on a log resolution of $\mathcal{X}$ it is the union of the images of the finitely many prime divisors with log discrepancy at most 0

[KM98, Corollary 2.32(2)]. We have $Z \cap X_0 = \emptyset$, so the image of Z in $\mathbb{A}^1$ is a closed subset not containing 0. Via Φ, $Z \cap \mathcal{X}|_{\mathbb{G}_m}$ corresponds to $Z_X \times \mathbb{G}_m$, where Z_X is the non-klt locus of X (if $\widetilde{X} \to X$ is a log resolution, then $\widetilde{X} \times \mathbb{G}_m \to X \times \mathbb{G}_m$ is a log resolution, and for any prime divisor F on $\widetilde{X}$, the log discrepancy of $F \times \mathbb{G}_m$ with respect to $X \times \mathbb{G}_m$ equals that of F with respect to X). If $Z_X \neq \emptyset$, then the image of Z would be $\mathbb{G}_m$, which is not closed in $\mathbb{A}^1$. Thus $Z_X = \emptyset$ and X is klt. Together with **Step 3**, X is a $\mathbb{Q}$-Fano variety of dimension d. □

4.2. **A plt blow-up.** In this subsection, we construct a plt blow-up $f : Y \to X$ of X at p. The exceptional divisor E of f satisfies $\frac{A_X(E)}{T_X(E)} = \frac{k}{e}$ (Proposition 4.11), and we use it to show that $(X, \frac{k}{e}B)$ is lc near p for any $\mathbb{Q}$-divisor $B \geq 0$ with $B \sim_{\mathbb{Q}} H$ (Proposition 4.13). With the weights $(ea_1/\ell, \dots, ea_d/\ell, k/\ell)$, where $\ell = \gcd(k, c)$ may be larger than 1 (it is 2 and 750 for the examples of dimensions 4 and 6 in Section 5), the lowest-weight part of the equation of X at p is $h + y^e$, so that $E \cong W$, $\Delta_E = (1 - \frac{1}{e})S$, and the ramification index along E is c/ℓ.

Construction-Proposition 4.9. Notation and conditions as in Set-up 4.1, and let $\ell := \gcd(k, c)$. Let $\rho_z : U_z \to (z \neq 0) \subset \mathbb{P}$ be the orbifold chart, and let $X_U := (h + y^e + G = 0) \subset U_z = \mathbb{A}^{d+1}$ be the preimage of X, on which μ_c acts with weights $(a_1, \dots, a_d, 1)$ on $(x_1, \dots, x_d, y)$. Let $f_U : Y_U \to X_U$ be the morphism from the strict transform Y_U of X_U in the weighted blow-up of $\mathbb{A}^{d+1}$ with weights $w := (ea_1/\ell, \dots, ea_d/\ell, k/\ell)$ on $(x_1, \dots, x_d, y)$, and let E_U be the exceptional divisor of f_U. Then the action of μ_c lifts to Y_U, and $Y_U/\mu_c \to X \cap (z \neq 0)$ extends to a plt blow-up $f : Y \to X$ of X at p with exceptional divisor E. Moreover, the following hold.

(1) The ramification index of $Y_U \to Y_U/\mu_c$ along E_U is c/ℓ, and $A_X(E) = k/c$.
(2) There exists an isomorphism $E \cong W$ under which $\Delta_E = (1 - \frac{1}{e})S$.
(3) The polynomial $F_U := h + y^e + G$ and the weights w satisfy the assumptions of Lemma 3.2, and the lowest-weight part of F_U with respect to w is $h + y^e$.

Proof. The proof is long, so we divide it into several steps.

Step 1. In this step, we describe Y_U and construct f.

We have $X_U = (F_U = 0)$. Let $\psi_X : X_U \to X \cap (z \neq 0)$ be the restriction of ρ_z, which is the quotient by μ_c. Since R is normal (Proposition 4.8), X_U is normal by Definition-Lemma 2.20(1)(2). By Lemma 4.5 with $\tau = 1$, ψ_X is étale outside a closed subset of codimension at least 2 in X_U, so $K_{X_U} = \psi_X^* K_X$ by the Hurwitz formula.

Since $\gcd(e, k) = \gcd(c, k) = \ell$ and $\gcd(a_1, \dots, a_d) = 1$, the gcd of the entries of w is 1. Any monomial of h and the monomial y^e have weight ek/ℓ, and any monomial of G has weight $e^2/\ell = ek/\ell + ec/\ell$. Thus, with the notation of Lemma 3.2, $(F_U)_w = h + y^e$ and $w(F_U) = ek/\ell$. The polynomial $h + y^e$ is quasismooth, since $\frac{\partial}{\partial y}(h + y^e) = ey^{e-1}$ and h is quasismooth. It is irreducible by Lemma 2.21, and it is not of the form $c'x_i$ or $c'y$ with $c' \in \mathbb{C}$, since it contains the term y^e with $e \geq 2$. Moreover,

$$\sum_{i=1}^{d+1} w_i - \frac{ek}{\ell} = \frac{e\sum_i a_i + k - ek}{\ell} = \frac{k}{\ell} > 0.$$

This proves (3). Possibly replacing X_U with a μ_c-invariant open neighborhood of the origin, by Lemma 3.2, f_U is a plt blow-up of X_U at the origin, $E_U \cong (h + y^e = 0) \subset \mathbb{P}(w)$, and

$$a(E_U, X_U, 0) = \frac{k}{\ell}.$$

We use the following description of Y_U. Let $\mathbb{G}_m$ act on $(\mathbb{A}^{d+1}\backslash\{0\}) \times \mathbb{A}^1$, with coordinates $x_1', \dots, x_d', y', u$, by

$$\lambda \cdot (x', y', u) = \left(\lambda^{ea_1/\ell} x_1', \dots, \lambda^{ea_d/\ell} x_d', \lambda^{k/\ell} y', \lambda^{-1} u\right).$$

The weighted blow-up of $\mathbb{A}^{d+1}$ with weights w is the quotient of $(\mathbb{A}^{d+1}\backslash\{0\}) \times \mathbb{A}^1$ by $\mathbb{G}_m$, the morphism to $\mathbb{A}^{d+1}$ is given by $x_i = u^{ea_i/\ell} x_i'$ and $y = u^{k/\ell} y'$, and the exceptional divisor is

$(u=0)/\mathbb{G}_m = \mathbb{P}(w)$. The charts of [Pro01, Section 3.2] are the open subsets $(x_i' \neq 0)/\mathbb{G}_m$ and $(y' \neq 0)/\mathbb{G}_m$. For $1 \leq i \leq d$, any $\mathbb{G}_m$-orbit in $(x_i' \neq 0)$ intersects the slice $(x_i' = 1)$ in one orbit of the stabilizer $\mu_{ea_i/\ell}$ of the slice. Thus $(x_i' \neq 0)/\mathbb{G}_m = (x_i' = 1)/\mu_{ea_i/\ell}$, with coordinates u, x_j' for $j \neq i$, and y', is the chart U_i of the weighted blow-up of $\mathbb{C}^{d+1}$ in [Pro01, Section 3.2] (where the cyclic group acting on $\mathbb{C}^n$ there is trivial), with $y_i = u$ and $y_j = x_j'$, where the action of $\mu_{ea_i/\ell}$ differs from the one there by the automorphism $\zeta \to \zeta^{-1}$, and the coordinate relations of [Pro01, Section 3.2] read

$$x_i = u^{ea_i/\ell}, \quad x_j = u^{ea_j/\ell} x_j' \text{ for } j \neq i, \quad y = u^{k/\ell} y'.$$

For $(y' \neq 0)$, the stabilizer of the slice $(y' = 1)$ is $\mu_{k/\ell}$, and $(y' \neq 0)/\mathbb{G}_m = (y' = 1)/\mu_{k/\ell}$, with coordinates u and $x_1', \dots, x_d'$, and with $y = u^{k/\ell}$ and $x_j = u^{ea_j/\ell} x_j'$ for $1 \leq j \leq d$, is the chart U_{d+1} of the weighted blow-up of $\mathbb{C}^{d+1}$ in [Pro01, Section 3.2], with $y_{d+1} = u$ and $y_j = x_j'$. Since

$$F_U\big(u^{ea_1/\ell} x_1', \dots, u^{ea_d/\ell} x_d', u^{k/\ell} y'\big) = u^{ek/\ell}\big(h(x') + y'^e + u^{ec/\ell} G(x')\big),$$

we have $Y_U = \widehat{Y}/\mathbb{G}_m$ and $E_U = \widehat{E}/\mathbb{G}_m$, where

$$\widehat{Y} := \big(h(x') + y'^e + u^{ec/\ell} G(x') = 0\big) \subset (\mathbb{A}^{d+1} \backslash \{0\}) \times \mathbb{A}^1 \quad \text{and} \quad \widehat{E} := \widehat{Y} \cap (u = 0).$$

Let μ_c act on $(\mathbb{A}^{d+1} \backslash \{0\}) \times \mathbb{A}^1$ by

$$\zeta \cdot (x', y', u) = (\zeta^{a_1} x_1', \dots, \zeta^{a_d} x_d', \zeta y', u).$$

The action of μ_c commutes with the action of $\mathbb{G}_m$, preserves $\widehat{Y}$, and is compatible with the action of μ_c on X_U. Thus the action of μ_c lifts to Y_U.

Let $\psi : Y_U \to Y_U/\mu_c$ be the quotient morphism. Since $Y_U \to X_U$ is an isomorphism over $X_U \backslash \{0\}$, $Y_U/\mu_c \to X_U/\mu_c = X \cap (z \neq 0)$ is an isomorphism over the complement of p. Gluing Y_U/μ_c and $X \backslash \{p\}$ over $X \cap (z \neq 0) \backslash \{p\}$, we obtain a proper birational morphism $f : Y \to X$ which is an isomorphism over $X \backslash \{p\}$. The exceptional locus of f is $E := \psi(E_U)$, which is a prime divisor with $f(E) = \{p\}$.

Step 2. In this step, we show that f is a plt blow-up and prove (1).

Let $\zeta \in \mu_c$. Since $\widehat{E}$ is irreducible and none of $x_1', \dots, x_d', y'$ vanishes identically on $\widehat{E}$, ζ acts trivially on $E_U = \widehat{E}/\mathbb{G}_m$ if and only if there exists $\lambda \in \mathbb{G}_m$ such that

$$\zeta^{a_i} = \lambda^{ea_i/\ell} \text{ for } 1 \leq i \leq d \quad \text{and} \quad \zeta = \lambda^{k/\ell}.$$

Since $\gcd(a_1, \dots, a_d) = 1$, the first condition is equivalent to $\zeta = \lambda^{e/\ell}$. Thus the conditions are equivalent to $\lambda^{c/\ell} = 1$ and $\zeta = \lambda^{k/\ell}$. Since $\gcd(k/\ell, c/\ell) = 1$, the homomorphism $\mu_{c/\ell} \to \mu_c$, $\lambda \to \lambda^{k/\ell}$, is injective. Hence the subgroup of μ_c acting trivially on E_U is of order c/ℓ, and the ramification index of ψ along E_U is c/ℓ.

Since $Y_U \backslash E_U \cong X_U \backslash \{0\}$, Lemma 4.5 with $\tau = 1$ implies that ψ is étale outside E_U and a closed subset of codimension at least 2. By the Hurwitz formula, $K_{Y_U} = \psi^* K_Y + (\frac{c}{\ell} - 1) E_U$ and $\psi^* E = \frac{c}{\ell} E_U$, so

$$K_{Y_U} + E_U = \psi^*(K_Y + E).$$

Since (Y_U, E_U) is plt near E_U, $K_{Y_U} + E_U$ is $\mathbb{Q}$-Cartier, and $K_Y + E$ is $\mathbb{Q}$-Cartier [KM98, Proposition 5.20(1)]. By [KM98, Definition 2.34], $\operatorname{discrep}(Y_U, E_U) > -1$ near E_U, where $\operatorname{discrep}(Y_U, E_U) + 1$ is the infimum of $a(F, Y_U, E_U)$ over exceptional prime divisors F over Y_U [KM98, Definition 2.28] (the discrepancy of [KM98] is $a(F, \cdot, \cdot) - 1$ in our notation), so

$$c\big(\operatorname{discrep}(Y, E) + 1\big) \geq \operatorname{discrep}(Y_U, E_U) + 1 > 0$$

near E [KM98, Proposition 5.20(2)], that is, (Y, E) is plt near E. The variety Y is normal, since Y_U is normal. Since $\psi^* E = \frac{c}{\ell} E_U$ is $\mathbb{Q}$-Cartier, E is $\mathbb{Q}$-Cartier, because the norm of $\psi^* E$ is cE and the norm of a $\mathbb{Q}$-Cartier divisor is $\mathbb{Q}$-Cartier [Kol13, Claim 2.40.1]. Let $V \subset E$ be an

irreducible subvariety of positive dimension, and let $V' \subset E_U$ be an irreducible subvariety with $\psi(V') = V$. Since $-E_U$ is ample over X_U and $f_U(V')$ is the origin, by the projection formula,

$$\deg(V' \to V) \cdot \big((-E)^{\dim V} \cdot V\big) = \big((-\psi^* E)^{\dim V} \cdot V'\big) = \left(\frac{c}{\ell}\right)^{\dim V} \big((-E_U)^{\dim V} \cdot V'\big) > 0.$$

Any positive-dimensional subvariety of Y contracted by f is contained in E, and f is proper (**Step 1**), so $-E$ is f-ample [Laz04, Corollary 1.7.9]. Thus f is projective, and f is a plt blow-up of X at p (Definition-Proposition 2.14).

The prime divisor E_U over X_U maps to the prime divisor E over X, with ramification index c/ℓ, and $K_{X_U} = \psi_X^* K_X$ by **Step 1**. Since the ramification is tame, [Kol13, 2.42, (2.42.2)] implies

$$\frac{k}{\ell} = a(E_U, X_U, 0) = \frac{c}{\ell} a(E, X, 0),$$

where a denotes the log discrepancy. Thus $A_X(E) = k/c$, which proves (1).

Step 3. In this step, we construct the isomorphism $E \cong W$.

Let

$$\Gamma := \{(t_1, t_2) \in \mathbb{G}_m^2 \mid t_2^e = t_1^k\},$$

acting on $\widehat{E}$ by

$$(t_1, t_2) \cdot (x', y') = (t_1^{a_1} x'_1, \dots, t_1^{a_d} x'_d, t_2 y').$$

The action of $\mathbb{G}_m \times \mu_c$ on $\widehat{E}$ factors through the homomorphism

$$\theta : \mathbb{G}_m \times \mu_c \to \Gamma, \quad (\lambda, \zeta) \to (\lambda^{e/\ell}\zeta, \lambda^{k/\ell}\zeta),$$

which is well defined since

$$(\lambda^{k/\ell}\zeta)^e (\lambda^{e/\ell}\zeta)^{-k} = \zeta^{e-k} = \zeta^c = 1.$$

The homomorphism θ is surjective: for $(t_1, t_2) \in \Gamma$, choose λ with $\lambda^{c/\ell} = t_1/t_2$ and let $\zeta := t_1 \lambda^{-e/\ell}$. Then $\zeta^c = t_1^c t_2^e t_1^{-e} = t_1^{-k} t_2^e = 1$ and $\lambda^{k/\ell}\zeta = t_1 \lambda^{-c/\ell} = t_2$. The kernel of θ is

$$I := \{(\lambda, \lambda^{-k/\ell}) \mid \lambda^{c/\ell} = 1\}.$$

Since E is normal (Definition-Proposition 2.14), the finite bijective morphism $E_U/\mu_c \to E$ is an isomorphism, so

$$E = \widehat{E}/(\mathbb{G}_m \times \mu_c) = \widehat{E}/\Gamma.$$

On $\widehat{E}$, we have $x' \neq 0$, because $x' = 0$ and $y'^e = -h(x')$ imply $y' = 0$. The morphism $\widehat{E} \to \mathbb{A}^d \backslash \{0\}$, $(x', y') \to x'$, is finite and surjective, and the fibers of $(x', y') \to x'$ are the orbits of the subgroup $\{1\} \times \mu_e \subset \Gamma$. Since $\mathbb{A}^d$ is normal, $\widehat{E}/\mu_e \cong \mathbb{A}^d \backslash \{0\}$. The homomorphism $\Gamma \to \mathbb{G}_m$, $(t_1, t_2) \to t_1$, is surjective with kernel $\{1\} \times \mu_e$, and t_1 acts on $\mathbb{A}^d$ with weights $a_1, \dots, a_d$. Hence $(x', y') \to x'$ induces an isomorphism

$$E = \widehat{E}/\Gamma \cong (\mathbb{A}^d \backslash \{0\})/\mathbb{G}_m = W.$$

Under $E \cong W$, the preimage of S in $\widehat{E}$ is $(y' = 0) \cap \widehat{E}$, because $y'^e = -h(x')$ on $\widehat{E}$.

Step 4. In this step, we compute Δ_E.

Fix $1 \leq j \leq d$. Let $\widetilde{Y}_j := \widehat{Y} \cap (x'_j = 1)$, let

$$\widetilde{E}_j := \widetilde{Y}_j \cap (u = 0) = \widehat{E} \cap (x'_j = 1),$$

and let

$$\Gamma_j := \{(\lambda, \zeta) \in \mathbb{G}_m \times \mu_c \mid \lambda^{ea_j/\ell} \zeta^{a_j} = 1\},$$

which is the finite subgroup of $\mathbb{G}_m \times \mu_c$ preserving $(x'_j = 1)$. Any point of $\widehat{Y} \cap (x'_j \neq 0)$ is moved into $\widetilde{Y}_j$ by some $\lambda \in \mathbb{G}_m$, uniquely up to Γ_j, so $Y_j := (\widehat{Y} \cap (x'_j \neq 0))/(\mathbb{G}_m \times \mu_c)$ is an open subset of Y with

$$Y_j = \widetilde{Y}_j/\Gamma_j \quad \text{and} \quad E \cap Y_j = \widetilde{E}_j/\Gamma_j.$$

Under $E \cong W$, $E \cap Y_j$ corresponds to $W \cap (x_j \neq 0)$, and the sets $E \cap Y_j$ with $1 \leq j \leq d$ cover E, since $x' \neq 0$ on $\widehat{E}$. Let $\psi_j : \widetilde{Y}_j \to Y_j$ be the quotient morphism and $\varphi_j : \widetilde{E}_j \to E \cap Y_j$ the restriction of ψ_j.

We first show that $\widetilde{Y}_j$ is smooth near $\widetilde{E}_j$. Let $v \in \widetilde{E}_j$. If the partial derivatives of $h + y'^e$ with respect to y' and the x'_l with $l \neq j$ vanished at v, then the Euler formula for $h + y'^e$ with the weights w and $x'_j(v) = 1$ would imply that all partial derivatives of $h + y'^e$ vanish at $v \neq 0$, which contradicts the quasismoothness of $h + y^e$. Thus $\widetilde{E}_j$ is smooth. Since $\widetilde{E}_j = (u = 0)$ is a Cartier divisor on $\widetilde{Y}_j$, $\widetilde{Y}_j$ is smooth near $\widetilde{E}_j$.

Next, we compute the ramification of ψ_j. The morphism

$$\mathbb{G}_m \times \widetilde{E}_j \to \widehat{E} \cap (x'_j \neq 0)$$

is finite and surjective, so any irreducible component $\widetilde{E}'$ of $\widetilde{E}_j$ is of dimension $d-1$ and $\mathbb{G}_m \cdot \widetilde{E}'$ is dense in $\widehat{E}$. Thus none of $x'_1, \dots, x'_d, y'$ vanishes identically on an irreducible component of $\widetilde{E}_j$. Hence $(\lambda, \zeta) \in \Gamma_j$ acts trivially on an irreducible component of $\widetilde{E}_j$ if and only if $(\lambda^{e/\ell}\zeta)^{a_i} = 1$ for $1 \leq i \leq d$ and $\lambda^{k/\ell}\zeta = 1$, that is, $(\lambda, \zeta) \in I$. Since I is of order c/ℓ, the ramification index of ψ_j along any irreducible component of $\widetilde{E}_j$ is c/ℓ. If $(\lambda, \zeta) \in \Gamma_j$ fixes a point (x', y', u) with $u \neq 0$, then $\lambda = 1$ and ζ fixes the image point $(u^{ea_i/\ell}x'_i, u^{k/\ell}y')$ of X_U. By Lemma 4.5 with $\tau = 1$, such points form a closed subset of codimension at least 2. Thus ψ_j is étale outside $\widetilde{E}_j$ and a closed subset of codimension at least 2.

The action of Γ_j on $\widetilde{E}_j$ factors through

$$\Gamma'_j := \theta(\Gamma_j) = \{(t_1, t_2) \in \Gamma \mid t_1^{a_j} = 1\},$$

which acts faithfully on $\widetilde{E}_j$, so φ_j is the quotient by Γ'_j. Let P be a prime divisor on $E \cap Y_j$, let $\widetilde{P}$ be a prime divisor on $\widetilde{E}_j$ with $\varphi_j(\widetilde{P}) = P$, and let r_P be the ramification index of φ_j along $\widetilde{P}$, which is the order of the stabilizer in Γ'_j of a general point (x', y') of $\widetilde{P}$. Since W is well formed, for any prime number r, at least two of $a_1, \dots, a_d$ are not divisible by r, so the points of W with non-trivial stabilizer form a closed subset of codimension at least 2 in W. Since $\dim P = d-2$ and the points of W with non-trivial stabilizer form a closed subset of dimension at most $d-3$, a general point (x', y') of $\widetilde{P}$ satisfies $\gcd\{a_i \mid x'_i \neq 0\} = 1$.

- If P corresponds to $S \cap (x_j \neq 0)$, then $y' = 0$, and $(t_1, t_2) \in \Gamma'_j$ fixes $(x', 0)$ if and only if $t_1 = 1$. Thus the stabilizer is $\{1\} \times \mu_e$, and $r_P = e$.
- Otherwise, $y' \neq 0$, and (t_1, t_2) fixes (x', y') if and only if $t_1 = 1$ and $t_2 = 1$. Thus $r_P = 1$.

Finally, we compare the different with the ramification indices r_P. Let m be a positive integer such that $m(K_Y + E)$ is Cartier near the generic point of P, and let σ be a local generator of $\mathcal{O}_Y(m(K_Y + E))$ there, regarded as a rational section of $\omega_Y^{[m]}$. By the definition of the different [Kol13, Definition 4.2], the residue $\operatorname{Res}\sigma$ is a rational section of $\omega_E^{[m]}$ with

$$\operatorname{mult}_P \operatorname{div}(\operatorname{Res}\sigma) = -m \operatorname{mult}_P \Delta_E.$$

By the Hurwitz formula for ψ_j and the ramification of ψ_j computed above, the coefficient of $\widetilde{E}_j$ in $\operatorname{div}(\psi_j^*\sigma)$ is $\frac{c}{\ell}(-m) + m(\frac{c}{\ell} - 1) = -m$, and $\psi_j^*\sigma$ has no other zero or pole near the generic point of $\widetilde{P}$. Since $\widetilde{Y}_j$ is smooth there, $\psi_j^*\sigma$ is a local generator of $\omega_{\widetilde{Y}_j}^{\otimes m}(m\widetilde{E}_j)$, and by adjunction for the smooth Cartier divisor $\widetilde{E}_j$, $\operatorname{Res}(\psi_j^*\sigma)$ is a local generator of $\omega_{\widetilde{E}_j}^{\otimes m}$ near the generic point of $\widetilde{P}$. If t and s are local equations of E and $\widetilde{E}_j$ near the generic point of $\widetilde{E}_j$, then $\psi_j^* t = s^{c/\ell} v$ for a unit v, and $\psi_j^*(dt/t) = \frac{c}{\ell} ds/s + dv/v$. Thus

$$\operatorname{Res}(\psi_j^*\sigma) = \left(\frac{c}{\ell}\right)^m \varphi_j^*(\operatorname{Res}\sigma)$$

as rational sections of $\omega_{\widetilde{E}_j}^{\otimes m}$. By the Hurwitz formula for φ_j along $\widetilde{P}$ [Kol13, 2.41],

$$0 = \operatorname{mult}_{\widetilde{P}} \operatorname{div}\left(\varphi_j^*(\operatorname{Res}\sigma)\right) = r_P \operatorname{mult}_P \operatorname{div}(\operatorname{Res}\sigma) + m(r_P - 1) = -mr_P \operatorname{mult}_P \Delta_E + m(r_P - 1).$$

Hence $\operatorname{mult}_P \Delta_E = 1 - \frac{1}{r_P}$. By the computation of r_P, $\Delta_E = (1-\frac{1}{e})S$ on $E \cap Y_j$. Since $1 \le j \le d$ is arbitrary, $\Delta_E = (1-\frac{1}{e})S$ under the isomorphism $E \cong W$, which proves (2). □

4.3. **Volumes and log canonicity near p.** In this subsection, we compute the quantities that enter the alpha and beta invariants of X: the complete volume function of E, which determines $T_X(E) = \frac{e}{c}$ and $\beta_X(E)$, and the largest coefficient $\frac{k}{e}$ such that $(X, \frac{k}{e}B)$ is lc near p for any $\mathbb{Q}$-divisor $B \ge 0$ with $B \sim_{\mathbb{Q}} H$ (Proposition 4.13).

The volumes in Proposition 4.11 are volumes of complete linear systems, so we need the order along E of any section of $\mathcal{O}_X(m)$.

Definition-Lemma 4.10. Notation and conditions as in Set-up 4.1, and let E be as in Construction-Proposition 4.9. For any non-zero homogeneous element $s \in R$, let $b(s)$ be the supremum, over all homogeneous polynomials representing s, of the minimal weight of the monomials of the polynomial with respect to the weights $ea_1, \dots, ea_d, k, 0$ of $x_1, \dots, x_d, y, z$. Then the following hold.

(1) The monomials $x_1^{\gamma_1} \cdots x_d^{\gamma_d} y^j z^i$ with $j < e$ form a basis of R, and for any non-zero homogeneous $s \in R$, $b(s)$ is the minimal weight of the basis monomials appearing in s with non-zero coefficient.

(2) For any integer $m \ge 0$ and any non-zero $s \in R_m = H^0(X, \mathcal{O}_X(m))$,

$$\operatorname{mult}_E(s = 0) = \frac{b(s)}{c},$$

where $(s = 0) \sim mH$ denotes the Weil divisor of zeros of s.

Proof. (1) Let $F := zh + y^e + G \in \mathbb{C}[x_1, \dots, x_d, z, y]$, and consider the setting of Lemma 3.3 with y as the last variable and with the weights $ea_1, \dots, ea_d, 0, k$ of $x_1, \dots, x_d, z, y$. The monomials of zh and of y^e have weight ke. Since $G \in \mathbb{C}[x_1, \dots, x_d]$ has degree e, the monomials of G have weight $e \cdot e > ke$, where the inequality holds because $c > 0$. Thus $F_w = zh + y^e$. Since $zh + y^e$ has degree one in z and the coefficients h and y^e are coprime, F_w is prime. By Lemma 3.3(1), the supremum $b(s)$ is attained for any non-zero homogeneous $s \in R$. Since F and F_w are both monic of degree e as polynomials in y, (1) follows from Lemma 3.3(2).

(2) Since $H \sim -K_X$ and X is a $\mathbb{Q}$-Fano variety (Proposition 4.8), $(s = 0)$ is $\mathbb{Q}$-Cartier, so $\operatorname{mult}_E(s = 0)$ is defined.

Let ℓ, w, X_U, $f_U : Y_U \to X_U$, E_U, and $F_U = h + y^e + G$ be as in Construction-Proposition 4.9. By Construction-Proposition 4.9(3), the hypotheses of Lemma 3.2 hold for F_U and the weights w, and the lowest-weight part of F_U is $h + y^e$. We use the notation $w(\varphi)$ and φ_w of Lemma 3.2 for the weights w.

Write $s = \sum \lambda_{\gamma,j,i} x^\gamma y^j z^i$ in the basis of (1), where $x^\gamma := x_1^{\gamma_1} \cdots x_d^{\gamma_d}$, $j < e$, and $\lambda_{\gamma,j,i} \in \mathbb{C}$, and let

$$\varphi := \sum \lambda_{\gamma,j,i} x^\gamma y^j \in \mathbb{C}[x_1, \dots, x_d, y].$$

Since $s \in R_m$, the exponent $i = (m - \sum_r \gamma_r a_r - j)/c$ is determined by (γ, j), so the monomials of φ correspond bijectively to the basis monomials appearing in s. The auxiliary weight $e\sum_r \gamma_r a_r + kj$ of $x^\gamma y^j z^i$ is ℓ times the weight of $x^\gamma y^j$ with respect to w. Thus, by (1),

$$w(\varphi) = \frac{b(s)}{\ell}.$$

All monomials of φ_w have y-exponent less than e, and $h + y^e$ is monic of degree e in y, so φ_w is not divisible by $h + y^e$. By Lemma 3.2(4),

$$\operatorname{mult}_{E_U}(\varphi|_{X_U} = 0) = \frac{b(s)}{\ell}.$$

On X_U, the section s is given by the function $\varphi|_{X_U}$. By Lemma 4.5, the quotient map $X_U \to X \cap (z \neq 0)$ by μ_c is étale in codimension one, so $(\varphi|_{X_U} = 0)$ is the pullback of $(s = 0)$. Let $\psi : Y_U \to Y$ be the composition of the quotient map $Y_U \to Y_U/\mu_c$ and the open immersion $Y_U/\mu_c = f^{-1}(X \cap (z \neq 0)) \subset Y$. Then $f \circ \psi$ is the composition of f_U and $X_U \to X$. The ramification index of $Y_U \to Y_U/\mu_c$ along E_U is $\frac{c}{\ell}$ (Construction-Proposition 4.9(1)), so $\psi^*E = \frac{c}{\ell}E_U$ near the generic point of E_U. Comparing the coefficients of E_U in $\psi^* f^*(s = 0) = f_U^*(\varphi|_{X_U} = 0)$ (cf. Lemma 2.8(1)), we obtain

$$\frac{c}{\ell}\operatorname{mult}_E(s = 0) = \frac{b(s)}{\ell}.$$

Thus $\operatorname{mult}_E(s = 0) = \frac{b(s)}{c}$. □

Proposition 4.11 shows that $\operatorname{vol}(-K_X - tE) = \operatorname{vol}(-K_X)\big(1 - (\frac{ct}{e})^d\big)$ for $0 \le t \le \frac{e}{c}$, as for the blow-up of the vertex of a cone. Thus $S_X(E)/T_X(E) = \frac{d}{d+1}$, which is the largest value allowed by [BJ20, Proposition 3.11], and the sign of $\beta_X(E)$ is the sign of $k - dc$.

Proposition 4.11. *Notation and conditions as in Set-up 4.1, and let E be as in Construction-Proposition 4.9. Then* $\operatorname{vol}(-K_X) = H^d = \frac{e}{ca_1 \cdots a_d}$ *and*

$$\operatorname{vol}(-K_X - tE) = \begin{cases} \operatorname{vol}(-K_X)\Big(1 - \big(\frac{ct}{e}\big)^d\Big) & \text{if } 0 \le t \le \frac{e}{c}, \\ 0 & \text{if } t \ge \frac{e}{c}. \end{cases}$$

In particular, $T_X(E) = \frac{e}{c}$, $S_X(E) = \frac{de}{c(d+1)}$, *and*

$$\beta_X(E) = \frac{k - dc}{c(d+1)}\operatorname{vol}(-K_X) < 0.$$

Proof. By Proposition 4.8, $-K_X \sim H$, and H is ample. Thus the positive integers m such that $-mK_X$ is Cartier are the positive integers m such that mH is Cartier. Let m be such an integer, and let $t \ge 0$. Then

$$\lfloor m(f^*H - tE) \rfloor = f^*(mH) - \lceil mt \rceil E.$$

For any non-zero $s \in R_m$, the divisor $(s = 0)$ is Cartier, so $\operatorname{mult}_E(s = 0)$ is an integer. Since $f_*\mathcal{O}_Y = \mathcal{O}_X$ and $H^0(X, \mathcal{O}_X(m)) = R_m$ (Proposition 4.8), we have

$$h^0\big(Y, \mathcal{O}_Y(\lfloor m(f^*H - tE) \rfloor)\big) = \dim V_m(t),$$

where

$$V_m(t) := \{s \in R_m \mid s = 0 \text{ or } \operatorname{mult}_E(s = 0) \ge mt\}.$$

By Definition-Lemma 4.10, $V_m(t)$ is spanned by the basis monomials $x^\gamma y^j z^i$ of degree m with $j < e$ and $b(x^\gamma y^j z^i) \ge cmt$, where $x^\gamma := x_1^{\gamma_1} \cdots x_d^{\gamma_d}$. Such a monomial satisfies $\sum_r \gamma_r a_r = m - j - ci$, so

$$b(x^\gamma y^j z^i) = e(m - j - ci) + kj = em - cj - cei.$$

Thus $b(x^\gamma y^j z^i) \ge cmt$ if and only if

$$ei + j \le m\Big(\frac{e}{c} - t\Big). \tag{4.1}$$

For an integer L, let N_L be the number of monomials x^γ with $\sum_r \gamma_r a_r = L$. Then

$$\dim V_m(t) = \sum_{j=0}^{e-1} \sum_{i \ge 0 \text{ satisfying (4.1)}} N_{m-j-ci}.$$

We estimate N_L. Since W is well formed and $d \ge 3$, $\gcd(a_1, \dots, a_d) = 1$. The generating function

$$\sum_{L \ge 0} N_L \xi^L = \prod_{r=1}^{d} \frac{1}{1 - \xi^{a_r}}$$

has a pole of order d at $\xi = 1$ with leading term $(a_1 \cdots a_d)^{-1}(1-\xi)^{-d}$, and a pole of order $\#\{r \mid \zeta^{a_r} = 1\} \le d-1$ at any root of unity $\zeta \ne 1$. By the partial fraction decomposition, for any integer $L \ge 1$,

$$N_L = \frac{L^{d-1}}{(d-1)!a_1 \cdots a_d} + O(L^{d-2})$$

(cf. [Wil94, Theorem 3.15.2]), where the constant in $O(L^{d-2})$ depends only on $a_1, \dots, a_d$.

Assume that $0 \le t \le \frac{e}{c}$. For $0 \le j < e$ and any $i \ge 0$ satisfying (4.1), we have $ci \le m(1 - \frac{ct}{e})$, so

$$0 \le \frac{cmt}{e} \le m - ci \le m.$$

Since $N_L \le 1$ for $L \le 0$ and $(m-ci)^{d-1} < e^{d-1}$ when $m - j - ci \le 0$, the estimate of N_L implies

$$N_{m-j-ci} = \frac{(m-ci)^{d-1}}{(d-1)!a_1 \cdots a_d} + O(m^{d-2}).$$

For any j, the number of $i \ge 0$ satisfying (4.1) differs from $m(\frac{1}{c} - \frac{t}{e})$ by at most 2, and the function $(m-cx)^{d-1}$ of $x \in [0, \frac{m}{c}]$ is decreasing and bounded by m^{d-1}. Thus

$$\begin{aligned}\dim V_m(t) &= \frac{e}{(d-1)!a_1 \cdots a_d} \int_0^{m(\frac{1}{c} - \frac{t}{e})} (m-cx)^{d-1} dx + O(m^{d-1}) \\ &= \frac{em^d}{cd!a_1 \cdots a_d} \Big(1 - \Big(\frac{ct}{e}\Big)^d\Big) + O(m^{d-1}).\end{aligned}$$

By Definition 2.11,

$$\operatorname{vol}(-K_X - tE) = \frac{e}{ca_1 \cdots a_d} \Big(1 - \Big(\frac{ct}{e}\Big)^d\Big).$$

For $t = 0$, we obtain $\operatorname{vol}(-K_X) = \frac{e}{ca_1 \cdots a_d}$, and $\operatorname{vol}(-K_X) = H^d$ by the asymptotic Riemann–Roch theorem [Laz04, Corollary 1.4.41] applied to the ample Cartier divisor mH.

Assume that $t \ge \frac{e}{c}$. Then (4.1) implies $i = j = 0$, so $\dim V_m(t) \le N_m = O(m^{d-1})$, and $\operatorname{vol}(-K_X - tE) = 0$.

Therefore $T_X(E) = \frac{e}{c}$ and

$$S_X(E) = \int_0^{e/c} \Big(1 - \Big(\frac{ct}{e}\Big)^d\Big) dt = \frac{e}{c} - \frac{e}{c(d+1)} = \frac{de}{c(d+1)}.$$

Since $A_X(E) = \frac{k}{c}$ (Construction-Proposition 4.9(1)),

$$\beta_X(E) = \operatorname{vol}(-K_X)\Big(\frac{k}{c} - \frac{de}{c(d+1)}\Big) = \frac{k(d+1) - d(k+c)}{c(d+1)} \operatorname{vol}(-K_X) = \frac{k - dc}{c(d+1)} \operatorname{vol}(-K_X),$$

which is negative since $c > \frac{k}{d}$ (Set-up 4.1(4)). □

Proposition 4.12 is the hypothesis of Lemma 3.1 at p, since $(E, \Delta_E) \cong (W, (1 - \frac{1}{e})S)$ by Construction-Proposition 4.9(2). The divisors in the definition of $\alpha(W, (1 - \frac{1}{e})S)$ are arbitrary and not only coordinate divisors, so we use Theorem 2.28, which holds for any subvariety of a weighted projective space.

Proposition 4.12. *Notation and conditions as in Set-up 4.1, and let* $\Delta := (1 - \frac{1}{e})S$. *Then* (W, Δ) *is klt,* $-(K_W + \Delta) \sim_{\mathbb{Q}} \frac{k}{e}\mathcal{O}_W(1)$, *and* $\alpha(W, \Delta) \ge 1$.

Proof. The proof is long, so we divide it into several steps. Let $\rho_j : U_j \to (x_j \ne 0) \subset W$, $1 \le j \le d$, be the orbifold charts of W, and let $S_j := \rho_j^{-1}(S) = (h = 0) \cap U_j$.

Step 1. In this step, we show that (W, Δ) is klt and $-(K_W + \Delta) \sim_{\mathbb{Q}} \frac{k}{e}\mathcal{O}_W(1)$.

Since W is well formed, the points of W with non-trivial stabilizer form a closed subset of codimension at least 2 in W (cf. [Tot22, Section 1]). Thus ρ_j is étale in codimension one for any j, and $\mathcal{O}_W(m)$ is the class of mH_W for any integer m (Definition-Lemma 2.16). Since h

is quasismooth, the hypersurface ring $\mathbb{C}[x_1,\dots,x_d]/(h)$ is regular outside the origin, which has codimension $d-1\geq 2$, so the ring is normal by Serre's criterion. By Lemma 2.21, h is irreducible. Thus $S\sim kH_W$. Since $\omega_W\cong\mathcal{O}_W(-k)$ (Lemma 2.22 for $\mathbb{P}=W$ and $r=0$),

$$K_W+\Delta\sim_{\mathbb{Q}}-kH_W+\Big(1-\frac{1}{e}\Big)kH_W=-\frac{k}{e}H_W.$$

Since h is quasismooth and $kh=\sum_r a_r x_r\partial h/\partial x_r$, the divisor S_j of U_j is smooth for any j. Since ρ_j is étale in codimension one, $\rho_j^*\Delta=(1-\frac{1}{e})S_j$, so $(U_j,\rho_j^*\Delta)$ is klt for any j. By Lemma 2.24, (W,Δ) is klt. In particular, $\alpha(W,\Delta)$ is defined.

Step 2. In this step, we show that for any rational number $0\leq t\leq 1$ and any $\mathbb{Q}$-divisor $D\geq 0$ on W such that $D\sim_{\mathbb{Q}}t\mathcal{O}_W(1)$, the multiplicity of D at any closed point of W is at most $\frac{t}{a_{(1)}}$, and (U_j,ρ_j^*D) is lc for any j.

Let m be a positive integer such that $mt\in\mathbb{Z}$ and $mD\sim mtH_W$. Since $\mathbb{C}[x_1,\dots,x_d]$ is normal and the vertex of $\mathbb{A}^d$ has codimension $d\geq 3$, the algebraic Hartogs lemma [Mat86, Theorem 11.5(ii)] implies that $mD=(\sigma=0)$ for some $\sigma\in\mathbb{C}[x_1,\dots,x_d]_{mt}$. The components of D are not contained in the locus of non-trivial stabilizers, so, by Definition-Lemma 2.20(1)(2), the coefficient of any component $V(\mathfrak{p})$ in mD is the order of σ at the height one prime $\mathfrak{p}$, that is, the length of $\mathbb{C}[x_1,\dots,x_d]_{\mathfrak{p}}/(\sigma)$. By Lemma 2.27,

$$\deg D=\frac{t}{a_1\cdots a_d}.$$

The product of the $d-1$ largest weights is $\frac{a_1\cdots a_d}{a_{(1)}}$, so by Corollary 2.29 for W with the weights $a_1,\dots,a_d$ in decreasing order and $r=d-2$, the multiplicity of D at any closed point is at most

$$\frac{a_1\cdots a_d}{a_{(1)}}\cdot\frac{t}{a_1\cdots a_d}=\frac{t}{a_{(1)}}\leq 1.$$

Since the components of D are not contained in the locus of non-trivial stabilizers, the multiplicity of D at $v\in(x_j\neq 0)$ (Definition 2.23) is the multiplicity of ρ_j^*D at any point of $\rho_j^{-1}(v)$. By Lemma 2.7, (U_j,ρ_j^*D) is lc.

Step 3. In this step, we show that for any rational number $0\leq t\leq 1$ and any $\mathbb{Q}$-divisor $C\geq 0$ on S such that $C\sim_{\mathbb{Q}}t\mathcal{O}_S(1)$, the pair (S_j,ρ_j^*C) is lc for any j.

We first show that the hypotheses of Definition-Lemma 2.16 hold for S. The ring $R_S:=\mathbb{C}[x_1,\dots,x_d]/(h)$ is normal (**Step 1**). For a prime number λ, let $I_\lambda:=\{r\mid \lambda\nmid a_r\}$ and $L_\lambda:=(x_r=0 \text{ for } r\in I_\lambda)\subset W$. The points of W with non-trivial stabilizer form the union of the L_λ, and $\#I_\lambda\geq 2$ since W is well formed. If $L_\lambda\not\subset S$, then $S\cap L_\lambda$ has codimension at least $\#I_\lambda\geq 2$ in S. Suppose that $L_\lambda\subset S$. Then $\dim L_\lambda=d-1-\#I_\lambda$, and we show that $\#I_\lambda\geq 3$, so that L_λ has codimension $\#I_\lambda-1\geq 2$ in S. Suppose that $I_\lambda=\{r,r'\}$. Since $k=\sum_{r''}a_{r''}\equiv a_r+a_{r'}\pmod\lambda$, no monomial of degree k is the product of x_r or $x_{r'}$ with a monomial in the variables $x_{r''}$, $r''\notin I_\lambda$. Thus $\partial h/\partial x_r$ and $\partial h/\partial x_{r'}$ vanish on the cone over L_λ. Since $L_\lambda\subset S$, we have $h\in(x_r,x_{r'})$, so $\partial h/\partial x_{r''}$ vanishes on the cone over L_λ for any $r''\notin I_\lambda$. Thus all partial derivatives of h vanish on the cone over L_λ, which is non-zero since $\dim L_\lambda=d-3\geq 0$. It contradicts the quasismoothness of h.

Let m be a positive integer such that $mt\in\mathbb{Z}$ and $mC\sim mtH_S$. Since R_S is normal and the vertex of the cone $\operatorname{Spec}R_S$ has codimension $d-1\geq 2$, the algebraic Hartogs lemma [Mat86, Theorem 11.5(ii)] implies that $mC=(\sigma=0)$ for some non-zero $\sigma\in(R_S)_{mt}$. Let $\tilde{\sigma}\in\mathbb{C}[x_1,\dots,x_d]_{mt}$ be a lift of σ. Since h is irreducible and $\tilde{\sigma}$ is not divisible by h, the polynomials $h,\tilde{\sigma}$ form a regular sequence. The minimal primes $\mathfrak{p}$ of $\mathbb{C}[x_1,\dots,x_d]/(h,\tilde{\sigma})$ correspond to the components $V(\mathfrak{p})$ of C, and the local ring $(R_S)_{\mathfrak{p}}$ is a discrete valuation ring since R_S is normal. The components of C are not contained in the locus of non-trivial stabilizers, so the length of

$(R_S)_{\mathfrak{p}}/(\sigma)$ is the coefficient of $V(\mathfrak{p})$ in mC. Thus the cycle in Lemma 2.27 for $h, \tilde{\sigma}$ is mC, and

$$\deg C = \frac{k \cdot mt}{ma_1 \cdots a_d} = \frac{tk}{a_1 \cdots a_d}.$$

The product of the $d-2$ largest weights is $\frac{a_1 \cdots a_d}{a_{(1)}a_{(2)}}$, so by Corollary 2.29 for W with the weights $a_1, \ldots, a_d$ in decreasing order and $r = d-3$, and by Set-up 4.1(3), the multiplicity of C at any closed point is at most

$$\frac{a_1 \cdots a_d}{a_{(1)}a_{(2)}} \cdot \frac{tk}{a_1 \cdots a_d} = \frac{tk}{a_{(1)}a_{(2)}} \leq t \leq 1.$$

As in **Step 2**, the multiplicity of C at $v \in (x_j \neq 0)$ is the multiplicity of $\rho_j^* C$ at any point of $\rho_j^{-1}(v)$. Since S_j is smooth (**Step 1**), $(S_j, \rho_j^* C)$ is lc by Lemma 2.7.

Step 4. In this step, we conclude the proof.

Let $B_W \geq 0$ be a $\mathbb{Q}$-divisor on W such that $B_W \sim_{\mathbb{Q}} \frac{k}{e}\mathcal{O}_W(1)$. Write $B_W = uS + B'$, where $u \geq 0$ and $S \not\subset \operatorname{Supp} B'$. Since $S \sim kH_W$, we have $B' \sim_{\mathbb{Q}} t\mathcal{O}_W(1)$, where $t := \frac{k}{e} - uk$. Since $B' \geq 0$, the first two sentences of the argument in **Step 2**, which do not use $0 \leq t \leq 1$, give a positive integer m and a non-zero $\sigma \in \mathbb{C}[x_1, \ldots, x_d]_{mt}$ with $mB' = (\sigma = 0)$, so $mt \geq 0$. Since moreover $u \geq 0$ and $e = k + c > k$, we have

$$u \leq \frac{1}{e} \quad \text{and} \quad 0 \leq t \leq \frac{k}{e} < 1.$$

Recall that $mB' = (\sigma = 0)$, where m is a positive integer and $\sigma \in \mathbb{C}[x_1, \ldots, x_d]_{mt}$ is non-zero. Since $S \not\subset \operatorname{Supp} B'$, σ is not divisible by h, and $C := \frac{1}{m}(\sigma|_S = 0)$ is a $\mathbb{Q}$-divisor $C \geq 0$ on S such that $C \sim_{\mathbb{Q}} t\mathcal{O}_S(1)$. Both $\rho_j^* C$ and $(\rho_j^* B')|_{S_j}$ are $\frac{1}{m}$ times the divisor of $\sigma|_{S_j}$, so $\rho_j^* C = (\rho_j^* B')|_{S_j}$. Since U_j and S_j are smooth, $\operatorname{Diff}_{S_j}(\rho_j^* B') = (\rho_j^* B')|_{S_j}$. By **Step 3**, $(S_j, \operatorname{Diff}_{S_j}(\rho_j^* B'))$ is lc. By Theorem 2.6, $(U_j, S_j + \rho_j^* B')$ is lc near S_j. Since

$$\rho_j^*(\Delta + B_W) = \Big(1 - \frac{1}{e} + u\Big) S_j + \rho_j^* B' \leq S_j + \rho_j^* B',$$

$(U_j, \rho_j^*(\Delta + B_W))$ is lc near S_j. Outside S_j, $\rho_j^*(\Delta + B_W) = \rho_j^* B'$, and $(U_j, \rho_j^* B')$ is lc by **Step 2**. Thus $(U_j, \rho_j^*(\Delta + B_W))$ is lc for any j.

By Lemma 2.24, $(W, \Delta + B_W)$ is lc. Since B_W is arbitrary, $\alpha(W, \Delta) \geq 1$. □

Proposition 4.13. *Notation and conditions as in Set-up 4.1. Then for any $\mathbb{Q}$-divisor $B \geq 0$ on X such that $B \sim_{\mathbb{Q}} H$, the pair $(X, \frac{k}{e}B)$ is lc near p. Moreover, let $(x_1 = 0)|_X$ denote the divisor of zeros of the section $x_1 \in R_{a_1}$ of $\mathcal{O}_X(a_1)$. Then $\operatorname{lct}\big(X; \frac{1}{a_1}(x_1 = 0)|_X\big) \leq \frac{k}{e}$. In particular, $\alpha(X) \leq \frac{k}{e}$.*

Proof. Recall that $f : Y \to X$ is the plt blow-up of X at p with exceptional divisor E from Construction-Proposition 4.9. Let $B \geq 0$ be a $\mathbb{Q}$-divisor on X such that $B \sim_{\mathbb{Q}} H$. Since $H \sim -K_X$ (Proposition 4.8), B is $\mathbb{Q}$-Cartier and $B \sim_{\mathbb{Q}} -K_X$. By Lemma 2.13 and Proposition 4.11, $\operatorname{mult}_E B \leq T_X(E) = \frac{e}{c}$. Since $A_X(E) = \frac{k}{c}$ (Construction-Proposition 4.9(1)),

$$\operatorname{mult}_E\Big(\frac{k}{e}B\Big) \leq \frac{k}{e} \cdot \frac{e}{c} = \frac{k}{c} = A_X(E).$$

The variety X is klt (Proposition 4.8), and $\alpha(E, \Delta_E) = \alpha(W, (1 - \frac{1}{e})S) \geq 1$ by Construction-Proposition 4.9(2) and Proposition 4.12. By Lemma 3.1, $(X, \frac{k}{e}B)$ is lc near p.

Since x_1 is a basis monomial of weight ea_1, $b(x_1) = ea_1$. By Definition-Lemma 4.10(2),

$$\operatorname{mult}_E \frac{1}{a_1}(x_1 = 0)|_X = \frac{1}{a_1} \cdot \frac{ea_1}{c} = \frac{e}{c}.$$

Let $D := \frac{1}{a_1}(x_1 = 0)|_X$. For any $\lambda > \frac{k}{e}$,

$$a(E, X, \lambda D) = A_X(E) - \lambda \operatorname{mult}_E D = \frac{k}{c} - \frac{\lambda e}{c} < 0,$$

so $(X, \lambda D)$ is not lc near p, since $f(E) = \{p\}$. Thus $\mathrm{lct}(X; D) \le \frac{k}{e}$. Since $D \ge 0$ and $D \sim_{\mathbb{Q}} H \sim -K_X$, Definition 2.10 implies $\alpha(X) \le \mathrm{lct}(X; D) \le \frac{k}{e}$. □

4.4. **Projection from the point p.** In this subsection, we prove three lemmas for the proof of Theorem 4.3, where it remains to show that (X, D) is lc outside p for any $\mathbb{Q}$-divisor $D \ge 0$ with $D \sim_{\mathbb{Q}} -K_X$. Since X may fail to be quasismooth over the singular points of $(G = 0)|_S$, we project $X\backslash\{p\}$ from p to $\mathbb{P}(a_1, \dots, a_d, 1)$. The projection is crepant for the pair formed by $\mathbb{P}(a_1, \dots, a_d, 1)$ and the cone over S (Lemma 4.17), and we prove log canonicity on $\mathbb{P}(a_1, \dots, a_d, 1)$ and on the cone over S outside the vertex by multiplicity bounds and inversion of adjunction (Lemmas 4.15 and 4.16).

Set-up 4.14. Notation and conditions as in Set-up 4.1, and let $\mathbb{P}'$, C_S, q, π, and $(f = 0)|_{C_S}$ be as follows.

(1) $\mathbb{P}' := \mathbb{P}(a_1, \dots, a_d, 1)$ with coordinates $x_1, \dots, x_d, y$, in which W is identified with the hyperplane $(y = 0)$.
(2) $C_S := (h = 0) \subset \mathbb{P}'$, so that $S = C_S \cap W$, and $q := [0 : \cdots : 0 : 1] \in \mathbb{P}'$.
(3) $\pi : X\backslash\{p\} \to \mathbb{P}'$ is the projection forgetting the coordinate z.
(4) For a weighted homogeneous polynomial $f \in \mathbb{C}[x_1, \dots, x_d, y]$ that is not divisible by h, $(f = 0)|_{C_S}$ is the $\mathbb{Q}$-divisor on C_S whose restriction to $C_S\backslash\{q\}$ is the $\mathbb{Q}$-divisor $(f = 0)|_{C_S\backslash\{q\}}$ of Definition-Lemma 2.20(3), where $C_S\backslash\{q\}$ is quasismooth because h is quasismooth; it is unique, since q has codimension $d - 1 \ge 2$ in C_S.

After clearing the denominators of z, a section of $\mathcal{O}_X(M)$ becomes a polynomial in $\mathbb{C}[x_1, \dots, x_d, y]$ of degree $M + jk$ for some integer $0 \le j \le M/c$, so the divisors on $\mathbb{P}'$ to be treated have class at most $\frac{e}{c}\mathcal{O}_{\mathbb{P}'}(1)$. Lemma 4.15(1) treats them using $a_{(1)} \ge e/c$, and Lemma 4.15(2)(3) treat divisors on C_S of class at most $\mathcal{O}_{C_S}(1)$ using $a_{(1)}a_{(2)} \ge k$.

Lemma 4.15. *Notation and conditions as in Set-up 4.14, and assume that $a_{(1)} \ge e/c$. Then the following hold.*

(1) Let $D \ge 0$ be a $\mathbb{Q}$-divisor on $\mathbb{P}'$ such that $D \sim_{\mathbb{Q}} t\mathcal{O}_{\mathbb{P}'}(1)$ for some $0 \le t \le e/c$ and W is not a component of D. Then $(\mathbb{P}', \lambda W + D)$ is lc for any $0 \le \lambda \le 1$.
(2) Let $D \ge 0$ be a $\mathbb{Q}$-divisor on C_S such that $D \sim_{\mathbb{Q}} t\mathcal{O}_{C_S}(1)$ for some $0 \le t \le 1$ and S is not a component of D. Then $(C_S\backslash\{q\}, \lambda S + D)$ is lc on orbifold charts for any $0 \le \lambda \le 1$.
(3) For any homogeneous polynomial $s \in \mathbb{C}[x_1, \dots, x_d, y]$ of degree $m > 0$ that is not divisible by h, the pair $\left(C_S\backslash\{q\}, \frac{1}{m}(s = 0)|_{C_S}\right)$ is lc on orbifold charts.

Proof. We first collect facts on $\mathbb{P}'$ and C_S. Since $e = k + c > c$, we have $a_{(1)} \ge e/c > 1$. Thus y is the only coordinate of $\mathbb{P}'$ of the smallest weight 1, and in Corollary 2.29 on $\mathbb{P}'$ the hyperplane $(x_N = 0)$ is W. A point of $\mathbb{P}'$ with $y \ne 0$ has trivial stabilizer, and so does a point with $x_1 \cdots x_d \ne 0$, because W is well formed and hence $\gcd(a_1, \dots, a_d) = 1$. Thus the points of $\mathbb{P}'$ with non-trivial stabilizer are contained in $W \cap (x_1 \cdots x_d = 0)$, which has codimension 2 in $\mathbb{P}'$, and $\mathbb{P}'$ is well formed.

Let $R_{C_S} := \mathbb{C}[x_1, \dots, x_d, y]/(h)$. Since h is quasismooth, the affine cone $(h = 0) \times \mathbb{A}^1_y$ of C_S is singular only along $x_1 = \cdots = x_d = 0$, whose codimension is $d - 1 \ge 2$. Since a hypersurface satisfies S_2, R_{C_S} is normal by Serre's criterion [Mat86, Theorem 23.8], and R_{C_S} is a domain because h is irreducible (Lemma 2.21). In particular, $C_S\backslash\{q\}$ and S are quasismooth. The points of C_S with non-trivial stabilizer are contained in $S \cap (x_1 \cdots x_d = 0)$, which has codimension at least 2 in C_S because h divides no x_i. Since $R_{C_S}/(y) \cong \mathbb{C}[x_1, \dots, x_d]/(h)$ is a domain, (y) is a prime ideal of R_{C_S} and $\mathrm{div}_{C_S}(y) = S$, where for $V' \in \{\mathbb{P}', C_S\backslash\{q\}, W, S\}$, respectively $V' = C_S$, and a homogeneous polynomial f that does not vanish identically on V', we write $\mathrm{div}_{V'}(f) := (f = 0)|_{V'}$ (Definition-Lemma 2.20(3), respectively Set-up 4.14(4)).

Write $x_{d+1} := y$ and $a_{d+1} := 1$. For $1 \le j \le d + 1$, let $\rho_j : U_j \to (x_j \ne 0) \subset \mathbb{P}'$ be the j-th orbifold chart of $\mathbb{P}'$, where $U_j = \mathbb{A}^d$ has coordinates x_i $(i \ne j)$, and for a polynomial f in $x_1, \dots, x_{d+1}$ let f_j be the polynomial on U_j obtained by setting $x_j = 1$. Since $C_S\backslash\{q\}$

and S are quasismooth, Definition-Lemma 2.20(2) implies that $V_j := \rho_j^{-1}(C_S\backslash\{q\})$ and $T_j := \rho_j^{-1}(S) = V_j \cap (y=0)$ are smooth. For $j \le d$, T_j is a divisor on V_j, and $T_{d+1} = \emptyset$. Moreover, for $V' \in \{\mathbb{P}', C_S\backslash\{q\}, W, S\}$ and a homogeneous polynomial f that does not vanish identically on V', Definition-Lemma 2.20(3) gives

$$\rho_j^* \operatorname{div}_{V'}(f) = \operatorname{div}_{\rho_j^{-1}(V')}(f_j).$$

By Set-up 4.14(4), the formula for $\rho_j^* \operatorname{div}_{V'}(f)$ also holds for $V' = C_S$, since $\rho_j^{-1}(q)$ is empty for $j \le d$ and is the origin of U_{d+1}, which has codimension $d-1 \ge 2$ in the normal hypersurface $\rho_{d+1}^{-1}(C_S) = (h=0) \subset U_{d+1}$. In particular, $\rho_j^* S = \operatorname{div}_{V_j}(y) = T_j$.

Let V' be $\mathbb{P}'$ in (1) and C_S in (2), let $R_{V'}$ be $\mathbb{C}[x_1,\dots,x_d,y]$ in (1) and R_{C_S} in (2), and let D be as in (1), respectively (2). We show that $ND = \operatorname{div}_{V'}(s)$ for a positive integer N and a non-zero element $s \in (R_{V'})_{Nt}$. In (1), since $\mathbb{C}[x_1,\dots,x_d,y]$ is a UFD and y has weight 1, any prime divisor of $\mathbb{P}'$ is $(\varphi = 0)$ for an irreducible homogeneous polynomial φ, and $(\varphi=0) - (\deg\varphi)W$ is the divisor of $\varphi/y^{\deg\varphi}$. Writing $D = \sum_i c_i(\varphi_i = 0)$ and choosing N with $Nc_i \in \mathbb{Z}$ for any i, the polynomial $s := \prod_i \varphi_i^{Nc_i}$ satisfies $ND - (\deg s)W \sim 0$. Since $\mathcal{O}_{\mathbb{P}'}(1)$ is the class of $W = \operatorname{div}(y)$ and W is not $\mathbb{Q}$-linearly trivial, $\deg s = Nt$. In (2), $\mathcal{O}_{C_S}(ND) \cong \mathcal{O}_{C_S}(Nt)$ for N divisible enough (Definition-Lemma 2.16). Since R_{C_S} is a normal domain of dimension $d \ge 3$, R_{C_S} is the intersection of the localizations at height one primes [Mat86, Theorem 11.5(ii)], and no height one prime of R_{C_S} contains the irrelevant ideal. Thus the sections of $\mathcal{O}_{C_S}(Nt)$ are the elements of $(R_{C_S})_{Nt}$, and s exists.

The components of $\operatorname{div}_{V'}(s)$ are not contained in the locus of points with non-trivial stabilizer, so the local ring of the affine cone of V' at the corresponding homogeneous prime is a discrete valuation ring (the cone is normal) whose valuation of s is, by Definition-Lemma 2.20(1)(2), the coefficient of the component in $\operatorname{div}_{V'}(s)$. Thus ND is the cycle Z of Lemma 2.27 for the regular sequence s in (1), respectively h, s in (2), and

$$\deg D = \frac{t}{a_1\cdots a_d} \text{ in (1)}, \qquad \deg D = \frac{kt}{a_1\cdots a_d} \text{ in (2)}.$$

Let $\mathfrak{p}$ be a minimal prime of $R_{V'}/(s)$, and let $(\mathfrak{p}_j = 0) \subset U_j$ be the closed subscheme defined by the f_j with $f \in \mathfrak{p}$. By Definition-Lemma 2.20(1)(2), the morphism $\mathbb{G}_m \times U_j \to (x_j \ne 0)$ is étale and the preimage of the affine cone of $V(\mathfrak{p})$ is $\mathbb{G}_m \times (\mathfrak{p}_j = 0)$. Thus the cycle of $\operatorname{div}_{\rho_j^{-1}(V')}(s_j) = N\rho_j^* D$ on U_j is $\sum_{\mathfrak{p}} n_{\mathfrak{p}}[(\mathfrak{p}_j = 0)]$, where $n_{\mathfrak{p}}$ is the coefficient of $V(\mathfrak{p})$ in ND. Since the multiplicity of a cycle at a point does not depend on the ambient variety [Ful98, Section 4.3], Definition 2.23 implies that the multiplicity of D at $\rho_j(u)$ is $\operatorname{mult}_u \rho_j^* D$ for any $u \in \rho_j^{-1}(V')$. The orbifold charts of W are the restrictions of $\rho_1,\dots,\rho_d$ to $(y=0)$, so the argument for D shows that for a cycle Z on W and $u \in (y=0) \subset U_j$, the multiplicity of Z at $\rho_j(u)$ is the multiplicity of $(\rho_j|_{(y=0)})^* Z$ at u.

(1) Since W is not a component of D, y does not divide s, and no component of D is contained in W. By the second bound of Corollary 2.29 on $\mathbb{P}'$ with $r = d-1$, the multiplicity of D at any point of $\mathbb{P}'$ is at most

$$\frac{a_1\cdots a_d}{a_{(1)}} \cdot 1 \cdot \deg D = \frac{t}{a_{(1)}} \le \frac{e}{ca_{(1)}} \le 1.$$

Thus $\operatorname{mult}_u \rho_j^* D \le 1$ for any $u \in U_j$, and $(U_j, \rho_j^* D)$ is lc by Lemma 2.7. Let $j \le d$ and $T := \rho_j^{-1}(W) = (y=0) \subset U_j$. Since $y \nmid s$, the polynomial $s(x_1,\dots,x_d,0)$ is non-zero, $N\rho_j^* D|_T = \operatorname{div}_T(s_j|_T)$, and by Lemma 2.27 the cycle Z_W of $(s(x_1,\dots,x_d,0)=0)$ on W satisfies $\deg Z_W = Nt/(a_1\cdots a_d)$. Since Z_W is a cycle on W, the multiplicity of Z_W at $\rho_j(u)$ is $N \operatorname{mult}_u(\rho_j^* D|_T)$ for $u \in T$. By the first bound of Corollary 2.29 on W with $r = d-2$,

$$\operatorname{mult}_u(\rho_j^* D|_T) \le \frac{a_1\cdots a_d}{a_{(1)}} \cdot \frac{t}{a_1\cdots a_d} = \frac{t}{a_{(1)}} \le 1.$$

Since U_j and T are smooth, $\mathrm{Diff}_T(\rho_j^*D) = \rho_j^*D|_T$, so $(U_j, T+\rho_j^*D)$ is lc near T by Lemma 2.7 and Theorem 2.6. Since $\rho_j^*(\lambda W+D) = \lambda T+\rho_j^*D \le T+\rho_j^*D$ and $\rho_{d+1}^{-1}(W)=\emptyset$, $(U_j, \rho_j^*(\lambda W+D))$ is lc for any j. Hence $(\mathbb{P}', \lambda W+D)$ is lc by Lemma 2.24.

(2) A closed subvariety of C_S of dimension $d-2$ contained in W is contained in $C_S\cap W = S$, which is irreducible of dimension $d-2$, so it is S. Hence no component of D is contained in W, and by the second bound of Corollary 2.29 on $\mathbb{P}'$ with $r=d-2$, the multiplicity of D at any point of C_S is at most

$$\frac{a_1\cdots a_d}{a_{(1)}a_{(2)}}\cdot 1\cdot \deg D = \frac{kt}{a_{(1)}a_{(2)}} \le 1.$$

Thus (V_j, ρ_j^*D) is lc by Lemma 2.7. Let $j\le d$. If $s\in yR_{C_S}$, then

$$\mathrm{div}_{C_S}(s) \ge \mathrm{div}_{C_S}(y) = S,$$

which is not the case. Thus $s(x_1,\dots,x_d,0)$ is not divisible by h, $N\rho_j^*D|_{T_j} = \mathrm{div}_{T_j}(s_j|_{T_j})$, and by Lemma 2.27 the cycle Z_S of $(h=s(x_1,\dots,x_d,0)=0)$ on W satisfies $\deg Z_S = kNt/(a_1\cdots a_d)$. Since Z_S is a cycle on W, the multiplicity of Z_S at $\rho_j(u)$ is $N\,\mathrm{mult}_u(\rho_j^*D|_{T_j})$ for $u\in T_j$. By the first bound of Corollary 2.29 on W with $r=d-3$,

$$\mathrm{mult}_u(\rho_j^*D|_{T_j}) \le \frac{a_1\cdots a_d}{a_{(1)}a_{(2)}}\cdot\frac{kt}{a_1\cdots a_d} = \frac{kt}{a_{(1)}a_{(2)}} \le 1.$$

As in (1), Lemma 2.7 and Theorem 2.6 imply that $(V_j, T_j+\rho_j^*D)$ is lc near T_j. Since

$$\rho_j^*(\lambda S+D) = \lambda T_j+\rho_j^*D \le T_j+\rho_j^*D$$

and $T_{d+1}=\emptyset$, $(V_j, \rho_j^*(\lambda S+D))$ is lc for any j, that is, $(C_S\backslash\{q\}, \lambda S+D)$ is lc on orbifold charts.

(3) The image $\bar{s}$ of s in R_{C_S} is non-zero. Let $n\ge 0$ be the largest integer such that $\bar{s}=y^ns'$ for some $s'\in R_{C_S}$. Since y has degree 1, we have $n\le m$ and $s'\in (R_{C_S})_{m-n}$. Then

$$\frac{1}{m}(s=0)|_{C_S} = \frac{n}{m}S+\frac{1}{m}\mathrm{div}_{C_S}(s'),$$

where

$$\frac{1}{m}\mathrm{div}_{C_S}(s') \sim_{\mathbb{Q}} \frac{m-n}{m}\mathcal{O}_{C_S}(1)$$

by Definition-Lemma 2.16, and S is not a component of $\mathrm{div}_{C_S}(s')$ because $s'\notin(y)$ and (y) is a prime ideal of R_{C_S}. Thus (3) follows from (2) with $\lambda=n/m$ and $t=(m-n)/m$. □

Since $z=-g/h$ on X, we have $\pi^*(g=0) = (z=0)|_X+\pi^*C_S$ on $X\backslash\{p\}$, where $(z=0)|_X$ is the divisor of zeros of the section $z\in R_c$, and $\frac{1}{c}(z=0)|_X \sim_{\mathbb{Q}} -K_X$ (Proposition 4.8).

Lemma 4.16. *Notation and conditions as in Set-up 4.14, and assume that $\left(S, \frac{1}{c}(G=0)|_S\right)$ is lc on orbifold charts. Then the pair $\left(C_S\backslash\{q\}, \frac{1}{c}(g=0)|_{C_S}\right)$ is lc on orbifold charts.*

Proof. Write $x_{d+1}:=y$ and $a_{d+1}:=1$. For $1\le j\le d+1$, let $\rho_j: U_j\to(x_j\ne 0)\subset\mathbb{P}'$ be the j-th orbifold chart of $\mathbb{P}'$, and for a polynomial f in $x_1,\dots,x_{d+1}$ let f_j be the polynomial on $U_j=\mathbb{A}^d$ obtained by setting $x_j=1$.

Let $V_j:=\rho_j^{-1}(C_S\backslash\{q\})$ and $T_j:=\rho_j^{-1}(S)=V_j\cap(y=0)$. Since h is quasismooth, $C_S\backslash\{q\}$ and S are quasismooth, so V_j and T_j are smooth by Definition-Lemma 2.20(2). For $j\le d$, T_j is a divisor on V_j and $\rho_j|_{T_j}$ is the j-th orbifold chart of S, and $T_{d+1}=\emptyset$. By Definition-Lemma 2.20(3) and Set-up 4.14(4), and since $g_j|_{T_j}=G_j|_{T_j}$, we have

$$\rho_j^*\big((g=0)|_{C_S}\big) = \mathrm{div}_{V_j}(g_j) \quad\text{and}\quad \rho_j^*\big((G=0)|_S\big) = \mathrm{div}_{T_j}(G_j) = \mathrm{div}_{V_j}(g_j)|_{T_j}.$$

Here T_j is not a component of $\mathrm{div}_{V_j}(g_j)$, because h does not divide G.

Let $j\le d$. By assumption, $\left(T_j, \frac{1}{c}\mathrm{div}_{T_j}(G_j)\right)$ is lc. Since V_j and T_j are smooth,

$$\mathrm{Diff}_{T_j}\left(\frac{1}{c}\mathrm{div}_{V_j}(g_j)\right) = \frac{1}{c}\mathrm{div}_{T_j}(G_j),$$

so $\big(V_j, T_j + \frac{1}{c}\operatorname{div}_{V_j}(g_j)\big)$ is lc near T_j by Theorem 2.6.

It remains to treat $V_j\backslash T_j$ for any j. Let $(x,y) \in \mathbb{A}^{d+1}$ be a point with $h(x) = g(x,y) = 0$ and $y \neq 0$. If $x = 0$, then $g(x,y) = y^e \neq 0$. Thus $x \neq 0$, and $\nabla_x h(x) \neq 0$ because h is quasismooth. The Jacobian matrix of (h,g) at (x,y) has rows $(\nabla_x h(x), 0)$ and $(\nabla_x G(x), ey^{e-1})$, which are linearly independent. Thus $(h = g = 0) \subset \mathbb{A}^{d+1}$ is smooth of codimension 2 along $(y \neq 0)$. By Definition-Lemma 2.20(1), $(h_j = g_j = 0) \cap (y \neq 0) \subset U_j$ is smooth, also for $j = d+1$, where $y = 1$. Hence $\operatorname{div}_{V_j}(g_j)$ is a smooth reduced divisor on $V_j\backslash T_j$, and since $c \geq 1$,

$$\operatorname{mult}_u \frac{1}{c}\operatorname{div}_{V_j}(g_j) \leq \frac{1}{c} \leq 1 \quad \text{for any } u \in V_j\backslash T_j.$$

By Lemma 2.7, the pair

$$\Big(V_j\backslash T_j, \frac{1}{c}\operatorname{div}_{V_j}(g_j)\Big)$$

is lc. Thus $\big(V_j, \frac{1}{c}\operatorname{div}_{V_j}(g_j)\big)$ is lc for any j, that is, $\big(C_S\backslash\{q\}, \frac{1}{c}(g=0)|_{C_S}\big)$ is lc on orbifold charts. □

Lemma 4.17. *Notation and conditions as in Set-up 4.14. Then π is an isomorphism over $\mathbb{P}'\backslash C_S$, the point q is not in the image of π, and $K_{X\backslash\{p\}} = \pi^*(K_{\mathbb{P}'} + C_S)$. In particular,*

$$a(F, X\backslash\{p\}, \pi^* D) = a(F, \mathbb{P}', C_S + D)$$

for any prime divisor F over X whose center on X is not p and any $\mathbb{Q}$-Cartier $\mathbb{Q}$-divisor D on $\mathbb{P}'$.

Proof. Since $x_1 = \cdots = x_d = y = 0$ defines the point p of $\mathbb{P}$, forgetting z defines a morphism $\mathbb{P}\backslash\{p\} \to \mathbb{P}'$, whose restriction to $X\backslash\{p\}$ is π. Over $\mathbb{P}'\backslash C_S$, the equation of X is $z = -g/h$, where $-g/h$ has degree $e - k = c$. Thus $[x_1 : \cdots : x_d : y] \to [x_1 : \cdots : x_d : y : -g/h]$ is the inverse of π over $\mathbb{P}'\backslash C_S$. If $[x_1 : \cdots : x_d : y : z] \in X$ and $h(x_1, \ldots, x_d) = 0$, then $g(x_1, \ldots, x_d, y) = 0$. Since $g(0, \ldots, 0, 1) = 1$, q is not in the image of π.

Write $x_{d+1} := y$ and $a_{d+1} := 1$. For $1 \leq j \leq d+1$, let

$$\rho_j : U_j \to (x_j \neq 0) \subset \mathbb{P}' \quad \text{and} \quad \tilde{\rho}_j : U_j \times \mathbb{A}^1_z \to (x_j \neq 0) \subset \mathbb{P}$$

be the j-th orbifold charts of $\mathbb{P}'$ and $\mathbb{P}$, where $U_j = \mathbb{A}^d$, and for a polynomial f let f_j be the polynomial obtained by setting $x_j = 1$. Let $\widetilde{X}_j := \tilde{\rho}_j^{-1}(X) = (zh_j + g_j = 0)$ and let $\pi_j : \widetilde{X}_j \to U_j$ be the projection, so that $\rho_j \circ \pi_j = \pi \circ \tilde{\rho}_j|_{\widetilde{X}_j}$. The open subsets $(x_j \neq 0)$, $1 \leq j \leq d+1$, cover $X\backslash\{p\}$. By Definition-Lemma 2.20(2) for $X \subset \mathbb{P}$, the morphism

$$\mathbb{G}_m \times \widetilde{X}_j \to \widehat{X} \cap (x_j \neq 0)$$

is étale and surjective, where $\widehat{X} \subset \mathbb{A}^{d+2}\backslash\{0\}$ is the preimage of X. Since R is normal (Proposition 4.8), $\widetilde{X}_j$ is normal.

Let du be the standard volume form on $U_j = \mathbb{A}^d$. By adjunction for the closure of $\widetilde{X}_j$ in $\mathbb{P}^{d+1} \supset U_j \times \mathbb{A}^1_z$ [KM98, Proposition 5.73], $\omega_{\widetilde{X}_j}$ is generated by the residue Ω_j of $du \wedge dz/(zh_j + g_j)$. The residue Ω_j satisfies $d(zh_j + g_j) \wedge \Omega_j = du \wedge dz$ along $\widetilde{X}_j$. Since $\partial(zh_j + g_j)/\partial z = h_j$, on $\widetilde{X}_j \cap (h_j \neq 0)$ we have $\Omega_j = (-1)^d \pi_j^* du/h_j$, because

$$d(zh_j + g_j) \wedge (-1)^d du/h_j = (-1)^d dz \wedge du = du \wedge dz.$$

Thus

$$\pi_j^* du = (-1)^d h_j \Omega_j$$

on $\widetilde{X}_j \cap (h_j \neq 0)$, hence on $\widetilde{X}_j$, because

$$\widetilde{X}_j \cap (h_j = 0) = (h_j = g_j = 0) \times \mathbb{A}^1_z \quad \text{and} \quad \dim(h_j = g_j = 0) = d - 2,$$

as h is irreducible and does not divide g.

Let ω be a rational d-form on $\mathbb{P}'$, $K_{\mathbb{P}'} := \operatorname{div}(\omega)$, and $K_{X\setminus\{p\}} := \operatorname{div}(\pi^*\omega)$. A point of $\mathbb{P}'$ with non-trivial stabilizer satisfies $y = 0$ and $x_i = 0$ for some i, because $\gcd(a_1, \dots, a_d) = 1$ (W is well formed). Thus ρ_j is étale in codimension one, and so is $\tilde{\rho}_j|_{\widetilde{X}_j}$ by Lemma 4.5 with $\tau = 1$. By Lemma 2.8(2) with $B = 0$,

$$\operatorname{div}(\rho_j^*\omega) = \rho_j^* K_{\mathbb{P}'} \quad \text{and} \quad \operatorname{div}(\tilde{\rho}_j^*\pi^*\omega) = \tilde{\rho}_j^* K_{X\setminus\{p\}} \text{ on } \widetilde{X}_j.$$

Moreover, since $C_S = (h = 0)|_{\mathbb{P}'}$, Definition-Lemma 2.20(3) gives $\rho_j^* C_S = \operatorname{div}(h_j)$. Writing $\rho_j^*\omega = \varphi du$ for a rational function φ on U_j, we obtain

$$\tilde{\rho}_j^*\pi^*\omega = \pi_j^*(\varphi du) = (-1)^d(\pi_j^*\varphi)h_j\Omega_j.$$

Since Ω_j generates $\omega_{\widetilde{X}_j}$,

$$\tilde{\rho}_j^* K_{X\setminus\{p\}} = \pi_j^*\big(\operatorname{div}(\varphi) + \operatorname{div}(h_j)\big) = \pi_j^*\rho_j^*(K_{\mathbb{P}'} + C_S) = \tilde{\rho}_j^*\pi^*(K_{\mathbb{P}'} + C_S)$$

on $\widetilde{X}_j$. Since the morphism

$$\tilde{\rho}_j|_{\widetilde{X}_j} : \widetilde{X}_j \to X \cap (x_j \neq 0)$$

is finite and surjective, pulling back $\mathbb{Q}$-divisors along $\tilde{\rho}_j|_{\widetilde{X}_j}$ is injective. Thus

$$K_{X\setminus\{p\}} = \pi^*(K_{\mathbb{P}'} + C_S).$$

Finally, let D be a $\mathbb{Q}$-Cartier $\mathbb{Q}$-divisor on $\mathbb{P}'$ and F a prime divisor over X whose center on X is not p, so that F is a prime divisor over $X\setminus\{p\}$. Then

$$K_{X\setminus\{p\}} + \pi^* D = \pi^*(K_{\mathbb{P}'} + C_S + D).$$

Let $\sigma : Y' \to X\setminus\{p\}$ be a proper birational morphism from a normal variety Y' on which F is a divisor, and let $K_{Y'} := \operatorname{div}(\sigma^*\pi^*\omega)$. Since $\pi \circ \sigma$ is a birational morphism and a log discrepancy only depends on a neighborhood in Y' of the generic point of F, both $a(F, X\setminus\{p\}, \pi^*D) - 1$ and $a(F, \mathbb{P}', C_S + D) - 1$ are the coefficient of F in $K_{Y'} - \sigma^*\pi^*(K_{\mathbb{P}'} + C_S + D)$. □

4.5. **Proof of Theorem 4.3.** In this subsection, we prove Theorem 4.3. **Steps 1–3** prepare the proof: **Step 2** reduces the theorem to divisors of sections that are not divisible by h in R, and **Step 3** writes such a divisor outside p as a pullback from $\mathbb{P}'$. **Steps 4 and 5** treat prime divisors over X whose centers on $\mathbb{P}'$ are not contained in, respectively contained in, C_S, and **Step 6** concludes the proof, using Proposition 4.13 near p.

Lemma 4.18. *Notation and conditions as in Set-up 4.14. Let $u \in \mathbb{C}[x_1, \dots, x_d, y]$ be a weighted homogeneous polynomial of positive degree that is not divisible by h, and let $\lambda > 0$ be a rational number. Assume that the pair $\big(C_S\setminus\{q\}, \lambda(u = 0)|_{C_S}\big)$ is lc on orbifold charts. Then $\big(\mathbb{P}', C_S + \lambda(u = 0)\big)$ is lc near $C_S\setminus\{q\}$.*

Proof. The weighted projective space $\mathbb{P}'$ is well formed: any d of the weights $a_1, \dots, a_d, 1$ either contain the weight 1 or are $a_1, \dots, a_d$, whose greatest common divisor is 1 because W is well formed, so $\mathbb{P}'$ satisfies the conditions of Lemma 2.24. Let $\rho_t : U_t \to \mathbb{P}'$, $1 \le t \le d+1$, be the orbifold charts of $\mathbb{P}'$ (Definition 2.15), and let $U_t^\circ := U_t\setminus\rho_t^{-1}(q)$. Since h is quasismooth, the preimage $(h = 0)$ of C_S in $\mathbb{A}^{d+1}\setminus\{0\}$ is smooth outside the y-axis, and the y-axis maps to q. Hence $\rho_t^* C_S = \rho_t^{-1}(C_S)$ is a smooth divisor on the smooth variety U_t°, and it is not a component of $\rho_t^*(u = 0)$. By Definition-Proposition 2.5(1), the different of $\lambda\rho_t^*(u = 0)$ on $\rho_t^{-1}(C_S) \cap U_t^\circ$ is $\rho_t^*\big(\lambda(u = 0)|_{C_S}\big)$. Thus Theorem 2.6 implies that $\big(U_t^\circ, \rho_t^*(C_S + \lambda(u = 0))\big)$ is lc near $\rho_t^{-1}(C_S) \cap U_t^\circ$ for any t. Let $v \in C_S\setminus\{q\}$ be a closed point, and choose t such that the t-th coordinate of v is non-zero. Then $\rho_t^{-1}(v) \subset \rho_t^{-1}(C_S) \cap U_t^\circ$, so Lemma 2.24, applied to the open subset $\mathbb{P}'\setminus\{q\}$, implies that $\big(\mathbb{P}', C_S + \lambda(u = 0)\big)$ is lc near v. Thus $\big(\mathbb{P}', C_S + \lambda(u = 0)\big)$ is lc near $C_S\setminus\{q\}$. □

Proof of Theorem 4.3. The proof is long, so we divide it into several steps. By Proposition 4.8, X is a $\mathbb{Q}$-Fano variety of dimension d, $-K_X \sim H$, and $H^0(X, \mathcal{O}_X(m)) = R_m$ for every integer m. For a non-zero $s \in R_m$, we denote by $(s=0)$ the divisor of zeros of s as a section of $\mathcal{O}_X(m)$, and we write $(s=0)|_X$ when s is given by a polynomial. Since the generic point of any prime divisor on X has trivial stabilizer (Lemma 4.5 with $\tau = 1$), the coefficients of $(s=0)$ can be computed on the affine cone $\operatorname{Spec} R$ by Definition-Lemma 2.20(1)(2), so $(st=0) = (s=0) + (t=0)$ for any non-zero homogeneous $s, t \in R$. For a prime divisor F over X such that $\operatorname{center}_X F \neq \{p\}$, we denote by $\operatorname{center}_{\mathbb{P}'} F$ the closure of $\pi(\operatorname{center}_X F \backslash \{p\})$, which is the center of F on $\mathbb{P}'$ because π is birational (Lemma 4.17).

Step 1. In this step we show that $(y=0)|_X$ is a prime divisor and that $\operatorname{mult}_{(y=0)|_X} D \leq 1$ for any $\mathbb{Q}$-divisor $D \geq 0$ on X such that $D \sim_{\mathbb{Q}} H$. We use $\operatorname{mult}_{(y=0)|_X} D \leq 1$ in **Step 4**.

Since $d \geq 3$ and h is quasismooth, h is irreducible (Lemma 2.21). Since h does not divide G, the polynomial $zh + G$ is primitive of degree one in z over $\mathbb{C}[x_1, \dots, x_d]$, hence irreducible. Thus

$$R/(y) \cong \mathbb{C}[x_1, \dots, x_d, z]/(zh + G)$$

is a domain, so yR is a prime ideal of the normal domain R. Hence the divisor of zeros of $y \in R_1$ is the prime divisor $(y=0)|_X$ with coefficient one, and $(y=0)|_X \sim H$.

Let $D \geq 0$ be a $\mathbb{Q}$-divisor on X such that $D \sim_{\mathbb{Q}} H$, and let $\gamma := \operatorname{mult}_{(y=0)|_X} D$. Then $D - \gamma(y=0)|_X \geq 0$ and $D - \gamma(y=0)|_X \sim_{\mathbb{Q}} (1-\gamma)H$. Since H is ample,

$$(1-\gamma)(H^d) = \big((D - \gamma(y=0)|_X) \cdot H^{d-1}\big) \geq 0 \quad \text{and} \quad (H^d) > 0.$$

Thus $\gamma \leq 1$.

Step 2. In this step we prove that $\big(X, \frac{1}{k}(h=0)|_X\big)$ is lc on $X\backslash\{p\}$, and we reduce the theorem to divisors of sections that are not divisible by h in R.

First we prove that $\big(X, \frac{1}{k}(h=0)|_X\big)$ is lc on $X\backslash\{p\}$. By Lemma 4.16, $\big(C_S\backslash\{q\}, \frac{1}{c}(g=0)|_{C_S}\big)$ is lc on orbifold charts. Since $c < \frac{k}{d-1} < k$ and the preimage of $C_S\backslash\{q\}$ in any orbifold chart of $\mathbb{P}'$ is smooth (Definition-Lemma 2.20(2)), Lemma 2.4 implies that $\big(C_S\backslash\{q\}, \frac{1}{k}(g=0)|_{C_S}\big)$ is lc on orbifold charts. The polynomial $h \in \mathbb{C}[x_1, \dots, x_d]$ is not constant, so h does not divide $g = y^e + G$. By Lemma 4.18, $\big(\mathbb{P}', C_S + \frac{1}{k}(g=0)\big)$ is lc near $C_S\backslash\{q\}$. Since $zh = -g$ in R,

$$(g=0)|_X = (h=0)|_X + (z=0)|_X.$$

For any weighted homogeneous $f \in \mathbb{C}[x_1, \dots, x_d, y]$ of positive degree, $\pi^*(f=0) = (f=0)|_X$ on $X\backslash\{p\}$ (on $(x_j \neq 0)$ both sides equal $\frac{1}{a_j}\operatorname{div}(f^{a_j}/x_j^{\deg f})$: the left side by Definition-Lemma 2.20(3) for the quasismooth $\mathbb{P}'$, and the right side because x_j is a unit of degree a_j in R_{x_j} and $(f^{a_j}=0)|_X = a_j(f=0)|_X$). Thus $\pi^*\big(\frac{1}{k}(g=0)\big) = \frac{1}{k}(h=0)|_X + \frac{1}{k}(z=0)|_X$ on $X\backslash\{p\}$. Let F be a prime divisor over X such that $\operatorname{center}_X F \neq \{p\}$. If $\operatorname{center}_{\mathbb{P}'} F \not\subset C_S$, then $a\big(F, X, \frac{1}{k}(h=0)|_X\big) = a(F, X, 0) \geq 0$, because π is an isomorphism over $\mathbb{P}'\backslash C_S$ (Lemma 4.17), $\pi\big((h=0)|_X\backslash\{p\}\big) \subset C_S$, and X is klt. If $\operatorname{center}_{\mathbb{P}'} F \subset C_S$, then $\operatorname{center}_{\mathbb{P}'} F$ intersects $C_S\backslash\{q\}$, because q is not in the image of π. By Lemma 4.17,

$$a\Big(F, X, \frac{1}{k}(h=0)|_X\Big) = a\Big(F, \mathbb{P}', C_S + \frac{1}{k}(g=0)\Big) + \frac{1}{k}\operatorname{mult}_F(z=0)|_X \geq 0.$$

Thus $\big(X, \frac{1}{k}(h=0)|_X\big)$ is lc on $X\backslash\{p\}$.

Next we reduce the theorem. Let $D \geq 0$ be a $\mathbb{Q}$-divisor on X such that $D \sim_{\mathbb{Q}} H$. Choose a positive integer M such that MD is a divisor and $MD \sim MH$. Since $H^0(X, \mathcal{O}_X(M)) = R_M$, there exists a non-zero $s \in R_M$ such that $D = \frac{1}{M}(s=0)$. Let $m \geq 0$ be the largest integer such that $s = h^m s'$ for some $s' \in R_{M-mk}$. The integer m exists because $R_n = 0$ for $n < 0$. If $M = mk$, then s' is a non-zero constant and $D = \frac{1}{k}(h=0)|_X$. If $M > mk$, then s' is not divisible by h in R, and

$$D = \frac{mk}{M} \cdot \frac{1}{k}(h=0)|_X + \frac{M-mk}{M} \cdot \frac{1}{M-mk}(s'=0).$$

By Lemma 2.4, applied on $X\backslash\{p\}$, it suffices to prove that $\big(X, \frac{1}{M}(s=0)\big)$ is lc on $X\backslash\{p\}$ for any positive integer M and any non-zero $s \in R_M$ that is not divisible by h in R.

Step 3. In this step we fix such M and s, let $D := \frac{1}{M}(s=0)$, and write D on $X\backslash\{p\}$ as a pullback from $\mathbb{P}'$ minus a multiple of $(h=0)|_X$.

Write s as the image of $\sum_{i=0}^{j} z^i s_i$, where $s_i \in \mathbb{C}[x_1, \dots, x_d, y]$ is weighted homogeneous of degree $M - ci$. If $j \geq 1$ and h divides s_j, then $z^j s_j = -z^{j-1} g \cdot (s_j/h)$ in R. Possibly replacing s_{j-1} with $s_{j-1} - g \cdot (s_j/h)$, we may decrease j. If the reduction ends with $j = 0$ and h dividing s_0, then h divides s in R, which is not the case. Thus we may assume that h does not divide s_j. Then $s_j \neq 0$, so $M - cj \geq 0$, that is, $j \leq \frac{M}{c}$. Let

$$\tilde{s} := \sum_{i=0}^{j} (-g)^i h^{j-i} s_i \in \mathbb{C}[x_1, \dots, x_d, y],$$

which is weighted homogeneous of degree $M + jk$. Since $-g = zh$ in R, we have $(-g)^i h^{j-i} s_i = z^i h^j s_i$ in R, hence

$$\tilde{s} = s h^j \quad \text{in } R.$$

Moreover, $\tilde{s} \equiv (-g)^j s_j$ modulo h. Since $\mathbb{C}[x_1, \dots, x_d, y]/(h)$ is a domain and h divides neither g nor s_j, h does not divide $\tilde{s}$, and $\tilde{s}|_{C_S} = (-g)^j s_j|_{C_S} \neq 0$. Since $\pi^*(\tilde{s}=0) = (\tilde{s}=0)|_X$ on $X\backslash\{p\}$, we obtain

$$D = \pi^*\Big(\frac{1}{M}(\tilde{s}=0)\Big) - \frac{j}{M}(h=0)|_X \quad \text{on } X\backslash\{p\}. \tag{4.2}$$

If $M > cj$, then on C_S,

$$\frac{1}{M}(\tilde{s}=0)|_{C_S} = \frac{cj}{M} \cdot \frac{1}{c}(g=0)|_{C_S} + \frac{M-cj}{M} \cdot \frac{1}{M-cj}(s_j=0)|_{C_S}. \tag{4.3}$$

If $M = cj$, then s_j is a non-zero constant and $\frac{1}{M}(\tilde{s}=0)|_{C_S} = \frac{1}{c}(g=0)|_{C_S}$.

Step 4. In this step we prove that $a(F, X, D) \geq 0$ for any prime divisor F over X such that $\operatorname{center}_X F \neq \{p\}$ and $\operatorname{center}_{\mathbb{P}'} F \not\subset C_S$.

Let r be the largest integer such that y^r divides $\tilde{s}$. Then

$$\frac{1}{M}(\tilde{s}=0) = \frac{r}{M} W + \frac{1}{M}(\tilde{s}/y^r = 0),$$

where $W = (y=0) \subset \mathbb{P}'$ (Set-up 4.14(1)) is not a component of $(\tilde{s}/y^r = 0)$, and

$$\frac{1}{M}(\tilde{s}/y^r = 0) \sim_{\mathbb{Q}} \Big(1 + \frac{jk}{M} - \frac{r}{M}\Big)\mathcal{O}_{\mathbb{P}'}(1).$$

Since π is an isomorphism over $\mathbb{P}'\backslash C_S$ (Lemma 4.17) and $\pi\big((h=0)|_X\backslash\{p\}\big) \subset C_S$, (4.2) implies that π restricts to an isomorphism from $\pi^{-1}(\mathbb{P}'\backslash C_S)$ onto $\mathbb{P}'\backslash C_S$ under which D is the pullback of $\frac{1}{M}(\tilde{s}=0)$. The element h is non-zero in $R/(y) \cong \mathbb{C}[x_1, \dots, x_d, z]/(zh + G)$, because $zh + G$ has degree one in z. Thus $\pi^{-1}(W\backslash C_S)$ is a dense open subset of $(y=0)|_X$, and **Step 1** implies

$$\frac{r}{M} = \operatorname{mult}_{(y=0)|_X} D \leq 1.$$

Since $0 \leq r \leq M + jk$ and $j \leq \frac{M}{c}$,

$$0 \leq 1 + \frac{jk}{M} - \frac{r}{M} \leq 1 + \frac{k}{c} = \frac{e}{c}.$$

Since $a_{(1)} \geq e/c$, Lemma 4.15(1) implies that $\big(\mathbb{P}', \frac{r}{M}W + \frac{1}{M}(\tilde{s}/y^r = 0)\big)$ is lc. Thus

$$a(F, X, D) = a\Big(F, \mathbb{P}', \frac{1}{M}(\tilde{s}=0)\Big) \geq 0.$$

Step 5. In this step we prove that $a(F, X, D) \geq 0$ for any prime divisor F over X such that $\operatorname{center}_X F \neq \{p\}$ and $\operatorname{center}_{\mathbb{P}'} F \subset C_S$.

By Lemma 4.16, $\big(C_S\backslash\{q\}, \frac{1}{c}(g=0)|_{C_S}\big)$ is lc on orbifold charts. If $M > cj$, then s_j has degree $M - cj > 0$ and is not divisible by h, so $\big(C_S\backslash\{q\}, \frac{1}{M-cj}(s_j=0)|_{C_S}\big)$ is lc on orbifold charts by Lemma 4.15(3). By (4.3) and Lemma 2.4, applied on the orbifold charts of $C_S\backslash\{q\}$, the pair $\big(C_S\backslash\{q\}, \frac{1}{M}(\tilde{s}=0)|_{C_S}\big)$ is lc on orbifold charts. If $M = cj$, then $\big(C_S\backslash\{q\}, \frac{1}{M}(\tilde{s}=0)|_{C_S}\big)$ is lc on orbifold charts by Lemma 4.16. Since h does not divide $\tilde{s}$, Lemma 4.18, applied to $u = \tilde{s}$ and $\lambda = \frac{1}{M}$, implies that $\big(\mathbb{P}', C_S + \frac{1}{M}(\tilde{s}=0)\big)$ is lc near $C_S\backslash\{q\}$.

Since q is not in the image of π, $\operatorname{center}_{\mathbb{P}'} F$ intersects $C_S\backslash\{q\}$. By (4.2) and Lemma 4.17,

$$a(F, X, D) = a\Big(F, \mathbb{P}', C_S + \frac{1}{M}(\tilde{s}=0)\Big) + \frac{j}{M}\operatorname{mult}_F(h=0)|_X \geq 0,$$

where $\frac{j}{M}\operatorname{mult}_F(h=0)|_X \geq 0$ because $(h=0)|_X \geq 0$.

Step 6. In this step we conclude the proof.

By **Steps 2**, **4**, and **5**, (X, D) is lc on $X\backslash\{p\}$ for any $\mathbb{Q}$-divisor $D \geq 0$ on X such that $D \sim_{\mathbb{Q}} H$. Let D be such a divisor. By Proposition 4.13, $\big(X, \frac{k}{e}D\big)$ is lc near p. Since X is klt and $\frac{k}{e} < 1$, Lemma 2.4 implies that $\big(X, \frac{k}{e}D\big)$ is lc on $X\backslash\{p\}$. Thus $\big(X, \frac{k}{e}D\big)$ is lc. Since $-K_X \sim H$, we have $\alpha(X) \geq \frac{k}{e}$, and Proposition 4.13 implies $\alpha(X) \leq \frac{k}{e}$. Hence $\alpha(X) = \frac{k}{e}$. Since $\frac{k}{d} < c < \frac{k}{d-1}$,

$$\frac{d+1}{d} < \frac{e}{k} = 1 + \frac{c}{k} < \frac{d}{d-1},$$

so $\frac{d-1}{d} < \alpha(X) < \frac{d}{d+1}$.

Let E be the exceptional divisor of the plt blow-up $f : Y \to X$ of Construction-Proposition 4.9. By Proposition 4.11,

$$\beta_X(E) = \frac{k - dc}{c(d+1)}\operatorname{vol}(-K_X) < 0,$$

because $c > \frac{k}{d}$. By Theorem 2.12, X is not K-semistable. □

4.6. **Two criteria.** In this subsection, we prove two criteria for the conditions of Theorem 4.3: Corollary 4.19 uses the multiplicity bound of Theorem 2.28 when $a_{(1)}a_{(2)} \geq ke/c$ and is used for Proposition 5.2 and Proposition 5.6 in Section 5, and Corollary 4.20 uses simple normal crossings when h is of Fermat type and G is a monomial with exponents at most c and is used for Proposition 5.3. The fourfold of Proposition 5.1 is in neither case, and the conditions of Theorem 4.3 are checked directly there.

Corollary 4.19. *Notation and conditions as in Set-up 4.1, and assume that $a_{(1)}a_{(2)} \geq ke/c$. Then X is a $\mathbb{Q}$-Fano variety of dimension d with $\alpha(X) = k/e$, and there exists a prime divisor E over X such that $\beta_X(E) < 0$.*

Proof. Since $d \geq 3$, we have $a_{(2)} < a_{(1)} + a_{(2)} < k$. Thus

$$a_{(1)} \geq \frac{ke}{ca_{(2)}} > \frac{e}{c}.$$

By Theorem 4.3, it suffices to show that $\big(S, \frac{1}{c}(G=0)|_S\big)$ is lc on orbifold charts.

Let $\rho_j : U_j \to (x_j \neq 0) \subset W$, $1 \leq j \leq d$, be the orbifold charts of W. Since S is quasismooth, $\rho_j^{-1}(S)$ is smooth and $\rho_j^*\big((G=0)|_S\big) = \operatorname{div}(G_j)$ on $\rho_j^{-1}(S)$, where G_j is the restriction of G to U_j (Definition-Lemma 2.20(2)(3)). By Lemma 2.7, it suffices to show that the multiplicity of $\rho_j^*\big((G=0)|_S\big)$ at any point of $\rho_j^{-1}(S)$ is at most c.

Since h is irreducible (Lemma 2.21) and does not divide G, the polynomials h, G form a regular sequence in $\mathbb{C}[x_1, \dots, x_d]$. Let Z be the cycle on W of Lemma 2.27 for h, G. Then

$$\deg Z = \frac{ke}{a_1 \cdots a_d}.$$

For a minimal prime ideal $\mathfrak{p}$ of $\mathbb{C}[x_1, \dots, x_d]/(h, G)$, the localization of $\mathbb{C}[x_1, \dots, x_d]/(h)$ at $\mathfrak{p}$ is a discrete valuation ring, because the affine cone $(h=0) \subset \mathbb{A}^d$ is a hypersurface smooth outside

the origin and $d \geq 3$, hence normal. So the length of the localization of $\mathbb{C}[x_1, \dots, x_d]/(h, G)$ at $\mathfrak{p}$ is the order of vanishing of G along the prime divisor $(\mathfrak{p} = 0)$ of $(h = 0) \subset \mathbb{A}^d$. The morphism

$$U_j \times \mathbb{G}_m \to (x_j \neq 0) \subset \mathbb{A}^d, \quad \big((x_i)_{i \neq j}, \theta\big) \to (\theta^{a_1} x_1, \dots, \theta^{a_d} x_d),$$

where $x_j = 1$ on U_j, is étale by Definition-Lemma 2.20(1), and it pulls back h and G to $\theta^k h$ and $\theta^e G$. Thus the coefficient of any component of $\rho_j^{-1}\big(V(\mathfrak{p})\big)$ in $\rho_j^*\big((G = 0)|_S\big)$ is the length of the localization of $\mathbb{C}[x_1, \dots, x_d]/(h, G)$ at $\mathfrak{p}$.

For $M = V(\mathfrak{p})$, the reduced closed subscheme $(I_M = 0) \cap U_j$ in Definition 2.23 is $\rho_j^{-1}\big(V(\mathfrak{p})\big)$, and every component of $\rho_j^*\big((G = 0)|_S\big)$ is a component of $\rho_j^{-1}\big(V(\mathfrak{p})\big)$ for some $\mathfrak{p}$. Since the multiplicity of a divisor at a point is the sum over its components of the coefficient times the multiplicity of the component at that point, Definition 2.23 implies that the multiplicity of $\rho_j^*\big((G = 0)|_S\big)$ at any point of $\rho_j^{-1}(v)$ is $\operatorname{mult}_v Z$ for any closed point $v \in S \cap (x_j \neq 0)$.

Every component of Z is of dimension $d - 3$. Ordering the weights $a_1, \dots, a_d$ decreasingly, the product of the first $d - 2$ of them is $\frac{a_1 \cdots a_d}{a_{(1)} a_{(2)}}$. By the first bound of Corollary 2.29, for any closed point $v \in W$,

$$\operatorname{mult}_v Z \leq \frac{a_1 \cdots a_d}{a_{(1)} a_{(2)}} \deg Z = \frac{ke}{a_{(1)} a_{(2)}} \leq c.$$

Thus $\big(S, \frac{1}{c}(G = 0)|_S\big)$ is lc on orbifold charts, and Theorem 4.3 implies the corollary. □

Corollary 4.20. *Notation and conditions as in Set-up 4.1, and assume that $a_{(1)} \geq e/c$, that a_i divides k for every i, that $h = \sum_{i=1}^d x_i^{k/a_i}$, and that $G = \prod_{i=1}^d x_i^{m_i}$ for some integers $0 \leq m_i \leq c$. Then X is a $\mathbb{Q}$-Fano variety of dimension d with $\alpha(X) = k/e$, and there exists a prime divisor E over X such that $\beta_X(E) < 0$.*

Proof. By Theorem 4.3, it suffices to show that $\big(S, \frac{1}{c}(G = 0)|_S\big)$ is lc on orbifold charts. Let $\rho_j : U_j \to (x_j \neq 0) \subset W$, $1 \leq j \leq d$, be the orbifold charts of W, where $U_j = \mathbb{A}^{d-1}$ has coordinates x_i, $i \neq j$, and x_j is the constant function 1 on U_j. Let $v' \in \rho_j^{-1}(S)$. Since

$$h|_{U_j} = 1 + \sum_{i \neq j} x_i^{k/a_i}$$

vanishes at v', there exists $i_0 \neq j$ such that $x_{i_0}(v') \neq 0$. Then

$$\frac{\partial h}{\partial x_{i_0}} = \frac{k}{a_{i_0}} x_{i_0}^{k/a_{i_0} - 1}$$

is non-zero at v'. Thus near v', the functions x_i, $i \notin \{j, i_0\}$, form local coordinates of $\rho_j^{-1}(S)$, and x_{i_0} is a unit. Hence near v',

$$\rho_j^*\Big(\frac{1}{c}(G = 0)|_S\Big) = \sum_{i \notin \{j, i_0\}} \frac{m_i}{c}(x_i = 0)|_{\rho_j^{-1}(S)},$$

whose support is a simple normal crossing divisor and whose coefficients satisfy $\frac{m_i}{c} \leq 1$. By [KM98, Corollary 2.31(3)], $\big(\rho_j^{-1}(S), \rho_j^*(\frac{1}{c}(G = 0)|_S)\big)$ is lc near v'. Thus $\big(S, \frac{1}{c}(G = 0)|_S\big)$ is lc on orbifold charts, and Theorem 4.3 implies the corollary. □

5. Examples in dimension at least four

In this section, we prove that explicit hypersurfaces as in Set-up 4.1 satisfy the conditions of Theorem 4.3 or of one of Corollaries 4.19 and 4.20. Besides the arithmetic conditions and quasismoothness, we check the lc condition on S in Theorem 4.3 directly for $d = 4$, and for $d \geq 5$ we use Corollary 4.19 ($d = 5, 6$ and $d \geq 8$) or Corollary 4.20 ($d = 7$), whose additional conditions are arithmetic. The data are the weights $(5, 6, 8, 11)$ of a quasismooth surface of degree 30 for $d = 4$, the weights of the quasismooth Calabi–Yau hypersurfaces of Esser, Totaro,

and Wang for $d = 5, 6$, Fermat weights dividing k for $d = 7$, and for $d \geq 8$ the weights of Construction-Proposition 5.4, built from Egyptian fractions (Subsection 5.2).

5.1. **Dimensions four to seven.** Table 2 lists the data and the invariants of Propositions 5.1, 5.2, and 5.3.

d	$(a_1, \dots, a_d)$	k	c	$\alpha(X)$	$\beta_X(E)/\operatorname{vol}(-K_X)$
4	$(5, 6, 8, 11)$	30	8	15/19	$-1/20$
5	$(867, 578, 102, 96, 91)$	1734	431	1734/2165	$-421/2586$
6	$(328125, 218750, 93750, 5250, 5208, 5167)$	656250	114750	875/1028	$-43/1071$
7	$(4620, 3080, 660, 280, 264, 231, 105)$	9240	1369	9240/10609	$-343/10952$

TABLE 2. The weights, the integers k and c, the alpha invariant, and the normalized beta invariant of the hypersurfaces X in Propositions 5.1, 5.2, and 5.3.

In dimension four, for the weights $(5, 6, 8, 11)$ and $c = 8$, the condition $a_{(1)}a_{(2)} \geq k$ of Set-up 4.1 holds with equality, and $a_{(1)} = 5$ is only slightly larger than $\frac{e}{c} = \frac{19}{4}$. We cannot use either corollary: $a_{(1)}a_{(2)} < \frac{ke}{c}$, and 8 and 11 do not divide $k = 30$. Thus we check the lc condition of Theorem 4.3 on S directly, using inversion of adjunction (Theorem 2.6) and Lemma 2.4 at $[0 : 0 : 1 : 0]$, where the support of $\frac{1}{c}(G = 0)|_S$ is not simple normal crossing.

Proposition 5.1. *Let*

$$X := \left(z(x_1^6 + x_2^5 + x_2x_3^3 + x_3x_4^2) + y^{38} + x_1^3x_2^2x_4 = 0\right) \subset \mathbb{P}(5, 6, 8, 11, 1, 8),$$

where x_1, x_2, x_3, x_4, y, z are the coordinates of degrees $5, 6, 8, 11, 1, 8$. Then X is a $\mathbb{Q}$-Fano fourfold with $\alpha(X) = \frac{15}{19}$, so that $\frac{3}{4} < \alpha(X) < \frac{4}{5}$, and there exists a prime divisor E over X such that $\beta_X(E) = -\frac{1}{20}\operatorname{vol}(-K_X)$, where $\operatorname{vol}(-K_X) = \frac{19}{10560}$.

Proof. We check the conditions of Set-up 4.1 and Theorem 4.3 for $d = 4$, $c = 8$, and

$$(a_1, a_2, a_3, a_4) = (5, 6, 8, 11), \quad h = x_1^6 + x_2^5 + x_2x_3^3 + x_3x_4^2, \quad G = x_1^3x_2^2x_4.$$

Any three of $5, 6, 8, 11$ have greatest common divisor 1, so $W = \mathbb{P}(5, 6, 8, 11)$ is well formed. We have $k = 30$, $a_{(1)} = 5$, $a_{(2)} = 6$, and $a_{(1)}a_{(2)} = 30 = k$. Since $4c = 32 > 30$ and $3c = 24 < 30$, we have $\frac{k}{4} < c < \frac{k}{3}$, and $e = 38$. The four monomials of h have degrees

$$6 \cdot 5 = 5 \cdot 6 = 6 + 3 \cdot 8 = 8 + 2 \cdot 11 = 30 = k,$$

and $\deg G = 3 \cdot 5 + 2 \cdot 6 + 11 = 38 = e$. The partial derivatives of h are

$$\frac{\partial h}{\partial x_1} = 6x_1^5, \quad \frac{\partial h}{\partial x_2} = 5x_2^4 + x_3^3, \quad \frac{\partial h}{\partial x_3} = 3x_2x_3^2 + x_4^2, \quad \frac{\partial h}{\partial x_4} = 2x_3x_4.$$

Suppose that they all vanish at a point (x_1, x_2, x_3, x_4) of $\mathbb{A}^4$. Then $x_1 = 0$ and $x_3x_4 = 0$. If $x_3 = 0$, then $\partial h/\partial x_2 = 0$ implies $x_2 = 0$, and $\partial h/\partial x_3 = 0$ implies $x_4 = 0$. If $x_4 = 0$ and $x_3 \neq 0$, then $\partial h/\partial x_3 = 0$ implies $x_2 = 0$, and hence $\partial h/\partial x_2 = x_3^3 \neq 0$, a contradiction. Thus the gradient of h vanishes only at the origin, and h is quasismooth. By Lemma 2.21, h is irreducible. Since $\deg h = k > a_i$ for any i, h is not a constant multiple of a variable, and hence h does not divide the monomial G. Thus the conditions of Set-up 4.1 hold. Moreover, $a_{(1)} = 5 > \frac{19}{4} = \frac{e}{c}$.

It remains to show that $\left(S, \frac{1}{8}(G = 0)|_S\right)$ is lc on orbifold charts. Let $1 \leq j \leq 4$, let $\rho_j : U_j \to (x_j \neq 0) \subset W$ be the j-th orbifold chart of W, where $U_j = (x_j = 1) \subset \mathbb{A}^4$, and let $S_j := \rho_j^{-1}(S) = (h = 0) \cap U_j$. By the Euler formula

$$\sum_{i=1}^{4} a_i x_i \frac{\partial h}{\partial x_i} = kh,$$

at any point of S_j, where $x_j = 1$ and the gradient of h is non-zero, there exists $m \neq j$ such that $\partial h/\partial x_m \neq 0$. Hence S_j is a smooth surface, and near such a point the coordinates x_i with $i \notin \{j, m\}$ form a system of local coordinates of S_j. For $i \neq j$, let D_i be the divisor of x_i on S_j, and let $D_j := 0$. Then

$$B_j := \rho_j^*\Big(\frac{1}{8}(G = 0)|_S\Big) = \frac{3}{8}D_1 + \frac{1}{4}D_2 + \frac{1}{8}D_4,$$

and we show that (S_j, B_j) is lc.

Setting $x_1 = 0$ in the computation above, the gradient of $x_2^5 + x_2x_3^3 + x_3x_4^2$ with respect to (x_2, x_3, x_4) vanishes only at the origin. The gradient of $x_1^6 + x_2^5 + x_2x_3^3$ with respect to (x_1, x_2, x_3) is $(6x_1^5, 5x_2^4 + x_3^3, 3x_2x_3^2)$. If it vanishes, then $x_1 = 0$ and $x_2x_3 = 0$, and then $5x_2^4 + x_3^3 = 0$ implies $x_2 = x_3 = 0$. The gradient of $x_1^6 + x_3x_4^2$ with respect to (x_1, x_3, x_4) is $(6x_1^5, x_4^2, 2x_3x_4)$, which vanishes exactly where $x_1 = x_4 = 0$. Applying the Euler formula to $x_2^5 + x_2x_3^3 + x_3x_4^2$, $x_1^6 + x_2^5 + x_2x_3^3$, and $x_1^6 + x_3x_4^2$, we see that D_1 and D_4 are smooth curves, and D_2 is a smooth curve outside $\rho_j^{-1}([0:0:1:0])$.

We determine the points of S on at least two of the curves $(x_1 = 0)$, $(x_2 = 0)$, and $(x_4 = 0)$. If $x_1 = x_2 = 0$, then $h = x_3x_4^2$, so the point is $[0:0:1:0]$ or $[0:0:0:1]$. If $x_2 = x_4 = 0$, then $h = x_1^6$, so the point is $[0:0:1:0]$. If $x_1 = x_4 = 0$, then $h = x_2(x_2^4 + x_3^3)$, so either the point is $[0:0:1:0]$, or $x_2x_3 \neq 0$ and $x_2^4 + x_3^3 = 0$.

The point $[0:0:0:1]$. The point $[0:0:0:1]$ is contained in $(x_j \neq 0)$ only for $j = 4$, and the preimage of $[0:0:0:1]$ in U_4 is the origin, where $\partial h/\partial x_3 = 3x_2x_3^2 + x_4^2 = 1$. Thus (x_1, x_2) is a system of local coordinates of S_4 at the origin, $D_1 = (x_1 = 0)$ and $D_2 = (x_2 = 0)$ near the origin, and $D_4 = 0$ on S_4.

Points with $x_1 = x_4 = 0$, $x_2x_3 \neq 0$, *and* $x_2^4 + x_3^3 = 0$. At such a point,

$$\frac{\partial h}{\partial x_2} = 5x_2^4 + x_3^3 = 4x_2^4 \neq 0 \quad \text{and} \quad \frac{\partial h}{\partial x_3} = 3x_2x_3^2 \neq 0.$$

Such a point is contained in $(x_j \neq 0)$ only for $j \in \{2, 3\}$. Taking $m \in \{2, 3\}\backslash\{j\}$ above, (x_1, x_4) is a system of local coordinates of S_j at any preimage of the point. Thus D_1 and D_4 are coordinate axes there, and D_2 does not pass through the preimage since $x_2 \neq 0$.

Therefore, outside $\rho_j^{-1}([0:0:1:0])$, the support of B_j is simple normal crossing and the coefficients of B_j are at most 1, so (S_j, B_j) is lc there [KM98, Corollary 2.31(3)].

The point $[0:0:1:0]$. The point $[0:0:1:0]$ is contained in $(x_j \neq 0)$ only for $j = 3$, and the preimage of $[0:0:1:0]$ in U_3 is the origin with coordinates (x_1, x_2, x_4). On U_3,

$$S_3 = (x_1^6 + x_2^5 + x_2 + x_4^2 = 0).$$

Since $\partial h/\partial x_2 = 5x_2^4 + 1$ is 1 at the origin, x_1 and x_4 are local analytic coordinates of S_3 at the origin. Since $x_2(1 + x_2^4) = -(x_1^6 + x_4^2)$ on S_3 and $1 + x_2^4$ is a unit near the origin, $\operatorname{div}(x_2) = \operatorname{div}(x_1^6 + x_4^2)$ near the origin. Thus it suffices to show that $\big(\mathbb{A}^2, \frac{3}{8}D_1 + \frac{1}{4}D_2 + \frac{1}{8}D_4\big)$ is lc near the origin, where $\mathbb{A}^2$ has coordinates (x_1, x_4), $D_1 = (x_1 = 0)$, $D_4 = (x_4 = 0)$, and $D_2 = \operatorname{div}(x_1^6 + x_4^2)$ (the projection $(x_1, x_4) : S_3 \to \mathbb{A}^2$ is étale at the origin, B_3 is, near the origin, the pullback of $\frac{3}{8}D_1 + \frac{1}{4}D_2 + \frac{1}{8}D_4$ on $\mathbb{A}^2$, and lc is preserved under étale pullback).

Since $\mathbb{A}^2$ and D_1 are smooth, $\operatorname{Diff}_{D_1}(\frac{1}{2}D_2) = \frac{1}{2}D_2|_{D_1}$ by Definition-Proposition 2.5(1). Since $(x_1^6 + x_4^2)|_{D_1} = x_4^2$, the $\mathbb{Q}$-divisor $\frac{1}{2}D_2|_{D_1}$ is the origin with coefficient 1, so $(D_1, \operatorname{Diff}_{D_1}(\frac{1}{2}D_2))$ is lc. By Theorem 2.6, $(\mathbb{A}^2, D_1 + \frac{1}{2}D_2)$ is lc near D_1, in particular near the origin. The pairs $(\mathbb{A}^2, D_4)$ and $(\mathbb{A}^2, 0)$ are lc, so $(\mathbb{A}^2, \frac{1}{4}D_4)$ is lc by Lemma 2.4, applied with $t = \frac{1}{4}$. By Lemma 2.4, applied to $D_1 + \frac{1}{2}D_2$ and $\frac{1}{4}D_4$ with $t = \frac{1}{2}$,

$$\Big(\mathbb{A}^2, \frac{1}{2}D_1 + \frac{1}{4}D_2 + \frac{1}{8}D_4\Big)$$

is lc near the origin. By the definition of log discrepancy, $a(F, \mathbb{A}^2, B') \geq a(F, \mathbb{A}^2, B)$ for any prime divisor F over $\mathbb{A}^2$ and any $\mathbb{R}$-divisors $0 \leq B' \leq B$. Decreasing the coefficient of D_1 from $\frac{1}{2}$ to $\frac{3}{8}$, we deduce that $\left(\mathbb{A}^2, \frac{3}{8}D_1 + \frac{1}{4}D_2 + \frac{1}{8}D_4\right)$ is lc near the origin.

Thus (S_j, B_j) is lc for any j, that is, $\left(S, \frac{1}{8}(G=0)|_S\right)$ is lc on orbifold charts. By Theorem 4.3, X is a $\mathbb{Q}$-Fano fourfold with $\alpha(X) = \frac{k}{e} = \frac{30}{38} = \frac{15}{19}$. The inequalities $\frac{3}{4} < \frac{15}{19} < \frac{4}{5}$ read $3 \cdot 19 = 57 < 60 = 4 \cdot 15$ and $5 \cdot 15 = 75 < 76 = 4 \cdot 19$.

Let E be the exceptional divisor of the plt blow-up $f : Y \to X$ at p in Construction-Proposition 4.9. By Proposition 4.11,

$$\operatorname{vol}(-K_X) = \frac{e}{ca_1a_2a_3a_4} = \frac{38}{8 \cdot 5 \cdot 6 \cdot 8 \cdot 11} = \frac{19}{10560}$$

and

$$\beta_X(E) = \frac{k - dc}{c(d+1)} \operatorname{vol}(-K_X) = \frac{30 - 32}{8 \cdot 5} \operatorname{vol}(-K_X) = -\frac{1}{20} \operatorname{vol}(-K_X).$$

□

The weights in Proposition 5.2 are those of the quasismooth Calabi–Yau hypersurfaces constructed by Esser, Totaro, and Wang [ETW21, Proposition 3.3] from the terms 7 and 43 of the Sylvester sequence. By computations of Brown and Kasprzyk recalled in [ETW21, Section 3.1], their smallest weights 91 and 5167 are the largest possible for quasismooth Calabi–Yau hypersurfaces of dimensions three and four respectively. Even so, the condition of Corollary 4.19 holds only with a small margin for $d = 5$, while the margin is large for $d = 6$.

Proposition 5.2. *Let $d \in \{5, 6\}$, and let $(a_1, \dots, a_d)$, c, h, and G be as follows, where $x_1, \dots, x_d$ have degrees $a_1, \dots, a_d$.*

(1) If $d = 5$, then $(a_1, \dots, a_5) = (867, 578, 102, 96, 91)$, $c = 431$, and

$$h = x_1^2 + x_2^3 + x_3^{17} + x_3x_4^{17} + x_4x_5^{18}, \quad G = x_2^2x_3^9x_5.$$

(2) If $d = 6$, then $(a_1, \dots, a_6) = (328125, 218750, 93750, 5250, 5208, 5167)$, $c = 114750$, and

$$h = x_1^2 + x_2^3 + x_3^7 + x_4^{125} + x_4x_5^{125} + x_5x_6^{126}, \quad G = x_1x_2x_3^2x_4^5x_5x_6.$$

Let $k := a_1 + \cdots + a_d$ and $e := k + c$, and let

$$X := (zh + y^e + G = 0) \subset \mathbb{P}(a_1, \dots, a_d, 1, c),$$

where y and z are the coordinates of degrees 1 and c. Then X is a $\mathbb{Q}$-Fano variety of dimension d with $\alpha(X) = \frac{k}{e}$, and there exists a prime divisor E over X such that

$$\beta_X(E) = \frac{k - dc}{c(d+1)} \operatorname{vol}(-K_X) < 0.$$

In particular, $\alpha(X) = \frac{1734}{2165}$ and $\frac{4}{5} < \alpha(X) < \frac{5}{6}$ if $d = 5$, and $\alpha(X) = \frac{875}{1028}$ and $\frac{5}{6} < \alpha(X) < \frac{6}{7}$ if $d = 6$.

Proof. We check the conditions of Set-up 4.1 and Corollary 4.19.

If $d = 5$, then $\gcd(96, 91) = 1$, $\gcd(867, 578, 102) = 17$, and 17 divides neither 96 nor 91. If $d = 6$, then $\gcd(5208, 5167) = 1$, $\gcd(328125, 218750, 93750, 5250) = 125$, and 5 divides neither 5208 nor 5167. Deleting one of $a_1, \dots, a_{d-2}$ leaves a_{d-1} and a_d, and deleting a_{d-1} or a_d leaves $a_1, \dots, a_{d-2}$ and one of a_{d-1}, a_d. In both cases the remaining weights have greatest common divisor 1, so W is well formed. We have $k = 1734$ if $d = 5$, and $k = 656250$ if $d = 6$. The smallest weights are $a_{(1)} = 91$ and $a_{(2)} = 96$ if $d = 5$, and $a_{(1)} = 5167$ and $a_{(2)} = 5208$ if $d = 6$. Since

$$5 \cdot 431 = 2155 > 1734 > 1724 = 4 \cdot 431 \quad \text{if } d = 5,$$
$$6 \cdot 114750 = 688500 > 656250 > 573750 = 5 \cdot 114750 \quad \text{if } d = 6,$$

we have $\frac{k}{d} < c < \frac{k}{d-1}$. Moreover, $e = 2165$ if $d = 5$, and $e = 771000$ if $d = 6$.

For $r \in \{17, 125\}$, the gradient of $u^r + uv^r + vw^{r+1}$ with respect to (u, v, w) is

$$\big(ru^{r-1} + v^r,\ ruv^{r-1} + w^{r+1},\ (r+1)vw^r\big).$$

Suppose that it vanishes. Then $vw = 0$. If $v = 0$, then $ru^{r-1} = 0$ implies $u = 0$, and then $w^{r+1} = 0$ implies $w = 0$. If $w = 0$ and $v \neq 0$, then $ruv^{r-1} = 0$ implies $u = 0$, and then the first component is $v^r \neq 0$, a contradiction. Thus the gradient vanishes only at the origin. The polynomial h is the sum of pure powers x_i^m with $m \geq 2$ and of $u^r + uv^r + vw^{r+1}$, where $r = 17$ and $(u, v, w) = (x_3, x_4, x_5)$ if $d = 5$, and $r = 125$ and $(u, v, w) = (x_4, x_5, x_6)$ if $d = 6$. The summands of h are polynomials in disjoint sets of variables, so the gradient of h vanishes only at the origin, and h is quasismooth. The degrees of the monomials of h are

$$2 \cdot 867 = 3 \cdot 578 = 17 \cdot 102 = 102 + 17 \cdot 96 = 96 + 18 \cdot 91 = 1734$$

if $d = 5$, and

$$2 \cdot 328125 = 3 \cdot 218750 = 7 \cdot 93750 = 125 \cdot 5250 = 5250 + 125 \cdot 5208 = 5208 + 126 \cdot 5167 = 656250$$

if $d = 6$, so $\deg h = k$. Moreover,

$$\deg G = 2 \cdot 578 + 9 \cdot 102 + 91 = 2165 = e$$

if $d = 5$, and $\deg G = k + a_3 + 4a_4 = 656250 + 93750 + 4 \cdot 5250 = 771000 = e$ if $d = 6$. By Lemma 2.21, h is irreducible. Since $\deg h = k > a_i$ for any i, h is not a constant multiple of a variable, and hence h does not divide the monomial G.

If $d = 5$, then

$$a_{(1)}a_{(2)}c = 91 \cdot 96 \cdot 431 = 3765216 > 3754110 = 1734 \cdot 2165 = ke.$$

If $d = 6$, then $6c = 688500 > k$, so $\frac{e}{c} = 1 + \frac{k}{c} < 7$, and hence

$$\frac{ke}{c} < 7k = 4593750 < 26909736 = 5167 \cdot 5208 = a_{(1)}a_{(2)}.$$

Thus $a_{(1)}a_{(2)} \geq \frac{ke}{c}$ in both cases. Since $\frac{e}{c} > 1$, we have $a_{(1)}a_{(2)} \geq \frac{ke}{c} > k$. Therefore the conditions of Set-up 4.1 and Corollary 4.19 hold.

By Corollary 4.19, X is a $\mathbb{Q}$-Fano variety of dimension d with $\alpha(X) = \frac{k}{e}$. If $d = 5$, then $\alpha(X) = \frac{1734}{2165}$, and

$$4 \cdot 2165 = 8660 < 8670 = 5 \cdot 1734, \quad 6 \cdot 1734 = 10404 < 10825 = 5 \cdot 2165.$$

If $d = 6$, then $\alpha(X) = \frac{656250}{771000} = \frac{875}{1028}$, and

$$5 \cdot 1028 = 5140 < 5250 = 6 \cdot 875, \quad 7 \cdot 875 = 6125 < 6168 = 6 \cdot 1028.$$

Thus $\frac{4}{5} < \alpha(X) < \frac{5}{6}$ if $d = 5$, and $\frac{5}{6} < \alpha(X) < \frac{6}{7}$ if $d = 6$.

Let E be the exceptional divisor of the plt blow-up $f : Y \to X$ at p in Construction-Proposition 4.9. By Proposition 4.11, $\beta_X(E) = \frac{k-dc}{c(d+1)} \operatorname{vol}(-K_X) < 0$. Explicitly,

$$\frac{k - 5c}{6c} = \frac{1734 - 2155}{6 \cdot 431} = -\frac{421}{2586} \quad \text{and} \quad \frac{k - 6c}{7c} = \frac{656250 - 688500}{7 \cdot 114750} = -\frac{43}{1071}$$

for $d = 5$ and $d = 6$ respectively. □

In dimension seven, we use Corollary 4.20: all weights divide k, and the exponents of G are at most c. Corollary 4.19 does not apply, as we check in the proof.

Proposition 5.3. *Let* $\mathbb{P} := \mathbb{P}(4620, 3080, 660, 280, 264, 231, 105, 1, 1369)$, *where* $x_1, \ldots, x_7, y, z$ *are the coordinates of degrees* $4620, 3080, 660, 280, 264, 231, 105, 1, 1369$, *and let*

$$X := \big(z(x_1^2 + x_2^3 + x_3^{14} + x_4^{33} + x_5^{35} + x_6^{40} + x_7^{88}) + y^{10609} + x_1x_2x_3^3x_4^2x_5x_7 = 0\big) \subset \mathbb{P}.$$

Then X *is a* $\mathbb{Q}$*-Fano variety of dimension* 7 *with* $\alpha(X) = \frac{9240}{10609}$, *so that* $\frac{6}{7} < \alpha(X) < \frac{7}{8}$, *and there exists a prime divisor* E *over* X *such that* $\beta_X(E) = -\frac{343}{10952} \operatorname{vol}(-K_X)$.

Proof. We check the conditions of Set-up 4.1 and Corollary 4.20 for $d = 7$, $c = 1369$, and

$$(a_1, \dots, a_7) = (4620, 3080, 660, 280, 264, 231, 105),$$
$$h = x_1^2 + x_2^3 + x_3^{14} + x_4^{33} + x_5^{35} + x_6^{40} + x_7^{88}, \quad G = x_1 x_2 x_3^3 x_4^2 x_5 x_7.$$

We have $k = 9240$, and

$$\left(\frac{k}{a_1}, \dots, \frac{k}{a_7}\right) = (2, 3, 14, 33, 35, 40, 88),$$

so a_i divides k for any i and $h = \sum_{i=1}^7 x_i^{k/a_i}$. Since $\gcd(105, 264) = 3$ and 3 does not divide 280, we have $\gcd(105, 264, 280) = 1$, so $\gcd(a_1, \dots, a_7) = 1$. Suppose that, for some j, a prime number l divides a_i for every $i \neq j$. Since a_i divides k for some $i \neq j$, l divides k, and hence l divides $a_j = k - \sum_{i \neq j} a_i$, a contradiction to $\gcd(a_1, \dots, a_7) = 1$. Thus W is well formed.

We have $a_{(1)} = 105$, $a_{(2)} = 231$, and $a_{(1)}a_{(2)} = 24255 \geq 9240 = k$. Since

$$7 \cdot 1369 = 9583 > 9240 > 8214 = 6 \cdot 1369,$$

we have $\frac{k}{7} < c < \frac{k}{6}$, and $e = 10609$. The partial derivatives $\partial h / \partial x_i = \frac{k}{a_i} x_i^{k/a_i - 1}$ vanish simultaneously only at the origin, so h is quasismooth, and $\deg h = k$. The exponents of G are $(1, 1, 3, 2, 1, 0, 1)$, which are at most c, and

$$\deg G = 4620 + 3080 + 3 \cdot 660 + 2 \cdot 280 + 264 + 105 = 10609 = e.$$

By Lemma 2.21, h is irreducible. Since $\deg h = k > a_i$ for any i, h is not a constant multiple of a variable, and hence h does not divide the monomial G. Thus the conditions of Set-up 4.1 hold. Since $8c = 10952 > 10609 = e$, we have $\frac{e}{c} < 8 < 105 = a_{(1)}$. Therefore the conditions of Corollary 4.20 hold. Note that

$$a_{(1)}a_{(2)}c = 24255 \cdot 1369 = 33205095 < 98027160 = 9240 \cdot 10609 = ke,$$

so the condition of Corollary 4.19 fails.

By Corollary 4.20, X is a $\mathbb{Q}$-Fano variety of dimension 7 with $\alpha(X) = \frac{k}{e} = \frac{9240}{10609}$. The inequalities $\frac{6}{7} < \alpha(X) < \frac{7}{8}$ read

$$6 \cdot 10609 = 63654 < 64680 = 7 \cdot 9240 \quad \text{and} \quad 8 \cdot 9240 = 73920 < 74263 = 7 \cdot 10609.$$

Let E be the exceptional divisor of the plt blow-up $f : Y \to X$ at p in Construction-Proposition 4.9. By Proposition 4.11,

$$\beta_X(E) = \frac{k - 7c}{8c} \operatorname{vol}(-K_X) = \frac{9240 - 9583}{8 \cdot 1369} \operatorname{vol}(-K_X) = -\frac{343}{10952} \operatorname{vol}(-K_X).$$

□

5.2. **Dimension at least eight.** The weights in this subsection are $a_i = k/p_i$, where $p_1, \dots, p_d$ are positive integers with $\sum_{i=1}^d 1/p_i = 1$ and $k = \operatorname{lcm}(p_1, \dots, p_d)$, so that $\sum_{i=1}^d a_i = k$, and for three indices u, v, w we take $G = x_u x_v \prod_{i \neq w} x_i$ and $c = a_u + a_v - a_w$, so that $\deg G = e = k + c$ and $d - 1 < k/c < d$. Corollary 4.19 requires $a_{(1)}a_{(2)}/k \geq e/c$, where $e/c = 1 + k/c < d + 1$ and $a_{(1)}a_{(2)}/k = k/(p_s p_t)$ if p_s and p_t are the largest and the second largest numbers among $p_1, \dots, p_d$. For $d \geq 17$, $k = \operatorname{lcm}(1, \dots, d)$ grows exponentially in d while $p_s p_t$ is a polynomial in d, and for $11 \leq d \leq 16$, we replace a few of the p_i and add a new prime factor 13 or 31 to k, so that $k/(p_s p_t)$ becomes large enough, and Table 3 lists the data for $d = 8, 9, 10$.

Construction-Proposition 5.4. Let $d \geq 8$ be an integer. We define positive integers $p_1, \dots, p_d$ and three distinct indices $u, v, w \in \{1, \dots, d\}$ as follows.

(1) If $d = 8$, then $(p_1, \dots, p_8) = (2, 4, 7, 20, 55, 66, 78, 91)$ and $(u, v, w) = (3, 6, 5)$.
(2) If $d = 9$, then $(p_1, \dots, p_9) = (2, 3, 9, 56, 72, 91, 123, 247, 1558)$ and $(u, v, w) = (3, 4, 5)$.
(3) If $d = 10$, then $(p_1, \dots, p_{10}) = (2, 6, 7, 10, 32, 72, 78, 90, 91, 96)$ and $(u, v, w) = (4, 6, 8)$.
(4) If $d \geq 11$, then $(u, v, w) := (1, d - 1, d)$, and $p_1, \dots, p_d$ are given as follows, where (ii) and (iii) are modifications of (i).

(i) $p_1 := d$, $p_2 := 2$, and $p_{j+1} := j(j+1)$ for $2 \le j \le d-1$.
(ii) If $d \in \{11, 12\}$, then $(p_5, p_6) := (13, 156)$.
(iii) If $13 \le d \le 16$, then $(p_3, p_4, p_6) := (4, 31, 930)$.

Let $k := \operatorname{lcm}(p_1, \dots, p_d)$, $a_i := k/p_i$ for any i, $c := a_u + a_v - a_w$, and $e := k + c$, and let $h := \sum_{i=1}^{d} x_i^{p_i}$ and $G := x_u x_v \prod_{i \ne w} x_i$ in $\mathbb{C}[x_1, \dots, x_d]$, graded by $\deg x_i = a_i$. Then c is a positive integer. More precisely,

$$\frac{c}{k} = \frac{1}{d} + \frac{2}{d(d-1)(d-2)} \quad \text{if } d \ge 11,$$

and $c = 8398$, 587366, 13468 if $d = 8$, 9, 10, respectively.

Proof. Suppose that $d \ge 11$. Since $d-1 \ge 10$ and the modifications change only p_i with $3 \le i \le 6$, we have $p_u = d$, $p_v = (d-2)(d-1)$, and $p_w = d(d-1)$. Thus

$$\frac{c}{k} = \frac{a_u + a_v - a_w}{k} = \frac{1}{d} + \frac{1}{(d-2)(d-1)} - \frac{1}{d(d-1)} = \frac{1}{d} + \frac{2}{d(d-1)(d-2)} > 0.$$

Suppose that $d \in \{8, 9, 10\}$. The data for $d = 8, 9, 10$ are listed in Table 3, and $c = a_u + a_v - a_w$ is computed from the columns a_u, a_v, and a_w. For example, if $d = 8$, then $k = 60060$, $a_u = k/7 = 8580$, $a_v = k/66 = 910$, $a_w = k/55 = 1092$, and $c = 8580 + 910 - 1092 = 8398$.

In any case, c is an integer, since a_u, a_v, and a_w are integers. □

The integers $p_1, \dots, p_d$ need not be distinct, so some of the weights a_i may coincide. The arguments below do not use that the p_i are distinct.

d	k	a_u	a_v	a_w	c	k/c	$a_{(1)}$	$a_{(2)}$	$a_{(1)}a_{(2)}/k$
8	$2^2 \cdot 3 \cdot 5 \cdot 7 \cdot 11 \cdot 13$	8580	910	1092	8398	2310/323	660	770	110/13
9	$2^3 \cdot 3^2 \cdot 7 \cdot 13 \cdot 19 \cdot 41$	567112	91143	70889	587366	252/29	3276	20664	252/19
10	$2^5 \cdot 3^2 \cdot 5 \cdot 7 \cdot 13$	13104	1820	1456	13468	360/37	1365	1440	15

TABLE 3. The data of Construction-Proposition 5.4 for $d = 8, 9, 10$.

d	m: two of $p_1, \dots, p_d$ divisible by m					
8	2^2: 4, 20	3: 66, 78	5: 20, 55	7: 7, 91	11: 55, 66	13: 78, 91
9	2^3: 56, 72	3^2: 9, 72	7: 56, 91	13: 91, 247	19: 247, 1558	41: 123, 1558
10	2^5: 32, 96	3^2: 72, 90	5: 10, 90	7: 7, 91	13: 78, 91	

TABLE 4. For $d = 8, 9, 10$ and any prime l dividing k, the largest power m of l dividing k, followed by two of $p_1, \dots, p_d$ divisible by m.

We check well-formedness by the following criterion.

Lemma 5.5. *Let $p_1, \dots, p_d$ be positive integers, $k := \operatorname{lcm}(p_1, \dots, p_d)$, and $a_i := k/p_i$ for any i. Assume that for any prime number l, the largest power of l dividing k divides at least two of $p_1, \dots, p_d$. Then $\mathbb{P}(a_1, \dots, a_d)$ is well formed.*

Proof. Let $1 \le i \le d$, and suppose that a prime l divides a_j for any $j \ne i$. By assumption, the largest power of l dividing k divides p_j for some $j \ne i$, so l does not divide $a_j = k/p_j$, a contradiction. □

Proposition 5.6. *Let $d \ge 8$, and let $a_1, \dots, a_d$, k, c, e, h, and G be as in Construction-Proposition 5.4. Let*

$$X := (zh + y^e + G = 0) \subset \mathbb{P}(a_1, \dots, a_d, 1, c),$$

where $x_1, \dots, x_d, y, z$ are the coordinates of degrees $a_1, \dots, a_d, 1, c$. Then X is a $\mathbb{Q}$-Fano variety of dimension d with

$$\alpha(X) = \frac{k}{e} = \begin{cases} \dfrac{2310}{2633} & \text{if } d = 8, \\ \dfrac{252}{281} & \text{if } d = 9, \\ \dfrac{360}{397} & \text{if } d = 10, \\ \dfrac{d(d-1)(d-2)}{(d+1)(d-1)(d-2)+2} & \text{if } d \geq 11, \end{cases}$$

so that $\frac{d-1}{d} < \alpha(X) < \frac{d}{d+1}$, and there exists a prime divisor E over X such that

$$\frac{\beta_X(E)}{\mathrm{vol}(-K_X)} = \frac{k - dc}{c(d+1)} = \begin{cases} -\dfrac{274}{2907} & \text{if } d = 8, \\ -\dfrac{9}{290} & \text{if } d = 9, \\ -\dfrac{10}{407} & \text{if } d = 10, \\ -\dfrac{2d}{(d+1)(d^2-3d+4)} & \text{if } d \geq 11. \end{cases}$$

Proof. The proof is long, so we divide it into several steps.

Step 1. In this step we show that $\sum_{i=1}^{d} 1/p_i = 1$.

Suppose that $d \geq 11$. Since $1/(j(j+1)) = 1/j - 1/(j+1)$, we have

$$\frac{1}{d} + \frac{1}{2} + \sum_{j=2}^{d-1} \frac{1}{j(j+1)} = \frac{1}{d} + \frac{1}{2} + \left(\frac{1}{2} - \frac{1}{d}\right) = 1.$$

If $d \geq 17$, then the left-hand side is $\sum_{i=1}^{d} 1/p_i$. If $d \in \{11, 12\}$ or $13 \leq d \leq 16$, then respectively

$$\sum_{i=1}^{d} \frac{1}{p_i} = 1 - \frac{1}{20} - \frac{1}{30} + \frac{1}{13} + \frac{1}{156} \quad \text{or} \quad \sum_{i=1}^{d} \frac{1}{p_i} = 1 - \frac{1}{6} - \frac{1}{12} - \frac{1}{30} + \frac{1}{4} + \frac{1}{31} + \frac{1}{930}.$$

Since

$$\frac{1}{20} + \frac{1}{30} = \frac{1}{12} = \frac{1}{13} + \frac{1}{156} \quad \text{and} \quad \frac{1}{6} + \frac{1}{12} + \frac{1}{30} = \frac{17}{60} = \frac{1}{4} + \frac{1}{31} + \frac{1}{930},$$

we have $\sum_{i=1}^{d} 1/p_i = 1$ for any $d \geq 11$.

Suppose that $d \in \{8, 9, 10\}$. Then

$$\begin{aligned} d = 8: \quad & a_1 + \cdots + a_8 = 30030 + 15015 + 8580 + 3003 + 1092 + 910 + 770 + 660 = 60060 = k, \\ d = 9: \quad & a_1 + \cdots + a_9 = 2552004 + 1701336 + 567112 + 91143 + 70889 + 56088 + 41496 \\ & \qquad + 20664 + 3276 = 5104008 = k, \\ d = 10: \quad & a_1 + \cdots + a_{10} = 65520 + 21840 + 18720 + 13104 + 4095 + 1820 + 1680 + 1456 \\ & \qquad + 1440 + 1365 = 131040 = k, \end{aligned}$$

that is, $\sum_{i=1}^{d} 1/p_i = 1$.

Thus $k = a_1 + \cdots + a_d$. Hence the integer $k = \mathrm{lcm}(p_1, \dots, p_d)$ of Construction-Proposition 5.4 is the integer k of Set-up 4.1(2).

Step 2. In this step we show that $W = \mathbb{P}(a_1, \dots, a_d)$ is well formed. By Lemma 5.5, it suffices to show that for any prime l, the largest power of l dividing k divides at least two of $p_1, \dots, p_d$.

If $d \in \{8, 9, 10\}$, then Table 4 lists, for any prime l dividing k, the largest power of l dividing k and two of $p_1, \dots, p_d$ divisible by it.

Suppose that $d \geq 11$, and let $L_d := \mathrm{lcm}(1, \dots, d)$. Let l be a prime dividing L_d, and let m be the largest power of l dividing L_d. Then $m \leq d$. Moreover, $m \geq 5$, since $m \geq l$ and the largest

powers of 2 and 3 dividing L_d are at least 8 and 9 respectively. Thus $m = 5$ or $7 \leq m \leq d$. The modifications change only p_i with $3 \leq i \leq 6$, so:

- if $m = 5$, then m divides $p_{10} = 90$ and $p_{11} = 110$;
- if $7 \leq m \leq d-1$, then m divides $p_m = (m-1)m$ and $p_{m+1} = m(m+1)$;
- if $m = d$, then m divides $p_1 = d$ and $p_d = (d-1)d$.

Therefore L_d divides k. For any $d \geq 11$, every p_i not changed by the modifications divides L_d, since $p_1 = d$, $p_2 = 2$, and $j(j+1)$ divides L_d for $2 \leq j \leq d-1$ as j and $j+1$ are coprime.

If $d \geq 17$, then no p_i is modified, so any p_i divides L_d. Thus $k = L_d$, and the criterion of Lemma 5.5 holds.

If $d \in \{11, 12\}$, then any p_i divides $13L_d$, since the modified $p_5 = 13$ and $p_6 = 156 = 12 \cdot 13$ divide $13L_d$. Since 13 does not divide L_d, $k = 13L_d = 13 \cdot 27720 = 360360$. Thus for any prime $l \neq 13$, the largest power of l dividing k is the largest power of l dividing L_d, and the largest power of 13 dividing k is 13, which divides $p_5 = 13$ and $p_6 = 156$.

If $13 \leq d \leq 16$, then any p_i divides $31L_d$, since the modified $p_3 = 4$, $p_4 = 31$, and $p_6 = 930 = 30 \cdot 31$ divide $31L_d$. Since 31 does not divide L_d, $k = 31L_d$. Thus for any prime $l \neq 31$, the largest power of l dividing k is the largest power of l dividing L_d, and the largest power of 31 dividing k is 31, which divides $p_4 = 31$ and $p_6 = 930$.

Step 3. In this step we show that $d - 1 < k/c < d$.

Suppose that $d \geq 11$. By Construction-Proposition 5.4,

$$\frac{k}{c} = \frac{d(d-1)(d-2)}{(d-1)(d-2)+2} = d - \frac{2d}{(d-1)(d-2)+2}.$$

Since $(d-1)(d-2) + 2 - 2d = (d-1)(d-4) > 0$, we have $0 < 2d < (d-1)(d-2) + 2$, and thus $d - 1 < k/c < d$.

Suppose that $d \in \{8, 9, 10\}$. By Table 3, k/c is 2310/323, 252/29, and 360/37 respectively, and

$$\begin{aligned} 7 \cdot 323 = 2261 < 2310 < 2584 = 8 \cdot 323, \\ 8 \cdot 29 = 232 < 252 < 261 = 9 \cdot 29, \\ 9 \cdot 37 = 333 < 360 < 370 = 10 \cdot 37. \end{aligned}$$

Thus $d - 1 < k/c < d$, that is, $k/d < c < k/(d-1)$.

Step 4. In this step we show that $a_{(1)}a_{(2)} \geq ke/c$.

Choose distinct indices s, t such that p_s and p_t are the largest and the second largest numbers among $p_1, \ldots, p_d$, counted with multiplicity. Then $a_{(1)} = a_s$ and $a_{(2)} = a_t$, so

$$\frac{a_{(1)}a_{(2)}}{k} = \frac{k}{p_s p_t}. \tag{5.1}$$

Since $e/c = 1 + k/c$, it suffices to show that $k/(p_s p_t) \geq 1 + k/c$.

Suppose that $d \in \{8, 9, 10\}$. By Table 3, $1 + k/c$ is 2633/323, 281/29, and 397/37 respectively, and

$$\frac{110}{13} > \frac{2633}{323}, \quad \frac{252}{19} > \frac{281}{29}, \quad 15 > \frac{397}{37},$$

since $110 \cdot 323 = 35530 > 34229 = 2633 \cdot 13$, $252 \cdot 29 = 7308 > 5339 = 281 \cdot 19$, and $15 \cdot 37 = 555 > 397$.

Suppose that $d \geq 11$. By **Step 3**, $1 + k/c < d + 1$, so it suffices to show that $k/(p_s p_t) > d + 1$. Recall that $L_d = \operatorname{lcm}(1, \ldots, d)$ and that k was computed in **Step 2**.

If $d \in \{11, 12\}$, then $\{p_s, p_t\} = \{156, d(d-1)\}$ and $k = 360360$, so

$$\frac{k}{p_s p_t} = \frac{360360}{156d(d-1)} \geq \frac{360360}{156 \cdot 132} = \frac{35}{2} > 13 \geq d + 1.$$

If $13 \leq d \leq 16$, then $\{p_s, p_t\} = \{930, d(d-1)\}$ and $k = 31L_d$ with $L_d \geq L_{13} = 360360$, so

$$\frac{k}{p_s p_t} = \frac{31L_d}{930d(d-1)} = \frac{L_d}{30d(d-1)} \geq \frac{360360}{30 \cdot 16 \cdot 15} = \frac{1001}{20} > 17 \geq d+1.$$

If $d \geq 17$, then $\{p_s, p_t\} = \{d(d-1), (d-1)(d-2)\}$ and $k = L_d$, so

$$\frac{k}{p_s p_t} = \frac{L_d}{d(d-1)^2(d-2)},$$

and it suffices to show that $L_d > (d+1)d(d-1)^2(d-2)$. The polynomial $(d+1)d(d-1)^2(d-2)$ is increasing in $d \geq 2$, and L_d is non-decreasing in d.

- If $17 \leq d \leq 24$, then $(d+1)d(d-1)^2(d-2) \leq 25 \cdot 24 \cdot 23^2 \cdot 22 = 6982800$, while
$$L_d \geq L_{17} = 2^4 \cdot 3^2 \cdot 5 \cdot 7 \cdot 11 \cdot 13 \cdot 17 = 12252240.$$
- If $25 \leq d \leq 31$, then $(d+1)d(d-1)^2(d-2) \leq 32 \cdot 31 \cdot 30^2 \cdot 29 = 25891200$, while
$$L_d \geq L_{19} = 2^4 \cdot 3^2 \cdot 5 \cdot 7 \cdot 11 \cdot 13 \cdot 17 \cdot 19 = 232792560.$$
- Suppose that $d \geq 32$. The binomial coefficient $\binom{d}{\lfloor d/2 \rfloor}$ divides L_d (cf. [Nai82, pp. 126–127]). Since 2^d is the sum of the $d+1$ binomial coefficients $\binom{d}{j}$ and $\binom{d}{\lfloor d/2 \rfloor}$ is the largest of them, $\binom{d}{\lfloor d/2 \rfloor} \geq 2^d/(d+1)$. Moreover, $2^d > (d+1)^6$: if $d = 32$, then $2^{32} = 4294967296 > 1291467969 = 33^6$, and the quotient of the values of $2^d/(d+1)^6$ at $d+1$ and d is
$$2\left(\frac{d+1}{d+2}\right)^6 \geq 2\left(\frac{33}{34}\right)^6 \geq 2\left(1 - \frac{6}{34}\right) > 1.$$
Since $d(d-2) < (d+1)^2$ and $(d-1)^2 < (d+1)^2$, we obtain
$$L_d \geq \binom{d}{\lfloor d/2 \rfloor} \geq \frac{2^d}{d+1} > (d+1)^5 > (d+1)d(d-1)^2(d-2).$$

Thus $a_{(1)}a_{(2)} \geq ke/c$ for any $d \geq 8$. Since $e/c > 1$, we also have $a_{(1)}a_{(2)} \geq k$, which is Set-up 4.1(3).

Step 5. In this step we show that h is quasismooth of degree k, that G is a monomial of degree e, and that h does not divide G.

Since $a_i p_i = k$ for any i, h is weighted homogeneous of degree k. Since $p_i \geq 2$ for any i, the partial derivatives $\partial h/\partial x_i = p_i x_i^{p_i - 1}$ have no common zero outside the origin of $\mathbb{A}^d$, so h is quasismooth. The monomial G has degree

$$\sum_{i \neq w} a_i + a_u + a_v = k - a_w + a_u + a_v = k + c = e$$

by **Step 1**. Since $d \geq 3$, h is irreducible (Lemma 2.21). Since $\deg h = k > a_i$ for any i (as $p_i \geq 2$), h is not a constant multiple of a variable. Since h is irreducible and any irreducible factor of the monomial G is a constant multiple of a variable, h does not divide G.

Step 6. In this step we conclude the proof.

The conditions of Set-up 4.1 hold: (1) by $d \geq 8$ and **Step 2**, (2) by **Step 1**, (3) by **Step 4**, (4) by Construction-Proposition 5.4 and **Step 3**, and (5) by **Step 5**. The variety X is the variety X of Set-up 4.1(6). By **Step 4**, $a_{(1)}a_{(2)} \geq ke/c$. Thus by Corollary 4.19, X is a $\mathbb{Q}$-Fano variety of dimension d with $\alpha(X) = k/e$.

Since $e = k + c$, we have

$$\frac{k}{e} = \frac{k/c}{k/c + 1}.$$

If $d \in \{8, 9, 10\}$, then substituting $k/c = 2310/323$, $252/29$, $360/37$ from Table 3 shows that k/e is $2310/2633$, $252/281$, $360/397$ respectively. If $d \geq 11$, then substituting the formula for k/c in **Step 3** shows that

$$\frac{k}{e} = \frac{d(d-1)(d-2)}{d(d-1)(d-2)+(d-1)(d-2)+2} = \frac{d(d-1)(d-2)}{(d+1)(d-1)(d-2)+2}.$$

Since the function $r \to r/(r+1)$ is strictly increasing for $r > 0$ and $d-1 < k/c < d$ by **Step 3**, we have

$$\frac{d-1}{d} < \alpha(X) = \frac{k}{e} < \frac{d}{d+1}.$$

Let E be the exceptional divisor of the plt blow-up $f : Y \to X$ at p in Construction-Proposition 4.9. By Proposition 4.11,

$$\frac{\beta_X(E)}{\mathrm{vol}(-K_X)} = \frac{k-dc}{c(d+1)} = \frac{k/c-d}{d+1}.$$

If $d \in \{8, 9, 10\}$, then substituting $k/c = 2310/323$, $252/29$, $360/37$ gives $\frac{2310-2584}{9\cdot 323} = -\frac{274}{2907}$, $\frac{252-261}{10\cdot 29} = -\frac{9}{290}$, and $\frac{360-370}{11\cdot 37} = -\frac{10}{407}$ respectively. If $d \geq 11$, then

$$\frac{k}{c} - d = \frac{d(d-1)(d-2)}{(d-1)(d-2)+2} - d = -\frac{2d}{d^2-3d+4},$$

so $\frac{\beta_X(E)}{\mathrm{vol}(-K_X)} = -\frac{2d}{(d+1)(d^2-3d+4)}$. □

6. Dimension three

In this section, we prove Theorem 6.4. In Subsection 6.1, we construct a plt blow-up of X at p and compute the volume of $-K_X - tE$ for its exceptional divisor E, and in Subsection 6.2, we prove that $(X, \frac{7}{10}D)$ is lc near the curves of quotient singularities of X for any $\mathbb{Q}$-divisor $D \geq 0$ with $D \sim_{\mathbb{Q}} -K_X$. In Subsection 6.3, the remaining non-lc centers of $(X, \frac{7}{10}D)$, a finite set of smooth points, are excluded by Kawamata–Viehweg vanishing, using $H^0(X, \mathcal{O}_X(1)) = 0$, and we conclude the proof.

Remark 6.1. Set-up 4.1 has no member with $d = 3$, since conditions (3) and (4) fail. In fact, let $d = 3$, and order the weights so that $a_1 \leq a_2 \leq a_3$. Since W is well formed, a_1, a_2, and a_3 are pairwise coprime. Since h is quasismooth, for each i the polynomial h contains a monomial $x_i^m x_j$, where j may equal i, since otherwise h and all partial derivatives of h vanish on the x_i-axis. For $i = 3$, the monomials $x_3^m x_3$, $x_3^m x_1$, and $x_3^m x_2$ give

$$ma_3 = a_1 + a_2, \quad (m-1)a_3 = a_2, \quad \text{and} \quad (m-1)a_3 = a_1$$

respectively. Since $0 < a_1 \leq a_2 \leq a_3$, the first equation gives $m = 1$ and $a_3 = a_1 + a_2$, or $m = 2$ and $a_1 = a_2 = a_3$. The second and the third equations give $m = 2$, since their right-hand sides are positive and at most a_3, and then $a_2 = a_3$ and $a_1 = a_3$ respectively. Since a_1, a_2, and a_3 are pairwise coprime, each of $a_1 = a_2 = a_3$, $a_2 = a_3$, and $a_1 = a_3$ implies $a_1 = a_2 = a_3 = 1$. Thus $a_1 = a_2 = a_3 = 1$ or $a_3 = a_1 + a_2$. If $a_3 = a_1 + a_2$, then $k = 2(a_1 + a_2)$, and for $i = 2$ the monomials $x_2^m x_2$, $x_2^m x_1$, and $x_2^m x_3$ give

$$(m-1)a_2 = 2a_1, \quad (m-2)a_2 = a_1, \quad \text{and} \quad (m-1)a_2 = a_1$$

respectively. Since $\gcd(a_1, a_2) = 1$, each of them implies $a_2 \mid 2$. Thus (a_1, a_2, a_3) is $(1, 1, 1)$, $(1, 1, 2)$, or $(1, 2, 3)$, so

$$a_{(1)}a_{(2)} = a_{(2)} \leq 2 < 3 \leq k,$$

and condition (3) fails. Moreover, $k = 3$, 4, or 6 respectively, and the open interval $(\frac{k}{3}, \frac{k}{2})$ is $(1, \frac{3}{2})$, $(\frac{4}{3}, 2)$, or $(2, 3)$ respectively, which contains no integer. Thus condition (4) fails as well, and Set-up 4.1 has no member with $d = 3$ even without condition (3).

Set-up 6.2. Let $\mathbb{P}$, R, X, H, p, Γ, and Λ be as follows.

(1) $\mathbb{P} := \mathbb{P}(3,4,4,3,3)$ with coordinates x, u, v, y, z of degrees $3, 4, 4, 3, 3$.
(2) $X := \operatorname{Proj} R \subset \mathbb{P}$, where
$$R := \mathbb{C}[x,u,v,y,z]/\big(zx^4 + 2x^5 + y^5 + (z+y)u^3 + (z+x)v^3\big).$$
(3) $H := \mathcal{O}_X(1)$ and $p := [0:0:0:0:1] \in X$.
(4) $\Gamma := (u = v = 0) \cap X$ and $\Lambda := (x = y = z = 0) \cap X$.

Remark 6.3. The equation of X is linear in z. In the orbifold chart of X at p, for the weights 15, 20, 20, and 12 on x, u, v, and y, the lowest-weight part of the equation is $x^4 + u^3 + v^3 + y^5$, of weight 60, and the terms $2x^5$, yu^3, and xv^3 have weights 75, 72, and 75. The polynomial $x^4 + u^3 + v^3 + y^5$ gives the plt blow-up of Construction-Proposition 6.6, as $h + y^e$ does for X of Set-up 4.1 in the orbifold chart at p. The terms yu^3 and xv^3 make X quasismooth along Λ: on Λ, the partial derivatives of the equation with respect to y and x are u^3 and v^3, which do not vanish simultaneously. Without yu^3 and xv^3, X is not quasismooth at the three points of Λ with $u^3 + v^3 = 0$. The term $2x^5$ makes X quasismooth at the points with $uv \neq 0$, and the coefficient 2 avoids the finitely many values of the coefficient of x^5 for which X is not quasismooth outside p.

Theorem 6.4. *Notation as in Set-up 6.2. Then X is a $\mathbb{Q}$-Fano threefold with $\alpha(X) = \frac{7}{10}$, and there exists a prime divisor E over X such that $\beta_X(E) = -\frac{1}{6}\operatorname{vol}(-K_X)$, where $\operatorname{vol}(-K_X) = \frac{5}{18}$. In particular, $\frac{2}{3} < \alpha(X) < \frac{3}{4}$ and X is not K-semistable.*

We prove Theorem 6.4 in Subsection 6.3.

6.1. **The plt blow-up and the volume.** In this subsection, we construct a plt blow-up of X at p, compute the volume of $-K_X - tE$ for the exceptional divisor E of the plt blow-up, and prove that $(X, \frac{7}{10}D)$ is lc near p for any $\mathbb{Q}$-divisor $D \geq 0$ with $D \sim_{\mathbb{Q}} -K_X$.

Proposition 6.5. *Notation as in Set-up 6.2. Then R is a normal domain, X is quasismooth outside p, the locus of points of $\mathbb{P}$ with non-trivial stabilizer intersects X in $\Gamma \cup \Lambda$, and X is smooth outside $\Gamma \cup \Lambda$. Moreover, $\mathcal{O}_X(12)$ is an ample invertible sheaf, $\omega_X \cong \mathcal{O}_X(-2)$, $H^3 = \frac{5}{144}$, and $H^0(X, \mathcal{O}_X(m)) = R_m$ for every integer m.*

Proof. Let $h := x^4 + u^3 + v^3$ and $g := 2x^5 + y^5 + yu^3 + xv^3$, so that $\Phi := zh + g$ is the equation of X.

The polynomial h is irreducible. In fact, h is a monic cubic polynomial in u with constant term $x^4 + v^3$. If h were reducible in $\mathbb{C}[x,u,v]$, then, since h is monic of degree 3 in u, h would have a factor $u - r$ with $r \in \mathbb{C}[x,v]$, and $x^4 + v^3 = (-r)^3$. Then r is non-constant, and for an irreducible factor r_1 of r, r_1^2 divides $x^4 + v^3$, so r_1 divides the partial derivatives $4x^3$ and $3v^2$. Thus the curve $(r_1 = 0) \subset \mathbb{A}^2$ is contained in $(x = v = 0)$, a contradiction. Hence h is irreducible in $\mathbb{C}[x,u,v]$. Moreover, h does not divide g, since h does not involve y and g is monic of degree 5 in y. Hence Φ is a primitive polynomial of degree one in z over $\mathbb{C}[x,u,v,y]$, so Φ is irreducible and R is a domain.

The partial derivatives of Φ are
$$\Phi_z = x^4 + u^3 + v^3, \quad \Phi_x = 4zx^3 + 10x^4 + v^3, \quad \Phi_y = 5y^4 + u^3,$$
$$\Phi_u = 3u^2(z+y), \quad \Phi_v = 3v^2(z+x).$$

Let $c = (x,u,v,y,z) \in \mathbb{A}^5$ be a point with $\Phi(c) = 0$ at which Φ_z, Φ_x, Φ_y, Φ_u, and Φ_v vanish.

- If $u \neq 0$ and $v \neq 0$, then $z = -y = -x$, so $\Phi(c) = 2x^5 = 0$ implies $x = y = z = 0$, and then $\Phi_x(c) = v^3 \neq 0$, a contradiction.
- If $u \neq 0$ and $v = 0$, then $z = -y$ and $u^3 = -x^4$, so $x \neq 0$ and $\Phi_x(c) = x^3(10x - 4y) = 0$ implies $x = \frac{2}{5}y$. Then $\Phi_y(c) = 5y^4 - x^4 = \frac{3109}{625}y^4 = 0$ implies $y = 0$, so $x = 0$, a contradiction.
- If $u = 0$ and $v \neq 0$, then $z = -x$ and $v^3 = -x^4$, so $\Phi_x(c) = -4x^4 + 10x^4 - x^4 = 5x^4 = 0$ implies $x = 0$ and $v = 0$, a contradiction.

- If $u = v = 0$, then $\Phi_z(c) = x^4 = 0$ and $\Phi_y(c) = 5y^4 = 0$, so $x = y = 0$.

Conversely, Φ and all of the partial derivatives vanish on the z-axis. Thus the singular locus of the affine cone $(\Phi = 0) \subset \mathbb{A}^5$ is the z-axis, and X is quasismooth outside p.

The weighted projective space $\mathbb{P}$ is well formed. A point of $\mathbb{P}$ has non-trivial stabilizer if and only if $u = v = 0$, where the stabilizer is μ_3, or $x = y = z = 0$, where the stabilizer is μ_4. Thus the locus of points of $\mathbb{P}$ with non-trivial stabilizer intersects X in $\Gamma \cup \Lambda$. Moreover, $\Gamma = (zx^4 + 2x^5 + y^5 = 0) \subset (u = v = 0) \cong \mathbb{P}(3,3,3)$ is a curve, and since every monomial of Φ is divisible by x, y, or z, $\Lambda = (x = y = z = 0) \cong \mathbb{P}(4,4)$ is a curve. Since $\dim X = 3$, $\Gamma \cup \Lambda$ is of codimension 2 in X. Since $p \in \Gamma$, X is quasismooth at any point $w \in X \backslash (\Gamma \cup \Lambda)$, and the stabilizer of w is trivial. If $\xi \in \{x, u, v, y, z\}$ is a coordinate of degree a_ξ with $\xi(w) \neq 0$, then μ_{a_ξ} acts freely near $\rho_\xi^{-1}(w)$, so ρ_ξ is étale there, and $\rho_\xi^{-1}(X)$ is smooth near $\rho_\xi^{-1}(w)$ (Definition 2.19 and Definition-Lemma 2.20(2)). Thus X is smooth outside $\Gamma \cup \Lambda$.

Since the localizations of $\mathbb{C}[x, u, v, y, z]$ at prime ideals are regular local rings and Φ is a non-zero-divisor, R is Cohen–Macaulay [Mat86, Theorems 17.8 and 17.3(ii)], so R satisfies (S_2). The singular locus of $\operatorname{Spec} R$ is the z-axis, which is of codimension 3 in the four-dimensional variety $\operatorname{Spec} R$, so R satisfies (R_1). Thus R is normal by [Mat86, Theorem 23.8].

By Lemma 2.22, applied to the equation Φ of degree 15,

$$\omega_X \cong \mathcal{O}_X(15 - 3 - 4 - 4 - 3 - 3) = \mathcal{O}_X(-2), \quad H^3 = \frac{15}{3 \cdot 4 \cdot 4 \cdot 3 \cdot 3} = \frac{5}{144},$$

and $H^0(X, \mathcal{O}_X(m)) = R_m$ for every integer m.

Finally, for each coordinate ξ of degree a_ξ, $\xi^{12/a_\xi} \in R_{12}$ is a unit of the localization R_ξ, so $\mathcal{O}_X(12)$ is generated by ξ^{12/a_ξ} on $X \cap (\xi \neq 0)$. Thus $\mathcal{O}_X(12)$ is invertible. The non-vanishing loci $X \cap (\xi \neq 0)$ of the five sections x^4, u^3, v^3, y^4, z^4 of $\mathcal{O}_X(12)$ are affine and cover X, so $\mathcal{O}_X(12)$ is ample by [Gro61, Théorème (4.5.2)(a') and Définition (4.5.3)]. □

Construction-Proposition 6.6. Notation as in Set-up 6.2. Let

$$X' := (x^4 + u^3 + v^3 + y^5 + 2x^5 + yu^3 + xv^3 = 0) \subset \mathbb{A}^4$$

be the preimage of X in the orbifold chart $\rho_z : U_z \to (z \neq 0) \subset \mathbb{P}$, on which μ_3 acts with weights $(0, 1, 1, 0)$ on (x, u, v, y). Let $f' : Y' \to X'$ be the morphism from the strict transform Y' of X' in the weighted blow-up of $\mathbb{A}^4$ with weights $(15, 20, 20, 12)$ on (x, u, v, y), and let E' be the exceptional divisor of f'. Then the action of μ_3 lifts to Y', and $Y'/\mu_3 \to X \cap (z \neq 0)$ extends to a plt blow-up $f : Y \to X$ of X at p. The exceptional divisor E of f satisfies the following.

(1) The ramification index of $Y' \to Y'/\mu_3$ along E' is 3, and $a(E, X, 0) = \frac{7}{3}$.
(2) There exists an isomorphism $E \cong \mathbb{P}(1,1,3)$ under which

$$\Delta_E = \frac{3}{4}(W + U^3 + V^3 = 0) + \frac{4}{5}(W = 0),$$

where U, V, W are the coordinates of $\mathbb{P}(1,1,3)$ of degrees $1, 1, 3$.
(3) The monomials $x^{\gamma_1} u^{\gamma_2} v^{\gamma_3} y^j z^i$ with $j < 5$ form a basis of R, and for every integer $m \geq 0$ and every non-zero $s \in R_m = H^0(X, \mathcal{O}_X(m))$,

$$\operatorname{mult}_E(s = 0) = \frac{1}{3} \min\{15\gamma_1 + 20\gamma_2 + 20\gamma_3 + 12j\},$$

where the minimum is taken over the basis monomials $x^{\gamma_1} u^{\gamma_2} v^{\gamma_3} y^j z^i$ that appear in s with non-zero coefficient, and $(s = 0) \sim mH$ denotes the Weil divisor of zeros of s.

Moreover, X is a $\mathbb{Q}$-Fano threefold.

Proof. The proof is long, so we divide it into several steps. Let

$$\Phi := zx^4 + 2x^5 + y^5 + (z + y)u^3 + (z + x)v^3 \quad \text{and} \quad \Psi := x^4 + u^3 + v^3 + y^5 + 2x^5 + yu^3 + xv^3,$$

so that $\Psi(x, u, v, y) = \Phi(x, u, v, y, 1)$ and $X' = (\Psi = 0)$.

Step 1. In this step, we show that the preimage of X in any orbifold chart of $\mathbb{P}$ is smooth outside the preimage of p, that f' is a plt blow-up of X' at 0 with $a(E', X', 0) = 7$, and that X is klt.

Let $\xi \in \{x, u, v, y, z\}$ be a coordinate of degree a_ξ, and let $X_\xi := (\Phi = 0) \cap (\xi = 1) \subset \mathbb{A}^5$ be the preimage of X in the orbifold chart U_ξ. Let $c \in X_\xi$ be a singular point of X_ξ. Then all of the partial derivatives of Φ other than Φ_ξ vanish at c, and the Euler formula

$$3x\Phi_x + 4u\Phi_u + 4v\Phi_v + 3y\Phi_y + 3z\Phi_z = 15\Phi$$

implies $a_\xi \Phi_\xi(c) = 0$. Thus c is a singular point of the affine cone $(\Phi = 0) \subset \mathbb{A}^5$, and c is on the z-axis since X is quasismooth outside p (Proposition 6.5). Hence X_ξ is smooth outside the preimage of p. In particular, $X' = X_z$ is smooth outside 0.

Let $w := (15, 20, 20, 12)$. The monomials $2x^5$, yu^3, and xv^3 have weights 75, 72, and 75, so $w(\Psi) = 60$ and $\Psi_w = x^4 + u^3 + v^3 + y^5$. We have $\gcd(15, 20, 20, 12) = 1$ and $15 + 20 + 20 + 12 = 67 > 60$. The partial derivatives $4x^3$, $3u^2$, $3v^2$, and $5y^4$ of Ψ_w vanish simultaneously only at 0, so Ψ_w is quasismooth. The polynomial Ψ_w is irreducible: Ψ_w is a monic cubic polynomial in u with constant term $x^4 + v^3 + y^5$, so if Ψ_w were reducible in $\mathbb{C}[x, u, v, y]$, then Ψ_w would have a factor $u - r$ with $r \in \mathbb{C}[x, v, y]$, and $x^4 + v^3 + y^5 = (-r)^3$. Then r is non-constant, and an irreducible factor r_1 of r divides the partial derivatives $4x^3$, $3v^2$, and $5y^4$, which is impossible, since the surface $(r_1 = 0) \subset \mathbb{A}^3$ is not contained in $(x = v = y = 0)$. Hence Ψ_w is irreducible in $\mathbb{C}[x, u, v, y]$. By Lemma 3.2, X' is normal and klt near 0, f' is a plt blow-up of X' at 0, $E' \cong (\Psi_w = 0) \subset \mathbb{P}(15, 20, 20, 12)$, and

$$a(E', X', 0) = 67 - 60 = 7.$$

Since X' is smooth outside 0 and f' is an isomorphism over $X' \backslash \{0\}$, X' is normal and klt (X' is irreducible: μ_3 permutes the components of X' transitively since $X \cap (z \neq 0)$ is irreducible, each component contains the fixed point 0, and X' is normal at 0), f' is a plt blow-up of X' at 0, and Lemma 3.2(4) holds, without shrinking X'.

For $\xi \in \{x, u, v, y, z\}$, let $g_\xi : X_\xi \to X \cap (\xi \neq 0)$ be the quotient morphism by μ_{a_ξ}. A point of U_ξ with non-trivial stabilizer in μ_{a_ξ} maps to a point of $\mathbb{P}$ with non-trivial stabilizer, so g_ξ is étale outside the preimage of $\Gamma \cup \Lambda$ (Proposition 6.5), which is of codimension 2 since Γ and Λ are curves. Since $K_X \sim -2H$ and $12H$ is Cartier (Proposition 6.5 and Definition-Lemma 2.16), K_X is $\mathbb{Q}$-Cartier, and Lemma 2.8(2), applied to the restriction of g_ξ to each connected component of X_ξ, which maps finitely onto $X \cap (\xi \neq 0)$, implies $K_{X_\xi} = g_\xi^* K_X$. X_ξ is smooth outside the preimage of p, $p \in (\xi \neq 0)$ only for $\xi = z$, and $X_z = X'$ is klt by the previous paragraph. Thus each X_ξ is klt, and X is klt by Lemma 2.8(3), applied to the restriction of g_ξ to each connected component of X_ξ.

Step 2. In this step, we construct f and prove (1).

Since Ψ is invariant under μ_3, the group μ_3 acts on X', and ρ_z restricts to the quotient morphism $g := g_z : X' \to X \cap (z \neq 0)$. By **Step 1**, g is étale outside a closed subset of codimension 2, so the ramification index of g along any prime divisor is 1, and $K_{X'} = g^* K_X$.

Let $\zeta \in \mu_3$ and $\lambda := \zeta^2$. Then

$$(\lambda^{15} x, \lambda^{20} u, \lambda^{20} v, \lambda^{12} y) = (x, \zeta u, \zeta v, y),$$

so ζ acts on $\mathbb{A}^4$ as the weighted scaling by λ with weights w. The weighted blow-up of $\mathbb{A}^4$ is equivariant for the weighted scaling, and the exceptional divisor of the weighted blow-up of $\mathbb{A}^4$ is the quotient of $\mathbb{A}^4 \backslash \{0\}$ by the weighted scaling. Thus the action of μ_3 lifts to the weighted blow-up of $\mathbb{A}^4$, preserves Y', and fixes E' pointwise. Let $q : Y' \to Y'/\mu_3$ be the quotient morphism. Since μ_3 acts faithfully on Y' and fixes E' pointwise, the ramification index of q along E' is 3 (in characteristic zero the ramification index along a prime divisor of the quotient by a finite group is the order of the subgroup fixing the divisor pointwise), and $q|_{E'}$ is injective. Since q coincides with g over $X' \backslash \{0\}$, the ramification index of q along any prime divisor other than E' is 1.

The morphism $Y'/\mu_3 \to X'/\mu_3 = X \cap (z \neq 0)$ induced by f' is an isomorphism over $X \cap (z \neq 0)\backslash\{p\}$. Let Y be the variety obtained by gluing Y'/μ_3 and $X\backslash\{p\}$ along $(Y'\backslash E')/\mu_3 \cong X \cap (z \neq 0)\backslash\{p\}$, let $f : Y \to X$ be the induced morphism, and let $E := q(E')$. Then Y is normal, f is an isomorphism outside the prime divisor E, and $f(E) = \{p\}$. Moreover, f is proper, since $g \circ f' = f \circ q$ is proper and q is surjective.

Since $q|_{E'}$ is injective, $E' \to E$ is of degree 1, so E is the norm of E' under q. Since E' is $\mathbb{Q}$-Cartier (Definition-Proposition 2.14), E is $\mathbb{Q}$-Cartier [Kol13, Claim 2.40.1]. We have $q^*E = 3E'$. Let $V \subset E$ be an irreducible subvariety of positive dimension, and let $V' \subset E'$ be an irreducible subvariety with $q(V') = V$. Since $-E'$ is f'-ample and $f'(V') = \{0\}$, the projection formula implies

$$\deg(V' \to V) \cdot \big((-E)^{\dim V} \cdot V\big) = \big((-q^*E)^{\dim V} \cdot V'\big) = 3^{\dim V}\big((-E')^{\dim V} \cdot V'\big) > 0.$$

Since the positive-dimensional subvarieties of Y contracted by f are contained in E, $-E$ is f-ample by [Laz04, Corollary 1.7.9]. In particular, f is projective.

Let $a := a(E, X, 0)$, so that $K_Y + (1-a)E = f^*K_X$. Since $g \circ f' = f \circ q$,

$$q^*\big(K_Y + (1-a)E\big) = f'^*g^*K_X = f'^*K_{X'} = K_{Y'} + \big(1 - a(E', X', 0)\big)E' = K_{Y'} - 6E'.$$

By Lemma 2.8(2) along E', $-6 = 1 - 3\big(1-(1-a)\big) = 1 - 3a$, so $a = \frac{7}{3}$. Similarly, $K_Y + E = f^*K_X + aE$ is $\mathbb{Q}$-Cartier, and Lemma 2.8(2) implies $K_{Y'} + E' = q^*(K_Y + E)$, since the coefficient of E' is $1-3(1-1) = 1$. The pair (Y', E') is plt, since (Y', E') is plt near E' and $Y'\backslash E' \cong X'\backslash\{0\}$ is smooth. Since (Y', E') is plt, $\operatorname{discrep}(Y', E') > -1$ [KM98, Definition 2.34]. Since q is finite of degree 3 and $K_{Y'} + E' = q^*(K_Y + E)$ over Y'/μ_3, [KM98, Proposition 5.20(2)] gives

$$3\big(\operatorname{discrep}(Y'/\mu_3, E) + 1\big) \geq \operatorname{discrep}(Y', E') + 1 > 0,$$

so $(Y'/\mu_3, E)$ is plt. Since $E \subset Y'/\mu_3$, (Y, E) is plt near E. By Definition-Proposition 2.14, f is a plt blow-up of X at p with exceptional divisor E, and (1) holds.

Step 3. In this step, we prove (2).

By Definition-Proposition 2.14, E is normal. Since $q|_{E'} : E' \to E$ is finite and injective, $q|_{E'}$ is a finite birational morphism onto E, and since E is normal, $(q|_{E'})_*\mathcal{O}_{E'} = \mathcal{O}_E$, so $q|_{E'}$ is an isomorphism. Let $R_{E'} := \mathbb{C}[x, u, v, y]/(\Psi_w)$ with x, u, v, y of degrees $15, 20, 20, 12$, so that $E' \cong \operatorname{Proj} R_{E'}$ (**Step 1**). Since Ψ_w is monic of degree 4 in x, the monomials $x^{\gamma_1}u^{\gamma_2}v^{\gamma_3}y^j$ with $\gamma_1 < 4$ form a basis of $R_{E'}$. Such a monomial has degree divisible by 20 if and only if $4 \mid \gamma_1$ and $5 \mid j$, since

$$15\gamma_1 + 20\gamma_2 + 20\gamma_3 + 12j \equiv 3\gamma_1 \pmod 4 \quad \text{and} \quad 15\gamma_1 + 20\gamma_2 + 20\gamma_3 + 12j \equiv 2j \pmod 5.$$

Thus $\gamma_1 = 0$, and the Veronese subring $R_{E'}^{(20)}$ is the polynomial ring $\mathbb{C}[u, v, y^5]$. Let $U := u$, $V := v$, and $W := y^5$, of degrees $1, 1, 3$ after dividing the degrees by 20. Then $E' \cong \operatorname{Proj} R_{E'}^{(20)} \cong \mathbb{P}(1, 1, 3)$ [Gro61, Proposition (2.4.7)(i)]. Since $x^4 = -(W + U^3 + V^3)$ in $R_{E'}$, under the isomorphism $E \cong E' \cong \mathbb{P}(1, 1, 3)$, the curves $q((x = 0) \cap E')$ and $q((y = 0) \cap E')$ correspond to $C_x := (W + U^3 + V^3 = 0)$ and $C_y := (W = 0)$, respectively.

Since (Y, E) is plt and E is normal, the coefficient of Δ_E along a prime divisor P of E is $\frac{n_P - 1}{n_P}$ if, near the generic point of P, Y is locally analytically isomorphic to $(\mathbb{C}^2/\mu_{n_P}) \times \mathbb{C}$, where μ_{n_P} acts on $\mathbb{C}^2$ diagonally with weights coprime to n_P, with E the image of a coordinate axis of $\mathbb{C}^2$ times $\mathbb{C}$ [Kol+92, (16.6.3)]. We compute n_P by the charts of the weighted blow-up of $\mathbb{A}^4$.

The u-chart is $\mathbb{A}^4/\mu_{20}$ with coordinates (x_1, τ, v_1, y_1), where

$$x = x_1\tau^{15}, \quad u = \tau^{20}, \quad v = v_1\tau^{20}, \quad y = y_1\tau^{12},$$

and μ_{20} acts with weights $(5, 1, 0, 8)$ (we use the inverse of the generator in the convention of the proof of Lemma 3.2). In the u-chart, the preimage of Y' is $(\widetilde{\Psi}_u = 0)$, where

$$\widetilde{\Psi}_u := x_1^4 + 1 + v_1^3 + y_1^5 + \tau^{12}y_1 + \tau^{15}(2x_1^5 + x_1v_1^3),$$

and the preimage of E' is $(\tau = 0)$. A general point of $(x = 0) \cap E'$ has a preimage $\tilde{c} = (0, 0, v_1, y_1)$ with $y_1 \neq 0$. Then

$$\partial\widetilde{\Psi}_u/\partial y_1(\tilde{c}) = 5y_1^4 \neq 0,$$

so (x_1, τ, v_1) are local analytic coordinates of $(\widetilde{\Psi}_u = 0)$ at $\tilde{c}$. Recall from **Step 2** that $\zeta \in \mu_3$ acts on $\mathbb{A}^4$ as the weighted scaling by $\lambda = \zeta^2$ with weights w. In the u-chart, this is $\tau \to \lambda\tau$ with x_1, v_1, and y_1 fixed, since $x = x_1\tau^{15}$, $u = \tau^{20}$, $v = v_1\tau^{20}$, and $y = y_1\tau^{12}$. The stabilizer of $\tilde{c}$ in μ_{20} is μ_4, which acts with weights $(1, 1, 0)$ on (x_1, τ, v_1). Taking the quotient by μ_3 first, Y is locally analytically isomorphic near the image of $\tilde{c}$ to $(\mathbb{C}^2/\mu_4) \times \mathbb{C}$, where μ_4 acts on $\mathbb{C}^2$ with coordinates (x_1, τ^3) with weights $(1, 3)$, and E is the image of $(\tau^3 = 0)$. Thus $n_P = 4$ for $P = C_x$.

The x-chart is $\mathbb{A}^4/\mu_{15}$ with coordinates (τ, u_1, v_1, y_1), where

$$x = \tau^{15}, \quad u = u_1\tau^{20}, \quad v = v_1\tau^{20}, \quad y = y_1\tau^{12},$$

and μ_{15} acts with weights $(1, 10, 10, 3)$ (again with the inverse of the generator in the convention of the proof of Lemma 3.2). In the x-chart, the preimage of Y' is $(\widetilde{\Psi}_x = 0)$, where

$$\widetilde{\Psi}_x := 1 + u_1^3 + v_1^3 + y_1^5 + \tau^{12}y_1u_1^3 + \tau^{15}(2 + v_1^3).$$

A general point of $(y = 0) \cap E'$ has a preimage $\tilde{c} = (0, u_1, v_1, 0)$ with $u_1v_1 \neq 0$. Then

$$\partial\widetilde{\Psi}_x/\partial v_1(\tilde{c}) = 3v_1^2 \neq 0,$$

so (τ, u_1, y_1) are local analytic coordinates of $(\widetilde{\Psi}_x = 0)$ at $\tilde{c}$. The stabilizer of $\tilde{c}$ in μ_{15} is μ_5, which acts with weights $(1, 0, 3)$ on (τ, u_1, y_1), and, similarly, μ_3 acts by $\tau \to \lambda\tau$ with u_1, v_1, and y_1 fixed. Taking the quotient by μ_3 first, Y is locally analytically isomorphic near the image of $\tilde{c}$ to $(\mathbb{C}^2/\mu_5) \times \mathbb{C}$, where μ_5 acts on $\mathbb{C}^2$ with coordinates (τ^3, y_1) with weights $(3, 3)$, and E is the image of $(\tau^3 = 0)$. Thus $n_P = 5$ for $P = C_y$.

A point of $E'\setminus\big((x = 0) \cup (y = 0)\big)$ has non-trivial stabilizer in $\mathbb{P}(15, 20, 20, 12)$ only if $u = v = 0$, and $E' \cap (u = v = 0)$ is finite. Thus a general point c of any curve $P' \subset E'$ other than $(x = 0) \cap E'$ and $(y = 0) \cap E'$ has trivial stabilizer. Near c, the chart morphism is étale, the preimage of E' is smooth since Ψ_w is quasismooth, and so the preimage of Y' is smooth. Since, similarly, μ_3 acts by $\tau \to \lambda\tau$ in the chart, Y is smooth near $q(c)$ and E is the image of the smooth divisor $(\tau^3 = 0)$. Thus $n_P = 1$ for $P = q(P')$. Hence

$$\Delta_E = \frac{3}{4}C_x + \frac{4}{5}C_y = \frac{3}{4}(W + U^3 + V^3 = 0) + \frac{4}{5}(W = 0),$$

which proves (2).

Step 4. In this step, we prove (3).

Let $h := x^4 + u^3 + v^3$. Order the variables as (x, u, v, z, y), and assign to them the auxiliary weights $(15, 20, 20, 0, 12)$. The monomials of Φ have auxiliary weights 60 for zx^4, y^5, zu^3, and zv^3, and 75, 72, and 75 for $2x^5$, yu^3, and xv^3. Thus the lowest-weight part of Φ is $zh + y^5$. The polynomials Φ and $zh + y^5$ are monic of degree 5 in y. Since y does not divide h, $\gcd(h, y^5) = 1$, so $zh + y^5$ is a primitive polynomial of degree one in z, and $zh + y^5$ is prime. By Lemma 3.3(2), the monomials $x^{\gamma_1}u^{\gamma_2}v^{\gamma_3}y^jz^i$ with $j < 5$ form a basis of R, and for any non-zero $s \in R_m$, the weight $b(s)$ of Lemma 3.3 is

$$b(s) = \min\{15\gamma_1 + 20\gamma_2 + 20\gamma_3 + 12j\},$$

where the minimum is taken over the basis monomials that appear in s.

Let $s \in R_m$ be non-zero, let $\tilde{s}$ be the expansion of s in the basis, and let $\varphi := \tilde{s}(x, u, v, y, 1)$, a polynomial in $\mathbb{C}[x, u, v, y]$. Since the degree m is fixed, the exponent i of z in a monomial of degree m is determined by $(\gamma_1, \gamma_2, \gamma_3, j)$. Thus the monomials of φ correspond bijectively to the basis monomials in s, and $w(\varphi) = b(s)$. Every monomial of φ_w has degree less than 5 in y, while Ψ_w is monic of degree 5 in y, so Ψ_w does not divide φ_w. By Lemma 3.2(4),

$$\mathrm{mult}_{E'}(\varphi|_{X'} = 0) = w(\varphi) = b(s).$$

The element s^3/z^m of degree 0 is a regular function on $X \cap (z \neq 0)$ with divisor $3(s=0)|_{X\cap(z\neq 0)}$, since z is a unit of degree 3 in R_z, $\mathcal{O}_X(3)$ is generated by z on $X \cap (z \neq 0)$, and $(s^3 = 0) = 3(s = 0)$ by Definition-Lemma 2.16. Moreover, $g^*(s^3/z^m) = \varphi^3|_{X'}$. Thus $g^*(s = 0) = (\varphi|_{X'} = 0)$. Since $g \circ f' = f \circ q$, $q^*E = 3E'$, and E' is the only prime divisor on Y' over E,

$$b(s) = \operatorname{mult}_{E'} f'^* g^*(s = 0) = \operatorname{mult}_{E'} q^* f^*(s = 0) = 3 \operatorname{mult}_E(s = 0).$$

This proves (3).

Step 5. In this step, we show that X is a $\mathbb{Q}$-Fano threefold.

By Proposition 6.5, X is a normal projective threefold. By **Step 1**, X is klt. Finally, $-K_X \sim 2H$ and $12H$ is an ample Cartier divisor (Proposition 6.5 and Definition-Lemma 2.16), so $-K_X$ is an ample $\mathbb{Q}$-Cartier divisor. Thus X is a $\mathbb{Q}$-Fano threefold. □

As for Proposition 4.11, we compute $\operatorname{vol}(-K_X - tE)$ from the monomial basis of Construction-Proposition 6.6(3).

Proposition 6.7. *Notation as in Set-up 6.2, and let E be as in Construction-Proposition 6.6. Then*

$$\operatorname{vol}(-K_X - tE) = \begin{cases} \frac{5}{18}\left(1 - \left(\frac{3t}{10}\right)^3\right) & \text{if } 0 \le t \le \frac{10}{3}, \\ 0 & \text{if } t \ge \frac{10}{3}. \end{cases}$$

In particular, $\operatorname{vol}(-K_X) = \frac{5}{18}$, $T_X(E) = \frac{10}{3}$, $S_X(E) = \frac{5}{2}$, *and* $\beta_X(E) = -\frac{1}{6}\operatorname{vol}(-K_X)$.

Proof. Let $f : Y \to X$ be the plt blow-up of X at p constructed in Construction-Proposition 6.6, with exceptional divisor E. Let $t \ge 0$ be a real number, and let m be a positive integer such that $-mK_X$ is Cartier. Since $-K_X \sim 2H$ (Proposition 6.5), Definition-Lemma 2.16 implies $\mathcal{O}_X(-mK_X) \cong \mathcal{O}_X(2m)$, so $H^0(X, \mathcal{O}_X(-mK_X)) \cong R_{2m}$ by Proposition 6.5, and the divisor of a section corresponds to the Weil divisor of zeros of the corresponding element of R_{2m}. Since $f^*(-mK_X)$ is Cartier and $f_*\mathcal{O}_Y = \mathcal{O}_X$,

$$\lfloor m\big(f^*(-K_X) - tE\big)\rfloor = f^*(-mK_X) - \lceil mt\rceil E,$$

and $H^0\big(Y, \mathcal{O}_Y(f^*(-mK_X) - \lceil mt\rceil E)\big)$ is isomorphic to the subspace

$$V_{m,t} := \{s \in R_{2m} \mid s = 0 \text{ or } \operatorname{mult}_E(s = 0) \ge mt\} \subset R_{2m}.$$

By Construction-Proposition 6.6(3), $V_{m,t}$ has a basis consisting of the monomials $x^{\gamma_1}u^{\gamma_2}v^{\gamma_3}y^j z^i$ with $j \le 4$ of degree $2m$ with $15\gamma_1 + 20\gamma_2 + 20\gamma_3 + 12j \ge 3mt$. For such a monomial, $3\gamma_1 + 4\gamma_2 + 4\gamma_3 = 2m - 3i - 3j$, so

$$15\gamma_1 + 20\gamma_2 + 20\gamma_3 + 12j = 5(2m - 3i - 3j) + 12j = 10m - 15i - 3j,$$

and the condition on the weight is

$$5i + j \le m\Big(\frac{10}{3} - t\Big).$$

For an integer N, let $c(N)$ be the number of $(\gamma_1, \gamma_2, \gamma_3) \in \mathbb{Z}^3_{\ge 0}$ with $3\gamma_1 + 4\gamma_2 + 4\gamma_3 = N$. Then

$$\dim V_{m,t} = \sum_{j=0}^{4} \sum_{i\in\mathbb{Z}_{\ge 0},\ 5i+j\le m(10/3-t)} c(2m - 3i - 3j).$$

We estimate $c(N)$. Let $N \ge 0$, and let $n_0 \in \{0, 1, 2, 3\}$ with $3n_0 \equiv N \pmod 4$. The integers $\gamma_1 \ge 0$ with $3\gamma_1 \le N$ and $4 \mid N - 3\gamma_1$ are $\gamma_1 = n_0 + 4n$ with $0 \le n \le K$, where $A := \frac{N-3n_0}{4}$ and $K := \lfloor A/3\rfloor$ (there is no such γ_1 if $A < 0$). For each such γ_1, there are $\frac{N-3\gamma_1}{4} + 1 = A - 3n + 1$

pairs (γ_2, γ_3). Since $A = \frac{N}{4} + O(1)$ and $K = \frac{N}{12} + O(1)$,

$$\begin{aligned} c(N) &= \sum_{n=0}^{K} (A - 3n + 1) = (K+1)(A+1) - \frac{3}{2}K(K+1) \\ &= \frac{N^2}{48} - \frac{N^2}{96} + O(N+1) = \frac{N^2}{96} + O(N+1). \end{aligned}$$

In the sum for $\dim V_{m,t}$, we have $i \le \frac{2m}{3} - \frac{j}{5}$, so $2m - 3i - 3j \ge -\frac{12j}{5} \ge -10$. Since $c(N) = 0$ for $N < 0$, we obtain $c(2m - 3i - 3j) = \frac{(2m-3i-3j)^2}{96} + O(m+1)$ for any (i, j) in the sum.

If $t > \frac{10}{3}$, then the sum is empty, and $\mathrm{vol}(-K_X - tE) = 0$. Assume that $0 \le t \le \frac{10}{3}$, and let $L := \frac{m}{5}(\frac{10}{3} - t) \le \frac{2m}{3}$. There are at most $5(m+1)$ pairs (i, j) in the sum. Replacing $(2m - 3i - 3j)^2$ with $(2m - 3i)^2$ and the bound $\frac{1}{5}(m(\frac{10}{3} - t) - j)$ for i with L changes the sum by $O(m^2)$. Since $(2m - 3i)^2 \le 4m^2$ is decreasing in $i \in [0, L]$, replacing the sum over i with an integral also changes the sum by $O(m^2)$. Since $2m - 3L = \frac{3mt}{5}$, we obtain

$$\begin{aligned} \dim V_{m,t} &= \frac{5}{96} \int_0^L (2m - 3i)^2 di + O(m^2) = \frac{5}{96} \cdot \frac{(2m)^3 - (\frac{3mt}{5})^3}{9} + O(m^2) \\ &= \frac{5m^3}{108} \Big(1 - \Big(\frac{3t}{10}\Big)^3\Big) + O(m^2). \end{aligned}$$

Thus, by Definition 2.11,

$$\mathrm{vol}(-K_X - tE) = \lim_{m \to +\infty} \frac{\dim V_{m,t}}{m^3/6} = \frac{5}{18}\Big(1 - \Big(\frac{3t}{10}\Big)^3\Big),$$

where m runs over the positive integers such that $-mK_X$ is Cartier.

In particular, $\mathrm{vol}(-K_X) = \frac{5}{18}$ and $T_X(E) = \frac{10}{3}$. Moreover,

$$\int_0^\infty \mathrm{vol}(-K_X - tE)dt = \frac{5}{18} \int_0^{10/3} \Big(1 - \Big(\frac{3t}{10}\Big)^3\Big) dt = \frac{5}{18} \cdot \frac{10}{3} \cdot \frac{3}{4} = \frac{5}{18} \cdot \frac{5}{2},$$

so $S_X(E) = \frac{5}{2}$. Since $A_X(E) = a(E, X, 0) = \frac{7}{3}$ (Definition 2.11 and Construction-Proposition 6.6(1)),

$$\beta_X(E) = \mathrm{vol}(-K_X)\big(A_X(E) - S_X(E)\big) = \mathrm{vol}(-K_X)\Big(\frac{7}{3} - \frac{5}{2}\Big) = -\frac{1}{6}\,\mathrm{vol}(-K_X).$$

□

By Construction-Proposition 6.6(2), the next lemma verifies the hypothesis $\alpha(E, \Delta_E) \ge 1$ of Lemma 3.1 for the plt blow-up of X at p. The curves $(W + U^3 + V^3 = 0)$ and $(W = 0)$ in the support of Δ_E meet at three points, where the multiplicity of Δ_E is $\frac{31}{20} > 1$, so Lemma 2.7 does not suffice. Instead, we pull back to $\mathbb{P}^2$ by a triple cover and use inversion of adjunction along a smooth cubic curve and along a line.

Lemma 6.8. *Let U, V, W be the coordinates of $\mathbb{P}(1,1,3)$ of degrees $1, 1, 3$, and let*

$$\Delta := \frac{3}{4}(W + U^3 + V^3 = 0) + \frac{4}{5}(W = 0).$$

Then $(\mathbb{P}(1,1,3), \Delta)$ is klt,

$$-(K_{\mathbb{P}(1,1,3)} + \Delta) \sim_{\mathbb{Q}} \frac{7}{20}\mathcal{O}_{\mathbb{P}(1,1,3)}(1),$$

and $\alpha(\mathbb{P}(1,1,3), \Delta) \ge 1$.

Proof. Let $C_1 := (W + U^3 + V^3 = 0)$ and $C_2 := (W = 0)$, so that $\Delta = \frac{3}{4}C_1 + \frac{4}{5}C_2$. By Lemma 2.22, applied to $\mathbb{P}(1,1,3)$ with no equations, $K_{\mathbb{P}(1,1,3)} \sim \mathcal{O}_{\mathbb{P}(1,1,3)}(-5)$. Since C_1 and C_2 are the zero divisors of elements of degree 3,

$$-(K_{\mathbb{P}(1,1,3)} + \Delta) \sim_{\mathbb{Q}} \Big(5 - 3 \cdot \frac{3}{4} - 3 \cdot \frac{4}{5}\Big)\mathcal{O}_{\mathbb{P}(1,1,3)}(1) = \frac{7}{20}\mathcal{O}_{\mathbb{P}(1,1,3)}(1).$$

Let $g : \mathbb{P}^2 \to \mathbb{P}(1,1,3)$ be the morphism $[s : t : w] \to [s : t : w^3]$, and let $\ell := (w = 0) \subset \mathbb{P}^2$. Then g is the quotient of $\mathbb{P}^2$ by μ_3 acting on w, and g is étale outside $\ell \cup \{[0 : 0 : 1]\}$. The ramification index of g along ℓ is 3, $g^{-1}(C_2) = \ell$, and $g^*C_1 = C_1'$, where $C_1' := (w^3 + s^3 + t^3 = 0)$ is a smooth cubic curve. Moreover, $C_1' \cap \ell$ consists of the three points $[s : t : 0]$ with $s^3 + t^3 = 0$, and $(C_1' \cdot \ell) = 3$, so C_1' and ℓ intersect transversally.

Let $B \geq 0$ be a $\mathbb{Q}$-divisor on $\mathbb{P}(1,1,3)$ with

$$B \sim_{\mathbb{Q}} \frac{7}{20}\mathcal{O}_{\mathbb{P}(1,1,3)}(1).$$

Then $K_{\mathbb{P}(1,1,3)} + \Delta + B \sim_{\mathbb{Q}} 0$ is $\mathbb{Q}$-Cartier. Let Θ_B be the $\mathbb{Q}$-divisor on $\mathbb{P}^2$ with $K_{\mathbb{P}^2} + \Theta_B = g^*(K_{\mathbb{P}(1,1,3)} + \Delta + B)$. By Lemma 2.8(2),

$$\operatorname{mult}_\ell \Theta_B = 1 - 3\Big(1 - \frac{4}{5} - \operatorname{mult}_{C_2} B\Big) = \frac{2}{5} + 3\operatorname{mult}_{C_2} B, \quad \operatorname{mult}_{C_1'} \Theta_B = \frac{3}{4} + \operatorname{mult}_{C_1} B,$$

and $\operatorname{mult}_{P'} \Theta_B = \operatorname{mult}_{g(P')} B$ for any prime divisor $P' \neq C_1', \ell$ on $\mathbb{P}^2$. Thus $\Theta_B = \frac{3}{4}C_1' + \frac{2}{5}\ell + M_B$ for a $\mathbb{Q}$-divisor $M_B \geq 0$. Since $K_{\mathbb{P}^2} + \Theta_B \sim_{\mathbb{Q}} 0$, we have $\deg \Theta_B = 3$, and so

$$\deg M_B = 3 - 3 \cdot \frac{3}{4} - \frac{2}{5} = \frac{7}{20}.$$

By Lemma 2.8(3), $(\mathbb{P}(1,1,3), \Delta + B)$ is lc if and only if $(\mathbb{P}^2, \Theta_B)$ is lc.

Claim 6.9. *For any $\mathbb{Q}$-divisor $M \geq 0$ on $\mathbb{P}^2$ with $\deg M = \frac{7}{20}$, the pair $(\mathbb{P}^2, \frac{3}{4}C_1' + \frac{2}{5}\ell + M)$ is lc.*

Proof. We first show that $(\mathbb{P}^2, N)$ is lc for any $\mathbb{Q}$-divisor $N \geq 0$ on $\mathbb{P}^2$ with $\deg N \leq 1$. Let $q \in \mathbb{P}^2$ be a closed point and L_q a line through q which is not a component of N. Then

$$\operatorname{mult}_q N \leq (N \cdot L_q)_q \leq (N \cdot L_q) = \deg N \leq 1,$$

where the first inequality holds by [Ful98, Corollary 12.4]. Thus $(\mathbb{P}^2, N)$ is lc by Lemma 2.7.

Next, assume that neither C_1' nor ℓ is a component of M. We show that the three pairs

$$\Big(\mathbb{P}^2, C_1' + \frac{20}{21}M\Big), \quad \Big(\mathbb{P}^2, \ell + \frac{20}{7}M\Big), \quad (\mathbb{P}^2, C_1' + \ell)$$

are lc. Since C_1' is a smooth Cartier divisor on $\mathbb{P}^2$, $\operatorname{Diff}_{C_1'}(\frac{20}{21}M) = \frac{20}{21}M|_{C_1'}$ [Kol13, Proposition 4.5(4)], and

$$\deg\Big(\frac{20}{21}M|_{C_1'}\Big) = \frac{20}{21}(M \cdot C_1') = \frac{20}{21} \cdot \frac{7}{20} \cdot 3 = 1.$$

Thus $\frac{20}{21}M|_{C_1'} \geq 0$ is a $\mathbb{Q}$-divisor of degree 1 on the smooth curve C_1', so each coefficient of $\frac{20}{21}M|_{C_1'}$ is at most 1, and $(\mathbb{P}^2, C_1' + \frac{20}{21}M)$ is lc near C_1' by Theorem 2.6. Outside C_1', the pair $(\mathbb{P}^2, C_1' + \frac{20}{21}M)$ coincides with $(\mathbb{P}^2, \frac{20}{21}M)$, which is lc since $\deg \frac{20}{21}M = \frac{1}{3}$. Since $(\frac{20}{7}M \cdot \ell) = 1$ and $\deg \frac{20}{7}M = 1$, the argument of the previous two sentences with ℓ and $\frac{20}{7}M$ in place of C_1' and $\frac{20}{21}M$ shows that $(\mathbb{P}^2, \ell + \frac{20}{7}M)$ is lc. Since $\operatorname{Diff}_{C_1'}(\ell) = \ell|_{C_1'}$ is the sum of three distinct points with coefficient 1, Theorem 2.6 implies that $(\mathbb{P}^2, C_1' + \ell)$ is lc near C_1', and outside C_1' the pair $(\mathbb{P}^2, C_1' + \ell)$ coincides with $(\mathbb{P}^2, \ell)$, which is lc since $\deg \ell = 1$. Since

$$\frac{3}{5}\Big(C_1' + \frac{20}{21}M\Big) + \frac{1}{4}\Big(\ell + \frac{20}{7}M\Big) + \frac{3}{20}(C_1' + \ell) = \frac{3}{4}C_1' + \frac{2}{5}\ell + \frac{9}{7}M,$$

Lemma 2.4, applied twice, implies that $(\mathbb{P}^2, \frac{3}{4}C_1' + \frac{2}{5}\ell + \frac{9}{7}M)$ is lc. Since $M \geq 0$, $(\mathbb{P}^2, \frac{3}{4}C_1' + \frac{2}{5}\ell + M)$ is lc.

Finally, let M be arbitrary, and write $M = aC_1' + b\ell + M_0$, where $a, b \geq 0$ and neither C_1' nor ℓ is a component of $M_0 \geq 0$. Let $d := \frac{7}{20}$. Then $3a + b + \deg M_0 = d$. If $M_0 \neq 0$, then

$$\frac{3}{4}C_1' + \frac{2}{5}\ell + M = \frac{3a}{d}\Big(\frac{13}{15}C_1' + \frac{2}{5}\ell\Big) + \frac{b}{d}\Big(\frac{3}{4}C_1' + \frac{3}{4}\ell\Big) + \frac{\deg M_0}{d}\Big(\frac{3}{4}C_1' + \frac{2}{5}\ell + \frac{d}{\deg M_0}M_0\Big),$$

and if $M_0 = 0$, then $\frac{3}{4}C_1' + \frac{2}{5}\ell + M$ is the sum of the first two terms on the right-hand side. The first two boundaries on the right-hand side are at most $C_1' + \ell$, so the corresponding pairs are lc, and the third pair is lc by the previous paragraph, since $\deg \frac{d}{\deg M_0}M_0 = d$. Thus $(\mathbb{P}^2, \frac{3}{4}C_1' + \frac{2}{5}\ell + M)$ is lc by Lemma 2.4. □

Proof of Lemma 6.8 continued. By Claim 6.9 applied to $M = M_B$, $(\mathbb{P}(1,1,3), \Delta + B)$ is lc for any $\mathbb{Q}$-divisor $B \geq 0$ with $B \sim_{\mathbb{Q}} -(K_{\mathbb{P}(1,1,3)} + \Delta)$.

It remains to show that $(\mathbb{P}(1,1,3), \Delta)$ is klt. By Lemma 2.8(2),

$$K_{\mathbb{P}^2} + \frac{3}{4}C_1' + \frac{2}{5}\ell = g^*(K_{\mathbb{P}(1,1,3)} + \Delta),$$

so by Lemma 2.8(3) it suffices to show that $(\mathbb{P}^2, \frac{3}{4}C_1' + \frac{2}{5}\ell)$ is klt. Let F be a prime divisor over $\mathbb{P}^2$. If F is a divisor on $\mathbb{P}^2$, then $a(F, \mathbb{P}^2, \frac{3}{4}C_1' + \frac{2}{5}\ell) \geq \frac{1}{4}$. Otherwise, the center of F on $\mathbb{P}^2$ is a closed point q. Let L be a line through q. By Claim 6.9 applied to $M = \frac{7}{20}L$,

$$a\Big(F, \mathbb{P}^2, \frac{3}{4}C_1' + \frac{2}{5}\ell\Big) = a\Big(F, \mathbb{P}^2, \frac{3}{4}C_1' + \frac{2}{5}\ell + \frac{7}{20}L\Big) + \frac{7}{20}\operatorname{mult}_F L \geq \frac{7}{20}\operatorname{mult}_F L > 0.$$

Thus $(\mathbb{P}(1,1,3), \Delta)$ is klt, and $\alpha(\mathbb{P}(1,1,3), \Delta) \geq 1$ by Definition 2.10. □

Corollary 6.10. *Notation as in Set-up 6.2. Then for every $\mathbb{Q}$-divisor $D \geq 0$ on X such that $D \sim_{\mathbb{Q}} -K_X$, the pair $(X, \frac{7}{10}D)$ is lc near p. Moreover, $\operatorname{lct}\big(X; \frac{2}{3}(x = 0)\big) \leq \frac{7}{10}$, where $(x = 0) \sim 3H$ denotes the Weil divisor of zeros of $x \in R_3 = H^0(X, \mathcal{O}_X(3))$, and in particular $\alpha(X) \leq \frac{7}{10}$.*

Proof. Let $f : Y \to X$ be the plt blow-up of X at p constructed in Construction-Proposition 6.6, with exceptional divisor E. By Construction-Proposition 6.6(2) and Lemma 6.8, $\alpha(E, \Delta_E) \geq 1$.

Let $D \geq 0$ be a $\mathbb{Q}$-divisor on X with $D \sim_{\mathbb{Q}} -K_X$. Then D is $\mathbb{Q}$-Cartier, and Lemma 2.13 and Proposition 6.7 imply

$$\operatorname{mult}_E\Big(\frac{7}{10}D\Big) \leq \frac{7}{10}T_X(E) = \frac{7}{10} \cdot \frac{10}{3} = \frac{7}{3} = a(E, X, 0),$$

where the last equality is Construction-Proposition 6.6(1). Since $\operatorname{mult}_E f^*(\frac{7}{10}D) = \operatorname{mult}_E(\frac{7}{10}D)$ (Definition 2.2) and X is klt (Construction-Proposition 6.6), Lemma 3.1 implies that $(X, \frac{7}{10}D)$ is lc near p.

The element $x \in R_3$ is the basis monomial of Construction-Proposition 6.6(3) with $(\gamma_1, \gamma_2, \gamma_3, j, i) = (1, 0, 0, 0, 0)$, so $\operatorname{mult}_E(x = 0) = \frac{15}{3} = 5$. Let $D_0 := \frac{2}{3}(x = 0)$. Then $D_0 \geq 0$, $D_0 \sim_{\mathbb{Q}} 2H \sim -K_X$, and $\operatorname{mult}_E D_0 = \frac{10}{3}$. For any real number $c > \frac{7}{10}$,

$$a(E, X, cD_0) = a(E, X, 0) - c\operatorname{mult}_E D_0 = \frac{7}{3} - \frac{10c}{3} < 0,$$

so (X, cD_0) is not lc. Thus $\operatorname{lct}(X; D_0) \leq \frac{7}{10}$, and $\alpha(X) \leq \operatorname{lct}(X; D_0) \leq \frac{7}{10}$ by Definition 2.10. □

6.2. **The curves of quotient singularities.** In this subsection, we prove that $(X, \frac{7}{10}D)$ is lc near Γ and near Λ for any $\mathbb{Q}$-divisor $D \geq 0$ on X such that $D \sim_{\mathbb{Q}} -K_X$ (Propositions 6.14 and 6.17).

Near Γ we apply inversion of adjunction to the surfaces T_κ of Construction-Proposition 6.13, which contain Γ, and near Λ to the surfaces S_l of Construction-Proposition 6.15, which contain Λ. The differents are $\frac{2}{3}\Gamma$ and $\frac{3}{4}\Lambda$ respectively, which leave only $\frac{1}{3}$ and $\frac{1}{4}$ for the restriction of $\frac{7}{10}D$, so bounds on multiplicities do not suffice (Remark 6.11) and we use Theorem 2.9.

Remark 6.11. Corollary 2.29 alone does not bound the multiplicity of $\frac{7}{10}D$ by 1. Indeed, let $D \geq 0$ be a $\mathbb{Q}$-divisor on X such that $D \sim_{\mathbb{Q}} -K_X$, and let m be a positive integer such that mD is the divisor of zeros of some $\sigma \in R_{2m}$ (Proposition 6.5). Since R is normal (Proposition 6.5), the localizations of R at the homogeneous prime ideals of the components of mD are discrete valuation rings, and since no component of mD is contained in the locus $\Gamma \cup \Lambda$ of points with non-trivial stabilizer, the valuation of σ in each of them is the coefficient of the corresponding component in mD (the coefficients of mD can be computed on the affine cone $\operatorname{Spec} R$ by Definition-Lemma 2.20(1)(2)), so the cycle Z of Lemma 2.27 for the equation of X and σ is the cycle of mD. Thus D has degree

$$\frac{1}{m} \cdot \frac{15 \cdot 2m}{3 \cdot 4 \cdot 4 \cdot 3 \cdot 3} = \frac{5}{72}$$

as a cycle on $\mathbb{P}$. Both bounds of Corollary 2.29 equal $48 \cdot \frac{5}{72} = \frac{10}{3}$, since the three largest weights have product $4 \cdot 4 \cdot 3 = 48$ and the smallest weight 3 is the weight of x, y, and z. Thus Corollary 2.29 only implies $\operatorname{mult}(\frac{7}{10}D) \leq \frac{7}{3}$, and $\frac{7}{3} > 1$.

Remark 6.12. The quintic curve $(zx^4 + 2x^5 + y^5 = 0) \subset \mathbb{P}^2$ is irreducible, since $zx^4 + 2x^5 + y^5$ is linear in z with coprime coefficients x^4 and $2x^5 + y^5$. It is smooth outside $[0:0:1]$, since the partial derivatives $4zx^3 + 10x^4$, $5y^4$, and x^4 vanish simultaneously only at $[0:0:1]$.

Construction-Proposition 6.13. Notation as in Set-up 6.2. Let κ be a complex number such that the line

$$L_\kappa := \big((\kappa^3 + 1)z + \kappa^3 y + x = 0\big) \subset \mathbb{P}^2$$

intersects the curve $(zx^4 + 2x^5 + y^5 = 0) \subset \mathbb{P}^2$ in five distinct points $b_1, \dots, b_5$, where x, y, z are the coordinates of $\mathbb{P}^2$. Let $T_\kappa := (u = \kappa v) \cap X$, and let $g_\kappa : Y_\kappa \to \mathbb{P}^2$ be the blow-up at $b_1, \dots, b_5$ with exceptional curves $F_1, \dots, F_5$. Then the following hold.

(1) T_κ is a prime divisor on X, and there exists an isomorphism from T_κ onto the hypersurface

$$\big(zx^4 + 2x^5 + y^5 + ((\kappa^3 + 1)z + \kappa^3 y + x)w = 0\big) \subset \mathbb{P}(1,1,1,4),$$

where x, y, z, w are the coordinates of degrees $1, 1, 1, 4$, which maps Γ onto $(w = 0)$ and under which the $\mathbb{Q}$-Cartier divisor $3H|_{T_\kappa} := \frac{1}{4}\big(12H|_{T_\kappa}\big)$ is $\mathbb{Q}$-linearly equivalent to the restriction of $\mathcal{O}_{\mathbb{P}(1,1,1,4)}(1)$.

(2) T_κ is normal, and it is smooth outside the point o_κ corresponding to $[0:0:0:1]$, which is a cyclic quotient singularity of type $\frac{1}{4}(1,1)$.

(3) There exists a birational morphism $\sigma_\kappa : Y_\kappa \to T_\kappa$ which contracts the strict transform of L_κ to o_κ and is an isomorphism over $T_\kappa \backslash \{o_\kappa\}$, such that $g_\kappa \circ \sigma_\kappa^{-1}$ is the rational map $T_\kappa \dashrightarrow \mathbb{P}^2$ given by $[x:y:z]$, $\sigma_\kappa^* \Gamma$ is the strict transform of $(zx^4 + 2x^5 + y^5 = 0)$, and

$$\sigma_\kappa^*\big(3H|_{T_\kappa}\big) \sim_{\mathbb{Q}} \frac{5}{4} g_\kappa^* L_\kappa - \frac{1}{4}(F_1 + \cdots + F_5).$$

Proof. Let $l_\kappa := (\kappa^3 + 1)z + \kappa^3 y + x$, $c := zx^4 + 2x^5 + y^5$, and $C := (c = 0) \subset \mathbb{P}^2$. By Remark 6.12, c is irreducible. Since $L_\kappa \cap C$ consists of the five distinct points $b_1, \dots, b_5$, [Har77, Chapter I, Corollary 7.8] implies that $(L_\kappa \cdot C)_{b_i} = 1$ for any i. Thus C is smooth at b_i, and on an affine chart of $\mathbb{P}^2$ containing b_i the dehomogenizations of l_κ and c have linearly independent differentials at b_i.

(1) Substituting $u = \kappa v$, the homogeneous coordinate ring of T_κ is

$$R/(u - \kappa v) \cong \mathbb{C}[x, v, y, z]/\big(c + l_\kappa v^3\big).$$

A monomial $x^i v^j y^k z^n$ has degree $3(i + k + n) + 4j$, which is divisible by 3 if and only if $3 \mid j$, and $c + l_\kappa v^3$ has degree 15. Thus the third Veronese subring of $R/(u - \kappa v)$ is $\mathbb{C}[x, y, z, w]/(c + l_\kappa w)$ with $w := v^3$, and after dividing all degrees by 3, the elements x, y, z, w have degrees $1, 1, 1, 4$. Moreover, if the degree of a monomial $x^i v^j y^k z^n$ is divisible by 12, then $3 \mid j$ and $4 \mid i + k + n$, so the monomial is a power of w times a product of quartic monomials in x, y, z. Thus the twelfth Veronese subring of $R/(u - \kappa v)$, which is the fourth Veronese subring of $\mathbb{C}[x, y, z, w]/(c + l_\kappa w)$,

is generated by its elements of degree 12, that is, by w and the quartic monomials in x, y, z. By [Gro61, Proposition (2.4.7)(i) and (2.5.16)], applied to $R/(u-\kappa v)$ with $d=12$ and to $\mathbb{C}[x,y,z,w]/(c+l_\kappa w)$ (with degrees $1,1,1,4$) with $d=4$, T_κ is isomorphic to the hypersurface $(c+l_\kappa w=0)\subset\mathbb{P}(1,1,1,4)$, and under the isomorphism $\mathcal{O}_{T_\kappa}(12)$ and the restriction of $\mathcal{O}_{\mathbb{P}(1,1,1,4)}(4)$ both correspond to $\mathcal{O}(1)$ on Proj of the twelfth Veronese subring of $R/(u-\kappa v)$. Since $\mathcal{O}_X(12)$ is invertible (Proposition 6.5), $12H|_{T_\kappa}$ is linearly equivalent to the restriction of $\mathcal{O}_{\mathbb{P}(1,1,1,4)}(4)$ under the isomorphism $T_\kappa\cong(c+l_\kappa w=0)$, so $3H|_{T_\kappa}=\frac{1}{4}(12H|_{T_\kappa})$ is $\mathbb{Q}$-linearly equivalent to the restriction of $\mathcal{O}_{\mathbb{P}(1,1,1,4)}(1)$. On T_κ we have $\Gamma=(v=0)\cap T_\kappa$, which is mapped onto $(w=0)$. The polynomial $c+l_\kappa w$ is linear in w with coprime coefficients c and l_κ, since c is irreducible of degree 5. Thus $c+l_\kappa w$ is irreducible, T_κ is integral, and T_κ is a prime divisor on X. In the rest of the proof we identify T_κ with the hypersurface $(c+l_\kappa w=0)$, and $o_\kappa=[0:0:0:1]$.

(2) The open subset $T_\kappa\backslash\{o_\kappa\}$ is covered by the subsets $(x\neq0)$, $(y\neq0)$, and $(z\neq0)$, and each of them is a hypersurface in an affine space $\mathbb{A}^3$, since x, y, and z have degree 1. Let $q\in T_\kappa\backslash\{o_\kappa\}$. If $l_\kappa(q)\neq0$, then near q the surface T_κ is the graph of $w=-c/l_\kappa$, so T_κ is smooth at q. If $l_\kappa(q)=0$, then $c(q)=0$, so $[x:y:z](q)=b_i$ for some i, and the differential of $c+l_\kappa w$ at q is $dc+w(q)dl_\kappa\neq0$, since dc and dl_κ are linearly independent at b_i. Thus T_κ is smooth outside o_κ. Near o_κ, the orbifold chart of $(w\neq0)$ is $\mathbb{A}^3/\mu_4$, where μ_4 acts on x,y,z with weights $1,1,1$, and the preimage of T_κ is $(c+l_\kappa=0)\subset\mathbb{A}^3$. Since c has order 5 and $l_\kappa\neq0$, the preimage is smooth at the origin with tangent plane $(l_\kappa=0)$, and the projection of the preimage to $(l_\kappa=0)$ along a μ_4-stable line is a μ_4-equivariant isomorphism of analytic germs. Since μ_4 acts on $(l_\kappa=0)$ with weights $1,1$, the point o_κ is a cyclic quotient singularity of type $\frac{1}{4}(1,1)$ (Definition 2.18). In particular, T_κ is normal.

(3) Let L' and C' be the strict transforms of L_κ and C on Y_κ. Since C is smooth at the points b_i, [Har77, Proposition V.3.6] implies

$$g_\kappa^*L_\kappa=L'+\sum_{i=1}^5F_i,\quad g_\kappa^*C=C'+\sum_{i=1}^5F_i,\quad\text{and}\quad(L'\cdot C')=5-5=0,$$

so $L'\cap C'=\emptyset$. For any quartic monomial μ in x,y,z, the quintic forms $l_\kappa\mu$ and c define sections of $\mathcal{O}_{Y_\kappa}(5g_\kappa^*L_\kappa-\sum_iF_i)$ with divisors of zeros $L'+g_\kappa^*(\mu=0)$ and C'. Since the quartic monomials have no common zero and $L'\cap C'=\emptyset$, the 16 sections have no common zero, and they define a morphism $\phi:Y_\kappa\to\mathbb{P}^{15}$. On the other hand, a monomial $x^iy^jz^kw^n$ has degree $i+j+k+4n$, so the fourth Veronese subring of $\mathbb{C}[x,y,z,w]/(c+l_\kappa w)$ is generated by the quartic monomials in x,y,z and w, and $\mathcal{O}_{T_\kappa}(4)$ embeds T_κ into $\mathbb{P}^{15}$ by $[\mu:w]$. On $T_\kappa\cap(l_\kappa\neq0)$ we have $w=-c/l_\kappa$, so $[\mu:w]=[l_\kappa\mu:-c]$. Thus, after changing the sign of the last coordinate, ϕ maps $g_\kappa^{-1}(\mathbb{P}^2\backslash L_\kappa)$ isomorphically onto $T_\kappa\cap(l_\kappa\neq0)$, with inverse given by $[x:y:z]$. Hence ϕ defines a birational morphism $\sigma_\kappa:Y_\kappa\to T_\kappa$ such that $g_\kappa\circ\sigma_\kappa^{-1}$ is given by $[x:y:z]$ and

$$\sigma_\kappa^*\mathcal{O}_{T_\kappa}(4)\cong\mathcal{O}_{Y_\kappa}\Big(5g_\kappa^*L_\kappa-\sum_{i=1}^5F_i\Big).$$

Together with (1), the last isomorphism implies the formula for $\sigma_\kappa^*(3H|_{T_\kappa})$ in the statement. The sections $l_\kappa\mu$ vanish on L' and c does not vanish on L', so $\sigma_\kappa(L')=o_\kappa$. On $F_i\backslash L'\cong\mathbb{A}^1$ the function $w=-c/l_\kappa$ restricts to minus the ratio of the differentials of c and l_κ along the tangent directions at b_i, which is a coordinate of $F_i\backslash L'$ since dc and dl_κ are linearly independent at b_i. Thus σ_κ maps $F_i\backslash L'$ bijectively onto $T_\kappa\cap(l_\kappa=0)\cap([x:y:z]=b_i)$, and σ_κ maps $Y_\kappa\backslash L'$ bijectively onto $T_\kappa\backslash\{o_\kappa\}$. Since T_κ is normal, Zariski's main theorem implies that σ_κ is an isomorphism over $T_\kappa\backslash\{o_\kappa\}$. Finally, $o_\kappa\notin\Gamma$, so $\sigma_\kappa^*\Gamma$ is the strict transform of Γ, which is mapped onto C by g_κ. Thus $\sigma_\kappa^*\Gamma=C'$. □

Propositions 6.14 and 6.17 are proved with Theorem 2.9, and both estimates have no room to spare (Remark 6.18).

Proposition 6.14. *Notation as in Set-up 6.2. Then for any $\mathbb{Q}$-divisor $D \geq 0$ on X such that $D \sim_{\mathbb{Q}} -K_X$, the pair $(X, \frac{7}{10}D)$ is lc near any point of Γ.*

Proof. Since $\mathcal{O}_X(12)$ is invertible and $D \sim_{\mathbb{Q}} 2H$ (Proposition 6.5), D is $\mathbb{Q}$-Cartier. By Corollary 6.10, $(X, \frac{7}{10}D)$ is lc near p. Fix a point $q \in \Gamma \backslash \{p\}$. We show that $(X, \frac{7}{10}D)$ is lc near q.

Let $C := (zx^4 + 2x^5 + y^5 = 0) \subset \mathbb{P}^2$ and $q_0 := [x(q) : y(q) : z(q)] \in C$. Since $q \neq p$ and $[0 : 0 : 1]$ is the image of p, Remark 6.12 implies that C is irreducible and smooth at q_0.

We choose κ. For $\kappa \in \mathbb{C}$, the line L_κ of Construction-Proposition 6.13 is $\big((x+z)+\kappa^3(y+z) = 0\big)$, so as κ varies, L_κ runs through all lines through $[1 : 1 : -1]$ except $(y+z = 0)$. Since $zx^4+2x^5+y^5$ takes the value 2 at $(1, 1, -1)$, the point $[1 : 1 : -1]$ is not on C, and the projection from $[1 : 1 : -1]$ restricts to a finite morphism of degree 5 from the integral curve C to $\mathbb{P}^1$. Thus, by [Har77, Proposition IV.2.2(a)] applied to the normalization of C, all but finitely many lines through $[1 : 1 : -1]$ intersect C in five distinct points. Moreover, $T_\kappa \cap \Lambda = (u = \kappa v) \cap \Lambda$ is one point which determines κ, so distinct κ define distinct prime divisors T_κ (Construction-Proposition 6.13(1)), and only finitely many of them are components of D. Thus we may choose κ such that L_κ intersects C in five distinct points $b_1, \dots, b_5$, $q_0 \notin L_\kappa$, and T_κ is not a component of D. We use the notation of Construction-Proposition 6.13 for κ. Since $\Gamma \subset T_\kappa$, we have $q \in T_\kappa$. Since $o_\kappa \in \Lambda$ and $\Gamma \cap \Lambda = \emptyset$, we have $q \neq o_\kappa$, so T_κ is smooth near q and σ_κ is an isomorphism over a neighborhood of q. Since $q_0 \notin L_\kappa$, g_κ is an isomorphism near $\sigma_\kappa^{-1}(q)$, and $g_\kappa(\sigma_\kappa^{-1}(q)) = q_0$.

We compute the different of T_κ near q. Since $q \in \Gamma$, one of x, y, z does not vanish at q. Let $\xi \in \{x, y, z\}$ be such a coordinate, let $\rho_\xi : U_\xi \to (\xi \neq 0)$ be the corresponding orbifold chart, let $X' := \rho_\xi^{-1}(X)$, $T' := \rho_\xi^{-1}(T_\kappa)$, and $h := \rho_\xi|_{X'} : X' \to X \cap (\xi \neq 0)$, and let $q' \in X'$ be a point over q. The morphism h is the quotient by μ_3, which acts on u and v with weight 1 and trivially on the other two of x, y, z. Thus h is unramified outside $X' \cap (u = v = 0)$, which has codimension 2 in X'. Since X is quasismooth outside p, X' is smooth near q'. Near q', the differential of the equation of X' is the differential of $zx^4 + 2x^5 + y^5$ with $\xi = 1$, which is non-zero at q' since C is smooth at q_0. Thus $T' = X' \cap (u = \kappa v)$ is smooth at q', and T' has local coordinates at q' given by v and a coordinate on which μ_3 acts trivially. Hence $h_T := h|_{T'} : T' \to T_\kappa$ is ramified with index 3 along $T' \cap (v = 0) = h_T^{-1}(\Gamma)$ and is unramified elsewhere near q'.

Since $K_X \sim -2H$ and $T_\kappa \sim 4H$, the divisor $K_X + T_\kappa \sim 2H$ is $\mathbb{Q}$-Cartier (Proposition 6.5). By Lemma 2.8(2), $K_{X'} + T' = h^*(K_X + T_\kappa)$. Since $K_X + T_\kappa$ is $\mathbb{Q}$-Cartier, restriction to T' commutes with h^*, so Definition-Proposition 2.5(1) for the smooth pair (X', T') implies $K_{T'} = h_T^*(K_{T_\kappa} + \mathrm{Diff}_{T_\kappa}(0))$ near q'. Let d be the coefficient of Γ in $\mathrm{Diff}_{T_\kappa}(0)$. Lemma 2.8(2) for h_T implies $0 = 1 - 3(1 - d)$, that is, $d = \frac{2}{3}$ (the different is local in codimension two, and at the generic point of Γ, T_κ is the image of a line through a $\frac{1}{3}(1, 1)$ point, so the coefficient is $1 - \frac{1}{3}$ by [Kol13, (3.36.1)]). Near q and outside Γ, both X and T_κ are smooth, so $\mathrm{Diff}_{T_\kappa}(0) = 0$ there by Definition-Proposition 2.5(1). Thus $\mathrm{Diff}_{T_\kappa}(0) = \frac{2}{3}\Gamma$ near q. Since T_κ and Γ are smooth at q, $(T_\kappa, \frac{2}{3}\Gamma)$ is klt near q, and (X, T_κ) is plt near q by Definition-Proposition 2.5(2).

Let $B_\kappa := \frac{7}{10}D|_{T_\kappa}$. Since T_κ is not a component of D, $B_\kappa \geq 0$ is a $\mathbb{Q}$-divisor. Since $K_X + T_\kappa$ and D are $\mathbb{Q}$-Cartier, [Kol13, (4.2.10)] implies that near q,

$$\mathrm{Diff}_{T_\kappa}\left(\frac{7}{10}D\right) = \mathrm{Diff}_{T_\kappa}(0) + B_\kappa = \frac{2}{3}\Gamma + B_\kappa.$$

By Theorem 2.6 on a neighborhood of q, if $(T_\kappa, \frac{2}{3}\Gamma + B_\kappa)$ is lc near q, then $(X, T_\kappa + \frac{7}{10}D)$ is lc near q, and so is $(X, \frac{7}{10}D)$. Thus it suffices to show that $(T_\kappa, \frac{2}{3}\Gamma + B_\kappa)$ is lc near q.

Let s be the coefficient of Γ in B_κ, and let $B'_\kappa := B_\kappa - s\Gamma$. Since

$$B_\kappa \sim_{\mathbb{Q}} \frac{7}{5}H|_{T_\kappa} = \frac{7}{15} \cdot 3H|_{T_\kappa},$$

Construction-Proposition 6.13(3) implies

$$\sigma_\kappa^* B_\kappa \sim_{\mathbb{Q}} \frac{7}{12} g_\kappa^* L_\kappa - \frac{7}{60}\sum_{i=1}^{5} F_i \quad \text{and} \quad \sigma_\kappa^*\Gamma \sim 5 g_\kappa^* L_\kappa - \sum_{i=1}^{5} F_i,$$

where the second relation holds since $\sigma_\kappa^*\Gamma$ is the strict transform of the quintic C, which passes through each b_i with multiplicity 1 [Har77, Proposition V.3.6]. Let $\delta := \frac{7}{15} - 4s$. Then

$$\sigma_\kappa^* B'_\kappa \sim_{\mathbb{Q}} \frac{5\delta}{4} g_\kappa^* L_\kappa - \frac{\delta}{4}\sum_{i=1}^{5} F_i.$$

Let $D' := (g_\kappa)_* \sigma_\kappa^* B'_\kappa$. Then $D' \geq 0$ is a $\mathbb{Q}$-divisor on $\mathbb{P}^2$ of degree

$$(\sigma_\kappa^* B'_\kappa \cdot g_\kappa^* L_\kappa) = \frac{5\delta}{4}.$$

In particular, $\delta \geq 0$, that is, $0 \leq s \leq \frac{7}{60}$. The $\mathbb{Q}$-divisor $\sigma_\kappa^* B'_\kappa - g_\kappa^* D'$ is supported on $F_1 \cup \cdots \cup F_5$ and is numerically equivalent to $-\frac{\delta}{4}\sum_i F_i$. Since $(F_i^2) = -1$ and $(F_i \cdot F_k) = 0$ for $i \neq k$,

$$\sigma_\kappa^* B'_\kappa = g_\kappa^* D' - \frac{\delta}{4}\sum_{i=1}^{5} F_i.$$

Since $\sigma_\kappa^* B'_\kappa \geq 0$, we obtain $\operatorname{mult}_{b_i} D' \geq \frac{\delta}{4}$ for any i. Since Γ is not a component of B'_κ and $\sigma_\kappa^*\Gamma$ is the strict transform of C, the curve C is not a component of D'. Write $D' = \lambda L_\kappa + \Omega$, where $\lambda \geq 0$ and L_κ is not a component of Ω. Then $\Omega \geq 0$, C is not a component of Ω,

$$\deg \Omega = \frac{5\delta}{4} - \lambda, \quad \text{and} \quad \operatorname{mult}_{b_i} \Omega \geq \max\left\{\frac{\delta}{4} - \lambda, 0\right\} \text{ for any } i.$$

Let $m := \operatorname{mult}_{q_0} \Omega$ and $I := (\Omega \cdot C)_{q_0}$. Intersecting Ω with a general line through q_0, [Ful98, Corollary 12.4] implies

$$m \leq \deg \Omega \leq \frac{5\delta}{4} \leq \frac{7}{12}.$$

Since $q_0 \notin L_\kappa$, the point q_0 is different from $b_1, \ldots, b_5$. Since C is smooth at b_i, [Ful98, Corollary 12.4] implies $(\Omega \cdot C)_{b_i} \geq \operatorname{mult}_{b_i} \Omega$. Thus Bézout's theorem [Har77, Chapter I, Corollary 7.8], with the contributions of all five points b_i kept, implies

$$I \leq 5 \deg \Omega - \sum_{i=1}^{5} \operatorname{mult}_{b_i} \Omega \leq 5\left(\frac{5\delta}{4} - \lambda\right) - 5\max\left\{\frac{\delta}{4} - \lambda, 0\right\} \leq 5\delta,$$

where the last inequality is an equality if $\lambda \leq \frac{\delta}{4}$, and otherwise the middle term is less than $\frac{25\delta}{4} - \frac{5\delta}{4} = 5\delta$.

Near $\sigma_\kappa^{-1}(q)$, we have $\sigma_\kappa^* B'_\kappa = g_\kappa^* D' = g_\kappa^*\Omega$ and $\sigma_\kappa^*\Gamma = g_\kappa^* C$. Thus $g_\kappa \circ \sigma_\kappa^{-1}$ is an isomorphism from a neighborhood of q onto a neighborhood of q_0 which maps $\frac{2}{3}\Gamma + B_\kappa = (\frac{2}{3} + s)\Gamma + B'_\kappa$ to $(\frac{2}{3} + s)C + \Omega$. Hence it suffices to show that $(\mathbb{P}^2, (\frac{2}{3} + s)C + \Omega)$ is lc near q_0. Since $s \leq \frac{7}{60}$,

$$\frac{2}{3} + s \leq \frac{2}{3} + \frac{7}{60} = \frac{47}{60} < 1.$$

Suppose that $I \leq 1$. By Definition-Proposition 2.5(1), $\operatorname{Diff}_C(\Omega) = \Omega|_C$ near q_0, and the coefficient of q_0 in $\Omega|_C$ is $I \leq 1$. Thus $(C, \operatorname{Diff}_C(\Omega))$ is lc near q_0, and Theorem 2.6 on a neighborhood of q_0 implies that $(\mathbb{P}^2, C + \Omega)$ is lc near q_0. Hence $(\mathbb{P}^2, (\frac{2}{3} + s)C + \Omega)$ is lc near q_0.

Suppose that $I > 1$. Since C is smooth at q_0, C is not a component of Ω, and $m \leq \frac{7}{12} < 1$, Theorem 2.9 implies that $(\mathbb{P}^2, \Omega + tC)$ is lc near q_0 for any $0 \leq t \leq \min\{1, 1 + \frac{m}{I} - m\}$. For $I > 1$, the expression $1 + \frac{m}{I} - m = 1 - m(1 - \frac{1}{I})$ is non-increasing in $m \geq 0$ and in $I > 1$. Since $m \leq \frac{5\delta}{4}$ and $1 < I \leq 5\delta$,

$$1 + \frac{m}{I} - m \geq 1 + \frac{5\delta/4}{5\delta} - \frac{5\delta}{4} = \frac{5}{4} - \frac{5\delta}{4} = \frac{2}{3} + 5s \geq \frac{2}{3} + s.$$

Since $\frac{2}{3}+s<1$, we may take $t=\frac{2}{3}+s$, and $(\mathbb{P}^2,(\frac{2}{3}+s)C+\Omega)$ is lc near q_0.

In both cases $(T_\kappa,\frac{2}{3}\Gamma+B_\kappa)$ is lc near q, so $(X,\frac{7}{10}D)$ is lc near q. Since $q\in\Gamma\backslash\{p\}$ is arbitrary, Proposition 6.14 follows. □

Construction-Proposition 6.15. Notation as in Set-up 6.2. Let $l\in\mathbb{C}[x,y,z]$ be a linear form such that the line $(l=0)\subset\mathbb{P}^2$ does not pass through $[1:1:-1]$ and intersects the curve $(zx^4+2x^5+y^5=0)\subset\mathbb{P}^2$ in five distinct points outside $(z+y=0)\cup(z+x=0)$, where x,y,z are the coordinates of $\mathbb{P}^2$. Let $S_l:=(l=0)\cap X$, let a and b be the restrictions of $z+y$ and $z+x$ to the line $(l=0)$, let $f_5\in\mathbb{C}[a,b]$ be the binary quintic form given by the restriction of $zx^4+2x^5+y^5$ to $(l=0)$, and let $(a_1:b_1),\dots,(a_5:b_5)$ be the values of $(a:b)$ at the five points of $(l=0)\cap(zx^4+2x^5+y^5=0)$.

Let $g_l:Y_l\to\mathbb{P}^1\times\mathbb{P}^1$ be the blow-up at the intersection points of the curves $(f_5(a,b)=0)$ and $(au^3+bv^3=0)$, where the first factor of $\mathbb{P}^1\times\mathbb{P}^1$ has coordinates a,b and the second factor has coordinates u,v. For $1\le i\le 5$, let F_{i1},F_{i2},F_{i3} be the exceptional curves over the intersection points of $\{(a_i:b_i)\}\times\mathbb{P}^1$ and $(au^3+bv^3=0)$, and let C_i be the strict transform of $\{(a_i:b_i)\}\times\mathbb{P}^1$. Let M_1 and M_2 be the pullbacks to Y_l of the class of a point under the projections to the first factor and to the second factor respectively, and let Λ_l be the strict transform of $(au^3+bv^3=0)$. Then the following hold.

(1) The form f_5 has the five distinct roots $(a_1:b_1),\dots,(a_5:b_5)$, none of which is a root of ab, the curves $(f_5(a,b)=0)$ and $(au^3+bv^3=0)$ intersect in fifteen points, and S_l is isomorphic to the hypersurface
$$(f_5(a,b)+au^3+bv^3=0)\subset\mathbb{P}(3,3,4,4),$$
where a,b,u,v are the coordinates of degrees $3,3,4,4$.

(2) S_l is normal, contains Λ, and is smooth along Λ.

(3) $(C_i^2)=-3$, $(\Lambda_l^2)=-9$, and $(C_i\cdot\Lambda_l)=0$ for any i.

(4) There exists a birational morphism $\sigma_l:Y_l\to S_l$ which contracts exactly $C_1,\dots,C_5$, each to a point of $S_l\cap\Gamma$, and is an isomorphism elsewhere, such that
$$12\sigma_l^*\big(H|_{S_l}\big)\sim 5M_1+3M_2-\sum_{i,j}F_{ij}.$$
Moreover, Λ_l is a smooth rational curve, and $\sigma_l|_{\Lambda_l}:\Lambda_l\to\Lambda$ is an isomorphism under which the second projection $\Lambda_l\to\mathbb{P}^1$ corresponds to $[u:v]$ on Λ.

Proof. (1) The point $[1:1:-1]$ is the common zero of $z+y$ and $z+x$ in $\mathbb{P}^2$. Since l does not vanish at $[1:1:-1]$, the linear forms l, $z+y$, and $z+x$ are linearly independent. Thus a and b are coordinates on the line $(l=0)$, and changing the coordinates x,y,z of degree 3 to $l,z+y,z+x$ implies
$$R/(l)\cong\mathbb{C}[a,b,u,v]/\big(f_5(a,b)+au^3+bv^3\big),$$
with a,b,u,v of degrees $3,3,4,4$. Hence S_l is isomorphic to the hypersurface in (1). The curve $(zx^4+2x^5+y^5=0)$ is irreducible of degree 5 (Remark 6.12), so $f_5\neq 0$, and the roots of f_5 are the five distinct points $(a_1:b_1),\dots,(a_5:b_5)$. None of them is on $(z+y=0)\cup(z+x=0)$, that is, none of them is a root of ab. For a root $(a_i:b_i)$ of f_5, we have $a_ib_i\neq 0$, so $a_iu^3+b_iv^3$ has three distinct roots $[u:v]$. Thus the curves $(f_5(a,b)=0)$ and $(au^3+bv^3=0)$ intersect in fifteen points.

(2) Since l vanishes on $(x=y=z=0)$, S_l contains Λ. A direct calculation of the partial derivatives $3au^2$, $3bv^2$, $\partial_af_5+u^3$, and $\partial_bf_5+v^3$, using that f_5 has five distinct roots and none of them is a root of ab, shows that the affine hypersurface $(f_5(a,b)+au^3+bv^3=0)\subset\mathbb{A}^4$ is smooth outside the origin. We omit the case analysis according to which of u,v vanish. Thus the ring $\mathbb{C}[a,b,u,v]/(f_5(a,b)+au^3+bv^3)$ is a complete intersection which is regular in codimension one, hence normal by Serre's criterion [Mat86, Theorem 23.8]. It is graded with degree 0 part $\mathbb{C}$, so it is a normal domain, and S_l is normal. Let $q\in\Lambda$, and assume that $v(q)\neq 0$ (the case $u(q)\neq 0$ is symmetric). The orbifold chart of $(v\neq 0)$ is $\mathbb{A}^3/\mu_4$ with coordinates a,b,u of weights $3,3,0$, and

the preimage of S_l is $S' := (f_5(a,b) + au^3 + b = 0) \subset \mathbb{A}^3$. The derivative of $f_5(a,b) + au^3 + b$ in b is 1 at $a = b = 0$, so S' is smooth along $(a = b = 0)$ with local coordinates a, u. Since μ_4 acts by $a \to \zeta^3 a$ and $u \to u$, the quotient is smooth with local coordinates a^4, u. Thus S_l is smooth along Λ.

(3) Each curve in $(f_5(a,b) = 0)$ is a fiber of the first projection, of class M_1 after pullback, and $(au^3 + bv^3 = 0)$ has bidegree $(1,3)$. The curve $(au^3 + bv^3 = 0)$ is the graph of the morphism $\mathbb{P}^1 \to \mathbb{P}^1$, $[u : v] \to [v^3 : -u^3]$, so it is a smooth rational curve, and each curve in $(f_5(a,b) = 0) \cup (au^3 + bv^3 = 0)$ is smooth at the fifteen points. By [Har77, Propositions V.3.2 and V.3.6],

$$C_i \sim M_1 - \sum_{j=1}^{3} F_{ij}, \quad \Lambda_l \sim M_1 + 3M_2 - \sum_{i,j} F_{ij},$$

where $(M_1^2) = (M_2^2) = 0$, $(M_1 \cdot M_2) = 1$, $(F_{ij}^2) = -1$, and F_{ij} is orthogonal to M_1, M_2, and $F_{i'j'}$ for $(i', j') \neq (i, j)$. Thus

$$(C_i^2) = -3, \quad (\Lambda_l^2) = 6 - 15 = -9, \quad (C_i \cdot \Lambda_l) = 3 - 3 = 0.$$

(4) Let μ run through the five quartic monomials in a, b and ν run through the four cubic monomials in u, v. The forms $(au^3 + bv^3)\mu$ and $-f_5\nu$ have bidegree $(5,3)$ and vanish at the fifteen blown-up points with multiplicity one. They define sections of $\mathcal{O}_{Y_l}(5M_1 + 3M_2 - \sum_{i,j} F_{ij})$ with divisors of zeros $\Lambda_l + g_l^*(\mu = 0)$ and $C_1 + \cdots + C_5 + g_l^*(\nu = 0)$. Since the quartic monomials in a, b have no common zero, the cubic monomials in u, v have no common zero, and $\Lambda_l \cap C_i = \emptyset$ by (3), the nine sections have no common zero and define a morphism $\phi : Y_l \to \mathbb{P}^8$. Since $3i + 4j$ divisible by 12 forces $4 \mid i$ and $3 \mid j$, the twelfth Veronese subring of $\mathbb{C}[a, b, u, v]/(f_5(a,b) + au^3 + bv^3)$ is generated by the degree 12 part, which is spanned by the monomials μ and ν. Thus $\mathcal{O}_{S_l}(12)$ embeds S_l into $\mathbb{P}^8$ by $[\mu : \nu]$. For $(a_0 : b_0)$ and $(u_0 : v_0)$ such that $f_5(a_0, b_0) \neq 0$ and $a_0u_0^3 + b_0v_0^3 \neq 0$, the points of S_l of the form $(\alpha a_0, \alpha b_0, \beta u_0, \beta v_0)$ with $\alpha\beta \neq 0$ are the points with

$$\frac{\alpha^4}{\beta^3} = -\frac{a_0u_0^3 + b_0v_0^3}{f_5(a_0, b_0)},$$

and such a point has coordinates

$$[\alpha^4\mu(a_0, b_0) : \beta^3\nu(u_0, v_0)] = [(a_0u_0^3 + b_0v_0^3)\mu(a_0, b_0) : -f_5(a_0, b_0)\nu(u_0, v_0)].$$

Thus ϕ maps a dense open subset of Y_l isomorphically onto a dense open subset of S_l, and ϕ defines a birational morphism $\sigma_l : Y_l \to S_l$ with

$$\sigma_l^*\mathcal{O}_{S_l}(12) \cong \mathcal{O}_{Y_l}\Big(5M_1 + 3M_2 - \sum_{i,j} F_{ij}\Big).$$

Since $\mathcal{O}_X(12)$ is invertible (Proposition 6.5) and $\mathcal{O}_{S_l}(12) = \mathcal{O}_X(12)|_{S_l}$, the formula for $12\sigma_l^*(H|_{S_l})$ follows.

On C_i, the sections $-f_5\nu$ vanish, so $\sigma_l(C_i)$ is the point $[\mu(a_i, b_i) : 0]$, which is the point of S_l with $u = v = 0$ over the root $(a_i : b_i)$ of f_5, and it is in Γ. On Λ_l, the sections $(au^3 + bv^3)\mu$ vanish, so σ_l maps a point of Λ_l with second projection $[u_0 : v_0]$ to the point $[0 : \nu(u_0, v_0)]$, which is the point of Λ with $[u : v] = [u_0 : v_0]$. Since Λ_l is the graph of a morphism from the second factor, the second projection is an isomorphism $\Lambda_l \to \mathbb{P}^1$, so $\sigma_l|_{\Lambda_l} : \Lambda_l \to \Lambda$ is an isomorphism under which the second projection corresponds to $[u : v]$. On F_{ij} minus $C_i \cup \Lambda_l$, the ratio α^4/β^3 above restricts to a coordinate, since the fiber and $(au^3 + bv^3 = 0)$ meet transversally at the fifteen points by (3). Thus σ_l maps $F_{ij}\backslash(C_i \cup \Lambda_l)$ bijectively onto the points $(\alpha a_i, \alpha b_i, \beta u_{ij}, \beta v_{ij})$ of S_l with $\alpha\beta \neq 0$, where $((a_i : b_i), (u_{ij} : v_{ij})) = g_l(F_{ij})$. A point of $S_l\backslash(\Gamma \cup \Lambda)$ determines both $(a : b)$ and $(u : v)$. Thus σ_l maps $Y_l\backslash(C_1 \cup \cdots \cup C_5)$ bijectively onto the complement in S_l of the five points $\sigma_l(C_i)$. Since S_l is normal, Zariski's main theorem implies that σ_l is an isomorphism there. Hence σ_l contracts exactly $C_1, \ldots, C_5$, each to a point of $S_l \cap \Gamma$, and is an isomorphism elsewhere. □

Lemma 6.16. *Notation as in Set-up 6.2. Then the linear forms $l \in \mathbb{C}[x,y,z]$ satisfying the hypotheses of Construction-Proposition 6.15 form a non-empty Zariski open subset of the space $\mathbb{C}[x,y,z]_1$ of linear forms. Moreover, for any two such linear forms l and l', we have $S_l = S_{l'}$ if and only if l and l' are proportional.*

Proof. Let $C := (zx^4 + 2x^5 + y^5 = 0) \subset \mathbb{P}^2$. By Remark 6.12, C is an integral quintic curve which is smooth outside $[0:0:1]$. For a non-zero linear form l, the hypotheses of Construction-Proposition 6.15 are open conditions on the coefficients of l, since they say that $l([1:1:-1]) \neq 0$, that the restriction of $zx^4 + 2x^5 + y^5$ to $(l=0)$ has non-zero discriminant, and that it has no common root with the restriction of $(z+y)(z+x)$. By Bertini's theorem [Har77, Chapter III, Corollary 10.9 and Remark 10.9.2], applied to $C\backslash\{[0:0:1]\}$, a general line intersects $C\backslash\{[0:0:1]\}$ transversally, and a general line does not pass through $[0:0:1]$. Thus Bézout's theorem [Har77, Chapter I, Corollary 7.8] implies that a general line intersects C in five distinct points. The lines $(z+y=0)$ and $(z+x=0)$ are not components of C, so $C \cap \big((z+y)(z+x)=0\big)$ is finite, and a general line avoids it and the point $[1:1:-1]$. Thus a general linear form satisfies the hypotheses, and the open subset is non-empty.

Let l satisfy the hypotheses. We show that the image of $S_l\backslash\Lambda$ under $[x:y:z]$ is the line $(l=0)$. Let $[x_0:y_0:z_0]$ be a point of $(l=0)$. Since $(l=0)$ does not pass through $[1:1:-1]$, we have $(z_0+y_0, z_0+x_0) \neq (0,0)$, so the equation

$$(z_0+y_0)u^3 + (z_0+x_0)v^3 = -(z_0x_0^4 + 2x_0^5 + y_0^5)$$

has a solution $(u_1,v_1) \in \mathbb{C}^2$. The point of $\mathbb{P}$ with coordinates (x_0,u_1,v_1,y_0,z_0) is in $S_l\backslash\Lambda$, and it is mapped to $[x_0:y_0:z_0]$. Thus S_l determines the line $(l=0)$. Hence, for l and l' satisfying the hypotheses, $S_l = S_{l'}$ implies that $(l=0) = (l'=0)$, that is, l and l' are proportional. The converse is clear. □

Proposition 6.17. *Notation as in Set-up 6.2. Then for any $\mathbb{Q}$-divisor $D \geq 0$ on X such that $D \sim_{\mathbb{Q}} -K_X$, the pair $(X, \frac{7}{10}D)$ is lc near any point of Λ.*

Proof. Since $\mathcal{O}_X(12)$ is invertible and $D \sim_{\mathbb{Q}} 2H$ (Proposition 6.5), D is $\mathbb{Q}$-Cartier. Fix a point $q \in \Lambda$ and let $[u_0:v_0] := [u(q):v(q)]$. We show that $(X, \frac{7}{10}D)$ is lc near q.

Let $C := (zx^4 + 2x^5 + y^5 = 0) \subset \mathbb{P}^2$, which is an irreducible quintic curve by Remark 6.12. We choose a linear form $l \in \mathbb{C}[x,y,z]$ such that the following hold.

(1) The line $(l=0)$ does not pass through $[1:1:-1]$ and intersects C in five distinct points outside $(z+y=0) \cup (z+x=0)$.
(2) The line $(l=0)$ does not pass through any point of $C \cap \big((z+y)u_0^3 + (z+x)v_0^3 = 0\big)$.
(3) $S_l := (l=0) \cap X$ is not a component of D.

By Lemma 6.16, a general l satisfies (1). The line $\big((z+y)u_0^3 + (z+x)v_0^3 = 0\big)$ is not a component of C, so it intersects C in a finite set, which a general line avoids. Thus a general l satisfies (2). By Lemma 6.16, non-proportional general l define distinct surfaces S_l, and only finitely many of them are components of D, so a general l also satisfies (3).

We use the notation of Construction-Proposition 6.15 for l. Let $q_Y \in \Lambda_l$ be the point with $\sigma_l(q_Y) = q$. By Construction-Proposition 6.15(4), the second projection maps q_Y to $[u_0:v_0]$. A point of $\Lambda_l \cap F_{ij}$ is mapped by g_l to a point $((a':b'),(u':v'))$ with $f_5(a',b') = 0$ and $a'u'^3 + b'v'^3 = 0$, and $(a':b')$ is the value of $(z+y : z+x)$ at a point of $(l=0) \cap C$. By (2), $a'u_0^3 + b'v_0^3 \neq 0$ for any root $(a':b')$ of f_5. Thus $q_Y \notin F_{ij}$ for any i,j. Moreover, $C_i \cap \Lambda_l = \emptyset$ by Construction-Proposition 6.15(3). Hence g_l is an isomorphism near q_Y, and σ_l is an isomorphism near q_Y by Construction-Proposition 6.15(4).

We compute the different of S_l near q. Let $\xi \in \{u,v\}$ be a coordinate which does not vanish at q, let $\rho_\xi : U_\xi \to (\xi \neq 0)$ be the corresponding orbifold chart, let $X' := \rho_\xi^{-1}(X)$, $S' := \rho_\xi^{-1}(S_l)$, and $h := \rho_\xi|_{X'} : X' \to X \cap (\xi \neq 0)$, and let $q' \in X'$ be a point over q. The morphism h is the quotient by μ_4, which acts on x,y,z with weight 3 and trivially on the other coordinate of u,v. Thus h is unramified outside $X' \cap (x=y=z=0)$, which has codimension 2 in X'. Since X

is quasismooth outside p, X' is smooth near q'. At q', the differential of the equation of X' is $u_1^3 d(z+y) + v_1^3 d(z+x)$, where u_1 and v_1 are the values of u and v at q'. The linear form $u_1^3(z+y) + v_1^3(z+x)$ is not proportional to l, since the line $(u_1^3(z+y) + v_1^3(z+x) = 0)$ passes through $[1:1:-1]$ and $(l=0)$ does not. Thus $S' = X' \cap (l=0)$ is smooth at q', and S' has local coordinates at q' given by a linear form in x, y, z, on which μ_4 acts with weight 3, and a coordinate on which μ_4 acts trivially. Hence $h_S := h|_{S'} : S' \to S_l$ is ramified with index 4 along $h_S^{-1}(\Lambda)$ and is unramified elsewhere near q'.

Since $K_X \sim -2H$ and $S_l \sim 3H$, the divisor $K_X + S_l \sim H$ is $\mathbb{Q}$-Cartier (Proposition 6.5). By Lemma 2.8(2), $K_{X'} + S' = h^*(K_X + S_l)$. Since $K_X + S_l$ is $\mathbb{Q}$-Cartier, restriction to S' commutes with h^*, so Definition-Proposition 2.5(1) for the smooth pair (X', S') implies $K_{S'} = h_S^*(K_{S_l} + \mathrm{Diff}_{S_l}(0))$ near q'. Let d be the coefficient of Λ in $\mathrm{Diff}_{S_l}(0)$. Lemma 2.8(2) for h_S implies $0 = 1 - 4(1-d)$, that is, $d = \frac{3}{4}$ ((X, S_l) is plt near q since (X', S') is smooth [KM98, Proposition 5.20(2) and Definition 2.34], and along the generic point of Λ, X is locally $(\mathbb{C}^2/\mu_4) \times \mathbb{C}$ and S_l is the image of an invariant line, so the coefficient is $1 - \frac{1}{4}$ by [Kol+92, (16.6.3)]). Near q and outside Λ, both X and S_l are smooth, so $\mathrm{Diff}_{S_l}(0) = 0$ there by Definition-Proposition 2.5(1). Thus $\mathrm{Diff}_{S_l}(0) = \frac{3}{4}\Lambda$ near q. Since S_l is smooth along Λ (Construction-Proposition 6.15(2)), $(S_l, \frac{3}{4}\Lambda)$ is klt near q, and (X, S_l) is plt near q by Definition-Proposition 2.5(2).

Let $B_l := \frac{7}{10} D|_{S_l}$. Since S_l is not a component of D, $B_l \geq 0$ is a $\mathbb{Q}$-divisor. Since $K_X + S_l$ and D are $\mathbb{Q}$-Cartier, [Kol13, (4.2.10)] implies that near q,

$$\mathrm{Diff}_{S_l}\left(\frac{7}{10} D\right) = \mathrm{Diff}_{S_l}(0) + B_l = \frac{3}{4}\Lambda + B_l.$$

By Theorem 2.6 on a neighborhood of q, it suffices to show that $(S_l, \frac{3}{4}\Lambda + B_l)$ is lc near q. Since σ_l is an isomorphism near q_Y and $\sigma_l^*\Lambda = \Lambda_l$ near q_Y, it suffices to show that $(Y_l, \frac{3}{4}\Lambda_l + \sigma_l^* B_l)$ is lc near q_Y.

Since $B_l \sim_{\mathbb{Q}} \frac{7}{5} H|_{S_l}$, Construction-Proposition 6.15(4) implies

$$\sigma_l^* B_l \sim_{\mathbb{Q}} \frac{7}{60}\Big(5M_1 + 3M_2 - \sum_{i,j} F_{ij}\Big).$$

Let s be the coefficient of Λ_l in $\sigma_l^* B_l$, and let $\Omega_Y := \sigma_l^* B_l - s\Lambda_l$. Then $\Omega_Y \geq 0$ and Λ_l is not a component of Ω_Y. Since $(au^3 + bv^3 = 0)$ has bidegree $(1,3)$ and passes through each of the fifteen blown-up points with multiplicity one, $(M_1 \cdot \Lambda_l) = 3$, $(M_2 \cdot \Lambda_l) = 1$, and $(F_{ij} \cdot \Lambda_l) = 1$, and $(\Lambda_l^2) = -9$ by Construction-Proposition 6.15(3). Since M_1 is nef and $\Omega_Y \geq 0$,

$$0 \leq (\Omega_Y \cdot M_1) = \frac{7}{60} \cdot 3 - 3s = \frac{7}{20} - 3s,$$

so $0 \leq s \leq \frac{7}{60}$. Moreover,

$$(\Omega_Y \cdot \Lambda_l) = \frac{7}{60}(15 + 3 - 15) + 9s = \frac{7}{20} + 9s.$$

Let $m := \mathrm{mult}_{q_Y} \Omega_Y$ and $I := (\Omega_Y \cdot \Lambda_l)_{q_Y}$. Since Λ_l is not a component of Ω_Y, $I \leq \frac{7}{20} + 9s$. Let $Q \subset \mathbb{P}^1 \times \mathbb{P}^1$ be a general curve of bidegree $(1,1)$ through $g_l(q_Y)$. The base locus of such curves is $\{g_l(q_Y)\}$, so Q does not pass through the fifteen blown-up points, $g_l^* Q \sim M_1 + M_2$, and $g_l^* Q$ contains no component of Ω_Y. By [Ful98, Corollary 12.4],

$$m \leq (\Omega_Y \cdot g_l^* Q)_{q_Y} \leq (\Omega_Y \cdot (M_1 + M_2)) = \frac{7}{60}(3+5) - 4s = \frac{14}{15} - 4s \leq \frac{14}{15} < 1.$$

Since $s \leq \frac{7}{60}$,

$$\frac{3}{4} + s \leq \frac{3}{4} + \frac{7}{60} = \frac{13}{15} < 1.$$

Suppose that $I \leq 1$. By Definition-Proposition 2.5(1), $\mathrm{Diff}_{\Lambda_l}(\Omega_Y) = \Omega_Y|_{\Lambda_l}$ near q_Y, and the coefficient of q_Y in $\Omega_Y|_{\Lambda_l}$ is $I \leq 1$. Thus Theorem 2.6 on a neighborhood of q_Y implies that $(Y_l, \Lambda_l + \Omega_Y)$ is lc near q_Y. Hence $(Y_l, (\frac{3}{4} + s)\Lambda_l + \Omega_Y)$ is lc near q_Y.

Suppose that $I > 1$. Since Λ_l is smooth, Λ_l is not a component of Ω_Y, and $m < 1$, Theorem 2.9 implies that $(Y_l, \Omega_Y + t\Lambda_l)$ is lc near q_Y for any $0 \le t \le \min\{1, 1 + \frac{m}{I} - m\}$. For $I > 1$, the expression $1 + \frac{m}{I} - m = 1 - m(1 - \frac{1}{I})$ is non-increasing in $m \ge 0$ and in $I > 1$. Since $0 \le m \le \frac{14}{15} - 4s$ and $1 < I \le \frac{7}{20} + 9s$, we obtain

$$1 + \frac{m}{I} - m \ge 1 - \Big(\frac{14}{15} - 4s\Big) + \frac{\frac{14}{15} - 4s}{\frac{7}{20} + 9s}.$$

The right-hand side is at least $\frac{3}{4} + s$ if and only if

$$\Big(\frac{1}{4} - s\Big)\Big(\frac{7}{20} + 9s\Big) - \Big(\frac{14}{15} - 4s\Big)\Big(9s - \frac{13}{20}\Big) \ge 0.$$

A direct calculation shows

$$\Big(\frac{1}{4} - s\Big)\Big(\frac{7}{20} + 9s\Big) - \Big(\frac{14}{15} - 4s\Big)\Big(9s - \frac{13}{20}\Big) = 27\Big(s - \frac{7}{60}\Big)\Big(s - \frac{119}{540}\Big),$$

which is non-negative since $0 \le s \le \frac{7}{60} < \frac{119}{540}$. Since $\frac{3}{4} + s < 1$, we may take $t = \frac{3}{4} + s$, and $(Y_l, (\frac{3}{4} + s)\Lambda_l + \Omega_Y)$ is lc near q_Y.

In both cases, the pair

$$(Y_l, \tfrac{3}{4}\Lambda_l + \sigma_l^* B_l) = (Y_l, (\tfrac{3}{4} + s)\Lambda_l + \Omega_Y)$$

is lc near q_Y. Thus $(S_l, \frac{3}{4}\Lambda + B_l)$ is lc near q, Theorem 2.6 implies that $(X, S_l + \frac{7}{10}D)$ is lc near q, and so is $(X, \frac{7}{10}D)$. Since $q \in \Lambda$ is arbitrary, Proposition 6.17 follows. □

Remark 6.18. The estimates in the proofs of Propositions 6.14 and 6.17 have no room to spare. In the proof of Proposition 6.14, for $s = 0$ the lower bound $\frac{2}{3} + 5s$ for $1 + \frac{m}{I} - m$ equals $\frac{2}{3}$, the coefficient of Γ in $\mathrm{Diff}_{T_\kappa}(0)$. If one ignores the points $b_1, \dots, b_5$ and uses only $I \le 5 \deg \Omega \le \frac{25\delta}{4}$, then for $s = 0$ one only obtains $1 + \frac{m}{I} - m \ge \frac{6}{5} - \frac{7}{12} = \frac{37}{60} < \frac{2}{3}$. In the proof of Proposition 6.17, the polynomial $27(s - \frac{7}{60})(s - \frac{119}{540})$ vanishes at the largest possible value $s = \frac{7}{60}$.

6.3. **The global threshold.** In this subsection, we prove Theorem 6.4. For any $\mathbb{Q}$-divisor $D \ge 0$ on X such that $D \sim_{\mathbb{Q}} -K_X$, Lemma 6.19 below, Proposition 6.14, and Proposition 6.17 restrict the points near which $(X, \frac{7}{10}D)$ is not lc to a finite set of smooth points of X. We exclude them by Kawamata–Viehweg vanishing applied to $H - (K_X + c_0 D)$, where $c_0 = \mathrm{lct}(X; D)$, using $H^0(X, \mathcal{O}_X(1)) = 0$ (cf. [Zhu20, Theorem 1.6]), which gives $\mathrm{lct}(X; D) \ge \frac{7}{10}$.

Lemma 6.19. *Notation as in Set-up 6.2, and let $D \ge 0$ be a $\mathbb{Q}$-divisor on X such that $D \sim_{\mathbb{Q}} -K_X$. Then for a general linear form $l \in \mathbb{C}[x, y, z]$, the pair $(X, \frac{7}{10}D)$ is lc near $(l = 0) \cap X$.*

Proof. By Lemma 6.16, a general l satisfies the hypotheses of Construction-Proposition 6.15, and we use the notation there. By Lemma 6.16, non-proportional general l define distinct surfaces S_l. Since D has finitely many components, we may also assume that S_l is not a component of D and that $p \notin S_l$.

Since X is a $\mathbb{Q}$-Fano variety, D is $\mathbb{Q}$-Cartier. Let $B_l := \frac{7}{10}D|_{S_l}$. Then $B_l \ge 0$ is a $\mathbb{Q}$-Cartier $\mathbb{Q}$-divisor on S_l since S_l is not a component of D. Since $K_X \sim -2H$ (Proposition 6.5), Construction-Proposition 6.15(4) implies

$$\sigma_l^* B_l \sim_{\mathbb{Q}} \frac{7}{5}\sigma_l^*\big(H|_{S_l}\big) \sim_{\mathbb{Q}} \frac{7}{60}\Big(5M_1 + 3M_2 - \sum_{i,j} F_{ij}\Big).$$

Let $B_0 := g_{l*}\sigma_l^* B_l$. Then $B_0 \ge 0$ is a $\mathbb{Q}$-divisor on $\mathbb{P}^1 \times \mathbb{P}^1$ such that $B_0 \sim_{\mathbb{Q}} \frac{7}{60}(5g_{l*}M_1 + 3g_{l*}M_2)$. The $\mathbb{Q}$-divisor $\sigma_l^* B_l - g_l^* B_0$ is supported on $\bigcup_{i,j} F_{ij}$. Since the curves F_{ij} are pairwise disjoint, $(F_{ij}^2) = -1$, $(\sigma_l^* B_l \cdot F_{ij}) = \frac{7}{60}$, and $(g_l^* B_0 \cdot F_{ij}) = 0$, we obtain

$$\sigma_l^* B_l = g_l^* B_0 - \frac{7}{60}\sum_{i,j} F_{ij}.$$

Let $m_{ij} := \operatorname{mult}_{g_l(F_{ij})} B_0$. Then $g_l^* B_0 = g_{l*}^{-1} B_0 + \sum_{i,j} m_{ij} F_{ij}$, so

$$\sigma_l^* B_l = g_{l*}^{-1} B_0 + \sum_{i,j} \Big(m_{ij} - \frac{7}{60} \Big) F_{ij}, \tag{6.1}$$

and $m_{ij} \geq \frac{7}{60}$ since $\sigma_l^* B_l \geq 0$.

We bound the multiplicities of B_0. Let $q \in \mathbb{P}^1 \times \mathbb{P}^1$ be a closed point. A general curve Q of bidegree $(1,1)$ through q is smooth and contains no component of B_0, and

$$(B_0 \cdot Q) = \frac{7}{60} \big(5(g_{l*} M_1 \cdot g_{l*} M_2) + 3(g_{l*} M_2 \cdot g_{l*} M_1) \big) = \frac{14}{15}.$$

By [Ful98, Corollary 12.4],

$$\operatorname{mult}_q B_0 = \operatorname{mult}_q B_0 \cdot \operatorname{mult}_q Q \leq (B_0 \cdot Q) = \frac{14}{15}.$$

In particular, $0 \leq m_{ij} - \frac{7}{60} \leq \frac{49}{60}$ for any i, j.

We show that $(Y_l, \sigma_l^* B_l)$ is lc. The morphism g_l is an isomorphism over

$$(\mathbb{P}^1 \times \mathbb{P}^1) \backslash \bigcup_{i,j} g_l(F_{ij}),$$

and $\sigma_l^* B_l = g_l^* B_0$ there. Thus $(Y_l, \sigma_l^* B_l)$ is lc outside $\bigcup_{i,j} F_{ij}$ by Lemma 2.7. Fix i, j. Then

$$(g_{l*}^{-1} B_0 \cdot F_{ij}) = m_{ij} \leq \frac{14}{15}.$$

Since Y_l and $F_{ij} \cong \mathbb{P}^1$ are smooth, by the adjunction formula $\operatorname{Diff}_{F_{ij}}(g_{l*}^{-1} B_0) = g_{l*}^{-1} B_0|_{F_{ij}}$, and $g_{l*}^{-1} B_0|_{F_{ij}} \geq 0$ is a $\mathbb{Q}$-divisor of degree $m_{ij} \leq \frac{14}{15}$ on F_{ij}. Hence $(F_{ij}, \operatorname{Diff}_{F_{ij}}(g_{l*}^{-1} B_0))$ is lc, and $(Y_l, F_{ij} + g_{l*}^{-1} B_0)$ is lc near F_{ij} by Theorem 2.6. Since the curves F_{ij} are pairwise disjoint and $m_{ij} - \frac{7}{60} \leq 1$, (6.1) implies that $(Y_l, \sigma_l^* B_l)$ is lc near F_{ij}. Therefore $(Y_l, \sigma_l^* B_l)$ is lc.

By Construction-Proposition 6.15(4), σ_l contracts the curves C_i to points of Γ and is an isomorphism elsewhere. Thus $S_l \backslash \Gamma$ is smooth, and $(S_l \backslash \Gamma, B_l|_{S_l \backslash \Gamma})$ is lc. Let $U := X \backslash (\Gamma \cup \Lambda)$, which is smooth by Proposition 6.5. Since U and $S_l \cap U$ are smooth, by the adjunction formula $\operatorname{Diff}_{S_l \cap U}(\frac{7}{10} D|_U) = B_l|_{S_l \cap U}$. By Theorem 2.6, $(U, S_l \cap U + \frac{7}{10} D|_U)$ is lc near $S_l \cap U$. Hence $(X, \frac{7}{10} D)$ is lc near any point of $S_l \backslash (\Gamma \cup \Lambda)$. By Proposition 6.14 and Proposition 6.17, $(X, \frac{7}{10} D)$ is lc near any point of $\Gamma \cup \Lambda$. Therefore $(X, \frac{7}{10} D)$ is lc near $S_l = (l = 0) \cap X$. □

Proof of Theorem 6.4. The proof is long, so we divide it into several steps. Let $D \geq 0$ be a $\mathbb{Q}$-divisor on X such that $D \sim_{\mathbb{Q}} -K_X$, and let $\operatorname{Nlc}(X, \frac{7}{10} D)$ be the union of the centers on X of the prime divisors P over X such that $a(P, X, \frac{7}{10} D) < 0$.

Step 1. In this step we show that $\operatorname{Nlc}(X, \frac{7}{10} D)$ is a finite set of smooth points of X outside $\Gamma \cup \Lambda$.

Fix a log resolution of (X, D) on which the strict transform of $\operatorname{Supp} D$ is smooth, which exists by [KM98, Corollary 2.32(2)] applied to the reduced divisor $\operatorname{Supp} D$. For any $t \geq 0$, the pair (X, tD) is lc (resp. klt) near a point $w \in X$ if and only if $a(P, X, tD) \geq 0$ (resp. $a(P, X, tD) > 0$) for any of the finitely many prime divisors P on the fixed log resolution whose image contains w. If a component of D through w has coefficient greater than 1 in tD, then both conditions fail, since the strict transform of that component has negative log discrepancy. Otherwise, the equivalence is [KM98, Corollary 2.32(2)] applied to a neighborhood of w. Thus $\operatorname{Nlc}(X, \frac{7}{10} D)$ is the union of the images of the prime divisors P on the fixed log resolution with $a(P, X, \frac{7}{10} D) < 0$, and it is closed. By Lemma 6.19, Proposition 6.14, and Proposition 6.17, for a general linear form $l \in \mathbb{C}[x, y, z]$,

$$\operatorname{Nlc}\Big(X, \frac{7}{10} D\Big) \cap (S_l \cup \Gamma \cup \Lambda) = \emptyset,$$

where $S_l := (l = 0) \cap X \sim 3H$. Suppose that $\mathrm{Nlc}(X, \frac{7}{10}D)$ contains an irreducible curve C'. Since $12H$ is an ample Cartier divisor (Proposition 6.5), $(4S_l \cdot C') = (12H \cdot C') > 0$, so $C' \cap S_l \neq \emptyset$, a contradiction. Hence $\mathrm{Nlc}(X, \frac{7}{10}D)$ is a finite subset of $X \backslash (\Gamma \cup \Lambda)$, which consists of smooth points of X by Proposition 6.5.

Step 2. In this step, assuming that $(X, \frac{7}{10}D)$ is not lc, we show that the non-klt locus of $(X, c_0 D)$, where $c_0 := \mathrm{lct}(X; D)$, is a non-empty finite set of smooth points of X.

Suppose that $(X, \frac{7}{10}D)$ is not lc. For any prime divisor P over X and any $t \geq 0$,

$$a(P, X, tD) = a(P, X, 0) - t \operatorname{mult}_P D,$$

and $a(P, X, 0) > 0$ since X is klt. By [KM98, Corollary 2.32(2)] applied to the log resolution of **Step 1**, c_0 is the minimum of $a(P, X, 0)/\operatorname{mult}_P D$ over the finitely many prime divisors P on it with $\operatorname{mult}_P D > 0$. Thus $(X, c_0 D)$ is lc, $c_0 < \frac{7}{10}$, and $a(P, X, c_0 D) = 0$ for some prime divisor P over X. Let Z be the union of the centers on X of the prime divisors P over X with $a(P, X, c_0 D) = 0$. Then $Z \neq \emptyset$, and $(X, c_0 D)$ is klt on $X \backslash Z$. For any prime divisor P over X with $a(P, X, c_0 D) = 0$, $\operatorname{mult}_P D > 0$ since $a(P, X, 0) > 0$, so

$$a\Big(P, X, \frac{7}{10}D\Big) = a(P, X, c_0 D) - \Big(\frac{7}{10} - c_0\Big) \operatorname{mult}_P D < 0.$$

Thus $Z \subset \mathrm{Nlc}(X, \frac{7}{10}D)$, and Z is a finite set of smooth points of X by **Step 1**.

Step 3. In this step we show that $H^0(X, \mathcal{O}_X(1)) = 0$.

By Proposition 6.5, $H^0(X, \mathcal{O}_X(1)) = R_1$. The coordinates x, u, v, y, z have degrees $3, 4, 4, 3, 3$, so no monomial in them has degree 1, and $R_1 = 0$.

Step 4. In this step we derive a contradiction from **Steps 2 and 3** by a vanishing theorem.

By Proposition 6.5, R is normal, and the points of X with non-trivial stabilizer form $\Gamma \cup \Lambda$, which is a closed subset of codimension 2 in X since Γ and Λ are curves (the equation of X restricts to $zx^4 + 2x^5 + y^5 \neq 0$ on $(u = v = 0) \cong \mathbb{P}(3, 3, 3)$ and vanishes on $(x = y = z = 0) \cong \mathbb{P}(4, 4)$). Thus the hypotheses of Definition-Lemma 2.16 hold. Fix a Weil divisor in the class H, still denoted by H. Then $\mathcal{O}_X(H) \cong \mathcal{O}_X(1)$, and $12H$ is Cartier. Since $K_X \sim -2H$, we may assume that $K_X = -2H$. Since Z consists of smooth points of X, H is Cartier near Z.

Let Θ be the reduced divisor with support $\operatorname{Supp} H \cup \operatorname{Supp} D$, let $g : X' \to X$ be a projective log resolution (cf. [KM98, Theorem 0.2]) of (X, Θ) on which the strict transform of Θ is smooth ([KM98, Corollary 2.32(2)]), and let $K_{X'}$ be the canonical divisor with $g_* K_{X'} = K_X$. Let

$$M := g^*(3H - c_0 D) \quad \text{and} \quad L := K_{X'} + \lceil M \rceil.$$

Since $3H - c_0 D \sim_{\mathbb{Q}} (3 - 2c_0)H$ and $c_0 < \frac{7}{10}$, M is nef and big. The $\mathbb{Q}$-divisor $\lceil M \rceil - M$ has coefficients in $[0, 1)$, and the support of $\lceil M \rceil - M$ is contained in the simple normal crossing divisor $g^{-1}(\Theta) \cup \mathrm{Exc}(g)$. By [KM98, Theorem 2.64 and Remark 2.66] applied to $\lceil M \rceil = M + (\lceil M \rceil - M)$, $H^i(X', \mathcal{O}_{X'}(L)) = 0$ for $i > 0$. In particular,

$$H^1(X', \mathcal{O}_{X'}(L)) = 0. \tag{6.2}$$

Let $\mathcal{F} := g_* \mathcal{O}_{X'}(L)$. Since $K_{X'} - g^*(K_X + c_0 D) = \sum_P (a(P, X, c_0 D) - 1)P$, where P runs over the prime divisors on X', and $g^*(3H - c_0 D) = g^* H - g^*(K_X + c_0 D)$, we have

$$L = \Big\lceil g^* H + \sum_P \big(a(P, X, c_0 D) - 1\big) P \Big\rceil. \tag{6.3}$$

We show the following.

(1) $\mathcal{F} \subset \mathcal{O}_X(H)$. By (6.3), $g_* L = H + \lceil -c_0 D \rceil \leq H$. If s is a section of $\mathcal{F}$ over an open subset $V \subset X$, then $\operatorname{div}(g^* s) + L \geq 0$ on $g^{-1}(V)$, so $\operatorname{div}(s) + H \geq \operatorname{div}(s) + g_* L \geq 0$ on V.

(2) $\mathcal{F} = \mathcal{O}_X(H)$ on $X\backslash Z$. Let s be a section of $\mathcal{O}_X(H)$ over an open subset $V \subset X\backslash Z$. Then $\operatorname{div}(s) + H \geq 0$, so $g^*(\operatorname{div}(s) + H) \geq 0$. By **Step 2**, $a(P, X, c_0D) > 0$ for any prime divisor P on X' with $g(P) \cap V \neq \emptyset$. Thus on $g^{-1}(V)$, any coefficient of

$$\operatorname{div}(g^*s) + L = \Big\lceil g^*\big(\operatorname{div}(s) + H\big) + \sum_P \big(a(P, X, c_0D) - 1\big)P \Big\rceil$$

is the round-up of a number greater than -1, hence is non-negative. Therefore s is a section of $\mathcal{F}$ over V.

(3) $\mathcal{F} \neq \mathcal{O}_X(H)$ near any point $z \in Z$. Since (X, c_0D) is lc and not klt near z, by [KM98, Corollary 2.32(2)] there exists a prime divisor P_0 on X' such that $z \in g(P_0)$ and $a(P_0, X, c_0D) = 0$. Since H is Cartier near z, g^*H is a Cartier divisor near $g^{-1}(z)$, so by (6.3) and the projection formula, near z,

$$\mathcal{F} = \mathcal{O}_X(H) \otimes g_*\mathcal{O}_{X'}\Big(\Big\lceil \sum_P \big(a(P, X, c_0D) - 1\big)P \Big\rceil\Big).$$

The coefficient of P_0 in $\lceil \sum_P (a(P, X, c_0D) - 1)P \rceil$ is $\lceil -1 \rceil = -1$, so 1 is not a section of the second factor over any neighborhood of z. Since $\mathcal{O}_X(H)$ is invertible near z, $\mathcal{F} \neq \mathcal{O}_X(H)$ near z.

By (1), (2), and (3), $\mathcal{O}_X(H)/\mathcal{F}$ is a non-zero coherent sheaf supported on the finite set Z, so $H^0(X, \mathcal{O}_X(H)/\mathcal{F}) \neq 0$. By the exact sequence of low degree terms of the Leray spectral sequence [Wei94, Theorem 5.8.3 and 5.8.6], the natural map $H^1(X, \mathcal{F}) \to H^1(X', \mathcal{O}_{X'}(L))$ is injective, so $H^1(X, \mathcal{F}) = 0$ by (6.2). The exact sequence

$$0 \to \mathcal{F} \to \mathcal{O}_X(H) \to \mathcal{O}_X(H)/\mathcal{F} \to 0$$

and $H^0(X, \mathcal{O}_X(H)) = H^0(X, \mathcal{O}_X(1)) = 0$ (**Step 3**) imply $H^0(X, \mathcal{O}_X(H)/\mathcal{F}) = 0$, a contradiction.

Step 5. In this step we conclude the proof.

By **Steps 2–4**, $(X, \frac{7}{10}D)$ is lc for any $\mathbb{Q}$-divisor $D \geq 0$ on X such that $D \sim_{\mathbb{Q}} -K_X$, so $\alpha(X) \geq \frac{7}{10}$ by Definition 2.10. By Corollary 6.10, $\alpha(X) \leq \frac{7}{10}$. Thus $\alpha(X) = \frac{7}{10} \in (\frac{2}{3}, \frac{3}{4})$. By Construction-Proposition 6.6, X is a $\mathbb{Q}$-Fano threefold and E is a prime divisor over X. By Proposition 6.7, $\operatorname{vol}(-K_X) = \frac{5}{18}$ and $\beta_X(E) = -\frac{1}{6}\operatorname{vol}(-K_X) < 0$. Therefore X is not K-semistable by Theorem 2.12. □

7. Dimension two

In this section, we prove Theorem 7.3. The family of Set-up 4.1, studied in Section 4, contains no surface: condition (1) of Set-up 4.1 requires $d \geq 3$, and the other conditions cannot hold for $d = 2$ either. Indeed, for $d = 2$ the only well formed weighted projective line $W = \mathbb{P}(a_1, a_2)$ is $\mathbb{P}(1, 1)$, so the sum $k = a_1 + a_2$ of the weights of W is 2 and $a_{(1)}a_{(2)} = 1 < k$, which contradicts condition (3), $a_{(1)}a_{(2)} \geq k$. Moreover, condition (4) on the weight c of z, namely $\frac{k}{d} < c < \frac{k}{d-1}$, reads $1 < c < 2$ and has no integer solution c.

Set-up 7.1. Let λ, X_λ, H, p, q, C_a, and C_c be as follows.

(1) $\lambda \in \mathbb{C}^*$ and

$$X_\lambda := (ac - b^2 + \lambda xy = y^2 - x^3 - bc = 0) \subset \mathbb{P}(3, 4, 5, 6, 7),$$

where a, x, b, y, c are the coordinates of degrees $3, 4, 5, 6, 7$.

(2) $H := \mathcal{O}_{X_\lambda}(1)$, $p := [1 : 0 : 0 : 0 : 0]$, and $q := [0 : 0 : 0 : 0 : 1]$.

(3) $C_a := (a = 0) \cap X_\lambda$ and $C_c := (c = 0) \cap X_\lambda$.

Remark 7.2. For $\lambda \in \mathbb{C}^*$, choose $\beta \in \mathbb{C}^*$ with $\beta^2 = \lambda$. The automorphism

$$(a, x, b, y, c) \to (\beta^3 a, x, \beta b, y, \beta^{-1}c)$$

of $\mathbb{P}(3, 4, 5, 6, 7)$ pulls back $ac - b^2 + \lambda xy$ to $\beta^2(ac - b^2 + xy)$ and preserves $y^2 - x^3 - bc$, so it maps X_1 onto X_λ. Thus the surfaces X_λ are isomorphic to each other. The condition $\lambda \neq 0$

is used in Proposition 7.5, where X_λ is disjoint from the line $(a=b=c=0)$, in Lemma 7.7, where the divisor $(c=0)$ on X_λ is reduced, and in Lemma 7.11, where the preimage of C_a in the orbifold chart at q has a node at the origin. Indeed, X_0 contains the point $[0:1:0:1:0]$, at which the gradient of $ac-b^2$ vanishes, so X_0 is not quasismooth there, and $(c=0)\cap X_0$ and $(a=0)\cap X_0$ are twice the curves $(b=c=y^2-x^3=0)$ and $(a=b=y^2-x^3=0)$ respectively. We keep the parameter λ in the rest of this section, so that the uses of $\lambda\neq 0$ in Proposition 7.5, Lemma 7.7, and Lemma 7.11 remain visible.

Theorem 7.3. *Notation as in Set-up 7.1. Then X_λ is a $\mathbb{Q}$-Fano surface with $\alpha(X_\lambda)=\frac{7}{12}$, and there exists a prime divisor F over X_λ such that $\beta_{X_\lambda}(F)=-\frac{1}{14}\operatorname{vol}(-K_{X_\lambda})$, where $\operatorname{vol}(-K_{X_\lambda})=\frac{3}{7}$. In particular, $\frac{1}{2}<\alpha(X_\lambda)<\frac{2}{3}$ and X_λ is not K-semistable.*

Remark 7.4. For $d\geq 4$, let X and E be as in Set-up 4.1 and Construction-Proposition 4.9, where $e=k+c$ is the degree of the equation of X and c is the weight of z. Then Construction-Proposition 4.9 and Proposition 4.11 show that

$$\operatorname{vol}(-K_X-tE)=\operatorname{vol}(-K_X)\bigl(1-(t/T_X(E))^d\bigr)\ \text{ for } 0\leq t\leq T_X(E),\quad T_X(E)=\frac{e}{c}=A_X(E)+1.$$

For the threefold X of Set-up 6.2 and the prime divisor E of Construction-Proposition 6.6, Proposition 6.7 shows that

$$\operatorname{vol}(-K_X-tE)=\operatorname{vol}(-K_X)\bigl(1-(t/T_X(E))^3\bigr)\ \text{ for } 0\leq t\leq T_X(E),\quad T_X(E)=\frac{10}{3}=A_X(E)+1,$$

where $A_X(E)=\frac{7}{3}$ by Construction-Proposition 6.6(1). For the surfaces X_λ of Set-up 7.1 and the prime divisor F over X_λ of Construction-Proposition 7.6, we have $A_{X_\lambda}(F)=1$ and $T_{X_\lambda}(F)=\frac{12}{7}<A_{X_\lambda}(F)+1$, and $\operatorname{vol}(-K_{X_\lambda}-tF)$ changes from one quadratic polynomial in t to another at $t=\frac{3}{2}$. Thus $\operatorname{vol}(-K_{X_\lambda}-tF)$ is not of the form $\operatorname{vol}(-K_{X_\lambda})\bigl(1-(t/T_{X_\lambda}(F))^2\bigr)$ for $0\leq t\leq T_{X_\lambda}(F)$, and its computation in Proposition 7.8 uses a curve on Y of negative self-intersection (Lemma 7.7).

We prove Theorem 7.3 in Subsection 7.2.

7.1. **The plt blow-up and the volume.** In this subsection, we construct a plt blow-up of X_λ at p and compute the volume function of the exceptional divisor.

Proposition 7.5. *Notation as in Set-up 7.1. Let $\rho_a: U_a\to(a\neq 0)$ and $\rho_c: U_c\to(c\neq 0)$ be the orbifold charts of $\mathbb{P}(3,4,5,6,7)$, and let $X':=(y^2=x^3+b^3-\lambda bxy)\subset\mathbb{A}^3_{x,b,y}$. Then the following hold.*

1. *X_λ is a normal projective surface with $\operatorname{Sing}X_\lambda=\{p,q\}$.*
2. *The projection to the coordinates (x,b,y) induces an isomorphism $\rho_a^{-1}(X_\lambda)\cong X'$, under which μ_3 acts on X' with weights $(1,2,0)$ on (x,b,y) and freely on $X'\setminus\{0\}$, and ρ_a induces an isomorphism $X'/\mu_3\cong X_\lambda\cap(a\neq 0)$ that maps the image of 0 to p.*
3. *The projection to the coordinates (x,y) induces an isomorphism $\rho_c^{-1}(X_\lambda)\cong\mathbb{A}^2_{x,y}$, under which μ_7 acts on $\mathbb{A}^2_{x,y}$ with weights $(4,6)$ and freely on $\mathbb{A}^2_{x,y}\setminus\{0\}$, and ρ_c induces an isomorphism $\mathbb{A}^2_{x,y}/\mu_7\cong X_\lambda\cap(c\neq 0)$ that maps the image of 0 to q. In particular, $(X_\lambda\ni q)$ is a cyclic quotient singularity of type $\frac{1}{7}(1,5)$.*
4. *X_λ is a $\mathbb{Q}$-factorial $\mathbb{Q}$-Fano surface, $\omega_{X_\lambda}\cong\mathcal{O}_{X_\lambda}(-3)$, and $H^2=\frac{1}{21}$.*

Proof. Let

$$R_\lambda:=\mathbb{C}[a,x,b,y,c]/(ac-b^2+\lambda xy,\ y^2-x^3-bc),$$

graded by $\deg(a,x,b,y,c)=(3,4,5,6,7)$, so that $X_\lambda=\operatorname{Proj}R_\lambda$.

We first show that R_λ is a normal domain. In $R_\lambda[c^{-1}]$, we have $b=(y^2-x^3)/c$ and $a=(b^2-\lambda xy)/c$, so $R_\lambda[c^{-1}]\cong\mathbb{C}[x,y,c,c^{-1}]$. The closed subset $(c=0)$ of $\operatorname{Spec}R_\lambda$ is defined by $b^2=\lambda xy$ and $y^2=x^3$ in $\mathbb{A}^4_{a,x,b,y}$, so $\dim(c=0)=2$. Since $\dim Z\geq 3$ for any irreducible component Z of $\operatorname{Spec}R_\lambda$ and $(c\neq 0)\subset\operatorname{Spec}R_\lambda$ is irreducible of dimension 3, $\operatorname{Spec}R_\lambda$ is

irreducible of dimension 3. Thus $ac-b^2+\lambda xy$ and y^2-x^3-bc form a regular sequence, and R_λ is Cohen–Macaulay. Hence the only associated prime of R_λ is the minimal prime, which does not contain c, so $R_\lambda \subset R_\lambda[c^{-1}]$ is a domain. The gradients of $ac-b^2+\lambda xy$ and y^2-x^3-bc with respect to (a,x,b,y,c) are

$$(c,\lambda y,-2b,\lambda x,a) \quad \text{and} \quad (0,-3x^2,-c,2y,-b).$$

A direct calculation shows that they are linearly dependent at a point of $\operatorname{Spec} R_\lambda$ if and only if $x=b=y=c=0$. Thus $\operatorname{Spec} R_\lambda$ is smooth outside the a-axis, which has codimension 2, so R_λ is normal by Serre's criterion [Mat86, Theorem 23.8].

Next we determine the points with non-trivial stabilizer. The weighted projective space $\mathbb{P} := \mathbb{P}(3,4,5,6,7)$ is well formed. The sets of weights with greatest common divisor larger than 1 are the singletons, $\{3,6\}$, and $\{4,6\}$, so the points of $\mathbb{P}$ with non-trivial stabilizer are the coordinate points and the points of the lines $(x=b=c=0)$ and $(a=b=c=0)$. On $(x=b=c=0)$, the equations of X_λ reduce to $y^2=0$. On $(a=b=c=0)$, they reduce to $\lambda xy=0$ and $y^2=x^3$, which have no solution in $\mathbb{P}$ since $\lambda\neq 0$. Among the coordinate points, only p and q are on X_λ. Thus p and q are the only points of X_λ with non-trivial stabilizer, and their stabilizers are μ_3 and μ_7.

Let $z\in X_\lambda\backslash\{p,q\}$, and choose j such that the j-th coordinate of z is non-zero. The morphism

$$(\operatorname{Spec} R_\lambda \cap U_j)\times \mathbb{G}_m \to \operatorname{Spec} R_\lambda \cap (x_j\neq 0)$$

given by the $\mathbb{G}_m$-action is the quotient by a free action of μ_{a_j}, hence étale. Since the $\mathbb{G}_m$-orbit over z is disjoint from the a-axis, $\rho_j^{-1}(X_\lambda)=\operatorname{Spec} R_\lambda\cap U_j$ is smooth over a neighborhood of z. Since the stabilizer of z is trivial, ρ_j is étale over a neighborhood of z. Thus X_λ is smooth at z.

On $U_c=(c=1)$, the equations of X_λ become $b=y^2-x^3$ and $a=b^2-\lambda xy$, so the projection to the coordinates (x,y) induces an isomorphism $\rho_c^{-1}(X_\lambda)\cong \mathbb{A}^2_{x,y}$, on which μ_7 acts with weights $(4,6)$. With the generator ζ^4 of μ_7 in place of ζ, the weights become $(1,5)$, since $\zeta^6=(\zeta^4)^5$. Thus $(X_\lambda\ni q)$ is a cyclic quotient singularity of type $\frac{1}{7}(1,5)$ (Definition 2.18). On $U_a=(a=1)$, the first equation becomes $c=b^2-\lambda xy$, and the second becomes $y^2=x^3+b^3-\lambda bxy$. Thus the projection to the coordinates (x,b,y) induces an isomorphism $\rho_a^{-1}(X_\lambda)\cong X'$, where μ_3 acts on (x,b,y,c) with weights $(4,5,6,7)\equiv(1,2,0,1) \bmod 3$, which is compatible with $c=b^2-\lambda xy$, and ρ_a induces $X'/\mu_3\cong X_\lambda\cap(a\neq 0)$. A non-trivial element of μ_3 fixes a point of X' only if $x=b=0$, and then $y=0$. Thus μ_3 acts freely on $X'\backslash\{0\}$, and μ_7 acts freely on $\mathbb{A}^2\backslash\{0\}$ because the weights 4 and 6 are prime to 7. The points p and q are singular: $X'\backslash\{0\}$ and $\mathbb{A}^2\backslash\{0\}$ are connected étale covers of degrees 3 and 7 of punctured neighborhoods of p and q (by the previous paragraph, X' is smooth outside 0, so small punctured neighborhoods of 0 in the hypersurface $X'\subset\mathbb{A}^3$ are connected [Mil68, Theorems 2.10 and 5.2]), while a small punctured neighborhood of a smooth point of a surface is simply connected. Thus $\operatorname{Sing} X_\lambda=\{p,q\}$. This proves (1), (2), and (3).

The points of X_λ with non-trivial stabilizer form a closed subset of codimension 2. By Definition-Lemma 2.16, H is the class of a Weil divisor with $\mathcal{O}_{X_\lambda}(m)\cong\mathcal{O}_{X_\lambda}(mH)$ for any integer m. By Lemma 2.22, applied to $ac-b^2+\lambda xy$ and y^2-x^3-bc, of degrees 10 and 12, we have

$$\omega_{X_\lambda}\cong\mathcal{O}_{X_\lambda}(10+12-25)=\mathcal{O}_{X_\lambda}(-3) \quad \text{and} \quad H^2=\frac{10\cdot 12}{3\cdot 4\cdot 5\cdot 6\cdot 7}=\frac{1}{21}.$$

Thus $-K_{X_\lambda}\sim 3H$. Since $\mathcal{O}_{\mathbb{P}}(420)$ is an ample line bundle on $\mathbb{P}$ whose restriction to X_λ is $\mathcal{O}_{X_\lambda}(420)$, $420H$ is an ample Cartier divisor, so $-K_{X_\lambda}$ is an ample $\mathbb{Q}$-Cartier divisor.

Finally we show that X_λ is klt. The lowest-weight part of the polynomial $y^2-x^3-b^3+\lambda bxy$ with respect to the weights $(2,2,3)$ on (x,b,y) is $y^2-x^3-b^3$, which is irreducible and quasismooth of weight $6<2+2+3$, so X' is klt near 0 by Lemma 3.2(1). The quotient morphisms $X'\to X_\lambda\cap(a\neq 0)$ and $\mathbb{A}^2_{x,y}\to X_\lambda\cap(c\neq 0)$ are étale outside the origins, hence étale in codimension one. By Lemma 2.8(2)(3) with $B=0$, X_λ is klt near p and near q. Since X_λ is smooth outside $\{p,q\}$, X_λ is klt. Thus X_λ is a $\mathbb{Q}$-Fano surface. Moreover, X_λ has only quotient

singularities [KM98, Proposition 4.18], so X_λ is $\mathbb{Q}$-factorial [KM98, Proposition 5.15]. This proves (4). □

Construction-Proposition 7.6. Notation as in Set-up 7.1. Let $X' := (y^2 = x^3+b^3-\lambda bxy) \subset \mathbb{A}^3$, and let $f' : Y' \to X'$ be the morphism from the strict transform Y' of X' in the weighted blow-up of $\mathbb{A}^3$ with weights $(2,2,3)$ on (x,b,y). Then the action of μ_3 on X' with weights $(1,2,0)$ lifts to Y', and $Y'/\mu_3 \to X'/\mu_3$ extends to a plt blow-up $f : Y \to X_\lambda$ of X_λ at p. The exceptional divisor F of f satisfies $a(F, X_\lambda, 0) = 1$, $(F^2) = -\frac{1}{6}$, and

$$(F, \Delta_F) \cong \left(\mathbb{P}^1, \frac{1}{2}P_1 + \frac{2}{3}P_2 + \frac{2}{3}P_3\right)$$

for three distinct points $P_1, P_2, P_3 \in \mathbb{P}^1$. Moreover, the Weil divisors $(x = 0)$, $(y = 0)$, and $(c = 0)$ on X_λ satisfy $\mathrm{mult}_F(x = 0) = 2$, $\mathrm{mult}_F(y = 0) = 3$, and $\mathrm{mult}_F(c = 0) = 4$. We let $L := f^*(-K_{X_\lambda})$ and $C'_c := f_*^{-1}C_c$.

Proof. By Proposition 7.5(2), the projection to the coordinates (x,b,y) induces an isomorphism $\rho_a^{-1}(X_\lambda) \cong X'$, under which μ_3 acts on X' with weights $(1,2,0)$ and freely on $X'\backslash\{0\}$, and ρ_a induces the quotient morphism $g : X' \to X'/\mu_3 \cong X_\lambda \cap (a \neq 0)$. Thus $g^{-1}(p) = \{0\}$, and g is étale over $X_\lambda \cap (a \neq 0)\backslash\{p\}$. By Proposition 7.5(4), X_λ is klt near p.

Let F' be the exceptional divisor of f'. The lowest-weight part of the polynomial $y^2 - x^3 - b^3 + \lambda bxy$ with respect to the weights $(2,2,3)$ is $y^2 - x^3 - b^3$, which is irreducible and quasismooth of weight $6 < 2+2+3$. By Lemma 3.2, there exists an open neighborhood $W \subset X'$ of 0 such that the restriction of f' over W is a plt blow-up of W at 0, $F' \cong (y^2 = x^3 + b^3) \subset \mathbb{P}(2,2,3)$, $a(F', X', 0) = 2+2+3-6 = 1$, and

$$\mathrm{mult}_{F'}(x|_{X'} = 0) = 2, \quad \mathrm{mult}_{F'}(y|_{X'} = 0) = 3, \quad \mathrm{mult}_{F'}\left((b^2 - \lambda xy)|_{X'} = 0\right) = 4,$$

where the last equality holds because the monomials b^2 and xy have weights 4 and 5. Since g is finite and $g^{-1}(p) = \{0\}$, the set

$$V := (X_\lambda \cap (a \neq 0))\backslash g(X'\backslash W)$$

is an open neighborhood of p with $g^{-1}(V) \subset W$. Since the statement is local near p, we replace X' with $g^{-1}(V)$ and Y' with $f'^{-1}(g^{-1}(V))$, which are μ_3-invariant.

The morphism $F' \to \mathbb{P}^1$, $[x : b : y] \to [x : b]$, is well defined since $x = b = 0$ implies $y = 0$ on F', and it is bijective since $[x : b : y] = [x : b : -y]$ in $\mathbb{P}(2,2,3)$. Since F' is normal (Definition-Proposition 2.14), $F' \cong \mathbb{P}^1$ with coordinate x/b. Since μ_3 acts diagonally on (x,b,y), the action commutes with the $\mathbb{G}_m$-action defining the weighted blow-up, so it lifts to Y'. A generator ζ of μ_3 acts on F' by $x/b \to \zeta^{-1}x/b$. Thus μ_3 has exactly two fixed points on F', namely $P'_2 := (x = 0) \cap F'$ and $P'_3 := (b = 0) \cap F'$, and μ_3 permutes the three points of $(y = 0) \cap F' = ((x/b)^3 = -1)$ transitively. Since μ_3 acts freely on $Y'\backslash F' \cong X'\backslash\{0\}$, the quotient morphism $h : Y' \to Y'/\mu_3$ is étale outside $\{P'_2, P'_3\}$. The morphism f' descends to a proper birational morphism $Y'/\mu_3 \to V$, which is an isomorphism over $V\backslash\{p\}$. Gluing Y'/μ_3 and $X_\lambda\backslash\{p\}$ along $V\backslash\{p\}$, we obtain a proper birational morphism $f : Y \to X_\lambda$ from a normal surface Y such that $f \circ h = g \circ f'$ over V. The exceptional locus of f is the prime divisor $F := h(F')$, $f(F) = \{p\}$, and $h^*F = F'$ since h is étale at the generic point of F'.

We now describe Y near F. The weighted blow-up of $\mathbb{A}^3$ is covered by three charts. The x-chart is $\mathbb{A}^3_{s,v,w}/\mu_2$, where μ_2 acts by $(s,v,w) \to (-s,v,-w)$ and $x = s^2$, $b = s^2v$, $y = s^3w$. Over the x-chart, Y' and F' are the images of

$$V_x := (w^2 = 1 + v^3 - \lambda svw) \subset \mathbb{A}^3_{s,v,w}$$

and of $V_x \cap (s = 0)$. The b-chart is obtained by exchanging x and b, and the polynomial $y^2 - x^3 - b^3 + \lambda bxy$ is symmetric in x and b. The y-chart is $\mathbb{A}^3_{s,u,v}/\mu_3$, where $\epsilon \in \mu_3$ acts by $(s,u,v) \to (\epsilon s, \epsilon u, \epsilon v)$ and $x = s^2u$, $b = s^2v$, $y = s^3$. Over the y-chart, Y' is the image of $V_y := (u^3 + v^3 - \lambda suv = 1) \subset \mathbb{A}^3_{s,u,v}$. A direct calculation shows that V_x and V_y are smooth along $(s = 0)$, and that $V_x \cap (s = 0)$ and $V_y \cap (s = 0)$ are smooth curves. The action of μ_3 on V_y is free.

The action of μ_2 on $V_x \cap (s=0)$ has exactly three fixed points, the points $(0,v,0)$ with $v^3=-1$, which map to the three points of $(y=0)\cap F'$. At each of them, (s,w) are local coordinates of V_x, and μ_2 acts by $(s,w)\to(-s,-w)$. Thus Y' is smooth along $F'\backslash(y=0)$, and at any point of $(y=0)\cap F'$, the germ of (Y',F') is the germ of $((s=0)\subset \mathbb{A}^2_{s,w}/\mu_2)$, with μ_2 acting with weights $(1,1)$.

The point $P_2'=[0:1:1]$ is the image of $(0,0,1)$ in the b-chart, where $x=s^2u$, $b=s^2$, $y=s^3w$, and Y' is the image of $(w^2=1+u^3-\lambda suw)$. At $(0,0,1)$, (s,u) are local coordinates, the generator of μ_2 maps $(0,0,1)$ to $(0,0,-1)$, and ζ lifts to $(s,u,w)\to(\zeta s,\zeta^2 u,w)$, which fixes $(0,0,1)$. Thus the germ of (Y',F') at P_2' is the germ of $(\mathbb{A}^2_{s,u},(s=0))$ at the origin, on which the stabilizer μ_3 of P_2' acts with weights $(1,2)$. Exchanging the roles of x and b, the germ of (Y',F') at P_3' is the germ of $(\mathbb{A}^2_{s,u},(s=0))$ at the origin, on which the stabilizer μ_3 of P_3' acts with weights $(2,1)$.

Let P_1, P_2, and P_3 be the images of $(y=0)\cap F'$, P_2', and P_3' in F. Since h is étale near $(y=0)\cap F'$, the germs of (Y,F) at P_1, P_2, and P_3 are the germs at the image of the origin of $((s=0)\subset \mathbb{A}^2/\mu_m)$, where $m=2,3,3$ and μ_m acts with weights $(1,1)$, $(1,2)$, and $(2,1)$, respectively. Moreover, Y is smooth along $F\backslash\{P_1,P_2,P_3\}$. Since $Y\backslash F\cong X_\lambda\backslash\{p\}$ and $X_\lambda\backslash\{p\}$ is smooth outside the cyclic quotient singularity q (Proposition 7.5), Y has only quotient singularities. Thus Y is $\mathbb{Q}$-factorial [KM98, Proposition 5.15].

Since h is étale in codimension one and $h^*F=F'$, we have $K_{Y'}+F'=h^*(K_Y+F)$, and (Y,F) is plt near F because (Y',F') is plt near F' [KM98, Proposition 5.20(2) and Definition 2.34]. Thus F is normal [KM98, Proposition 5.51], and since F is the image of $F'\cong\mathbb{P}^1$, we have $F\cong\mathbb{P}^1$. By [Kol+92, (16.6.3)], the coefficient of Δ_F at a point where the germ of (Y,F) is $((s=0)\subset\mathbb{A}^2/\mu_m)$ is $1-\frac{1}{m}$. Thus

$$\Delta_F=\frac{1}{2}P_1+\frac{2}{3}P_2+\frac{2}{3}P_3.$$

Let $a:=a(F,X_\lambda,0)$, so that $K_Y+(1-a)F=f^*K_{X_\lambda}$. Since g and h are étale in codimension one, $g^*K_{X_\lambda}=K_{X'}$ and $h^*K_Y=K_{Y'}$. Pulling back by h and using $f\circ h=g\circ f'$ and $h^*F=F'$, we obtain $K_{Y'}+(1-a)F'=f'^*K_{X'}$, so $a=a(F',X',0)=1$. Thus $K_Y=f^*K_{X_\lambda}$, and

$$(F^2)=((K_Y+F)\cdot F)=\deg(K_F+\Delta_F)=-2+\frac{1}{2}+\frac{2}{3}+\frac{2}{3}=-\frac{1}{6}.$$

Since F is the only curve contracted by f and $((-F)\cdot F)=\frac{1}{6}>0$, $-F$ is f-ample [Laz04, Corollary 1.7.9]. Since f is proper and $-nF$ is an f-ample Cartier divisor for some positive integer n, f is projective. Thus f is a plt blow-up of X_λ at p.

Finally, let $\varphi\in\{x,y,c\}$ and let $D_\varphi:=(\varphi=0)$ on X_λ. Then $D_\varphi\sim(\deg\varphi)H$ is $\mathbb{Q}$-Cartier. Since g is étale over $V\backslash\{p\}$, we have $g^*D_\varphi=(\varphi|_{X'}=0)$, where $c|_{X'}=b^2-\lambda xy$. Since $h^*f^*D_\varphi=f'^*g^*D_\varphi$ and $h^*F=F'$, we have $\operatorname{mult}_F D_\varphi=\operatorname{mult}_{F'}(\varphi|_{X'}=0)$, which equals 2, 3, and 4 for $\varphi=x$, y, and c, respectively. □

For a rational number t with $\frac{3}{2}<t\le\frac{12}{7}$ and a sufficiently divisible positive integer m, any member of $|m(L-tF)|$ contains $m(2t-3)C_c'$, and the next lemma computes the intersection numbers of C_c' that determine $\operatorname{vol}(-K_{X_\lambda}-tF)$ for such t.

Lemma 7.7. *Notation as in Set-up 7.1 and Construction-Proposition 7.6. Then C_c is a prime divisor, $C_c\sim 7H$, $f^*C_c=C_c'+4F$, and*

$$C_c'\sim_{\mathbb{Q}}\frac{7}{3}L-4F,\quad (L\cdot C_c')=1,\quad (F\cdot C_c')=\frac{2}{3},\quad (C_c'^2)=-\frac{1}{3}.$$

Proof. We first show that the divisor $(c=0)$ of the section $c\in H^0(X_\lambda,\mathcal{O}_{X_\lambda}(7))$ is C_c with multiplicity one. The ideal $(c,\ ac-b^2+\lambda xy,\ y^2-x^3-bc)$ equals $(c,\ b^2-\lambda xy,\ y^2-x^3)$. Let

$$\Gamma:=(b^2-\lambda xy=y^2-x^3=0)\subset\mathbb{A}^3_{x,b,y}.$$

Then $\Gamma \cap (x=0) = \{0\}$, and the morphism

$$\mathbb{G}_m \to \Gamma \cap (x \neq 0), \quad u \to (u^4, \sqrt{\lambda}u^5, u^6)$$

is bijective. For $(x,b,y) \in \Gamma$ with $x \neq 0$, we have $x = \tau^2$ and $y = \tau^3$ for $\tau := y/x$, and $b^2 = \lambda\tau^5$, so exactly one square root u of τ satisfies $b = \sqrt{\lambda}u^5$. Since Γ is a complete intersection of two hypersurfaces in $\mathbb{A}^3$, any irreducible component Z of Γ satisfies $\dim Z \geq 1$, so $\{0\}$ is not an irreducible component of Γ, and Γ is the closure of $\Gamma \cap (x \neq 0)$, which is the image of $\mathbb{G}_m$. Thus Γ is an irreducible curve (the origin is the limit of $(u^4, \sqrt{\lambda}u^5, u^6)$ as $u \to 0$). At the point $(u^4, \sqrt{\lambda}u^5, u^6)$, the minor of the Jacobian matrix of $(b^2 - \lambda xy, y^2 - x^3)$ with respect to (b,y) is $4\sqrt{\lambda}u^{11} \neq 0$, so the complete intersection Γ is generically reduced, hence reduced. Since $\operatorname{Spec} R_\lambda/(c) \cong \Gamma \times \mathbb{A}^1_a$, the divisor $(c=0)$ is C_c with multiplicity one (the generic point of C_c has trivial stabilizer, so by Definition-Lemma 2.20(1)(2) the multiplicity of $(c=0)$ along C_c can be computed on $\operatorname{Spec} R_\lambda$), and C_c is a prime divisor.

Thus $C_c \sim 7H$ (Definition-Lemma 2.16) and $\operatorname{mult}_F C_c = 4$ (Construction-Proposition 7.6). Since F is the only exceptional curve of f, $f^*C_c = C_c' + 4F$. Since $-K_{X_\lambda} \sim 3H$ (Proposition 7.5), we have $L = 3f^*H$, and

$$C_c' = f^*C_c - 4F \sim_{\mathbb{Q}} 7f^*H - 4F = \frac{7}{3}L - 4F.$$

Since $(L \cdot F) = 0$, $(L^2) = ((-K_{X_\lambda})^2) = 9H^2 = \frac{3}{7}$, and $(F^2) = -\frac{1}{6}$, we obtain

$$(L \cdot C_c') = \frac{7}{3} \cdot \frac{3}{7} = 1, \quad (F \cdot C_c') = -4(F^2) = \frac{2}{3}, \quad (C_c'^2) = \frac{7}{3}(L \cdot C_c') - 4(F \cdot C_c') = \frac{7}{3} - \frac{8}{3} = -\frac{1}{3}.$$

□

Proposition 7.8. *Notation as in Set-up 7.1 and Construction-Proposition 7.6. Then*

$$\operatorname{vol}(-K_{X_\lambda} - tF) = \begin{cases} \frac{3}{7} - \frac{t^2}{6} & \text{if } 0 \leq t \leq \frac{3}{2}, \\ \frac{7}{6}\left(\frac{12}{7} - t\right)^2 & \text{if } \frac{3}{2} \leq t \leq \frac{12}{7}, \\ 0 & \text{if } t \geq \frac{12}{7}. \end{cases}$$

In particular, $\operatorname{vol}(-K_{X_\lambda}) = \frac{3}{7}$, $T_{X_\lambda}(F) = \frac{12}{7}$, $S_{X_\lambda}(F) = \frac{15}{14}$, *and* $\beta_{X_\lambda}(F) = -\frac{1}{14}\operatorname{vol}(-K_{X_\lambda})$.

Proof. By [Fuj19a, Definition 1.1(2)], $\operatorname{vol}(-K_{X_\lambda} - tF) = \operatorname{vol}_Y(L - tF)$, the limit superior in Definition 2.11 is a limit, and $\operatorname{vol}(-K_{X_\lambda} - tF)$ is a non-increasing continuous function of $t \geq 0$. Thus it suffices to prove the formula for rational t. For a rational number $t \geq 0$ and a positive integer n such that $n(L - tF)$ is Cartier (such n exists since L is the pullback of the $\mathbb{Q}$-Cartier divisor $-K_{X_\lambda}$ and F is $\mathbb{Q}$-Cartier by Definition-Proposition 2.14), we have $\operatorname{vol}_Y(L - tF) = \operatorname{vol}_Y(n(L - tF))/n^2$ [Laz04, Remark 2.2.39]. If moreover $L - tF$ is nef, then $\operatorname{vol}_Y(L - tF) = ((L - tF)^2)$ [Laz04, Corollary 1.4.41].

We show that $L - \frac{3}{2}F$ is nef. The sections x^{105}, y^{70}, and c^{60} of $\mathcal{O}_{X_\lambda}(420)$ define divisors $D_1 := 105(x=0)$, $D_2 := 70(y=0)$, and $D_3 := 60(c=0)$ with $D_i \geq 0$ and $D_i \sim 420H \sim_{\mathbb{Q}} -140K_{X_\lambda}$. By Construction-Proposition 7.6, $\operatorname{mult}_F D_1 = \operatorname{mult}_F D_2 = 210$ and $\operatorname{mult}_F D_3 = 240$, so

$$f^*D_i - 210F \geq 0 \quad \text{and} \quad f^*D_i - 210F \sim_{\mathbb{Q}} 140\left(L - \frac{3}{2}F\right) \quad \text{for } i = 1,2,3.$$

On X_λ, the equations $x = y = c = 0$ imply $b = 0$ by $ac - b^2 + \lambda xy = 0$, so $\operatorname{Supp} D_1 \cap \operatorname{Supp} D_2 \cap \operatorname{Supp} D_3 = \{p\}$. Let $C \neq F$ be a curve on Y. Then $f(C)$ is a curve, so $C \not\subset \operatorname{Supp}(f^*D_i - 210F)$ for some i, and $((L - \frac{3}{2}F) \cdot C) \geq 0$. Moreover, $((L - \frac{3}{2}F) \cdot F) = \frac{1}{4} > 0$. Thus $L - \frac{3}{2}F$ is nef, and by Lemma 7.7,

$$\left(\left(L - \frac{3}{2}F\right)^2\right) = \frac{3}{7} - \frac{9}{4} \cdot \frac{1}{6} = \frac{3}{56}, \quad \left(\left(L - \frac{3}{2}F\right) \cdot C_c'\right) = 1 - \frac{3}{2} \cdot \frac{2}{3} = 0.$$

Let $0 \leq t \leq \frac{3}{2}$ be a rational number. Then

$$L - tF = \left(1 - \frac{2t}{3}\right)L + \frac{2t}{3}\left(L - \frac{3}{2}F\right)$$

is nef, since L is the pullback of an ample $\mathbb{Q}$-divisor. Thus $\mathrm{vol}(-K_{X_\lambda}-tF)=((L-tF)^2)=\frac{3}{7}-\frac{t^2}{6}$.

Let $\frac{3}{2}\le t\le\frac{12}{7}$ be a rational number, and let

$$P_t:=\frac{14}{3}\Big(\frac{12}{7}-t\Big)\Big(L-\frac{3}{2}F\Big)\quad\text{and}\quad N_t:=(2t-3)C_c'.$$

By Lemma 7.7, $L-tF\sim_{\mathbb{Q}}P_t+N_t$, and P_t is nef. Let n be a positive integer such that $n(L-tF)$, nP_t, and nN_t are Cartier and $n(L-tF)\sim nP_t+nN_t$. Let m be a positive integer and $B\in|mn(L-tF)|$. Write $B=rC_c'+B'$, where $r\ge0$ and $B'\ge0$ is a $\mathbb{Q}$-Cartier divisor whose support does not contain C_c'. Then $(B'\cdot C_c')\ge0$, and by Lemma 7.7,

$$mn\Big(1-\frac{2}{3}t\Big)=(B\cdot C_c')=-\frac{r}{3}+(B'\cdot C_c')\ge-\frac{r}{3}.$$

Thus $r\ge mn(2t-3)$, that is, $B\ge mnN_t$. Hence multiplication by the section of $\mathcal{O}_Y(mnN_t)$ defining mnN_t is an isomorphism

$$H^0(Y,\mathcal{O}_Y(mnP_t))\to H^0(Y,\mathcal{O}_Y(mn(L-tF)))$$

for any m. Thus

$$\mathrm{vol}(-K_{X_\lambda}-tF)=\frac{\mathrm{vol}_Y(nP_t)}{n^2}=(P_t^2)=\frac{196}{9}\Big(\frac{12}{7}-t\Big)^2\cdot\frac{3}{56}=\frac{7}{6}\Big(\frac{12}{7}-t\Big)^2.$$

Let $t>\frac{12}{7}$ be a rational number. Then

$$\Big((L-tF)\cdot\Big(L-\frac{3}{2}F\Big)\Big)=\frac{3}{7}-\frac{t}{4}<0.$$

If $H^0(Y,\mathcal{O}_Y(mn(L-tF)))\ne0$ for some m, let B be a member of $|mn(L-tF)|$. Since $L-\frac{3}{2}F$ is nef,

$$0\le\Big(B\cdot\Big(L-\frac{3}{2}F\Big)\Big)=mn\Big(\frac{3}{7}-\frac{t}{4}\Big)<0,$$

a contradiction. Thus $\mathrm{vol}(-K_{X_\lambda}-tF)=0$.

In particular, $\mathrm{vol}(-K_{X_\lambda})=\frac{3}{7}$ and $T_{X_\lambda}(F)=\frac{12}{7}$. Moreover,

$$\int_0^\infty\mathrm{vol}(-K_{X_\lambda}-tF)dt=\int_0^{3/2}\Big(\frac{3}{7}-\frac{t^2}{6}\Big)dt+\int_{3/2}^{12/7}\frac{7}{6}\Big(\frac{12}{7}-t\Big)^2dt=\frac{51}{112}+\frac{3}{784}=\frac{45}{98},$$

so

$$S_{X_\lambda}(F)=\frac{45}{98}\cdot\frac{7}{3}=\frac{15}{14}.$$

Since X_λ is a $\mathbb{Q}$-Fano surface (Proposition 7.5) and $A_{X_\lambda}(F)=a(F,X_\lambda,0)=1$ (Construction-Proposition 7.6), we have

$$\beta_{X_\lambda}(F)=\mathrm{vol}(-K_{X_\lambda})\big(A_{X_\lambda}(F)-S_{X_\lambda}(F)\big)=-\frac{1}{14}\,\mathrm{vol}(-K_{X_\lambda}).$$

□

Corollary 7.9. *Notation as in Set-up 7.1. Then for any $\mathbb{Q}$-divisor $D\ge0$ on X_λ such that $D\sim_{\mathbb{Q}}-K_{X_\lambda}$, the pair $(X_\lambda,\frac{7}{12}D)$ is lc near p. Moreover, $\mathrm{lct}\big(X_\lambda;\frac{3}{7}C_c\big)\le\frac{7}{12}$, and in particular $\alpha(X_\lambda)\le\frac{7}{12}$.*

Proof. We use the plt blow-up $f:Y\to X_\lambda$ with exceptional divisor F of Construction-Proposition 7.6. Then

$$\deg\big(-(K_F+\Delta_F)\big)=2-\frac{1}{2}-\frac{2}{3}-\frac{2}{3}=\frac{1}{6}.$$

Let $D_F\ge0$ be a $\mathbb{Q}$-divisor on F such that $D_F\sim_{\mathbb{Q}}-(K_F+\Delta_F)$. Then $\deg D_F=\frac{1}{6}$, so any coefficient of Δ_F+D_F is at most $\frac{2}{3}+\frac{1}{6}<1$, and (F,Δ_F+D_F) is lc. Thus $\alpha(F,\Delta_F)\ge1$.

Let $D \geq 0$ be a $\mathbb{Q}$-divisor on X_λ such that $D \sim_{\mathbb{Q}} -K_{X_\lambda}$. Then D is $\mathbb{Q}$-Cartier. By Lemma 2.13 and Proposition 7.8, $\operatorname{mult}_F D \leq T_{X_\lambda}(F) = \frac{12}{7}$, so

$$\operatorname{mult}_F f^*\Big(\frac{7}{12}D\Big) \leq 1 = a(F, X_\lambda, 0).$$

Since X_λ is klt near p (Proposition 7.5), Lemma 3.1 implies that $(X_\lambda, \frac{7}{12}D)$ is lc near p.

By Lemma 7.7 and Proposition 7.5, $C_c \sim 7H$, $-K_{X_\lambda} \sim 3H$, and $f^*C_c = C_c' + 4F$. Thus $\frac{3}{7}C_c \sim_{\mathbb{Q}} -K_{X_\lambda}$ and $\operatorname{mult}_F(\frac{3}{7}C_c) = \frac{12}{7}$, and for any real number $s > \frac{7}{12}$,

$$a\Big(F, X_\lambda, \frac{3s}{7}C_c\Big) = 1 - \frac{12}{7}s < 0.$$

Thus $\operatorname{lct}(X_\lambda; \frac{3}{7}C_c) \leq \frac{7}{12}$, and $\alpha(X_\lambda) \leq \frac{7}{12}$ by Definition 2.10. □

7.2. **The global threshold.** In this subsection, we prove Theorem 7.3. By Propositions 7.5 and 7.8, Corollary 7.9, and Theorem 2.12, it remains to show that $(X_\lambda, \frac{7}{12}D)$ is lc away from p for any $\mathbb{Q}$-divisor $D \geq 0$ with $D \sim_{\mathbb{Q}} -K_{X_\lambda}$. As test curves, we use the curve C_a through q and the members of the pencil in $|\mathcal{O}_{X_\lambda}(6)|$ spanned by y and a^2, which give the coefficients $\frac{2}{3}$ near q and 1 elsewhere.

Lemma 7.10. *Notation as in Set-up 7.1. Let C be a prime divisor on X_λ, $Z \subset X_\lambda$ a closed subset, and r, s positive rational numbers. Assume that $rC \sim_{\mathbb{Q}} -K_{X_\lambda}$, that (X_λ, srC) is lc near Z, and that (X_λ, sD_1) is lc near Z for any $\mathbb{Q}$-divisor $D_1 \geq 0$ on X_λ such that $D_1 \sim_{\mathbb{Q}} -K_{X_\lambda}$ and $C \not\subset \operatorname{Supp} D_1$. Then (X_λ, sD) is lc near Z for any $\mathbb{Q}$-divisor $D \geq 0$ on X_λ such that $D \sim_{\mathbb{Q}} -K_{X_\lambda}$.*

Proof. Let $t \geq 0$ be the coefficient of C in D, and write $D = tC + D'$, where $D' \geq 0$ and $C \not\subset \operatorname{Supp} D'$. Then $D' \sim_{\mathbb{Q}} (1 - \frac{t}{r})(-K_{X_\lambda})$. Since $-K_{X_\lambda} \sim 3H$ is ample (Proposition 7.5), we have $(D' \cdot H) = 3(1 - \frac{t}{r})H^2 \geq 0$, with equality if and only if $D' = 0$. Thus $t \leq r$. If $t = r$, then $D = rC$, and (X_λ, sD) is lc near Z by assumption. If $t < r$, let $D_1 := (1 - \frac{t}{r})^{-1}D'$. Then $D_1 \geq 0$, $D_1 \sim_{\mathbb{Q}} -K_{X_\lambda}$, and $C \not\subset \operatorname{Supp} D_1$, so (X_λ, sD_1) is lc near Z. Since

$$sD = \frac{t}{r} \cdot srC + \Big(1 - \frac{t}{r}\Big) \cdot sD_1,$$

Lemma 2.4 implies that (X_λ, sD) is lc near Z. □

Lemma 7.11. *Notation as in Set-up 7.1. Then C_a is a prime divisor with $C_a \sim_{\mathbb{Q}} -K_{X_\lambda}$ and $p \notin C_a$, the curve C_a is smooth outside q, and the preimage of C_a in the orbifold chart at q has a node at the origin. Moreover, for any $\mathbb{Q}$-divisor $D \geq 0$ on X_λ such that $D \sim_{\mathbb{Q}} -K_{X_\lambda}$, the pair $(X_\lambda, \frac{2}{3}D)$ is lc near q, and the pair (X_λ, D) is lc near any point of $C_a \backslash \{q\}$.*

Proof. The proof is long, so we divide it into several steps.

Step 1. In this step we describe the curve C_a and the orbifold chart at q.

Since $a(p) \neq 0$, $p \notin C_a$. Let

$$V := (b^2 - \lambda xy = y^2 - x^3 - bc = 0) \subset \mathbb{A}^4_{x,b,y,c}$$

be the affine cone over C_a. On $V \cap (b \neq 0)$, the first equation implies $xy \neq 0$ and $y = b^2/(\lambda x)$, and the second equation then implies $c = (y^2 - x^3)/b$. Thus $V \cap (b \neq 0) \cong (\mathbb{A}^1 \backslash \{0\})^2$ with coordinates x, b. On $V \cap (b = 0)$, we have $\lambda xy = 0$ and $y^2 = x^3$, so $x = y = 0$ and $V \cap (b = 0)$ is the c-axis. Since V is a complete intersection in $\mathbb{A}^4$, any irreducible component of V is a surface and V has no embedded components. Thus V is the closure of $V \cap (b \neq 0)$, and V is reduced and irreducible. Therefore C_a is a prime divisor, and C_a is the divisor of the section a of $\mathcal{O}_{X_\lambda}(3)$ (the generic point of C_a has trivial stabilizer, so by Definition-Lemma 2.20(1)(2) the multiplicity of $(a = 0)$ along C_a can be computed on the affine cone over X_λ). Since $\omega_{X_\lambda} \cong \mathcal{O}_{X_\lambda}(-3)$ (Proposition 7.5), $C_a \sim 3H \sim -K_{X_\lambda}$.

The Jacobian matrix of the equations of V with respect to (x, b, y, c) is

$$\begin{pmatrix} -\lambda y & 2b & -\lambda x & 0 \\ -3x^2 & -c & 2y & -b \end{pmatrix},$$

and the minor in the columns of b and c equals $-2b^2$. Thus V is smooth at any point with $b \neq 0$. At such a point $xb \neq 0$, and since $\gcd(4, 5) = 1$, the point has trivial stabilizer under the $\mathbb{C}^*$-action. Hence C_a is smooth at any point with $b \neq 0$. The only point of C_a with $b = 0$ is q. Thus C_a is smooth outside q. Since $\operatorname{Sing} X_\lambda = \{p, q\}$ (Proposition 7.5) and $p \notin C_a$, any point of $C_a \backslash \{q\}$ is a smooth point of X_λ.

Let $U := X_\lambda \cap (c \neq 0)$. By Proposition 7.5(3), the projection to the coordinates (x, y) induces an isomorphism from $\rho_c^{-1}(U)$ onto $\mathbb{A}^2_{x,y}$, under which μ_7 acts with weights $(4, 6)$ and freely on $\mathbb{A}^2 \backslash \{0\}$, and ρ_c induces an isomorphism $\mathbb{A}^2_{x,y}/\mu_7 \cong U$. Let $g : \mathbb{A}^2 \to U$ be the quotient map. Then $g^{-1}(q) = \{0\}$, and g is étale over $U \backslash \{q\}$. On $\rho_c^{-1}(U)$, where $c = 1$, the equations of X_λ imply $b = y^2 - x^3$ and $a = (y^2 - x^3)^2 - \lambda xy$. The preimage of C_a is

$$C_a^\sharp := g^{-1}(C_a) = \big((y^2 - x^3)^2 - \lambda xy = 0\big) \subset \mathbb{A}^2.$$

The lowest-degree term of $(y^2 - x^3)^2 - \lambda xy$ is $-\lambda xy$, a product of two distinct linear forms since $\lambda \neq 0$. Thus $C_a^\sharp$ has a node at the origin: near the origin, $C_a^\sharp$ is the union of two smooth branches C_1 and C_2 with tangent lines $(x = 0)$ and $(y = 0)$ respectively. In particular, $\operatorname{mult}_0 C_a^\sharp = 2$. Since g is étale over $U \backslash \{q\}$ and C_a is reduced, $C_a^\sharp$ is reduced and $g^* C_a = C_a^\sharp$.

Since X_λ is $\mathbb{Q}$-factorial (Proposition 7.5(4)), $K_U + B$ is $\mathbb{Q}$-Cartier for any $\mathbb{Q}$-divisor B on U. Since g is étale over $U \backslash \{q\}$,

$$K_{\mathbb{A}^2} + g^* B = g^*(K_U + B),$$

so by Lemma 2.8, for $B \geq 0$, (U, B) is lc near q if and only if $(\mathbb{A}^2, g^* B)$ is lc near the origin.

Step 2. In this step we prove that (X_λ, C_a) and $(X_\lambda, \frac{2}{3} C_a)$ are lc near any point of C_a.

Let $X^\circ := X_\lambda \backslash \{p, q\}$. By **Step 1**, X° is smooth and $\operatorname{mult}_w C_a \leq 1$ for any closed point $w \in X^\circ$. Thus $(X^\circ, C_a|_{X^\circ})$ is lc by Lemma 2.7.

Let $\sigma : S \to \mathbb{A}^2$ be the blow-up of the origin, with exceptional curve E_0. Since $\operatorname{mult}_0 C_a^\sharp = 2$,

$$K_S + \sigma_*^{-1} C_a^\sharp + E_0 = \sigma^*\big(K_{\mathbb{A}^2} + C_a^\sharp\big),$$

where the coefficient of E_0 is $2 \cdot 1 - 1 = 1$. The strict transforms of the smooth branches C_1 and C_2 are smooth and intersect E_0 transversally at two distinct points, which correspond to the tangent lines $(x = 0)$ and $(y = 0)$. Thus there exists an open neighborhood W of the origin such that $\sigma_*^{-1} C_a^\sharp + E_0$ is an snc divisor over W with all coefficients equal to 1. By [KM98, Corollary 2.31(3)] and [KM98, Lemma 2.30], $(W, C_a^\sharp|_W)$ is lc. By **Step 1**, (X_λ, C_a) is lc near q.

Since X_λ is klt (Proposition 7.5), $(X_\lambda, 0)$ is lc. By Lemma 2.4 with $B_1 = C_a$, $B_2 = 0$, and $t = \frac{2}{3}$, $(X_\lambda, \frac{2}{3} C_a)$ is lc near any point of C_a.

Step 3. In this step we prove the lemma when $C_a \not\subset \operatorname{Supp} D$.

Let $C \subset X_\lambda$ be a curve with $C \not\subset \operatorname{Supp} D$. Since X_λ is $\mathbb{Q}$-factorial (Proposition 7.5(4)), D is $\mathbb{Q}$-Cartier. Let m be a positive integer such that mD is Cartier. Then $mD|_C$ is a Cartier divisor on C with $mD|_C \geq 0$. For a closed point $z \in C$, let $(D \cdot C)_z$ be $\frac{1}{m}$ times the length at z of $mD|_C$. Then $(D \cdot C)_z \geq 0$ for any z, and

$$\sum_{z \in C} (D \cdot C)_z = (D \cdot C).$$

Moreover, if z is a smooth point of X_λ, then

$$\operatorname{mult}_z D \cdot \operatorname{mult}_z C \leq (D \cdot C)_z$$

by [Ful98, Corollary 12.4], applied to the components of D and extended by linearity.

Assume that $C_a \not\subset \operatorname{Supp} D$. By **Step 1** and Proposition 7.5,

$$(D \cdot C_a) = (3H \cdot 3H) = 9H^2 = \frac{3}{7}.$$

Let $z \in C_a \backslash \{q\}$. By **Step 1**, z is a smooth point of X_λ and $\operatorname{mult}_z C_a = 1$. Thus

$$\operatorname{mult}_z D \le (D \cdot C_a)_z \le (D \cdot C_a) = \frac{3}{7}.$$

Since multiplicity is upper semicontinuous and D has finitely many components, the set $\{w \in X_\lambda : \operatorname{mult}_w D > 1\}$ is closed and does not contain z. Thus $\operatorname{mult}_w D \le 1$ for any closed point w in a smooth open neighborhood of z, and Lemma 2.7 implies that (X_λ, D) is lc near z.

Now consider q. Since g is a finite morphism of degree 7 and $g^* C_a = C_a^\sharp$, the restriction $g|_{C_a^\sharp} : C_a^\sharp \to C_a \cap U$ is finite of degree 7, and $g^*(mD)|_{C_a^\sharp} = (g|_{C_a^\sharp})^*(mD|_{C_a})$. By the projection formula [Ful98, Proposition 2.3(c)], the push-forward of the 0-cycle $(g|_{C_a^\sharp})^*(mD|_{C_a})$ equals 7 times the 0-cycle $mD|_{C_a}$. Since $g^{-1}(q) = \{0\}$, comparing the coefficients of q yields

$$(g^* D \cdot C_a^\sharp)_0 = 7(D \cdot C_a)_q \le 7(D \cdot C_a) = 3.$$

Since $\operatorname{mult}_0 C_a^\sharp = 2$, [Ful98, Corollary 12.4] implies

$$2 \operatorname{mult}_0 g^* D \le (g^* D \cdot C_a^\sharp)_0 \le 3.$$

Thus $\operatorname{mult}_0 \left(\frac{2}{3} g^* D\right) \le 1$. By Lemma 2.7 (applied on a neighborhood of the origin on which the multiplicity of $\frac{2}{3} g^* D$ is at most 1 at any closed point, which exists by upper semicontinuity of multiplicity), $(\mathbb{A}^2, \frac{2}{3} g^* D)$ is lc near the origin. By **Step 1**, $(X_\lambda, \frac{2}{3} D)$ is lc near q.

Step 4. In this step we prove the lemma for arbitrary D.

By **Steps 2** and **3**, and since $C_a \sim_{\mathbb{Q}} -K_{X_\lambda}$ by **Step 1**, the assumptions of Lemma 7.10 hold for $C = C_a$ and $r = 1$, with $s = \frac{2}{3}$ and $Z = \{q\}$, and with $s = 1$ and $Z = \{z\}$ for any $z \in C_a \backslash \{q\}$. Thus Lemma 7.10 implies that $(X_\lambda, \frac{2}{3} D)$ is lc near q and that (X_λ, D) is lc near any point of $C_a \backslash \{q\}$. □

To treat the points outside $C_a \cup \{p\}$, we use as test curves the members $(y - \kappa a^2 = 0) \cap X_\lambda$, $\kappa \in \mathbb{C}$, of the pencil in $|\mathcal{O}_{X_\lambda}(6)|$ spanned by y and a^2. Their preimages in X' (Proposition 7.5(2)) are plane cubic curves.

Lemma 7.12. *Notation as in Set-up 7.1. Then for any $\mathbb{Q}$-divisor $D \ge 0$ on X_λ such that $D \sim_{\mathbb{Q}} -K_{X_\lambda}$, the pair (X_λ, D) is lc near any closed point of $X_\lambda \backslash (C_a \cup \{p\})$.*

Proof. The proof is long, so we divide it into several steps. Let ζ be a primitive cube root of unity.

Step 1. In this step we describe the chart $(a \ne 0)$ and a pencil of curves C_κ on X_λ.

Let $U := X_\lambda \cap (a \ne 0)$ and $X' := (y^2 = x^3 + b^3 - \lambda b x y) \subset \mathbb{A}^3_{x,b,y}$. By Proposition 7.5(2), the projection to the coordinates (x, b, y) induces an isomorphism from $\rho_a^{-1}(U)$ onto X', under which μ_3 acts on X' with weights $(1, 2, 0)$ and freely on $X' \backslash \{0\}$, and ρ_a induces an isomorphism $X'/\mu_3 \cong U$. Let $g : X' \to U$ be the quotient map. Then $g^{-1}(p) = \{0\}$, and g is étale over $U \backslash \{p\}$.

For $\kappa \in \mathbb{C}$, let $C_\kappa := (y - \kappa a^2 = 0) \cap X_\lambda$. If $a = y = 0$ at a point of X_λ, then the equations of X_λ imply $b = 0$ and then $x = 0$. Thus $C_\kappa \backslash U = \{q\}$. On U_a, the section $y - \kappa a^2$ is the function $y - \kappa$, so the projection to the coordinates (x, b) induces an isomorphism from $g^{-1}(C_\kappa) = X' \cap (y = \kappa)$ onto the plane cubic curve

$$(P_\kappa = 0) \subset \mathbb{A}^2_{x,b}, \qquad P_\kappa := x^3 + b^3 - \lambda \kappa x b - \kappa^2,$$

and μ_3 acts on $\mathbb{A}^2_{x,b}$ by $(x, b) \to (\zeta x, \zeta^2 b)$, which preserves P_κ. Moreover, for any closed point $z \in U$ and any $z' \in g^{-1}(z)$, $z \in C_{y(z')}$, and $y(z')$ does not depend on z' since y has weight 0.

We show that for any κ, P_κ is square-free, μ_3 permutes the irreducible components of $(P_\kappa = 0)$ transitively, and any point of $(P_\kappa = 0)$ other than the origin is a smooth point or a node of $(P_\kappa = 0)$.
Case 1: $\kappa = 0$. Then

$$P_0 = x^3 + b^3 = \prod_{i=0}^{2}(x + \zeta^i b).$$

Let $\ell_i := (x + \zeta^i b = 0)$. The lines ℓ_0, ℓ_1, ℓ_2 are distinct and intersect only at the origin. Since $\zeta x + \zeta^{i-1} \cdot \zeta^2 b = \zeta(x + \zeta^i b)$, μ_3 maps ℓ_i onto ℓ_{i-1}, where the indices are taken modulo 3.
Case 2: $\kappa \notin \{0, -\frac{27}{\lambda^3}\}$. Since $\kappa \neq 0$, $P_\kappa(0,0) = -\kappa^2 \neq 0$. At a singular point (x, b) of $(P_\kappa = 0)$,

$$3x^2 = \lambda\kappa b, \qquad 3b^2 = \lambda\kappa x.$$

If $x = 0$ or $b = 0$, then $x = b = 0$, which is impossible since $P_\kappa(0,0) \neq 0$. Thus $xb \neq 0$ and $3x^3 = \lambda\kappa xb = 3b^3$, so $b = \zeta^i x$ for some i. Then $x = \frac{\lambda\kappa\zeta^i}{3}$, $b = \frac{\lambda\kappa\zeta^{2i}}{3}$, and

$$P_\kappa(x, b) = \frac{2(\lambda\kappa)^3}{27} - \frac{(\lambda\kappa)^3}{9} - \kappa^2 = -\kappa^2\Big(1 + \frac{\lambda^3\kappa}{27}\Big) \neq 0.$$

Thus $(P_\kappa = 0)$ is smooth. The closure of $(P_\kappa = 0)$ in $\mathbb{P}^2$ is $(x^3 + b^3 - \lambda\kappa xbv - \kappa^2 v^3 = 0)$, where v is the third homogeneous coordinate, and at a point with $v = 0$ the partial derivatives with respect to x and b are $3x^2$ and $3b^2$, which do not vanish simultaneously. Hence the closure is a smooth plane cubic curve, so it is irreducible by Bézout's theorem. Thus $(P_\kappa = 0)$ is smooth and irreducible, so P_κ is irreducible. In particular, P_κ is square-free and $(P_\kappa = 0)$ has exactly one irreducible component.
Case 3: $\kappa = -\frac{27}{\lambda^3}$. Then $\lambda\kappa = -\frac{27}{\lambda^2}$ and $\kappa^2 = \frac{729}{\lambda^6}$. By the identity

$$x^3 + b^3 + u^3 - 3xbu = \prod_{i=0}^{2}(x + \zeta^i b + \zeta^{2i} u)$$

with $u = -\frac{9}{\lambda^2}$, we obtain

$$P_\kappa = x^3 + b^3 - \frac{729}{\lambda^6} + \frac{27}{\lambda^2}xb = \prod_{i=0}^{2}\Big(x + \zeta^i b - \frac{9}{\lambda^2}\zeta^{2i}\Big).$$

Let $\ell_i := \big(x + \zeta^i b = \frac{9}{\lambda^2}\zeta^{2i}\big)$. The lines ℓ_0, ℓ_1, ℓ_2 have distinct slopes. If a point is contained in $\ell_0 \cap \ell_1 \cap \ell_2$, then summing the equations of ℓ_0, ℓ_1, and ℓ_2 yields $3x = 0$, and then $b = \frac{9}{\lambda^2}\zeta^i$ for $i = 0, 1, 2$, which is impossible. Thus the three pairwise intersection points of the lines ℓ_i are distinct, and $(P_\kappa = 0)$ has only nodes as singularities. Since

$$\zeta x + \zeta^{i-1} \cdot \zeta^2 b = \zeta(x + \zeta^i b) \quad \text{and} \quad \zeta \cdot \frac{9}{\lambda^2}\zeta^{2i} = \frac{9}{\lambda^2}\zeta^{2(i-1)},$$

μ_3 maps ℓ_i onto ℓ_{i-1}.

In all cases, $g(g^{-1}(C_\kappa)) = C_\kappa \cap U$ is irreducible, and $C_\kappa \backslash U = \{q\}$. Since g is étale over $U\backslash\{p\}$ and P_κ is square-free, C_κ is a prime divisor and C_κ is the divisor of the section $y - \kappa a^2$ of $\mathcal{O}_{X_\lambda}(6)$. Thus $C_\kappa \sim 6H$. Moreover, near any point $z \in C_\kappa\backslash\{p, q\}$, C_κ is étale locally isomorphic to $(P_\kappa = 0)$ near a point other than the origin, so $\operatorname{mult}_z C_\kappa \leq 2$.

Step 2. In this step we prove that $(X_\lambda, \frac{1}{2}C_\kappa)$ and (X_λ, D_1) are lc near a point $z \in C_\kappa\backslash(C_a \cup \{p\})$, where $D_1 \geq 0$ is a $\mathbb{Q}$-divisor such that $D_1 \sim_{\mathbb{Q}} -K_{X_\lambda}$ and $C_\kappa \not\subset \operatorname{Supp} D_1$.

Let $z \in X_\lambda\backslash(C_a \cup \{p\})$ be a closed point. Then $a(z) \neq 0$, so $z \in U\backslash\{p\}$, and z is a smooth point of X_λ since $\operatorname{Sing} X_\lambda = \{p, q\}$ (Proposition 7.5) and $q \in C_a$. By **Step 1**, there exists $\kappa \in \mathbb{C}$ such that $z \in C_\kappa$. Let $D_1 \geq 0$ be a $\mathbb{Q}$-divisor such that $D_1 \sim_{\mathbb{Q}} -K_{X_\lambda}$ and $C_\kappa \not\subset \operatorname{Supp} D_1$. Then

$$(D_1 \cdot C_\kappa) = (3H \cdot 6H) = 18H^2 = \frac{6}{7}.$$

Since $\operatorname{mult}_z C_\kappa \geq 1$ and the local intersection numbers $(D_1 \cdot C_\kappa)_w \geq 0$ at the points $w \in C_\kappa$ sum to $(D_1 \cdot C_\kappa)$, [Ful98, Corollary 12.4] implies

$$\operatorname{mult}_z D_1 \leq \operatorname{mult}_z D_1 \cdot \operatorname{mult}_z C_\kappa \leq (D_1 \cdot C_\kappa)_z \leq (D_1 \cdot C_\kappa) = \frac{6}{7}.$$

By Lemma 2.7, applied on a smooth open neighborhood of z on which $\operatorname{mult}_w D_1 \leq 1$ for any closed point w, (X_λ, D_1) is lc near z.

Moreover, by **Step 1**, $\operatorname{mult}_w \left(\frac{1}{2}C_\kappa\right) \leq 1$ for any closed point $w \in X_\lambda \backslash \{p, q\}$. By Lemma 2.7, $(X_\lambda, \frac{1}{2}C_\kappa)$ is lc near z.

Step 3. In this step we prove the lemma for arbitrary D.

Let z and κ be as in **Step 2**. By **Step 1** and Proposition 7.5, $\frac{1}{2}C_\kappa \sim_{\mathbb{Q}} -K_{X_\lambda}$, and by **Step 2**, the assumptions of Lemma 7.10 hold for $C = C_\kappa$, $r = \frac{1}{2}$, $s = 1$, and $Z = \{z\}$. Thus (X_λ, D) is lc near z by Lemma 7.10. □

Proof of Theorem 7.3. By Proposition 7.5, X_λ is a $\mathbb{Q}$-Fano surface. In particular, $(X_\lambda, 0)$ is lc.

Let $D \geq 0$ be a $\mathbb{Q}$-divisor on X_λ such that $D \sim_{\mathbb{Q}} -K_{X_\lambda}$. We show that $(X_\lambda, \frac{7}{12}D)$ is lc near any closed point $z \in X_\lambda$.

- If $z = p$, then $(X_\lambda, \frac{7}{12}D)$ is lc near p by Corollary 7.9.
- If $z = q$, then $(X_\lambda, s_z D)$ is lc near q with $s_z = \frac{2}{3}$ by Lemma 7.11.
- If $z \in C_a \backslash \{q\}$, then $(X_\lambda, s_z D)$ is lc near z with $s_z = 1$ by Lemma 7.11.
- If $z \in X_\lambda \backslash (C_a \cup \{p\})$, then $(X_\lambda, s_z D)$ is lc near z with $s_z = 1$ by Lemma 7.12.

In the last three cases, Lemma 2.4 with $B_1 = s_z D$, $B_2 = 0$, and $t = \frac{7}{12 s_z}$ implies that $(X_\lambda, \frac{7}{12}D)$ is lc near z. Since log canonicity is a local property, $(X_\lambda, \frac{7}{12}D)$ is lc. Since D is arbitrary, $\alpha(X_\lambda) \geq \frac{7}{12}$. By Corollary 7.9, $\alpha(X_\lambda) \leq \frac{7}{12}$. Thus $\alpha(X_\lambda) = \frac{7}{12}$, and $\frac{1}{2} < \frac{7}{12} < \frac{2}{3}$.

Let F be the prime divisor over X_λ in Construction-Proposition 7.6. By Proposition 7.8, $\operatorname{vol}(-K_{X_\lambda}) = \frac{3}{7}$ and

$$\beta_{X_\lambda}(F) = -\frac{1}{14}\operatorname{vol}(-K_{X_\lambda}) = -\frac{3}{98} < 0.$$

By Theorem 2.12, X_λ is not K-semistable. □

8. Proof of the main theorem

In this section, we prove Theorem 1.2.

Proof of Theorem 1.2. Let $d \geq 2$ be an integer. We consider six cases, which cover every integer $d \geq 2$. In each case, we give a $\mathbb{Q}$-Fano variety X of dimension d and a prime divisor E over X with $\frac{d-1}{d} < \alpha(X) < \frac{d}{d+1}$ and $\beta_X(E) < 0$ (a negative multiple of $\operatorname{vol}(-K_X)$ is negative, since $\operatorname{vol}(-K_X) = (-K_X)^d > 0$ for the ample $-K_X$ [Laz04, Corollary 1.4.41]). The values of $\alpha(X)$ and of $\beta_X(E)/\operatorname{vol}(-K_X)$ are also listed in Table 1.

Case 1. $d = 2$. Pick any $\lambda \in \mathbb{C}^*$, and let $X := X_\lambda$ be the surface in Set-up 7.1. By Theorem 7.3, X is a $\mathbb{Q}$-Fano surface with

$$\frac{1}{2} < \alpha(X) = \frac{7}{12} < \frac{2}{3},$$

and there exists a prime divisor E over X (denoted by F in Theorem 7.3) such that

$$\beta_X(E) = -\frac{1}{14}\operatorname{vol}(-K_X) < 0.$$

Case 2. $d = 3$. Let X be the threefold in Set-up 6.2. By Theorem 6.4, X is a $\mathbb{Q}$-Fano threefold with

$$\frac{2}{3} < \alpha(X) = \frac{7}{10} < \frac{3}{4},$$

and there exists a prime divisor E over X such that

$$\beta_X(E) = -\frac{1}{6}\operatorname{vol}(-K_X) < 0.$$

Case 3. $d = 4$. Let X be the fourfold in Proposition 5.1. By Proposition 5.1, X is a $\mathbb{Q}$-Fano fourfold with

$$\frac{3}{4} < \alpha(X) = \frac{15}{19} < \frac{4}{5},$$

and there exists a prime divisor E over X such that

$$\beta_X(E) = -\frac{1}{20}\operatorname{vol}(-K_X) < 0.$$

Case 4. $d \in \{5, 6\}$. Let X be the hypersurface of dimension d in Proposition 5.2. By Proposition 5.2, X is a $\mathbb{Q}$-Fano variety of dimension d with

$$\frac{4}{5} < \alpha(X) = \frac{1734}{2165} < \frac{5}{6} \quad \text{if } d = 5, \qquad \frac{5}{6} < \alpha(X) = \frac{875}{1028} < \frac{6}{7} \quad \text{if } d = 6,$$

and there exists a prime divisor E over X such that

$$\beta_X(E) = -\frac{421}{2586}\operatorname{vol}(-K_X) < 0 \quad \text{if } d = 5, \qquad \beta_X(E) = -\frac{43}{1071}\operatorname{vol}(-K_X) < 0 \quad \text{if } d = 6,$$

where $-\frac{421}{2586}$ and $-\frac{43}{1071}$ are the values of $\frac{k-dc}{c(d+1)}$ for the integers k and c of Proposition 5.2.

Case 5. $d = 7$. Let X be the sevenfold in Proposition 5.3. By Proposition 5.3, X is a $\mathbb{Q}$-Fano variety of dimension 7 with

$$\frac{6}{7} < \alpha(X) = \frac{9240}{10609} < \frac{7}{8},$$

and there exists a prime divisor E over X such that

$$\beta_X(E) = -\frac{343}{10952}\operatorname{vol}(-K_X) < 0.$$

Case 6. $d \geq 8$. Let X be the hypersurface of dimension d in Proposition 5.6. By Proposition 5.6, X is a d-dimensional $\mathbb{Q}$-Fano variety with

$$\frac{d-1}{d} < \alpha(X) < \frac{d}{d+1},$$

and there exists a prime divisor E over X such that $\beta_X(E) < 0$.

Thus, for any integer $d \geq 2$, we obtain a $\mathbb{Q}$-Fano variety X of dimension d and a prime divisor E over X such that $\frac{d-1}{d} < \alpha(X) < \frac{d}{d+1}$ and $\beta_X(E) < 0$. By Theorem 2.12, X is not K-semistable. □

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